
Los Angeles Regional Water Quality Control Board

November 20, 2013

Mr. Daniel E. Pfeifer
Wastewater Utility Manager
City of Ventura
Ventura Water Reclamation Plant
P.O. Box 99
Ventura, CA 93002-0099

Dear Mr. Pfeifer:

ADOPTED WASTE DISCHARGE REQUIREMENTS (WDRs) AND NATIONAL POLLUTANT DISCHARGE ELIMINATION SYSTEM (NPDES) PERMIT – CITY OF VENTURA, VENTURA WATER RECLAMATION FACILITY (NPDES NO. CA0053651, CI NO. 1822)

Our email, dated October 25, 2013, transmitted the revised tentative waste discharge requirements for renewal of your permit to discharge wastes under the NPDES.

Pursuant to Division 7 of the California Water Code, this Regional Water Board at a public hearing held on November 7, 2013, reviewed the revised tentative requirements, considered all the factors in the case, and adopted Order No. R4-2013-0174. Order No. R4-2013-0174 serves as your NPDES permit and expires on December 31, 2018. Section 13376 of the California Water Code requires that an application and Report of Waste Discharge for a new permit must be filed at least 180 days before the expiration date. A copy of the adopted order and attachments are enclosed.

The Order includes modifications listed in a change sheet mailed on November 5, 2013 and those made as an outgrowth of the deliberations at the meeting and read into the record at the public hearing on November 7, 2013.

Change Sheet November 5, 2013.

1. Replace Attachment N-6 with Attachment N-2 and N-3
2. Add Attachment K-6
3. Order: Table 1 Effluent Limitations
Winter Average Monthly Effluent Limit (AMEL) 1.3 mg/L and 98 lbs/day
4. Attachment E: Section VII.C. Table E-6
Add weekly observation of breaching (estuary flow to ocean)
5. Attachment F: Section III.C.1.c.iv.
Replace "on" with "in"
6. Attachment F: Section III.C.8

MARIA MEHRANIAN, CHAIR | SAM UNGER, EXECUTIVE OFFICER

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Move statement "nor does this Order require any such act if authorization cannot be obtained from the appropriate agency as specified by the Acts" to the preceding sentence.

7. Attachment F: Section IV.C.2.g.iii. and following Tables F-11 and F-12b
Revised the water quality based effluent limits (WQBEL): summer AMEL to 2.47 mg/L, summer maximum daily effluent limit (MDEL) to 3.53 mg/L, winter AMEL to 1.47 mg/L, and winter MDEL to 2.05 mg/L in text and tables. Revise the final effluent limit for the winter AMEL to 1.3 mg/L.

Results of Public Hearing Deliberations Nov. 7, 2013.

1. Order: Section II. New subsection E.
This permit is not intended as evidence of intent by the Regional Board to sanction a nuisance for purposes of providing immunity to the Discharger from liability otherwise authorized by law for flooding caused by the discharge.
2. Order: Section VI.C.1.new subsection i. and resulting changes in numbering
This Order may be reopened upon completion of the flood management plan described in VI.C.2.b.vi.
3. Order: Section V. A.
"Receiving water limitations are based on water quality objectives (WQOs) contained in the Basin Plan and are a required part of this Order. Determinations of compliance with receiving water limitations will take into account data collected from the receiving water monitoring locations set forth in Attachment E, as well as data collected from the discharge point monitoring point, location M-001. The discharge shall not cause the following in receiving..."
4. Order: Section VI. C.2.b. new subsection vi.
Flood Management Plan: The Discharger shall work with the Regional Board to convene a stakeholder group of all interested parties, including resource agencies, to recommend interim management solutions to the flooding at McGrath State Beach. This recommendation shall be presented to the Regional Board at a regularly convened meeting by December 31, 2014.
5. Order: Section VI. C.2. b. ii. and various
Rename Nutrient and Toxicity Special Study to Nutrient, Dissolved Oxygen and Toxicity Special Study and add language:
"The Discharger must perform a special study to identify the cause of nutrient, dissolved oxygen and toxicity impairments in the Estuary. The Dissolved Oxygen Study will include sufficient monitoring, including diurnal monitoring, to determine the suitability of DO levels for the Estuary's aquatic life. If it is determined....."
6. Attachment B
Update and add map provided by Discharger
7. Attachment E: Section VII. D. 1.
Remove the requirement to monitor sediment at receiving water monitoring location M-001A.

The complete adopted Order and attachments will be sent only to the Discharger. However, these documents are available on the Regional Water Board's website for your review. The Regional Water Board's web address is www.waterboards.ca.gov/losangeles/.

If you have any questions, please contact Elizabeth Erickson at (213) 576-6665 or the undersigned at (213) 576-6664.

Sincerely,



Brandt Outwin-Beals, P.E., Chief
Municipal Permitting Unit (NPDES)

Enclosures

cc: See Mailing List

Mailing List

Environmental Protection Agency, Region 9, Permits Branch (WTR-5)
NOAA, National Marine Fisheries Service
Department of Interior, U.S. Fish and Wildlife Service
Nichole Kuenzi, State Water Resources Control Board, Office of Chief Counsel
Department of Fish and Game, Region 5
California State Parks and Recreation
State Coastal Conservancy
Ventura County Planning Commission
Ventura County Department of Environmental Health
Ventura County Department of Public Health
Ventura County Department of Public Works, Flood Control and Drainage
Ventura Regional Sanitation District
Ventura Coast Keeper
Wishtoyo Foundation
Heal the Bay
Environment Now
Los Angeles Waterkeeper
Natural Resources Defense Council
Southern California Coastal Water Research Project

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
LOS ANGELES REGION**

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**ORDER R4-2013-0174
NPDES NO. CA0053651**

**WASTE DISCHARGE REQUIREMENTS
FOR THE CITY OF VENTURA, VENTURA WATER RECLAMATION FACILITY
DISCHARGE TO THE SANTA CLARA RIVER ESTUARY VIA DISCHARGE OUTFALL NO. 001**

The following Discharger is subject to waste discharge requirements (WDRs) set forth in this Order:

Table 1. Discharger Information

Discharger	City of Ventura
Name of Facility	Ventura Water Reclamation Facility
Facility Address	1400 Spinnaker Drive
	Ventura, CA 93001
	Ventura County

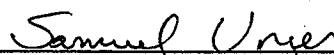
Table 2. Discharge Location

Discharge Point	Effluent Description	Discharge Point Latitude (North)	Discharge Point Longitude (West)	Receiving Water
001	Tertiary-treated wastewater	34 ° , 14' , 21.72" N	119 ° , 15' , 31.33" W	Santa Clara River Estuary via Wildlife Ponds

Table 3. Administrative Information

This Order was adopted on:	November 7, 2013
This Order shall become effective on:	January 1, 2014
This Order shall expire on:	December 31, 2018
The Discharger shall file a Report of Waste Discharge as an application for reissuance of WDRs in accordance with title 23, California Code of Regulations, and an application for reissuance of a National Pollutant Discharge Elimination System (NPDES) permit no later than:	180 days prior to the Order expiration date
The U.S. Environmental Protection Agency (USEPA) and the California Regional Water Quality Control Board, Los Angeles Region have classified this discharge as follows:	Major

I, Samuel Unger, Executive Officer, do hereby certify that this Order with all attachments is a full, true, and correct copy of the Order adopted by the California Regional Water Quality Control Board, Los Angeles Region, on November 7, 2013.



Samuel Unger, P.E., Executive Officer

ADOPTED 11/7/13

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I. FACILITY INFORMATION

Information describing the Ventura Water Reclamation Facility (VWRF or Facility) is summarized in Table 1 and in sections I and II of the Fact Sheet (Attachment F). Section I of the Fact Sheet also includes information regarding the Facility's permit application.

II. FINDINGS

The California Regional Water Quality Control Board, Los Angeles Region (Regional Water Board), finds:

- A. Legal Authorities.** This Order serves as WDRs pursuant to article 4, chapter 4, division 7 of the California Water Code (CWC)(commencing with section 13260). This Order is also issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the USEPA and chapter 5.5, division 7 of the CWC (commencing with section 13370). It shall serve as an NPDES permit for point source discharges from this facility to surface waters.
- B. Background and Rationale for Requirements.** The Regional Water Board developed the requirements in this Order based on information submitted as part of the application, through monitoring and reporting programs, and other available information. The Fact Sheet (Attachment F), which contains background information and rationale for the requirements in this Order, is hereby incorporated into and constitutes Findings for this Order. Attachments A through E and G through N are also incorporated into this Order.
- C. Notification of Interested Parties.** The Regional Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe WDRs for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Details of the notification are provided in the Fact Sheet.
- D. Consideration of Public Comment.** The Regional Water Board, in a public meeting, heard and considered all comments pertaining to the discharge. Details of the Public Hearing are provided in the Fact Sheet.
- E.** This permit is not intended as evidence of intent by the Regional Water Board to sanction a nuisance for purposes of providing immunity to the Discharger from liability otherwise authorized by law for flooding caused by the discharge.

THEREFORE, IT IS HEREBY ORDERED that this Order supersedes Order R4-2008-0011 as amended by Order No. R4-2008-0208, except for enforcement purposes, and, in order to meet the provisions contained in division 7 of the CWC (commencing with section 13000) and regulations adopted thereunder, and the provisions of the CWA and regulations and guidelines adopted thereunder, the Discharger shall comply with the requirements in this Order.

III. DISCHARGE PROHIBITIONS

- A. The discharge from the Facility may not exceed its current annual average of 9 million gallons per day (mgd)¹. This prohibition shall remain in effect pending the results of studies specified in section VI.C.1.i.vi. of this Order.
- B. Discharge of wastewater at a location different from that described in this Order is prohibited.
- C. The bypass or overflow of untreated wastewater or wastes to surface waters or surface water drainage courses is prohibited, except as allowed in Standard Provision I.G. of Attachment D, Standard Provisions.
- D. The maximum daily flow of influent from the collection system to the headworks of the Facility shall not exceed the design capacity of 14 mgd. This prohibition is not applicable during wet weather storm events.
- E. The Discharger shall not cause degradation of any water supply, except as consistent with State Water Board Resolution No. 68-16.
- F. The treatment or disposal of wastes from the Facility shall not cause pollution or nuisance as defined in section 13050, subdivision (l) and (m) of the CWC.
- G. The discharge of any substances in concentrations toxic to animal or plant is prohibited.
- H. The discharge of any radiological, chemical, or biological warfare agent or high level radiological waste is prohibited.

IV. EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

A. Effluent Limitations – Discharge Point Effluent Transfer Station M-001

1. Final Effluent Limitations – Discharge Point Effluent Transfer Station M-001

- j. The Discharger shall maintain compliance with the following effluent limitations at Discharge Point Effluent Transfer Station M-001, with compliance measured at Monitoring Location M-001 as described in the Monitoring and Reporting Program, Attachment E:

Table 4. Effluent Limitations

Parameter	Units	Effluent Limitations ²				
		Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum
Major Wastewater Constituents						
Biochemical Oxygen Demand 5-day @ 20°C	mg/L	20	30	45	--	--
	lbs/day ³	2,340	3,500	5,260	--	--
Total Suspended Solids (TSS)	mg/L	15	40	45	--	--
	lbs/day	1,750	4,670	5,250	--	--
pH	standard units	--	--	--	6.5	8.5
Oil and Grease	mg/L	10	--	15	--	--

¹ Footnote deleted

² Effluent limitations of all parameters, except summer and winter ammonia nitrogen, apply all year around

³ The mass emission rates are based on the flowrate of 14 mgd for TBELs BOD and TSS and capped at 9 mgd, and are calculated as follows: Flow (mgd) x Concentration (mg/L) x 8.34 (L-lbs/mg-mg) = lbs/day. During wet-weather storm events in which the flow exceeds the design capacity, the mass discharge rate limitations shall not apply, and concentration limitations will provide the only applicable effluent limitations.

Parameter	Units	Effluent Limitations ²				
		Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum
Major Wastewater Constituents						
	lbs/day	750	--	1,100	--	--
Settleable Solids	ml/L	0.1	--	0.3	--	--
Total Residual Chlorine	mg/L		--	0.1	--	--
	lbs/day		--	7.5	--	--
MBAS	mg/L	0.5	--	--	--	--
	lbs/day	38	--	--	--	--
Aquatic Life Toxicants						
Summer Total Ammonia ⁴ (May to October)	mg/L	1.07	--	1.17	--	--
	lbs/day	80	--	88	--	--
Winter Total Ammonia (November to April)	mg/L	1.3	--	1.4	--	--
	lbs/day	98	--	105	--	--
Nitrate + Nitrite as Nitrogen	mg/L	10 ⁵	--	--	--	--
	lbs/day	750	--	--	--	--
Nitrite as Nitrogen	mg/L	1	--	--	--	--
	lbs/day	75	--	--	--	--
Nitrate as Nitrogen	mg/L	10	--	--	--	--
	lbs/day	750	--	--	--	--
Copper	µg/L	6.1	--	14	--	--
	lbs/day	0.45	--	1.1	--	--
Selenium	µg/L	2.9	--	8.2	--	--
	lbs/day	0.22	--	0.62	--	--
Lead	µg/L	7.0	--	14	--	--
	lbs/day	0.52	--	1.1	--	--
Nickel	µg/L	7.2	--	18.8	--	--
	lbs/day	0.54	--	1.4	--	--

- k. The average monthly percent removal of BOD 5-day 20°C and total suspended solids (TSS) shall not be less than 85 percent.
- l. The temperature of wastes discharged shall not exceed 86°F, except as a result of external ambient temperature.
- m. Radioactivity of the wastes discharged shall not exceed the limits specified in title 22, chapter 15, article 5, section 64443, of the California Code of Regulations (CCR), or subsequent revisions.

⁴ Total ammonia means sum of unionized ammonia (NH₃) and ionized ammonia (NH₄⁺). The total ammonia nitrogen effluent limitation is the translated effluent limitation based on the water quality objective for ammonia in the current Basin Plan, Chapter 3, which resulted from Resolution 2004-022 *Saltwater Ammonia Objectives for Inland Surface Waters* and 2008-2013 salinity receiving water evidence that the estuary salinity is greater than 1 but less than 10 parts per thousand. Therefore, the applicable objectives are the more stringent of the freshwater or saltwater.

⁵ The 10 mg/L nitrate plus nitrite average monthly concentration is prescribed by the Basin Plan.

- n. The wastes discharged to water courses shall at all times be adequately disinfected. For the purpose of this requirement, the wastes shall be considered adequately disinfected if the median number of total coliform bacteria in the disinfected effluent does not exceed a Most Probable Number (MPN) or Colony Forming Units (CFU) of 2.2 per 100 milliliters, and the number of total coliform bacteria does not exceed an MPN or CFU of 23 per 100 milliliters in more than one sample within any 30-day period. No sample shall exceed an MPN or CFU of 240 total coliform bacteria per 100 milliliters. The median value shall be determined from the bacteriological results of the last 7 days for which an analysis has been completed. Samples shall be collected at a time when wastewater flow and characteristics are most demanding on treatment facilities and disinfection processes.
- o. For the protection of the water contact recreation beneficial use, the wastes discharged to water courses shall have received adequate treatment, so that the turbidity of the wastewater does not exceed any of the following: (a) an average of 2 Nephelometric turbidity units (NTUs) within a 24-hour period; (b) 5 NTUs more than 5 percent of the time (72 minutes) within a 24-hour period; and (c) 10 NTU at any time.
- p. To protect the underlying groundwater basins, pollutants shall not be present in the wastes discharged at concentrations that might degrade groundwater quality.
- q. Acute Toxicity Limitation
 - i. The acute toxicity of the effluent shall be such that:
 - (a) The average survival in the undiluted effluent for any 3 consecutive 96-hour static or continuous flow bioassay tests shall be at least 90 percent, and,
 - (b) No single test produces less than 70 percent survival.
 - ii. If either of the above requirements, is not met, the Discharger shall conduct 6 additional tests, approximately every 2 weeks, over a 12-week period. The Discharger shall ensure that results of a failing acute toxicity test are received by the Discharger within 24 hours of completion of the test and the additional tests shall begin within 5 business days of receipt of the result. If the additional tests indicate compliance with acute toxicity limitation, the Discharger may resume regular testing. However, if the results of any 2 of the 6 accelerated tests are less than 90 percent survival, then the Discharger shall begin a Toxicity Identification Evaluation (TIE). The TIE shall include all reasonable steps to identify the sources of toxicity. Once the sources are identified, the Discharger shall take all reasonable steps to reduce toxicity to meet the objective.
 - iii. If the initial test and any of the additional 6 acute toxicity bioassay tests results are less than 70 percent survival, the Discharger shall immediately implement an Initial Investigation Toxicity Reduction Evaluation (TRE) Work plan.
 - iv. The Discharger shall conduct acute toxicity monitoring as specified in the MRP.
- r. Chronic Toxicity Trigger and Requirements:
 - i. The chronic toxicity of the effluent shall be expressed and reported in toxic units(TU_c), where:

$$TU_c = \frac{100}{NOEC}$$

The No Observable Effect Concentration (NOEC) is expressed as the maximum percent effluent concentration that causes no observable effect on test organisms, as determined by the results of a critical life stage toxicity test.

- ii. There shall be no chronic toxicity in the effluent discharge.
- iii. If the chronic toxicity of the effluent exceeds the monthly trigger median of 1.0 TU_c , the Discharger shall immediately implement accelerated chronic toxicity testing according to MRP section V.B.3. If any 3 out of the initial test and the 6 accelerated tests results exceed 1.0 TU_c , the Discharger shall initiate a TIE and implement the Initial Investigation TRE Work Plan, as specified in MRP section V.D.
- iv. The Discharger shall conduct chronic toxicity monitoring as specified in Attachment E, the Monitoring and Reporting Program (MRP).

B. Recycling Specifications

Ventura Water Reclamation Facility is recycling wastewater under separate Waste Discharge Requirements (WDRs) and Water Recycling Requirements (WRRs) contained in Order No. 87-45, CI number 6190. The Discharger is in the process of planning an expansion of the recycled water system. Once the plans are completed the California Department of Public Health (CDPH) has approved of the expansion, the WDR/WRR will be renewed.

V. RECEIVING WATER LIMITATIONS

A. Surface Water Limitations

Receiving water limitations are based on water quality objectives (WQOs) contained in the Basin Plan and are a required part of this Order. Determinations of compliance with receiving water limitations will take into account data collected from the receiving water monitoring locations set forth in Attachment E, as well as data collected from the discharge point monitoring point, location M-001. The discharge shall not cause the following in receiving water:

1. For waters designated with a warm freshwater habitat (WARM) beneficial use, the temperature of the receiving water at any time or place and within any given 24-hour period shall not be altered by more than 5°F above the natural temperature (or above 70°F if the ambient receiving water temperature is less than 60°F) due to the discharge of effluent at the receiving water station located downstream of the discharge. Natural conditions shall be determined on a case-by-case basis. While the Santa Clara Estuary is not designated WARM, it does support EST (estuarine habitat) and MAR (marine habitat) so protection of aquatic life is necessary.
2. If the receiving water temperature, downstream of the discharge, exceeds 86°F as a result of any of the following, then the exceedance shall not be considered a violation:
 - a. High temperature in the ambient air; or,
 - b. High temperature in the receiving water upstream of the discharge.
3. The pH of bays and estuaries shall not be depressed below 6.5 or raised above 8.5 as a result of wastes discharged. Ambient pH levels shall not be changed more than 0.2 units from natural conditions as a result of wastes discharged. Natural conditions shall be determined on a case-by-case basis.

4. The dissolved oxygen in the receiving water shall not be depressed below 5 mg/L as a result of the wastes discharged.
5. The bacteria concentration in the receiving water shall not exceed the following as a result of wastes discharged:
 - a. State/Regional Water Board Contact Standards
 - i. 30-day Geometric Mean Limits
 - (a) Total coliform density shall not exceed 1,000 per 100 mL.
 - (b) Fecal coliform density shall not exceed 200 per 100 mL.
 - (c) Enterococcus density shall not exceed 35 per 100 mL.
 - ii. Single Sample Limits
 - (a) Total coliform density shall not exceed 10,000 per 100 mL.
 - (b) Fecal coliform density shall not exceed 400 per 100 mL.
 - (c) Enterococcus density shall not exceed 104 per 100 mL.
 - (d) Total coliform density shall not exceed 1,000 per 100 mL, if the ratio of fecal-to-total coliform exceeds 0.1.

These limits are based on Resolution 01-018 *Bacteria Objectives* for REC-1 in marine waters.

- b. Implementation Provisions for Bacterial Characteristics
 - i. At a minimum, weekly samples shall be collected from each site. The geometric mean values should be calculated using the five most recent sample results. If sampling occurs more frequently than weekly, all samples taken during the previous 30-day period shall be used to calculate the geometric mean.
 - ii. If a single sample exceeds any of the single sample median (SSM) standards, repeat sampling at that location shall be conducted to determine the extent and persistence of the exceedance. Repeat sampling shall be conducted within 24 hours of receiving analytical results and continued until the sample result is less than the SSM standard or until a sanitary survey is conducted to determine the source of the high bacterial densities.
 - iii. When repeat sampling is required because of an exceedance of any one single sample density, values from all samples collected during that 30-day period will be used to calculate the geometric mean.
 - iv. It is state policy that the geometric mean bacterial objectives are strongly preferred for use in water body assessment decisions (for example, in developing the CWA section 303(d) list of impaired waters) because the geometric mean objectives are a more reliable measure of long-term water body conditions. In making assessment decisions on bacterial quality, SSM data must be considered together with any available geometric mean data. The use of only SSM bacterial data is generally inappropriate unless there is a limited data set, the water is subject to short-term spikes in bacterial concentrations, or other circumstances justify the use of only SSM data.
 - v. For monitoring stations outside of the defined water-contact recreation zone (REC-1), samples will be analyzed for total coliform only.

6. Waters shall be free of changes in turbidity that cause nuisance or adversely affect beneficial uses. Increases in natural turbidity attributable to controllable water quality factors shall not exceed the following limits, as a result of wastes discharged:
 - a. Where natural turbidity is between 0 and 50 NTU, increases shall not exceed 20 percent; and,
 - b. Where natural turbidity is greater than 50 NTU, increases shall not exceed 10 percent.
7. The wastes discharged shall not produce concentrations of toxic substances in the receiving water that are toxic to or cause adverse physiological responses in human, animal, or aquatic life.
8. The wastes discharged shall not cause concentrations of pollutants to occur at levels that are harmful to human health in waters that are existing or potential sources of drinking water.
9. The concentrations of toxic pollutants in the water column, sediments, or biota shall not adversely affect beneficial uses as a result of the wastes discharged.
10. The wastes discharged shall not contain substances that result in increases in BOD, which adversely affect the beneficial uses of the receiving waters.
11. Waters shall not contain biostimulatory substances in concentrations that promote aquatic growth to the extent that such growth causes nuisance or adversely affects beneficial uses.
12. The wastes discharged shall not cause the receiving waters to contain any substance in concentrations that adversely affect any designated beneficial use.
13. The wastes discharged shall not alter the natural taste, odor, or color of fish, shellfish, or other surface water resources used for human consumption.
14. The wastes discharged shall not result in problems due to breeding of mosquitoes, gnats, black flies, midges, or other pests.
15. The wastes discharged shall not result in visible floating particulates, foams, or oil and grease in the receiving waters.
16. The wastes discharged shall not alter the color of the receiving waters; create a visual contrast with the existing appearance of the water; nor cause aesthetically undesirable discoloration of the receiving waters.
17. The wastes discharged shall not contain any individual pesticide or combination of pesticides in concentrations that adversely affect beneficial uses of the receiving waters. There shall be no increase in pesticide concentrations found in bottom sediments or aquatic life as a result of the wastes discharged.

B. Acute Toxicity Receiving WQO

1. There shall be no acute toxicity in ambient waters as a result of wastes discharged.
2. Receiving water and effluent toxicity testing shall be performed on the same day, as close to concurrently as possible.
3. The acute toxicity of the receiving water at the Stations R-005 and R-003 located upstream and downstream, respectively, of the discharge, shall be such that (i) the average survival in the undiluted receiving water for any 3 consecutive 96-hour static, static-renewal, or continuous flow bioassay tests shall be at least 90 percent, and (ii) no single test producing less than 70 percent survival. Static-renewal bioassay tests may

be used, as allowed by the most current USEPA test method for measuring acute toxicity.

4. If the upstream acute toxicity of the receiving water is greater than the downstream acute toxicity but the effluent acute toxicity is in compliance, the acute toxicity accelerated monitoring in the receiving water specified in MRP section V.A.2. does not apply.

C. Chronic Toxicity Receiving WQO

1. There shall be no chronic toxicity in ambient waters as a result of wastes discharged.
2. Receiving water and effluent toxicity testing shall be performed on the same day, as close to concurrently as possible.
3. If the chronic toxicity in the receiving water at the monitoring station(s) immediately downstream of the discharge exceeds the monthly median of 1.0 TUC trigger in a critical life stage test and the toxicity cannot be attributed to upstream toxicity, as assessed by the Discharger, the Discharger shall immediately implement an accelerated chronic toxicity testing according to MRP section V.B.3. If 2 of the 6 tests exceed a monthly median of 1.0 TUC trigger, the Discharger shall initiate a TIE and implement the Initial Investigation TRE Work Plan.
4. If the chronic toxicity of the receiving water upstream of the discharge is greater than the downstream and the TUC of the effluent chronic toxicity test is less than or equal to a monthly median of 1 TUC trigger, accelerated monitoring need not be implemented.

D. Groundwater Limitations

The discharge shall not cause the underlying groundwater to be degraded, exceed WQOs, unreasonably affect beneficial uses, or cause a condition of pollution or nuisance.

Groundwater limits are not included in this Order. This Order contains a study to better characterize the hydrological connection between the effluent, the Estuary and the groundwater and to identify any existing MUN beneficial uses of the perched and unconfined aquifers of the Oxnard Groundwater Basin in the vicinity of the Estuary. The information will be used by the Regional Water Board to determine if additional groundwater protection is necessary.

VI. PROVISIONS

A. Standard Provisions

1. The Discharger shall comply with all Standard Provisions included in Attachment D.
2. The Discharger shall comply with the following provisions. In the event that there is any conflict, duplication, or overlap between provisions specified by this Order, the more stringent provision shall apply:
 - a. Neither the treatment nor the discharge of pollutants shall create a pollution, contamination, or nuisance as defined by section 13050 of the CWC.
 - b. Odors, vectors, and other nuisances of sewage or sludge origin beyond the limits of the treatment plant site or the sewage collection system due to improper operation of facilities, as determined by the Regional Water Board, are prohibited.
 - c. All facilities used for collection, transport, treatment, or disposal of "wastes" shall be adequately protected against damage resulting from overflow, washout, or inundation from a storm or flood having a recurrence interval of once in 100 years.
 - d. Collection, treatment, and disposal systems shall be operated in a manner that precludes public contact with wastewater.

- e. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer of the Regional Water Board.
- f. The provisions of this Order are severable. If any provision of this Order is found invalid, the remainder of this Order shall not be affected.
- g. Nothing in this Order shall be construed to preclude the institution of any legal action or relieve the discharger from any responsibilities, liabilities or penalties established pursuant to any applicable state law or regulation under authority preserved by section 510 of the CWA.
- h. Nothing in this Order shall be construed to preclude the institution of any legal action or relieve the discharger from any responsibilities, liabilities or penalties to which the discharger is or may be subject to under section 311 of the CWA.
- i. The Discharger must comply with the lawful requirements of municipalities, counties, drainage districts, and other local agencies regarding discharges of storm water to storm drain systems or other water courses under their jurisdiction; including applicable requirements in municipal storm water management program developed to comply with NPDES permits issued by the Regional Water Board to local agencies.
- j. Discharge of wastes to any point other than specifically described in this Order is prohibited, and constitutes a violation thereof.
- k. The Discharger shall comply with all applicable effluent limitations, national standards of performance, toxic effluent standards, and all federal regulations established pursuant to sections 301, 302, 303(d), 304, 306, 307, 316, 403, and 405 of the CWA and amendments thereto.
- l. These requirements do not exempt the operator of the waste disposal facility from compliance with any other laws, regulations, or ordinances which may be applicable; they do not legalize this waste disposal facility, and they leave unaffected any further restraints on the disposal of wastes at this site which may be contained in other statutes or required by other agencies.
- m. Oil or oily material, chemicals, refuse, or other materials which may cause pollution shall not be stored or deposited in areas where they may be picked up by rainfall and carried off of the property and/or discharged to surface waters. Any such spill of such materials shall be contained and removed immediately.
- n. A copy of these waste discharge specifications shall be maintained at the Facility so as to be available at all times to operating personnel.
- o. If there is any storage of hazardous or toxic materials or hydrocarbons at this Facility and if the Facility is not manned at all times, a 24-hour emergency response telephone number shall be prominently posted where it can easily be read from the outside.
- p. The Discharger shall file with the Regional Water Board a ROWD at least 120 days before making any material change or proposed change in the character, location or volume of the discharge.
- q. In the event of any change in name, ownership, or control of these waste disposal facilities, the discharger shall notify the Regional Water Board of such change and shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to the Regional Water Board.

- r. The discharge of any waste resulting from the combustion of toxic or hazardous wastes to any waste stream that ultimately discharges to waters of the United States is prohibited, unless specifically authorized elsewhere in this Order.
- s. The Discharger shall notify the Executive Officer in writing no later than 6 months prior to planned discharge of any chemical, other than the products previously reported to the Executive Officer, which may be toxic to aquatic life. Such notification shall include:
 - i. Name and general composition of the chemical;
 - ii. Frequency of use;
 - iii. Quantities to be used;
 - iv. Proposed discharge concentrations; and,
 - v. USEPA registration number, if applicable.
- t. Violation of any of the provisions of this Order may subject the Discharger to any of the penalties described herein or in Attachment D of this Order, or any combination thereof, at the discretion of the prosecuting authority; except that only one kind of penalty may be applied for each kind of violation.
- u. Failure to comply with provisions or requirements of this Order, or violation of other applicable laws or regulations governing discharges from this facility, may subject the Discharger to administrative or civil liabilities, criminal penalties, and/or other enforcement remedies to ensure compliance. Additionally, certain violations may subject the Discharger to civil or criminal enforcement from appropriate local, state, or federal law enforcement entities.
- v. The CWC provides that any person who violates a waste discharge requirement or a provision of the CWC is subject to civil penalties of up to \$5,000 per day, \$10,000 per day, or \$25,000 per day of violation, or when the violation involves the discharge of pollutants, is subject to civil penalties of up to \$10 per gallon per day or \$25 per gallon per day of violation, or some combination thereof, depending on the violation, or upon the combination of violations.
- w. CWC section 13385(h)(i) requires the Regional Water Board to assess a mandatory minimum penalty of three-thousand dollars (\$3,000) for each serious violation. Pursuant to CWC section 13385(h)(2), a "serious violation" is defined as any waste discharge that violates the effluent limitations contained in the applicable waste discharge requirements for a Group II pollutant by 20 percent or more, or for a Group I pollutant by 40 percent or more. Appendix A of 40 CFR part 123.45 specifies the Group I and II pollutants. Pursuant to CWC section 13385.1(a)(1), a "serious violation" is also defined as "a failure to file a discharge monitoring report required pursuant to section 13383 for each complete period of 30 days following the deadline for submitting the report, if the report is designed to ensure compliance with limitations contained in waste discharge requirements that contain effluent limitations."
- x. CWC section 13385(i) requires the Regional Water Board to assess a mandatory minimum penalty of three-thousand dollars (\$3,000) for each violation whenever a person violates a waste discharge requirement effluent limitation in any period of six consecutive months, except that the requirement to assess the mandatory minimum penalty shall not be applicable to the first three violations within that time period.

- y. Pursuant to CWC section 13385.1(d), for the purposes of section 13385.1 and subdivisions (h), (i), and (j) of section 13385, "effluent limitation" means a numeric restriction or a numerically expressed narrative restriction, on the quantity, discharge rate, concentration, or toxicity units of a pollutant or pollutants that may be discharged from an authorized location. An effluent limitation may be final or interim, and may be expressed as a prohibition. An effluent limitation, for these purposes, does not include a receiving water limitation, a compliance schedule, or a best management practice.
- z. CWC section 13387(e) provides that any person who knowingly makes any false statement, representation, or certification in any record or other document submitted or required to be maintained under this order, including monitoring reports or reports of compliance or noncompliance, or who knowingly falsifies, tampers with, or renders inaccurate any monitoring device or method required to be maintained in this order shall be punished by a fine of not more than twenty-five thousand dollars (\$25,000), imprisonment pursuant to subdivision (h) of Section 1170 of the Penal Code for 16, 20, or 24 months, or by both that fine and imprisonment. For a subsequent conviction, such a person shall be punished by a fine of not more than twenty-five thousand dollars (\$25,000) per day of violation, by imprisonment pursuant to subdivision (h) of Section 1170 of the Penal Code for two, three, or four years, or by both that fine and imprisonment.
- aa. In the event the Discharger does not comply or will be unable to comply for any reason, with any prohibition, effluent limitation, or receiving water limitation of this Order, the Discharger shall notify the Watershed Regulatory Section Chief at the Regional Water Board by telephone (213) 576-6616, electronically at dhung@waterboards.ca.gov, or by fax at (213) 576-6660 within 24 hours of having knowledge of such noncompliance, and shall confirm this notification in writing within five days, unless the Regional Water Board waives confirmation. The written notification shall state the nature, time, duration, and cause of noncompliance, and shall describe the measures being taken to remedy the current noncompliance and prevent recurrence including, where applicable, a schedule of implementation. The written notification shall also be submitted via email with reference to CI-1822 to losangeles@waterboards.ca.gov. Other noncompliance requires written notification as above at the time of the normal monitoring report.
- bb. Prior to making any change in the point of discharge, place of use, or purpose of use of treated wastewater that results in a decrease of flow in any portion of a watercourse, the Discharger must file a petition with the State Water Board, Division of Water Rights, and receive approval for such a change. (CWC section 1211).

B. Monitoring and Reporting Program (MRP) Requirements

The Discharger shall comply with the MRP, and future revisions thereto, in Attachment E.

C. Special Provisions

1. Reopener Provisions

- a. This Order may be reopened for modification, or revocation and reissuance, as a result of the detection of a reportable priority pollutant generated by special conditions included in this Order. These special conditions may be, but are not limited to, fish tissue sampling, whole effluent toxicity, monitoring requirements on internal waste stream(s), and monitoring for surrogate parameters. Additional requirements may be included in this Order as a result of the special condition monitoring data

- b. This Order may be modified, revoked and reissued, or terminated for cause, including, but not limited to, the following:
 - i. Violation of any term or condition contained in this Order;
 - ii. Obtaining this Order by misrepresentation or by failure to disclose fully all relevant facts; or,
 - iii. A change in any condition that requires either a temporary or permanent reduction or elimination of the authorized discharge.

The filing of a request by the Discharger for an Order modification, revocation and issuance, or termination, or a notification of planned changes or anticipated noncompliances does not stay any condition of this Order.

- c. If an applicable toxic effluent standard or prohibition (including any schedule of compliance specified in such effluent standard or prohibition) is promulgated under section 307(a) of the CWA for a toxic pollutant and that standard or prohibition is more stringent than any limitation on the pollutant in this Order, the Regional Water Board may institute proceedings under these regulations to modify or revoke and reissue the Order to conform to the toxic effluent standard or prohibition.
- d. This Order may be reopened and modified to incorporate new limits based on future reasonable potential analyses to be conducted based on on-going monitoring data collected by the Discharger and evaluated by the Regional Water Board.
- e. This Order may be modified, in accordance with the provisions set forth in section 40 of the Code of Federal Regulations (40 CFR) parts 122 and 124 to include requirements for the implementation of the watershed protection management approach.
- f. This Order may be modified, in accordance with the provisions set forth in 40 CFR parts 122 and 124, to include new Minimum Levels (ML).
- g. The Regional Water Board may modify, or revoke and reissue, this Order if present or future investigations demonstrate that the discharge(s) governed by this Order will cause, have the potential to cause, or contribute to adverse impacts on water quality and/or beneficial uses of the receiving waters.
- h. This Order may also be modified, revoked and reissued, or terminated in accordance with the provisions of 40 CFR parts 122.44, 122.62 to 122.64, 125.62, and 125.64. Causes for taking such actions include, but are not limited to, failure to comply with any condition of this Order, endangerment to human health or the environment resulting from the permitted activity, or acquisition of newly obtained information which would have justified the application of different conditions if known at the time of Order adoption. The filing of a request by the Discharger for an Order modification, revocation and issuance, or termination or a notification of planned changes or anticipated noncompliance does not stay any condition of this Order.
- i. This Order may be reopened upon completion of the flood management plan described in VI.C.2.b.vi.
- j. This Order may be reopened and modified, to add or revise effluent limitations as a result of future Basin Plan Amendments, such as an update of a WQO, or the adoption of a total maximum daily load (TMDL).
- k. This Order may be reopened and modified, to revise effluent limitations as a result of the delisting of a pollutant from the 303(d) list.

- l. This Order may be reopened and modified to revise the chronic toxicity effluent limitation, to the extent necessary, to be consistent with State Water Board precedential decisions, new policies, new laws, or new regulations.
- m. This Order may be reopened to modify final effluent limits, if at the conclusion of necessary studies conducted by the Discharger, the Regional Water Board determines that dilution credits, attenuation factors, water effects ratios (WERs), or metal translators are warranted.
- n. This Order may be reopened upon completion of special studies specified in section VI.C.2.b. of this Order.
- o. This Order may be reopened upon completion the Estuary Phase 3 special study specified in section VI.C.2.b.i. of this Order, should groundwater impact be identified and groundwater limits found to be necessary to protect this water of the State, upon review by the Regional Water Board
- p. This Order may be reopened upon completion a special study specified in section VI.C.2.b.i. of this Order if the special study demonstrates that a revision of the 9 mgd discharge limit, is necessary upon review by the Regional Water Board.
- q. This Order may be reopened upon completion of the Nutrient, Dissolved Oxygen and Toxicity Special Study specified in section VI.C.2.b.ii. of this Order or if those studies demonstrates that a revision of the nutrient limits is necessary upon review by the Regional Water Board.
- r. This Order may be reopened to incorporate the requirement to develop a pretreatment program pursuant to 40 CFR 403.8(a) when the Regional Water Board Executive Officer determines that a pretreatment program is necessary due to any new introduction of pollutants into the POTW or any substantial change in the volume or character of pollutants being introduced.

2. Special Studies, Technical Reports and Additional Monitoring Requirements

- a. Constituents of Emerging Concern (CECs) in the Effluent
 - i. The Discharger shall conduct a special study to investigate the CECs in the effluent discharge. Within 90 days of the effective date of this Order, the Discharger shall submit to the Executive Officer a CEC special study work plan for approval. Upon approval, the Discharger shall implement the work plan.
 - ii. The Discharger shall follow the requirements of the work plan, as discussed in the MRP and the Fact Sheet.
- b. Estuary Studies
 - i. Phase 3 Estuary Studies – The Discharger shall perform additional estuary studies to provide sufficient information to allow the Regional Water Board to determine whether or not the continued discharge of effluent enhances the Estuary. The study will clarify the City's water budget analysis for the Santa Clara River Estuary, to determine whether any effluent discharge is needed to sustain the Estuary's native species, and if so how much.
 - ii. Nutrient, Dissolved Oxygen and Toxicity Special Study - The Discharger must perform a special study to identify the cause of nutrient, dissolved oxygen and toxicity impairments in the Estuary. The Dissolved Oxygen Study will include sufficient monitoring, including diurnal monitoring, to determine the suitability of DO levels for the Estuary's aquatic life. If it is determined that the effluent from the Facility is causing the impairments, the Facility must propose a plan for

reducing nutrient loading, including ammonia, nitrogen and phosphorus loading and toxicity impairments.

- iii. Groundwater Special Study – the discharger must perform a special study to document the interaction between the estuary, discharge and groundwater and determine if the beneficial use of MUN applies to the water impacted by the discharge.
- iv. The work plans for the Phase 3 Estuary, Nutrient, Dissolved Oxygen and Toxicity and Groundwater special studies shall be submitted 180 days following permit adoption for approval by the Executive Officer. The scope of the work of the Phase 3 Estuary study must be sufficient to support a staff finding concerning enhancement of the estuary in conformance with the Enclosed Bays and Estuary Policy.
- v. The Phase 3 Estuary, Special Studies must be completed with final reports submitted no later than January 1, 2018. The Nutrient, Dissolved Oxygen and Toxicity, and Groundwater Special Studies must be completed with final reports submitted no later than 180 days before the expiration of this order.
- vi. Flood Management Plan: The Discharger shall work with the Regional Water Board to convene a stakeholder group of all interested parties, including resource agencies, to recommend interim management solutions to the flooding at McGrath State Beach. This recommendation shall be presented to the Regional Water Board at a regularly convened meeting by December 31, 2014.

c. Toxicity Reduction Requirements

The Discharger shall prepare and submit a copy of the Discharger's initial investigation TRE work plan to the Executive Officer of the Regional Water Board for approval within 90 days of the effective date of this permit. If the Executive Officer does not disapprove the work plan within 60 days, the work plan shall become effective. The Discharger shall use USEPA manual EPA/833B-99/002 (municipal), or the most current version, as guidance. At a minimum, the initial investigation TRE work plan must contain the provisions in Attachment G. This work plan shall describe the steps the Discharger intends to follow if toxicity is detected, and should include, at a minimum:

- i. A description of the investigation and evaluation techniques that will be used to identify potential causes and sources of toxicity, effluent variability, and treatment system efficiency;
- ii. A description of the Facility's methods of maximizing in-house treatment efficiency and good housekeeping practices, and a list of all chemicals used in the operation of the Facility; and,
- iii. If a TIE is necessary, an indication of the person who would conduct the TIEs (i.e., an in-house expert or an outside contractor).

If the effluent toxicity test result exceeds the limitation, then the Discharger shall immediately implement accelerated toxicity testing that consists of six additional tests, approximately every two weeks, over a 12-week period. Effluent sampling for the first test of the six additional tests shall commence within five days of receipt of the test results exceeding the toxicity limitation.

If the results of any two of the six tests (any two tests in a 12-week period) exceed the limitation, the Discharger shall initiate a TRE.

If results of the implementation of the Facility's initial investigation TRE work plan (as described above) indicate the need to continue the TRE/TIE, the Discharger shall expeditiously develop a more detailed TRE work plan for submittal to the Executive Officer within 15 days of completion of the initial investigation TRE.

Detailed toxicity testing and reporting requirements are contained in section V of the MRP (Attachment E).

d. Treatment Plant Capacity

The Discharger shall submit a written report to the Executive Officer of the Regional Water Board within 90 days after the "30-day (monthly) average" daily dry-weather flow equals or exceeds 75 percent of the design capacity of waste treatment and/or disposal facilities. The Discharger's senior administrative officer shall sign a letter, which transmits that report and certifies that the discharger's policy-making body is adequately informed of the report's contents. The report shall include the following:

- i. The average daily flow for the month, the date on which the peak flow occurred, the rate of that peak flow, and the total flow for the day;
- ii. The best estimate of when the monthly average daily dry-weather flow rate will equal or exceed the design capacity of the facilities; and,
- iii. A schedule for studies, design, and other steps needed to provide additional capacity for waste treatment and/or disposal facilities before the waste flow rate equals the capacity of present units.

This requirement is applicable to those facilities which have not reached 75 percent of capacity as of the effective date of this Order. For those facilities that have reached 75 percent of capacity by that date but for which no such report has been previously submitted, such a report shall be filed within 90 days of the issuance of this Order.

3. Best Management Practices and Pollution Prevention

a. Storm Water Pollution Prevention Plan (SWPPP)

Within 90 days of the effective date of this Order, the Discharger shall submit an updated SWPPP to the Regional Water Board that describes site-specific management practices for minimizing pollution of storm water runoff and for preventing polluted storm water from being discharged directly to waters of the State. The SWPPP shall be developed in accordance with the requirements in Storm Water Pollution Prevention Plan Requirements (Attachment H). If all storm water is captured and treated on-site and no storm water is discharged or allowed to run off-site from the Facility, the Discharge shall provide certification with descriptions of on-site storm water management to the Regional Water Board.

b. Spill Clean-up Contingency Plan (SCCP)

- i. Within 90 days, the Discharger is required to submit a SCCP, which describes the activities and protocols to address clean-up of spills, overflows, and bypasses of untreated or partially treated wastewater from the Discharger's collection system or treatment facilities that reach water bodies, including dry channels and beach sands. At a minimum, the plan shall include sections on spill clean-up and containment measures, public notification, and monitoring. The Discharger shall review and amend the plan as appropriate after each spill from the Facility or in the service area of the Facility. The Discharger shall

include a discussion in the annual summary report of any modifications to the SCCP and the application of the SCCP to all spills during the year.

- ii. The updated SCCP shall include a conceptual monitoring protocol for spills greater than 10,000 gallons to beach sands (1) to define the extent of waste discharged to beach sands and adjacent surface waters, and (2) to confirm the conclusion and effectiveness of the clean-up and/or mitigation measures. The SCCP shall include a protocol for coordination with the local health department during such an event. This component of the SCCP shall be posted on the Regional Water Board website for stakeholder review and comment for 30 days prior to Executive Officer approval.

c. Pollutant Minimization Program (PMP)

Reporting protocols in the MRP (Attachment E) section IX.B.4 describe sample results that are to be reported as Detected but Not Quantified (DNQ) or Not Detected (ND). Definitions for a reported Minimum Level (ML) and Method Detection Limit (MDL) are provided in Attachment A.

These reporting protocols and definitions are used in determining the need to conduct a PMP as follows:

- i. The Discharger shall develop and conduct a PMP as further described below when there is evidence (e.g., sample results reported as DNQ when the effluent limitation is less than the MDL; sample results from analytical methods more sensitive than those methods required by this Order; presence of whole effluent toxicity; health advisories for fish consumption; or, results of benthic or aquatic organism tissue sampling) that a pollutant is present in the effluent above an effluent limitation and either of the following is true:
 - (a) The concentration of the pollutant is reported as DNQ and the effluent limitation is less than the reported ML; or,
 - (b) The concentration of the pollutant is reported as ND and the effluent limitation is less than the MDL, using definitions described in Attachment A and reporting protocols described in the MRP.
- ii. The goal of the PMP shall be to reduce all potential sources of a pollutant through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The Regional Water Board may consider cost-effectiveness when establishing the requirements of a PMP. The completion and implementation of a Pollution Prevention Plan (PPP), if required pursuant to CWC section 13263.3(d), shall be considered to fulfill the PMP requirements.
- iii. The PMP shall include, but not be limited to, the following actions and submittals acceptable to the Regional Water Board:
 - (a) An annual review and semi-annual monitoring of potential sources of the reportable priority pollutant(s), which may include fish tissue monitoring and other bio-uptake sampling;
 - (b) Quarterly monitoring for the reportable priority pollutant(s) in the influent to the wastewater treatment system;

- (c) Submittal of a control strategy designed to proceed toward the goal of maintaining concentrations of the reportable priority pollutant(s) in the effluent at or below the effluent limitation;
- (d) Implementation of appropriate cost-effective control measures for the reportable priority pollutant(s), consistent with the control strategy; and
- (e) An annual status report that shall be sent to the Regional Water Board including:
 - (1) All PMP monitoring results for the previous year;
 - (2) A list of potential sources of the reportable priority pollutant(s);
 - (3) A summary of all actions undertaken pursuant to the control strategy; and
 - (4) A description of actions to be taken in the following year.

4. Construction, Operation and Maintenance Specifications

- a. Wastewater treatment facilities subject to this Order shall be supervised and operated by persons possessing certificates of appropriate grade pursuant to CCR, title 23, division 3, chapter 26 (CWC sections 13625 – 13633).
- b. The Discharger shall maintain in good working order a sufficient alternate power source for operating the wastewater treatment and disposal facilities. All equipment shall be located to minimize failure due to moisture, liquid spray, flooding, and other physical phenomena. The alternate power source shall be designed to permit inspection and maintenance and shall provide for periodic testing. If such alternate power source is not in existence, the discharger shall halt, reduce, or otherwise control all discharges upon the reduction, loss, or failure of the primary source of power.
- c. The Discharger shall provide standby or emergency power facilities and/or storage capacity or other means so that in the event of plant upset or outage due to power failure or other cause, discharge of raw or inadequately treated sewage does not occur.

5. Special Provisions for Municipal Facilities (POTWs Only)

- a. Sludge Disposal Requirements
 - i. All sludge generated at the wastewater treatment plant must be disposed of, treated, or applied to land in accordance with federal regulations contained in 40 CFR part 503. These requirements are enforceable by USEPA.
 - ii. The Discharger shall ensure compliance with the requirements in State Water Board Order No. 2004-10-DWQ, General WDRs for the Discharge of Biosolids to Land for Use as a Soil Amendment in Agricultural, Silvicultural, Horticultural and Land Reclamation Activities for those sites receiving the Discharger's biosolids which a Regional Water Board has placed under this general order, and with the requirements in individual WDRs issued by a Regional Water Board for sites receiving the Discharger's biosolids.
 - iii. The Discharger shall comply, if applicable, with WDRs issued by other Regional Water Boards to which jurisdiction the biosolids are transported and applied.

- iv. The Discharger shall furnish this Regional Water Board with a copy of any report submitted to USEPA, the State Water Board or other Regional Water Board, with respect to municipal sludge or biosolids.
- b. Pretreatment Requirements
 - i. This Order includes the Discharger's Pretreatment Program as previously submitted to this Regional Water Board. Any change to the program shall be reported to the Regional Water Board in writing and shall not become effective until approved by the Executive Officer in accordance with procedures established in 40 CFR part 403.18.
 - ii. Additional information pertaining to the pretreatment requirements is contained in Attachment P – Pretreatment Reporting Requirements.
- c. Collection System Requirements
 - i. The Discharger's collection system is part of the system that is subject to this Order. As such, the Discharger must properly operate and maintain its collection system (40 CFR § 122.41(e)). The Discharger must report any non-compliance (40 CFR § 122.41(l)(6) and (7)) and mitigate any discharge from the collection system in violation of this Order (40 CFR § 122.41(d)). See the Order at Standard Provision VI.A.2.c and Attachment D, subsections I.D, V.E, V.H, and I.C, and the following section of this order.
- d. Spill Reporting Requirements

Initial Notification: Although State and Regional Water Board staffs do not have duties as first responders, this requirement is an appropriate mechanism to ensure that the agencies that do have first responder duties are notified in a timely manner in order to protect public health and beneficial uses. For certain spills, overflows and bypasses, the Discharger shall make notifications as required below:

- i. In accordance with the requirements of Health and Safety Code section 5411.5, the Discharger shall provide notification to the local health officer or the director of environmental health with jurisdiction over the affected water body of any unauthorized release of sewage or other waste that causes, or probably will cause, a discharge to any waters of the state as soon as possible, but no later than two hours after becoming aware of the release.
- ii. In accordance with the requirements of CWC section 13271, the Discharger shall provide notification to the California Emergency Management Agency (Cal EMA) of the release of reportable amounts of hazardous substances or sewage that causes, or probably will cause, a discharge to any waters of the state as soon as possible, but not later than two hours after becoming aware of the release. The CCR, Title 23, section 2250, defines a reportable amount of sewage as being 1,000 gallons. The phone number for reporting these releases to the Cal EMA is (800) 852-7550.
- iii. The Discharger shall notify the Regional Water Board of any unauthorized release of sewage from its POTW that causes, or probably will cause, a discharge to a water of the state as soon as possible, but not later than two hours after becoming aware of the release. This initial notification does not need to be made if the Discharger has notified Cal EMA and the local health officer or the director of environmental health with jurisdiction over the affected waterbody. The phone number for reporting these releases of sewage to the Regional Water Board is (213) 576-6657. The phone numbers for after hours

and weekend reporting of releases of sewage to the Regional Water Board are (213) 305-2284 and (213) 305-2253.

iv. At a minimum, the following information shall be provided to the Regional Water Board:

- (a) The location, date, and time of the release;
- (b) The water body that received or will receive the discharge;
- (c) An estimate of the amount of sewage or other waste released and the amount that reached a surface water at the time of notification;
- (d) If ongoing, the estimated flow rate of the release at the time of the notification;
- (e) The name, organization, phone number and email address of the reporting representative; and,
- (f) A certification that the State Office of Emergency Services and the local health officer or directors of environmental health with jurisdiction over the affected water bodies have been notified of the discharge.

e. Monitoring

For spills, overflows and bypasses reported under section VI.C.5.d, the Discharger shall monitor as required below:

To define the geographical extent of the spill's impact, the Discharger shall obtain grab samples (if feasible, accessible, and safe) for all spills, overflows or bypasses of any volume that reach any waters of the State (including surface and ground waters). The Discharger shall analyze the samples for total coliform, fecal coliform, enterococci coliform and E. coli (if fecal coliform test shows positive), and relevant pollutants of concern, upstream and downstream of the point of entry of the spill (if feasible, accessible, and safe). This monitoring shall be done on a daily basis from the time the spill is known until the results of two consecutive sets of bacteriological monitoring indicate the return to the background level or the CDPH authorizes cessation of monitoring.

f. Reporting

The initial notification required under section VI.C.5.d shall be followed by:

- i. As soon as possible, but not later than 24 hours after becoming aware of an unauthorized discharge of sewage or other waste from its wastewater treatment plant to a water of the state, the Discharger shall submit a statement to the Regional Water Board by email at aanijielo@waterboards.ca.gov. If the discharge is 1,000 gallons or more, this statement shall certify that Cal EMA has been notified of the discharge in accordance with CWC section 13271. The statement shall also certify that the local health officer or director of environmental health with jurisdiction over the affected water bodies has been notified of the discharge in accordance with Health and Safety Code section 5411.5. The statement shall also include at a minimum the following information:

- (a) Agency, NPDES No., Order No., and MRP CI No.1822, if applicable;
- (b) The location, date, and time of the discharge;
- (c) The water body that received the discharge;

- (d) A description of the level of treatment of the sewage or other waste discharged;
 - (e) An initial estimate of the amount of sewage or other waste released and the amount that reached a surface water;
 - (f) The Cal EMA control number and the date and time that notification of the incident was provided to Cal EMA; and,
 - (g) The name of the local health officer or director of environmental health representative notified (if contacted directly); the date and time of notification; and the method of notification (e.g., phone, fax, email).
 - ii. A written preliminary report five working days after disclosure of the incident is required. Submission to the Regional Water Board of the California Integrated Water Quality System (CIWQS) Sanitary Sewer Overflow (SSO) event number shall satisfy this requirement. Within 30 days after submitting the preliminary report, the Discharger shall submit the final written report to this Regional Water Board. (A copy of the final written report, for a given incident, already submitted pursuant to a statewide General WDRs for Wastewater Collection System Agencies, may be submitted to the Regional Water Board to satisfy this requirement.) The written report shall document the information required, as described below, monitoring results and any other information required in provisions of the Standard Provisions document including corrective measures implemented or proposed to be implemented to prevent/minimize future occurrences. The Executive Officer for just cause can grant an extension for submittal of the final written report.
 - iii. The Discharger shall include a certification in the annual summary report (due according to the schedule in the MRP) that states that the sewer system emergency equipment, including alarm systems, backup pumps, standby power generators, and other critical emergency pump station components were maintained and tested in accordance with the Discharger's preventive maintenance plan. Any deviations from or modifications to the Plan shall be discussed.
- g. Records
- i. The Discharger shall develop and maintain a record of all spills, overflows or bypasses of raw or partially treated sewage from its collection system or treatment plant. This record shall be made available to the Regional Water Board upon request and a spill summary shall be included in the annual summary report. The records shall contain:
 - (a) The date and time of each spill, overflow, or bypass;
 - (b) The location of each spill, overflow, or bypass;
 - (c) The estimated volume of each spill, overflow, and bypass including gross volume, amount recovered and amount not recovered, monitoring results as required by section VI.C.5.d;
 - (d) The cause of each spill, overflow, or bypass;
 - (e) Whether each spill, overflow, or bypass entered a receiving water and, if so, the name of the water body and whether it entered via storm drains or other man-made conveyances;
 - (f) Any mitigation measures implemented;

- (g) Any corrective measures implemented or proposed to be implemented to prevent/minimize future occurrences; and,
- (h) The mandatory information included in SSO online reporting for finalizing and certifying the SSO report for each spill, overflow, or bypass under the SSO WDR.

h. Activities Coordination

In addition, Regional Water Board expects that the POTW's owners/operators will coordinate their compliance activities for consistency and efficiency with other entities that have responsibilities to implement: (i) this NPDES permit, including the Pretreatment Program; (ii) a Municipal Separate Storm Sewer System (MS4) NPDES permit that may contain spill prevention, sewer maintenance, reporting requirements; and, (iii) the Sanitary Sewer Overflow (SSO) WDR.

i. Consistency with SSO WDRs

The CWA prohibits the discharge of pollutants from point sources to surface waters of the United States unless authorized under an NPDES permit. (33 United States Code sections 1311, 1342). The State Water Board adopted General Waste Discharge Requirements for Sanitary Sewer Systems, (WQ Order No. 2006-0003-DWQ; SSO WDR) on May 2, 2006, to provide a consistent, statewide regulatory approach to address sanitary sewer overflows. The SSO WDR requires public agencies that own or operate sanitary sewer systems to apply for coverage under the SSO WDR, develop and implement sewer system management plans, and report all SSO to the State Water Board's online SSOs database. Regardless of the coverage obtained under the SSO WDR, the Discharger's collection system is part of the POTW that is subject to this NPDES permit. As such, pursuant to federal regulations, the Discharger must properly operate and maintain its collection system (40 CFR part 122.41 (e)), report any non-compliance (40 CFR part 122.41(1)(6) and (7)), and mitigate any discharge from the collection system in violation of this NPDES permit (40 CFR part 122.41(d)).

The requirements contained in this Order in sections VI.C.3.b (SCCP Plan section), VI.C.4 (Construction, Operation and Maintenance Specifications section), and VI.C.5.d. (Spill Reporting Requirements section) are intended to be consistent with the requirements of the SSO WDR. The Regional Water Board recognizes that there may be some overlap between these NPDES permit provisions and SSO WDR requirements, related to the collection systems. The requirements of the SSO WDR are considered the minimum thresholds (see finding 11 of State Water Board Order No. 2006-0003-DWQ). To encourage efficiency, the Regional Water Board will accept the documentation prepared by the Permittees under the SSO WDR for compliance purposes as satisfying the requirements in sections VI.C.3.b, VI.C.4, and VI.C.5 provided the more stringent provisions contained in this NPDES permit are also addressed. Pursuant to SSO WDR, section D, provision 2(iii) and (iv), the provisions of this NPDES permit supersede the SSO WDR, for all purposes, including enforcement, to the extent the requirements may be deemed duplicative.

VII. COMPLIANCE DETERMINATION

Compliance with the effluent limitations contained in section IV of this Order will be determined as specified below:

A. General

Compliance with effluent limitations for priority pollutants shall be determined using sample reporting protocols defined in the MRP and Attachment A of this Order. For purposes of reporting and administrative enforcement by the Regional and State Water Boards, the Discharger shall be deemed out of compliance with effluent limitations if the concentration of the priority pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the RL.

B. Multiple Sample Data

When determining compliance with an average monthly effluent limitation (AMEL), for priority pollutants and more than one sample result is available, the Discharger shall compute the arithmetic mean unless the data set contains one or more reported determinations of DNQ or ND. In those cases, the Discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:

1. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
2. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.

C. Average Monthly Effluent Limitation (AMEL)

If the average (or when applicable, the median determined by subsection B above for multiple sample data) of daily discharges over a calendar month exceeds the AMEL for a given parameter, this will represent a single violation, though the Discharger may be considered out of compliance for each day of that month for that parameter (e.g., resulting in 31 days of non-compliance in a 31-day month). If only a single sample is taken during the calendar month and the analytical result for that sample exceeds the AMEL, the Discharger will be considered out of compliance for that calendar month. The Discharger will only be considered out of compliance for days when the discharge occurs. For any 1 calendar month during which no sample (daily discharge) is taken, no compliance determination can be made for that calendar month.

If the analytical result of a single sample, monitored monthly, quarterly, semiannually, or annually, does not exceed the AMEL for a given parameter, the Discharger will have demonstrated compliance with the AMEL for each day of that month for that parameter.

If the analytical result of any single sample, monitored monthly, quarterly, semiannually, or annually, exceeds the AMEL for any parameter, the Discharger shall collect up to 4 additional samples within the same calendar month. All analytical results shall be reported in the monitoring report for that month. The concentration of pollutant (an arithmetic mean or a median) in these samples estimated from the "Multiple Sample Data Reduction" section above, will be used for compliance determination.

In the event of noncompliance with an AMEL, the sampling frequency for that parameter shall be increased to weekly and shall continue at this level until compliance with the AMEL has been demonstrated.

D. Average Weekly Effluent Limitation (AWEL)

If the average (or when applicable, the median determined by subsection B above for multiple sample data) of daily discharges over a calendar week exceeds the AWEL for a given parameter, this will represent a single violation, though the Discharger may be considered out

of compliance for each day of that week for that parameter, resulting in 7 days of non-compliance. If only a single sample is taken during the calendar week and the analytical result for that sample exceeds the AWEL, the Discharger will be considered out of compliance for that calendar week. The Discharger will only be considered out of compliance for days when the discharge occurs. For any 1 calendar week during which no sample (daily discharge) is taken, no compliance determination can be made for that calendar week.

A calendar week will begin on Sunday and end on Saturday. Partial calendar weeks at the end of calendar month will be carried forward to the next month in order to calculate and report a consecutive seven-day average value on Saturday.

E. Maximum Daily Effluent Limitation (MDEL)

If a daily discharge exceeds the MDEL for a given parameter, an alleged violation will be flagged and the discharger will be considered out of compliance for that parameter for that day only within the reporting period. For any day during which no sample is taken, no compliance determination can be made for that day with respect to the MDEL.

F. Instantaneous Minimum Effluent Limitation

If the analytical result of a single grab sample is lower than the instantaneous minimum effluent limitation for a parameter, the Discharger may be considered out of compliance for that parameter for that single sample. Non-compliance for each sample will be considered separately (e.g., the results of 2 grab samples taken within a calendar day that both are lower than the instantaneous minimum effluent limitation would result in 2 instances of non-compliance with the instantaneous minimum effluent limitation).

G. Instantaneous Maximum Effluent Limitation

If the analytical result of a single grab sample is higher than the instantaneous maximum effluent limitation for a parameter, the Discharger may be considered out of compliance for that parameter for that single sample. Non-compliance for each sample will be considered separately (e.g., the results of 2 grab samples taken within a calendar day that both exceed the instantaneous maximum effluent limitation would result in 2 instances of non-compliance with the instantaneous maximum effluent limitation).

H. Six-month Median Effluent Limitation

If the median of daily discharges over any 180-day period exceeds the six-month median effluent limitation for a given parameter, an alleged violation will be flagged and the discharger will be considered out of compliance for each day of that 180-day period for that parameter. The next assessment of compliance will occur after the next sample is taken. If only a single sample is taken during a given 180-day period and the analytical result for that sample exceeds the six-month median, the discharger will be considered out of compliance for the 180-day period. For any 180-period during which no sample is taken, no compliance determination can be made for the six-month median effluent limitation.

I. Percent Removal

The average monthly percent removal is the removal efficiency expressed in percentage across a treatment plant for a given pollutant parameter, as determined from the 30-day average values of pollutant concentrations (C in mg/L) of influent and effluent samples collected at about the same time using the following equation:

$$\text{Percent Removal (percent)} = [1 - (C_{\text{Effluent}}/C_{\text{Influent}})] \times 100 \text{ percent}$$

When preferred, the Discharger may substitute mass loadings and mass emissions for the concentrations.

J. Mass and Concentration Limitations

Compliance with mass and concentration effluent limitations for the same parameter shall be determined separately with their respective limitations. When the concentration of a constituent in an effluent sample is determined to be ND or DNQ, the corresponding mass emission rate determined from that sample concentration shall also be reported as ND or DNQ.

K. Compliance with Single Constituent Effluent Limitations

Dischargers may be considered out of compliance with the effluent limitation if the concentration of the pollutant (see section B "Multiple Sample Data Reduction" above) in the monitoring sample is greater than the effluent limitation and greater than or equal to the RL.

L. Compliance with Effluent Limitations Expressed as a Sum of Several Constituents

Dischargers may be considered out of compliance with an effluent limitation which applies to the sum of a group of chemicals (e.g., PCBs) if the sum of the individual pollutant concentrations is greater than the effluent limitation. Individual pollutants of the group will be considered to have a concentration of zero if the constituent is reported as ND or DNQ.

M. Mass Emission Rate

The mass emission rate shall be obtained from the following calculation for any calendar day:

$$\text{Mass emission rate (lb/day)} = \frac{8.34}{N} \sum_{i=1}^N Q_i C_i$$

$$\text{Mass emission rate (kg/day)} = \frac{3.79}{N} \sum_{i=1}^N Q_i C_i$$

in which N is the number of samples analyzed in any calendar day. Q_i and C_i are the flow rate (mgd) and the constituent concentration (mg/L), respectively, which are associated with each of the N grab samples, which may be taken in any calendar day. If a composite sample is taken, C_i is the concentration measured in the composite sample and Q_i is the average flow rate occurring during the period over which samples are composited.

The daily concentration of all constituents shall be determined from the flow-weighted average of the same constituents in the combined waste streams as follows:

$$\text{Daily concentration} = \frac{1}{Q_t} \sum_{i=1}^N Q_i C_i$$

in which N is the number of component waste streams. Q_i and C_i are the flow rate (mgd) and the constituent concentration (mg/L), respectively, which are associated with each of the N waste streams. Q_t is the total flow rate of the combined waste streams.

N. Bacterial Standards and Analysis

1. The geometric mean used for determining compliance with bacterial standards is calculated with the following equation:

$$\text{Geometric Mean} = (C_1 \times C_2 \times \dots \times C_n)^{1/n}$$

where n is the number of days samples were collected during the period and C is the concentration of bacteria (MPN/100 mL or CFU/100 mL) found on each day of sampling.

2. For bacterial analyses, sample dilutions should be performed so the expected range of values is bracketed (for example, with multiple tube fermentation method or membrane filtration method, 2 to 16,000 per 100 ml for total and fecal coliform, at a minimum, and 1 to 1000 per 100 ml for enterococcus). The detection methods used for each analysis shall be reported with the results of the analyses.
3. Detection methods used for coliforms (total and fecal) shall be those presented in Table 1A of 40 CFR part 136 (revised May 18, 2012), unless alternate methods have been approved by USEPA pursuant to 40 CFR part 136, or improved methods have been determined by the Executive Officer and/or USEPA.
4. Detection methods used for enterococcus shall be those presented in the USEPA publication EPA 600/4-85/076, Test Methods for Escherichia coli and Enterococci in Water By Membrane Filter Procedure or any improved method determined by the Executive Officer and/or USEPA to be appropriate.

O. Single Operational Upset (SOU)

A SOU that leads to simultaneous violations of more than one pollutant parameter shall be treated as a single violation and limits the Discharger's liability in accordance with the following conditions:

1. A SOU is broadly defined as a single unusual event that temporarily disrupts the usually satisfactory operation of a system in such a way that it results in violation of multiple pollutant parameters.
2. A Discharger may assert SOU to limit liability only for those violations which the Discharger submitted notice of the upset as required in provision V.E.2(b) of Attachment D.
3. For purpose outside of CWC section 13385, subdivisions (h) and (i), determination of compliance and civil liability (including any more specific definition of SOU, the requirements for Dischargers to assert the SOU limitation of liability, and the manner of counting violations) shall be in accordance with USEPA Memorandum Issuance of Guidance Interpreting Single Operational Upset (September 27, 1989).
4. For purpose of CWC section 13385 (h) and (i), determination of compliance and civil liability (including any more specific definition of SOU, the requirements for Dischargers to assert the SOU limitation of liability, and the manner of counting violations) shall be in accordance with CWC section 13385 (f)(2).

ATTACHMENT A – DEFINITIONS

Acute Toxicity

Acute Toxicity (TUa) as expressed in Toxic Units Acute (TUa)

$$TUa = \frac{100}{96\text{-hr LC } 50\%}$$

Lethal Concentration 50% (LC 50): LC 50 (percent waste giving 50% survival of test organisms) shall be determined by static or continuous flow bioassay. When it is not possible to measure the 96-hour LC 50 due to greater than 50 percent survival of the test species in 100 percent waste, the toxicity concentration shall be calculated by the expression:

$$TUa = \frac{\log (100 - S)}{1.7}$$

where: S = percentage survival in 100% waste. If S > 99, TUa shall be reported as zero.

Arithmetic Mean (μ)

Also called the average, is the sum of measured values divided by the number of samples. For ambient water concentrations, the arithmetic mean is calculated as follows:

$$\text{Arithmetic mean} = \mu = \Sigma x / n \quad \text{where: } \Sigma x \text{ is the sum of the measured ambient water concentrations, and } n \text{ is the number of samples.}$$

Average Monthly Effluent Limitation (AMEL)

The highest allowable average of daily discharges over a calendar month, calculated as the sum of all daily discharges measured during a calendar month divided by the number of daily discharges measured during that month.

Average Weekly Effluent Limitation (AWEL)

The highest allowable average of daily discharges over a calendar week (Sunday through Saturday), calculated as the sum of all daily discharges measured during a calendar week divided by the number of daily discharges measured during that week.

Best Management Practice (BMP)

BMPs are schedules of activities, prohibitions of practices, maintenance procedures, and other management practices to prevent or reduce the pollution of waters of the United States. BMPs may include, but are not limited to, treatment requirements, operating procedures, or practices to control plant site runoff, spillage or leaks, sludge or waste disposal, or drainage from raw material storage.

Best Professional Judgment (BPJ)

BPJ is the method used by permit writers to develop technology-based NPDES permit conditions on a case-by-case basis using all reasonably available and relevant data.

Bioaccumulative

Those substances taken up by an organism from its surrounding medium through gill membranes, epithelial tissue, or from food and subsequently concentrated and retained in the body of the organism.

Bioassay

A test used to evaluate the relative potency of a chemical or a mixture of chemicals by comparing its effect on a living organism with the effect of a standard preparation on the same type of organism.

Biochemical Oxygen Demand (BOD)

A measurement of the amount of oxygen utilized by the decomposition of organic material, over a specified time period (usually 5 days) in a wastewater sample; it is used as a measurement of the readily decomposable organic content of a wastewater.

Biosolids

Sewage sludge that is used or disposed through land application, surface disposal, incineration, or disposal in a municipal solid waste landfill.

Bypass

Bypass means that the intentional diversion of waste streams from any portion of a treatment (or pretreatment) facility whose operation is necessary to maintain compliance with the terms and conditions of this order and permit.

Carcinogenic

Pollutants are substances that are known to cause cancer in living organisms.

Chlordane

Shall mean the sum of chlordane-alpha, chlordane-gamma, chlordene-alpha, chlordene-gamma, nonachlor-alpha, nonachlor-gamma, and oxychlordane.

Chlorinated Phenolic Compounds

Chlorinated phenolic compounds shall mean the sum of 2-chlorophenol, 2,4-dichlorophenol, 4-chloro-3-methylphenol, 2,4,6-trichlorophenol, and pentachlorophenol.

Clean Water Act (CWA)

The Clean Water Act is an act passed by the United States congress to control water pollution. It was formerly referred to as the Federal Water Pollution control Act of 1972 or Federal Water Pollution Control Act Amendments of 1972 (Public Law 92-500), 33 United States Code 1251 et seq., as amended by: Public Law 96-483; Public Law 97-117; Public Laws 95-217, 97-117, 97-440, and 100-04.

Chronic Toxicity

This parameter shall be used to measure the acceptability of waters for supporting a healthy marine biota until improved methods are developed to evaluate biological response. Chronic Toxicity (TUc) is expressed as Toxic Units Chronic (TUc)

$$TUc = \frac{100}{NOEL}$$

No Observed Effect Level (NOEL): The NOEL is expressed as the maximum percent effluent or receiving water that causes no observable effect on a test organism, as determined by the result of a critical life stage toxicity test.

Code of Federal Regulation (CFR): A codification of the final rules published daily in the Federal Register. Title 40 of the CFR contains the environmental regulations.

Coefficient of Variation (CV)

CV is a measure of the data variability and is calculated as the estimated standard deviation divided by the arithmetic mean of the observed values.

Composite sample means, for other than flow rate measurement,

- a. A combination of at least eight individual portions obtained at equal time intervals for 24 hours, or the duration of the discharge, whichever is shorter. The volume of each individual portion shall be directly proportional to the discharge flow rate at the time of sampling; or,
- b. A combination of at least eight individual portions of equal volume obtained over a 24-hour period. The time interval will vary such that the volume of wastewater discharged between sampling remains constant.

The compositing period shall equal the specified sampling period, or 24 hours, if no period is specified.

For 24-hour composite samples, if the duration of the discharge is less than 24 hours but greater than 8 hours, at least eight flow-weighted samples shall be obtained during the discharge period and composited. For discharge durations of less than eight hours, individual "grab samples" may be substituted. A grab sample is an individual sample collected in less than 15 minutes.

Conventional Pollutants

Pollutants typical of municipal sewage, and for which municipal secondary treatment plants are typically designed; defined at 40 CFR part 401.16 as BOD, TSS, fecal coliform, bacteria, oil and grease, and pH.

Daily Discharge

Daily Discharge is defined as either: (1) the total mass of the constituent discharged over the calendar day (12:00 am through 11:59 pm) or any 24-hour period that reasonably represents a calendar day for purposes of sampling (as specified in the permit), for a constituent with limitations expressed in units of mass or; (2) the unweighted arithmetic mean measurement of the constituent over the day for a constituent with limitations expressed in other units of measurement (e.g., concentration).

The daily discharge may be determined by the analytical results of a composite sample taken over the course of one day (a calendar day or other 24-hour period defined as a day) or by the arithmetic mean of analytical results from one or more grab samples taken over the course of the day.

For composite sampling, if 1 day is defined as a 24-hour period other than a calendar day, the analytical result for the 24-hour period will be considered as the result for the calendar day in which the 24-hour period ends.

DDT (Dichlorodiphenyltrichloroethane)

Shall mean the sum of 4,4'DDT, 2,4'DDT, 4,4'DDE, 2,4'DDE, 4,4'DDD, and 2,4'DDD.

Degrade

Degradation shall be determined by comparison of the waste field and reference site(s) for characteristic species diversity, population density, contamination, growth anomalies, debility, or supplanting of normal species by undesirable plant and animal species. Degradation occurs if there are significant differences in any of three major biotic groups, namely, demersal fish, benthic invertebrates, or attached algae. Other groups may be evaluated where benthic species are not affected, or are not the only ones affected.

Detected, but Not Quantified (DNQ)

DNQ are those sample results less than the RL, but greater than or equal to the laboratory's MDL. Sample results reported as DNQ are estimated concentrations.

Dilution Credit

Dilution Credit is the amount of dilution granted to a discharge in the calculation of a water quality-based effluent limitation, based on the allowance of a specified mixing zone. It is calculated from the

dilution ratio or determined through conducting a mixing zone study or modeling of the discharge and receiving water.

Effluent Concentration Allowance (ECA)

ECA is a value derived from the water quality criterion/objective, dilution credit, and ambient background concentration that is used, in conjunction with the coefficient of variation for the effluent monitoring data, to calculate a long-term average (LTA) discharge concentration. The ECA has the same meaning as waste load allocation (WLA) as used in USEPA guidance (Technical Support Document For Water Quality-based Toxics Control, March 1991, second printing, EPA/505/2-90-001).

Enclosed Bays

Enclosed Bays means indentations along the coast that enclose an area of oceanic water within distinct headlands or harbor works. Enclosed bays include all bays where the narrowest distance between the headlands or outermost harbor works is less than 75 percent of the greatest dimension of the enclosed portion of the bay. Enclosed bays include, but are not limited to, Humboldt Bay, Bodega Harbor, Tomales Bay, Drake's Estero, San Francisco Bay, Morro Bay, Los Angeles-Long Beach Harbor, Upper and Lower Newport Bay, Mission Bay, and San Diego Bay. Enclosed bays do not include inland surface waters or ocean waters.

Endosulfan

The sum of endosulfan-alpha and -beta and endosulfan sulfate.

Estimated Chemical Concentration

The estimated chemical concentration that results from the confirmed detection of the substance by the analytical method below the ML value.

Estuaries

Estuaries means waters, including coastal lagoons, located at the mouths of streams that serve as areas of mixing for fresh and ocean waters. Coastal lagoons and mouths of streams that are temporarily separated from the ocean by sandbars shall be considered estuaries. Estuarine waters shall be considered to extend from a bay or the open ocean to a point upstream where there is no significant mixing of fresh water and seawater. Estuarine waters included, but are not limited to, the Sacramento-San Joaquin Delta, as defined in CWC section 12220, Suisun Bay, Carquinez Strait downstream to the Carquinez Bridge, and appropriate areas of the Smith, Mad, Eel, Noyo, Russian, Klamath, San Diego, and Otay rivers. Estuaries do not include inland surface waters or ocean waters.

Grab Sample

Grab Sample is defined as any individual sample collected in a short period of time not exceeding 15 minutes. Grab samples shall be collected during normal peak loading conditions for the parameter of interest, which may or may not be during hydraulic peaks. It is used primarily in determining compliance with the maximum daily effluent limitations and the instantaneous maximum effluent limitations.

Halomethanes

The sum of bromoform, bromomethane (methyl bromide) and chloromethane (methyl chloride).

Hazardous Substance means any substance designated under 40 CFR part 116 pursuant to section 311 of the Clean Water Act and/or a hazardous waste, as defined in 40 CFR part 116 pursuant to section 311 of the Clean Water Act and/or a hazardous waste, as defined in 40 CFR part 261.3

HCH

The sum of the alpha, beta, gamma (lindane) and delta isomers of hexachlorocyclohexane.

Inland Surface Waters

All surface waters of the state that do not include the ocean, enclosed bays, or estuaries.

Instantaneous Maximum Effluent Limitation

The highest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous maximum limitation).

Instantaneous Minimum Effluent Limitation

The lowest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous minimum limitation).

Maximum Daily Effluent Limitation (MDEL)

The highest allowable daily discharge of a pollutant, over a calendar day (or 24-hour period). For pollutants with limitations expressed in units of mass, the daily discharge is calculated as the total mass of the pollutant discharged over the day. For pollutants with limitations expressed in other units of measurement, the daily discharge is calculated as the arithmetic mean measurement of the pollutant over the day.

Median

The middle measurement in a set of data. The median of a set of data is found by first arranging the measurements in order of magnitude (either increasing or decreasing order). If the number of measurements (n) is odd, then the median = $X_{(n+1)/2}$. If n is even, then the median = $(X_{n/2} + X_{(n/2)+1})/2$ (i.e., the midpoint between the $n/2$ and $n/2+1$).

Method Detection Limit (MDL)

MDL is the minimum concentration of a substance that can be measured and reported with 99 percent confidence that the analyte concentration is greater than zero, as defined in 40 CFR part 136, Attachment B, revised as of July 3, 1999.

Minimum Level (ML)

ML is the concentration at which the entire analytical system must give a recognizable signal and acceptable calibration point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

Mixing Zone

Mixing Zone is a limited volume of receiving water that is allocated for mixing with a wastewater discharge where water quality criteria can be exceeded without causing adverse effects to the overall water body.

Not Detected (ND)

Sample results which are less than the laboratory's MDL.

Ocean Waters

The territorial marine waters of the State as defined by California law to the extent these waters are outside of enclosed bays, estuaries, and coastal lagoons. Discharges to ocean waters are regulated in accordance with the State Water Board's California Ocean Plan.

PAHs (polynuclear aromatic hydrocarbons)

The sum of acenaphthylene, anthracene, 1,2-benzanthracene, 3,4-benzofluoranthene, benzo[k]fluoranthene, 1,12-benzoperylene, benzo[a]pyrene, chrysene, dibenzo[ah]anthracene, fluorene, indeno[1,2,3-cd]pyrene, phenanthrene and pyrene.

PCBs (polychlorinated biphenyls)

The sum of chlorinated biphenyls whose analytical characteristics resemble those of Aroclor-1016, Aroclor-1221, Aroclor-1232, Aroclor-1242, Aroclor-1248, Aroclor-1254 and Aroclor-1260.

PCB derivatives

At a minimum, PCB congeners whose analytical characteristics resemble those of PCB-18, 28, 37, 44, 49, 52, 66, 70, 74, 77, 81, 87, 99, 101, 105, 110, 114, 118, 119, 123, 126, 128, 138, 149, 151, 153, 156, 157, 158, 167, 168, 169, 170, 177, 180, 183, 187, 189, 194, 201, and 206 shall be individually quantified.

Persistent Pollutants

Persistent pollutants are substances for which degradation or decomposition in the environment is nonexistent or very slow.

Pollutant Minimization Program (PMP)

PMP means waste minimization and pollution prevention actions that include, but are not limited to, product substitution, waste stream recycling, alternative waste management methods, and education of the public and businesses. The goal of the PMP shall be to reduce all potential sources of a priority pollutant(s) through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the water quality-based effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The Regional Water Board may consider cost effectiveness when establishing the requirements of a PMP. The completion and implementation of a Pollution Prevention Plan, if required pursuant to CWC section 13263.3(d), shall be considered to fulfill the PMP requirements.

Pollution Prevention

Pollution Prevention means any action that causes a net reduction in the use or generation of a hazardous substance or other pollutant that is discharged into water and includes, but is not limited to, input change, operational improvement, production process change, and product reformulation (as defined in CWC section 13263.3). Pollution prevention does not include actions that merely shift a pollutant in wastewater from one environmental medium to another environmental medium, unless clear environmental benefits of such an approach are identified to the satisfaction of the State Water Resources Control Board (State Water Board) or Regional Water Board.

Pretreatment: The reduction of the amount of pollutants, the elimination of pollutants, or the alteration of the nature of pollutant properties in wastewater prior to or in lieu of discharging or otherwise introducing such pollutants into a publicly owned treatment works (40 CFR part 403.3(q)).

Priority Pollutants are those constituents referred to in 40 CFR part 401.15; a list of these pollutants is provided as Appendix A to 40 CFR part 423.

Publicly Owned Treatment Works (POTW): A treatment works, as defined by section 212 of the CWA that is owned by the State or municipality. This definition includes any devices and systems used in the storage, treatment, recycling, and reclamation of municipal sewage or industrial wastes of a liquid nature. It also includes sewers, pipes, and other conveyances only if they convey wastewater to a POTW treatment plant (40 CFR part 403.3).

Rare, threatened or Endangered species (RARE) means that one or more rare species utilize all ocean, bays, estuaries, and coastal wetlands for foraging and/or nesting.

Reporting Level (RL)

The RL is the ML (and its associated analytical method) chosen by the Discharger for reporting and compliance determination from the MLs included in this Order, including an additional factor if applicable as discussed herein. The MLs included in this Order correspond to approved analytical methods for reporting a sample result that are selected by the Regional Water Board either from Appendix 4 of the SIP in accordance with section 2.4.2 of the SIP or established in accordance with section 2.4.3 of the SIP. The ML is based on the proper application of method-based analytical procedures for sample preparation and the absence of any matrix interferences. Other factors may be applied to the ML depending on the specific sample preparation steps employed. For example, the treatment typically applied in cases where there are matrix-effects is to dilute the sample or sample aliquot by a factor of ten. In such cases, this additional factor must be applied to the ML in the computation of the RL.

Satellite Collection System

The portion, if any, of a sanitary sewer system owned or operated by a different public agency than the agency that owns and operates the wastewater treatment facility that a sanitary sewer system is tributary to.

Shellfish

Organisms identified by the California Department of Health Services as shellfish for public health purposes (i.e., mussels, clams and oysters).

Significant Difference

Defined as a statistically significant difference in the means of two distributions of sampling results at the 95 percent confidence level.

Sludge

The solids, semi-liquid suspensions of solids, residues, screenings, grit, scum, and precipitates separated from, or created in, wastewater by the unit processes of a treatment system. It also includes, but is not limited to, all supernatant, filtrate, centrate, decantate, and thickener overflow/underflow in the solids handling parts of the wastewater treatment system.

Source of Drinking Water

Any water designated as municipal or domestic supply (MUN) in a Regional Water Board Basin Plan.

Standard Deviation (σ)

Standard Deviation is a measure of variability that is calculated as follows:

$$\sigma = (\sum[(x - \mu)^2]/(n - 1))^{0.5}$$

where:

x is the observed value;

μ is the arithmetic mean of the observed values; and

n is the number of samples.

Toxicity Reduction Evaluation (TRE)

TRE is a study conducted in a step-wise process designed to identify the causative agents of effluent or ambient toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in toxicity. The first steps of the TRE consist of the collection of data relevant to the toxicity, including additional toxicity testing, and an evaluation of facility operations and

maintenance practices, and best management practices. A Toxicity Identification Evaluation (TIE) may be required as part of the TRE, if appropriate. (A TIE is a set of procedures to identify the specific chemical(s) responsible for toxicity. These procedures are performed in three phases (characterization, identification, and confirmation) using aquatic organism toxicity tests.)

Water Effect Ratio (WER)

A criteria adjustment factor accounting for the effect of site-specific water characteristics for certain metals from national and state aquatic life criteria that were originally developed using laboratory toxicity data. These aquatic life criteria for metals are intended to protect the aquatic life in most, but not all, surface waters. In some cases, such a criterion might not adequately protect the aquatic life at a given site. More commonly, though, these criteria are overprotective because most surface waters have greater hardness and often higher pH than the laboratory water which was used in toxicity tests which formed the basis for the standard.

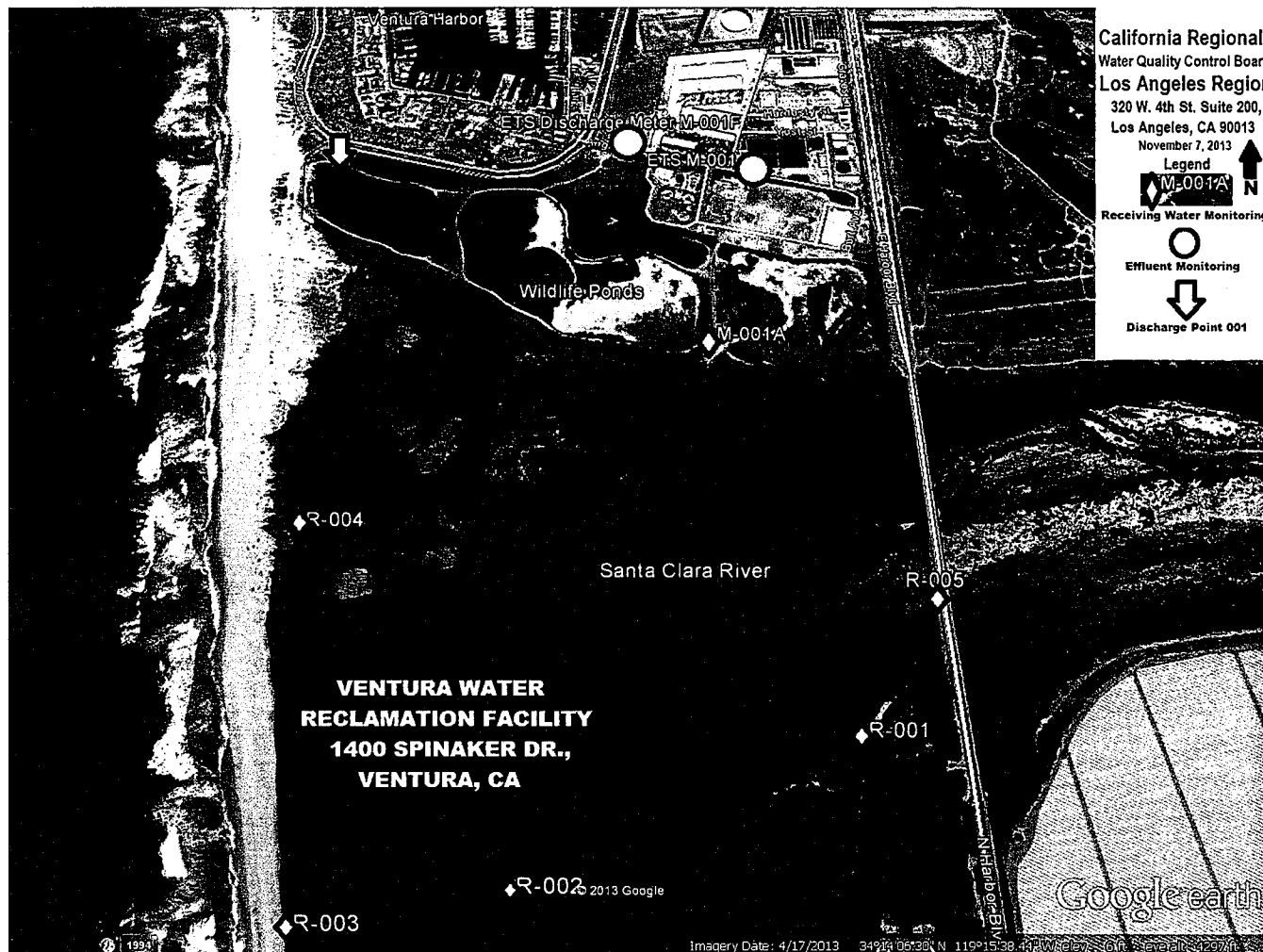
Water Reclamation

The treatment of wastewater to render it suitable for reuse, the transportation of treated wastewater to the place of use, and the actual use of treated wastewater for a direct beneficial use or controlled use that would not otherwise occur.

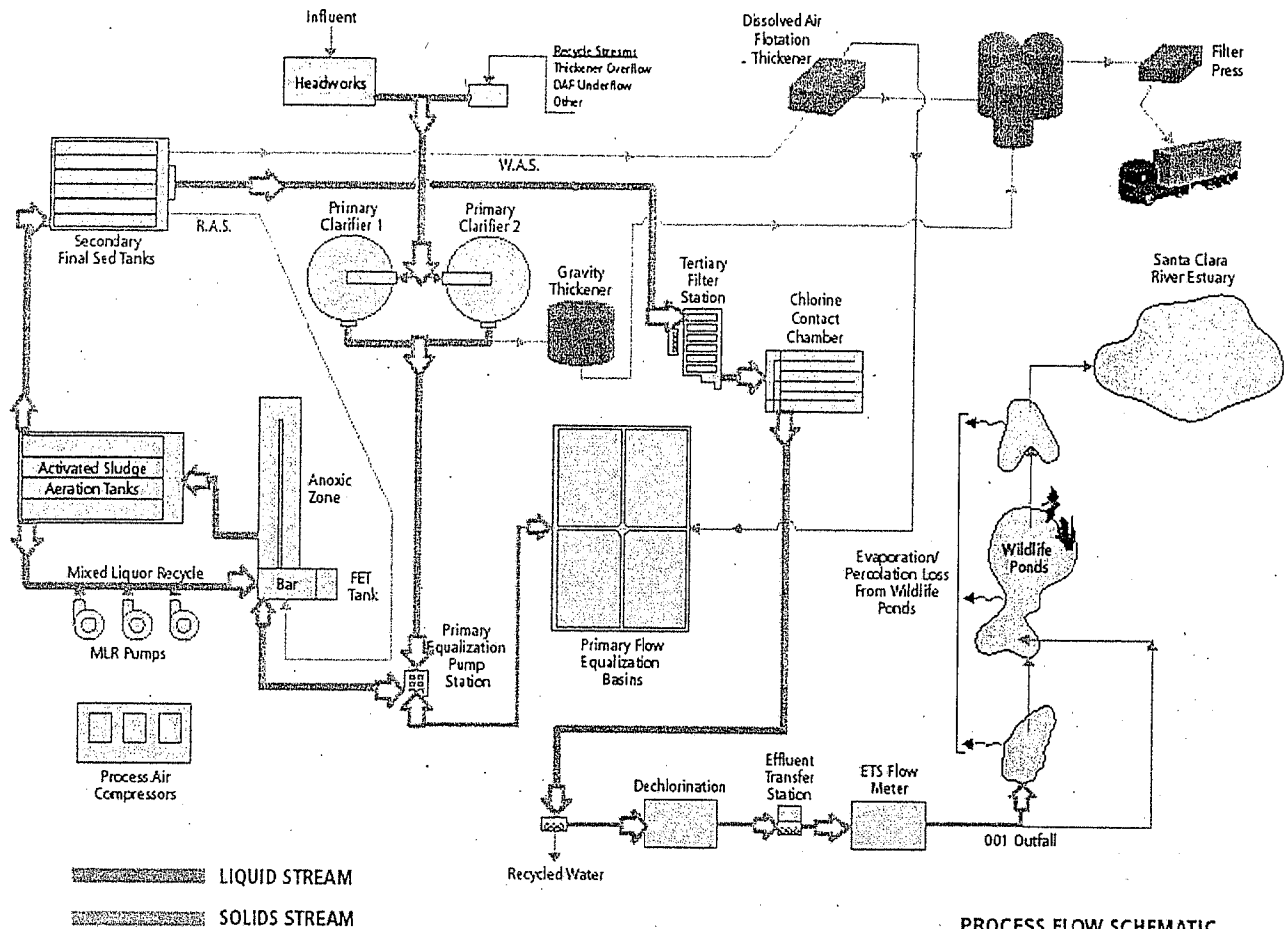
Whole Effluent Toxicity (WET)

The total toxic effect of an effluent measured directly with a toxicity test.

ATTACHMENT B – MAP



ATTACHMENT C – FLOW SCHEMATIC



PROCESS FLOW SCHEMATIC
VENTURA WATER RECLAMATION FACILITY

ATTACHMENT D – STANDARD PROVISIONS

I. STANDARD PROVISIONS – PERMIT COMPLIANCE

A. Duty to Comply

1. The Discharger must comply with all of the terms, requirements, and conditions of this Order. Any noncompliance constitutes a violation of the Clean Water Act (CWA), its regulations, and the California Water Code (CWC) and is grounds for enforcement action, for permit termination, revocation and reissuance, or modification; denial of a permit renewal application; or a combination thereof. (Title 40, Code of Federal Regulations (40 CFR) part 122.41(a); California Water Code (CWC) sections 13261, 13263, 13264, 13265, 13268, 13000, 13001, 13304, 13350, 13385.)
2. The Discharger shall comply with effluent standards or prohibitions established under Section 307(a) of the CWA for toxic pollutants and with standards for sewage sludge use or disposal established under Section 405(d) of the CWA within the time provided in the regulations that establish these standards or prohibitions, even if this Order has not yet been modified to incorporate the requirement. (40 CFR § 122.41(a)(1).)

B. Need to Halt or Reduce Activity Not a Defense

It shall not be a defense for a Discharger in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this Order. (40 CFR § 122.41(c).)

C. Duty to Mitigate

The Discharger shall take all reasonable steps to minimize or prevent any discharge or sludge use or disposal in violation of this Order that has a reasonable likelihood of adversely affecting human health or the environment. (40 CFR § 122.41(d).)

D. Proper Operation and Maintenance

The Discharger shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) which are installed or used by the Discharger to achieve compliance with the conditions of this Order. Proper operation and maintenance also includes adequate laboratory controls and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by a Discharger only when necessary to achieve compliance with the conditions of this Order. (40 CFR § 122.41(e).)

E. Property Rights

1. This Order does not convey any property rights of any sort or any exclusive privileges. (40 CFR § 122.41(g).)
2. The issuance of this Order does not authorize any injury to persons or property or invasion of other private rights, or any infringement of state or local law or regulations. (40 CFR § 122.5(c).)

F. Inspection and Entry

The Discharger shall allow the Regional Water Board, State Water Board, USEPA, and/or their authorized representatives (including an authorized contractor acting as their representative), upon the presentation of credentials and other documents, as may be required by law, , to (33 USC section 1318(a)(4)(B); 40 CFR part 122.41(i); CWC sections 13267 and 13383):

1. Enter upon the Discharger's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this Order (33 USC section 1318(a)(4)(B)(i); 40 CFR part 122.41(i)(1); CWC sections 13267 and 13383);
2. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order (33 USC section 1318(a)(4)(B)(ii); 40 CFR part 122.41(i)(2); CWC sections 13267 and 13383)
3. Inspect and photograph, at reasonable times, any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order (33 USC section 1318(a)(4)(B)(ii); 40 CFR part 122.41(i)(3); CWC sections 13267 and 13383);
4. Sample or monitor, at reasonable times, for the purposes of assuring Order compliance or as otherwise authorized by the CWA or the Water Code, any substances or parameters at any location (33 USC section 1318(a)(4)(B); 40 CFR part 122.41(i)(4); CWC sections 13267 and 13383)

G. Bypass

1. Definitions
 - a. "Bypass" means the intentional diversion of waste streams from any portion of a treatment facility. (40 CFR § 122.41(m)(1)(i).)
 - b. "Severe property damage" means substantial physical damage to property, damage to the treatment facilities, which causes them to become inoperable, or substantial and permanent loss of natural resources that can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production. (40 CFR § 122.41(m)(1)(ii).)
2. Bypass not exceeding limitations. The Discharger may allow any bypass to occur which does not cause exceedances of effluent limitations, but only if it is for essential maintenance to assure efficient operation. These bypasses are not subject to the provisions listed in Standard Provisions – Permit Compliance I.G.3, I.G.4, and I.G.5 below. (40 CFR § 122.41(m)(2).)
3. Prohibition of bypass. Bypass is prohibited, and the Regional Water Board may take enforcement action against a Discharger for bypass, unless (40 CFR § 122.41(m)(4)(i)):
 - a. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage (40 CFR § 122.41(m)(4)(i)(A));
 - b. There were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate back-up equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that occurred during normal periods of equipment downtime or preventive maintenance (40 CFR § 122.41(m)(4)(i)(B)); and
 - c. The Discharger submitted notice to the Regional Water Board as required under Standard Provisions – Permit Compliance I.G.5 below. (40 CFR § 122.41(m)(4)(i)(C).)
4. The Regional Water Board may approve an anticipated bypass, after considering its adverse effects, if the Regional Water Board determines that it will meet the three

conditions listed in Standard Provisions – Permit Compliance I.G.3 above. (40 CFR § 122.41(m)(4)(ii).)

5. Notice

- a. Anticipated bypass. If the Discharger knows in advance of the need for a bypass, it shall submit a notice, if possible at least 10 days before the date of the bypass. (40 CFR § 122.41(m)(3)(i).)
- b. Unanticipated bypass. The Discharger shall submit notice of an unanticipated bypass as required in Standard Provisions - Reporting V.E below (24-hour notice). (40 CFR § 122.41(m)(3)(ii).)

H. Upset

Upset means an exceptional incident in which there is unintentional and temporary noncompliance with technology based permit effluent limitations because of factors beyond the reasonable control of the Discharger. An upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation. (40 CFR § 122.41(n)(1).)

1. Effect of an upset. An upset constitutes an affirmative defense to an action brought for noncompliance with such technology based permit effluent limitations if the requirements of Standard Provisions – Permit Compliance I.H.2 below are met. No determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review. (40 CFR § 122.41(n)(2).)
2. Conditions necessary for a demonstration of upset. A Discharger who wishes to establish the affirmative defense of upset shall demonstrate, through properly signed, contemporaneous operating logs or other relevant evidence that (40 CFR § 122.41(n)(3)):
 - a. An upset occurred and that the Discharger can identify the cause(s) of the upset (40 CFR § 122.41(n)(3)(i));
 - b. The permitted facility was, at the time, being properly operated (40 CFR § 122.41(n)(3)(ii));
 - c. The Discharger submitted notice of the upset as required in Standard Provisions – Reporting V.E.2.b below (24-hour notice) (40 CFR § 122.41(n)(3)(iii)); and
 - d. The Discharger complied with any remedial measures required under Standard Provisions – Permit Compliance I.C above. (40 CFR § 122.41(n)(3)(iv).)
3. Burden of proof. In any enforcement proceeding, the Discharger seeking to establish the occurrence of an upset has the burden of proof. (40 CFR § 122.41(n)(4).)

II. STANDARD PROVISIONS – PERMIT ACTION

A. General

This Order may be modified, revoked and reissued, or terminated for cause. The filing of a request by the Discharger for modification, revocation and reissuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any Order condition. (40 CFR § 122.41(f).)

B. Duty to Reapply

If the Discharger wishes to continue an activity regulated by this Order after the expiration date of this Order, the Discharger must apply for and obtain a new permit. (40 CFR § 122.41(b).)

C. Transfers

This Order is not transferable to any person except after notice to the Los Angeles Regional Water Board. The Los Angeles Regional Water Board may require modification or revocation and reissuance of the Order to change the name of the Discharger and incorporate such other requirements as may be necessary under the CWA and the Water Code. (40 CFR § 122.41(l)(3) and; § 122.61.)

III. STANDARD PROVISIONS – MONITORING

- A. Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity. (40 CFR § 122.41(j)(1).)
- B. Monitoring results must be conducted according to test procedures under 40 CFR part 136 or, in the case of sludge use or disposal, approved under 40 CFR part 136 unless otherwise specified in 40 CFR part 503 unless other test procedures have been specified in this Order. (40 CFR § 122.41(j)(4); § 122.44(i)(1)(iv).)

IV. STANDARD PROVISIONS – RECORDS

- A. Except for records of monitoring information required by this Order related to the Discharger's sewage sludge use and disposal activities, which shall be retained for a period of at least five years (or longer as required by 40 CFR part 503), the Discharger shall retain records of all monitoring information, including all calibration and maintenance records and all original strip chart recordings for continuous monitoring instrumentation, copies of all reports required by this Order, and records of all data used to complete the application for this Order, for a period of at least three (3) years from the date of the sample, measurement, report or application. This period may be extended by request of the Regional Water Board Executive Officer at any time. (40 CFR § 122.41(j)(2).)
- B. Records of monitoring information shall include:
 - 1. The date, exact place, and time of sampling or measurements (40 CFR § 122.41(j)(3)(i));
 - 2. The individual(s) who performed the sampling or measurements (40 CFR § 122.41(j)(3)(ii));
 - 3. The date(s) analyses were performed (40 CFR § 122.41(j)(3)(iii));
 - 4. The individual(s) who performed the analyses (40 CFR § 122.41(j)(3)(iv));
 - 5. The analytical techniques or methods used (40 CFR § 122.41(j)(3)(v)); and
 - 6. The results of such analyses. (40 CFR § 122.41(j)(3)(vi).)
- C. Claims of confidentiality for the following information will be denied (40 CFR § 122.7(b)):
 - 1. The name and address of any permit applicant or Discharger (40 CFR § 122.7(b)(1)); and
 - 2. Permit applications and attachments, permits and effluent data. (40 CFR § 122.7(b)(2).)

V. STANDARD PROVISIONS – REPORTING

A. Duty to Provide Information

The Discharger shall furnish to the Regional Water Board, State Water Board, or USEPA within a reasonable time, any information which the Regional Water Board, State Water Board, or USEPA may request to determine whether cause exists for modifying, revoking and

reissuing, or terminating this Order or to determine compliance with this Order. Upon request, the Discharger shall also furnish to the Regional Water Board, State Water Board, or USEPA copies of records required to be kept by this Order. (40 CFR § 122.41(h); Water Code, § 13267, and 13383)

B. Signatory and Certification Requirements

1. All applications, reports, or information submitted to the Regional Water Board, State Water Board, and/or USEPA shall be signed and certified in accordance with Standard Provisions – Reporting V.B.2, V.B.3, V.B.4, and V.B.5 below. (40 CFR part 122.41(k).)
2. *Signatory requirements for a municipality, State, Federal, or other public agency.* All applications submitted to the Regional Water Board shall be signed by either a principal executive officer or ranking elected official. For purposes of this provision, a principal executive officer of a federal agency includes: (i) the chief executive officer of the agency, or (ii) a senior executive officer having responsibility for the overall operations of a principal geographic unit of the agency (e.g., Regional Administrators of USEPA). (40 CFR part 122.22(a)(3).)
3. All reports required by this Order and other information requested by the Regional Water Board, State Water Board, or USEPA shall be signed by a person described in Standard Provisions – Reporting V.B.2 above, or by a duly authorized representative of that person. A person is a duly authorized representative only if:
 - a. The authorization is made in writing by a person described in Standard Provisions – Reporting V.B.2 above (40 CFR part 122.22(b)(1));
 - b. The authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the company. (A duly authorized representative may thus be either a named individual or any individual occupying a named position.) (40 CFR part 122.22(b)(2)); and
 - c. The written authorization is submitted to the Regional Water Board and State Water Board. (40 CFR part 122.22(b)(3).)
4. If an authorization under Standard Provisions – Reporting V.B.3 above is no longer accurate because a different individual or position has responsibility for the overall operation of the facility, a new authorization satisfying the requirements of Standard Provisions – Reporting V.B.3 above must be submitted to the Regional Water Board and State Water Board prior to or together with any reports, information, or applications, to be signed by an authorized representative. (40 CFR part 122.22(c).)
5. Any person signing a document under Standard Provisions – Reporting V.B.2 or V.B.3 above shall make the following certification:

"I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations." (40 CFR § 122.22(d).)

C. Monitoring Reports

1. Monitoring results shall be reported at the intervals specified in the Monitoring and Reporting Program (Attachment E) in this Order. (40 CFR § 122.41(l)(4).)
2. Monitoring results must be reported on a Discharge Monitoring Report (DMR) form or forms provided or specified by the Regional Water Board or State Water Board for reporting results of monitoring of sludge use or disposal practices. (40 CFR § 122.41(l)(4)(i).)
3. If the Discharger monitors any pollutant more frequently than required by this Order using test procedures approved under 40 CFR part 136, or another method required for an industry-specific waste stream under 40 CFR subchapters N or O, the results of such monitoring shall be included in the calculation and reporting of the data submitted in the DMR or sludge reporting form specified by the Regional Water Board. (40 CFR § 122.41(l)(4)(ii).)
4. Calculations for all limitations, which require averaging of measurements, shall utilize an arithmetic mean unless otherwise specified in this Order. (40 CFR § 122.41(l)(4)(iii).)

D. Compliance Schedules

Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule of this Order, shall be submitted no later than 14 days following each schedule date. (40 CFR § 122.41(l)(5).)

E. Twenty-Four Hour Reporting

1. The Discharger shall report any noncompliance that may endanger health or the environment. Any information shall be provided orally within 24 hours from the time the Discharger becomes aware of the circumstances. A written submission shall also be provided within five (5) days of the time the Discharger becomes aware of the circumstances. The written submission shall contain a description of the noncompliance and its cause; the period of noncompliance, including exact dates and times, and if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent reoccurrence of the noncompliance. (40 CFR § 122.41(l)(6)(i).)
2. The following shall be included as information that must be reported within 24 hours under this paragraph (40 CFR § 122.41(l)(6)(ii)):
 - a. Any unanticipated bypass that exceeds any effluent limitation in this Order. (40 CFR § 122.41(l)(6)(ii)(A).)
 - b. Any upset that exceeds any effluent limitation in this Order. (40 CFR § 122.41(l)(6)(ii)(B).)
3. The Regional Water Board may waive the above-required written report under this provision on a case-by-case basis if an oral report has been received within 24 hours. (40 CFR § 122.41(l)(6)(iii).)

F. Planned Changes

The Discharger shall give notice to the Regional Water Board as soon as possible of any planned physical alterations or additions to the permitted facility. Notice is required under this provision only when (40 CFR § 122.41(l)(1)):

1. The alteration or addition to a permitted facility may meet one of the criteria for determining whether a facility is a new source in section 122.29(b) (40 CFR § 122.41(l)(1)(i)); or

2. The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants that are not subject to effluent limitations in this Order. (40 CFR § 122.41(l)(1)(ii).)
3. The alteration or addition results in a significant change in the Discharger's sludge use or disposal practices, and such alteration, addition, or change may justify the application of permit conditions that are different from or absent in the existing permit, including notification of additional use or disposal sites not reported during the permit application process or not reported pursuant to an approved land application plan. (40 CFR § 122.41(l)(1)(iii).)

G. Anticipated Noncompliance

The Discharger shall give advance notice to the Regional Water Board or State Water Board of any planned changes in the permitted facility or activity that may result in noncompliance with this Order's requirements. (40 CFR § 122.41(l)(2).)

H. Other Noncompliance

The Discharger shall report all instances of noncompliance not reported under Standard Provisions – Reporting V.C, V.D, and V.E above at the time monitoring reports are submitted. The reports shall contain the information listed in Standard Provision – Reporting V.E above. (40 CFR § 122.41(l)(7).)

I. Other Information

When the Discharger becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Regional Water Board, State Water Board, or USEPA, the Discharger shall promptly submit such facts or information. (40 CFR § 122.41(l)(8).)

VI. STANDARD PROVISIONS – ENFORCEMENT

- A. The Regional Water Board is authorized to enforce the terms of this permit under several provisions of the Water Code, including, but not limited to, sections 13268, 13385, 13386, and 13387.
- B. The CWA provides that any person who violates section 301, 302, 306, 307, 308, 318 or 405 of the CWA, or any permit condition or limitation implementing any such sections in a permit issued under section 402, or any requirement imposed in a pretreatment program approved under sections 402(a)(3) or 402(b)(8) of the CWA, is subject to a civil penalty not to exceed \$25,000 per day for each violation. The CWA provides that any person who negligently violates sections 301, 302, 306, 307, 308, 318, or 405 of the CWA, or any condition or limitation implementing any of such sections in a permit issued under section 402 of the CWA, or any requirement imposed in a pretreatment program approved under section 402(a)(3) or 402(b)(8) of the CWA, is subject to criminal penalties of \$2,500 to \$25,000 per day of violation, or imprisonment of not more than one year, or both. In the case of a second or subsequent conviction for a negligent violation, a person shall be subject to criminal penalties of not more than \$50,000 per day of violation, or by imprisonment of not more than two years, or both. Any person who knowingly violates such conditions or limitations is subject to criminal penalties of \$5,000 to \$50,000 per day of violation, or imprisonment for not more than three years, or both. In the case of a second or subsequent conviction for a knowing violation, a person shall be subject to criminal penalties of not more than \$100,000 per day of violation, or imprisonment of not more than 6 years, or both. Any person who knowingly violates section 301, 302, 303, 306, 307, 308, 318 or 405 of the CWA, or any permit condition or limitation implementing any of such sections in a permit issued under section 402 of the CWA, and who knows at that time that he thereby places another person in imminent danger of death or

serious bodily injury, shall, upon conviction, be subject to a fine of not more than \$250,000 or imprisonment of not more than 15 years, or both. In the case of a second or subsequent conviction for a knowing endangerment violation, a person shall be subject to a fine of not more than \$500,000 or by imprisonment of not more than 30 years, or both. An organization, as defined in section 309(c)(3)(B)(iii) of the CWA, shall, upon conviction of violating the imminent danger provision, be subject to a fine of not more than \$1,000,000 and can be fined up to \$2,000,000 for second or subsequent convictions (40 CFR part 122.41(a)(2)); CWC section 13385 and 13387).

- C. Any person may be assessed an administrative penalty by the Administrator of USEPA, the Regional Water Board, or State Water Board for violating section 301, 302, 306, 307, 308, 318 or 405 of this CWA, or any permit condition or limitation implementing any of such sections in a permit issued under section 402 of the CWA. Administrative penalties for Class I violations are not to exceed \$10,000 per violation, with the maximum amount of any Class I penalty assessed not to exceed \$25,000. Penalties for Class II violations are not to exceed \$10,000 per day for each day during which the violation continues, with the maximum amount of any Class II penalty not to exceed \$125,000. (40 CFR part 122.41(a)(3)).
- D. The CWA provides that any person who falsifies, tampers with, or knowingly renders inaccurate any monitoring device or method required to be maintained under this permit shall, upon conviction, be punished by a fine of not more than \$10,000, or by imprisonment for not more than two years, or both. If a conviction of a person is for a violation committed after a first conviction of such person under this paragraph, punishment is a fine of not more than \$20,000 per day of violation, or by imprisonment of not more than four years, or both. (40 CFR part 122.41(j)(5)).
- E. The CWA provides that any person who knowingly makes any false statement, representation, or certification in any record or other document submitted or required to be maintained under this permit, including monitoring reports or reports of compliance or non-compliance shall, upon conviction, be punished by a fine of not more than \$10,000 per violation, or by imprisonment for not more than six months per violation, or by both. (40 CFR part 122.41(k)(2)).

VII. ADDITIONAL PROVISIONS – NOTIFICATION LEVELS

A. Publicly-Owned Treatment Works (POTWs)

All POTWs shall provide adequate notice to the Regional Water Board of the following (40 CFR § 122.42(b)):

1. Any new introduction of pollutants into the POTW from an indirect discharger that would be subject to sections 301 or 306 of the CWA if it were directly discharging those pollutants (40 CFR § 122.42(b)(1)); and
2. Any substantial change in the volume or character of pollutants being introduced into that POTW by a source introducing pollutants into the POTW at the time of adoption of the Order. (40 CFR § 122.42(b)(2)).
3. Adequate notice shall include information on the quality and quantity of effluent introduced into the POTW as well as any anticipated impact of the change on the quantity or quality of effluent to be discharged from the POTW. (40 CFR § 122.42(b)(3)).

ATTACHMENT E – MONITORING AND REPORTING PROGRAM, CI NO. 1822

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ATTACHMENT E – MONITORING AND REPORTING PROGRAM (MRP)

Section 308(a) of the federal Clean Water Act and sections 122.41(h), (j)-(l), 122.44(i), and 122.48 of Title 40 of the Code of Federal Regulations (40 CFR) require that all NPDES permits specify monitoring and reporting requirements. California Water Code (CWC) sections 13267 and 13383 also authorizes the Regional Water Quality Control Board (Regional Water Board) to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. This MRP establishes monitoring, reporting, and recordkeeping requirements that implement the federal and California laws and/or regulations.

I. GENERAL MONITORING PROVISIONS

- A. All samples shall be representative of the waste discharge under conditions of peak load. Quarterly effluent analyses shall be performed during the months of February, May, August, and November. Semiannual analyses shall be performed during the months of February and August. Annual analyses shall be performed during the month of August. Should there be instances when monitoring could not be done during these specified months, the Discharger must notify the Regional Water Board, state the reason why monitoring could not be conducted, and obtain approval from the Executive Officer for an alternate schedule. Results of quarterly, semiannual, and annual analyses shall be reported in the monthly monitoring report following the analysis.
- B. Laboratories analyzing monitoring samples shall be certified by the California Department of Public Health (CDPH), in accordance with the provision of Water Code section 13176, and must include quality assurance/quality control data with their reports.
- C. Pollutants shall be analyzed using the analytical methods described in 40 CFR parts 136.3, 136.4, and 136.5 (revised May 18, 2012), which establish test procedures for the analysis of pollutants; or where no methods are specified for a given pollutant, by methods approved by this Regional Water Board or the State Water Board. Laboratories analyzing effluent samples and receiving water samples shall be certified by the California Department of Public Health (CDPH) Environmental Laboratory Accreditation Program (ELAP) or approved by the Executive Officer and must include quality assurance/quality control (QA/QC) data in their reports. A copy of the laboratory certification shall be provided each time a new certification and/or renewal of the certification is obtained from ELAP.
- D. Water/wastewater samples must be analyzed within allowable holding time limits as specified in 40 CFR part 136.3 (revised May 18, 2012). All QA/QC analyses must be run on the same dates that samples are actually analyzed. The Discharger shall retain the QA/QC documentation in its files and make available for inspection and/or submit them when requested by the Regional Water Board. Proper chain of custody procedures must be followed and a copy of that documentation shall be submitted with the monthly report.
- E. The Discharger shall calibrate and perform maintenance procedures on all monitoring instruments and to ensure accuracy of measurements, or shall ensure that both equipment activities will be conducted.
- F. For any analyses performed for which no procedure is specified in the United States Environmental Protection Agency (USEPA) guidelines, or in the MRP, the constituent or parameter analyzed and the method or procedure used must be specified in the monitoring report.
- G. Each monitoring report must affirm in writing that "all analyses were conducted at a laboratory certified for such analyses by the CDPH or approved by the Executive Officer and in accordance with current USEPA guideline procedures or as specified in this MRP."

- H. The monitoring report shall specify the USEPA analytical method used, the method detection limit (MDL) and the reporting level (RL) [either the minimum level (ML) or the reported minimum level (RML)] for each pollutant. The MLs are those published by the State Water Board in the *Policy for the Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California (SIP)*, February 9, 2005, Appendix 4. The ML represents the lowest quantifiable concentration in a sample based on the proper application of all method-based analytical procedures and the absence of any matrix interference. When all specific analytical steps are followed and after appropriate application of method specific factors, the ML also represents the lowest standard in the calibration curve for that specific analytical technique. When there is deviation from the method analytical procedures, such as dilution or concentration of samples, other factors may be applied to the ML depending on the sample preparation. The resulting value is the reported ML.
- I. The Discharger shall select the analytical method that provides a ML lower than the National Pollutant Discharge Elimination System (NPDES) permit limit established for a given parameter, unless the Discharger can demonstrate that a particular ML is not attainable, in accordance with procedures set forth in 40 CFR part 136, and obtains approval for a higher ML from the Executive Officer, as provided for in section J. below. If the effluent limitation is lower than all the MLs in Appendix 4 of the SIP, the Discharger must select the method with the lowest ML for compliance purposes. The Discharger shall include in the annual summary report a list of the analytical methods employed for each test.
- J. The Discharger shall instruct its laboratories to establish calibration standards so that the ML (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Discharger to use analytical data derived from extrapolation beyond the lowest point of the calibration curve. In accordance with Attachments K, L, M and N, the Discharger's laboratory may employ a calibration standard lower than the ML in Appendix 4 of the SIP.
- K. In accordance with section 2.4.3 of the SIP, the Regional Water Board Executive Officer, in consultation with the State Water Board's Quality Assurance Program Manager, may establish an ML that is not contained in Appendix 4 of the SIP to be included in this Order in any of the following situations:
1. When the pollutant under consideration is not included in Appendix 4 of the SIP;
 2. When the Discharger and the Regional Water Board agree to include in the NPDES permit a test method that is more sensitive than those specified in 40 CFR part 136 (revised as of May 18, 2012);
 3. When the Discharger agrees to use an ML that is lower than those listed in Appendix 4 of the SIP;
 4. When the Discharger demonstrates that the calibration standard matrix is sufficiently different from that used to establish the ML in Appendix 4 of the SIP and proposes an appropriate ML for the matrix; or,
 5. When the Discharger uses a method with quantification practices are not consistent with the definition of the ML. Examples of such methods are USEPA-approved method 1613 for dioxins, method 1624 for volatile organic substances, and method 1625 for semi-volatile organic substances. In such cases, the discharger, the Regional Water Board, and the State Water Board shall agree on a lowest quantifiable limit and that limit will substitute for the ML for reporting and compliance determination purposes.

If there is any conflict between foregoing provisions and the SIP, the provisions stated in the SIP (section 2.4) shall prevail.

- L. If the Discharger samples and performs analyses (other than for process/operational control, startup, research, or equipment testing) on any influent, effluent, or receiving water constituent more frequently than required by this MRP using approved analytical methods, the results of those analyses shall be included in the monitoring reports. These results shall be reflected in the calculation of the average used in demonstrating compliance with average effluent, receiving water, etc., limitations.
- M. The Discharger shall develop and maintain a record of all spills or bypasses of raw or partially treated sewage from its collection system or treatment plant according to the requirements in this Order. This record shall be made available to the Regional Water Board upon request and a spill summary shall be included in the annual summary report.
- N. For all bacteriological analyses, sample dilutions should be performed so the expected range of values is bracketed (for example, with multiple tube fermentation method or membrane filtration method, 2 to 16,000 per 100 ml for total and fecal coliform, at a minimum, and 1 to 1000 per 100 ml for enterococcus). The detection methods used for each analysis shall be reported with the results of the analyses.
 - 1. Detection methods used for coliforms (total and fecal) shall be those presented in Table 1A of 40 CFR part 136 (revised May 18, 2012), unless alternate methods have been approved in advance by the USEPA pursuant to 40 CFR part 136.
 - 2. Detection methods used for enterococcus shall be those presented in Table 1A of 40 CFR part 136 (revised May 18, 2012) or in the USEPA publication EPA 600/4-85/076, *Test Methods for Escherichia coli and Enterococci in Water By Membrane Filter Procedure*, or any improved method determined by the Regional Water Board to be appropriate.

II. MONITORING LOCATIONS

The Discharger shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements in this Order:

Table E-1. Monitoring Station Locations

Discharge Point Name	Monitoring Location Name	Monitoring Location Description
Influent Monitoring Station		
Inf-001	Influent Pump Station	Located in the main stream of the influent channel prior to the headworks adjacent to Harbor Boulevard. Latitude: 34° 14' 25.44" N Longitude: 119° 15' 25.53" W
Effluent Monitoring Station		
M-001	Effluent Transfer Station (All constituents but flow)	Located downstream of all treated effluent passing through this station, including the final disinfection process, and above the wildlife ponds. Latitude: 34° 14' 21.72" N Longitude: 119° 15' 31.33" W
M-001F	Effluent Transfer Station – Flow Meter	Located downstream from M-001A. Latitude: 34° 14' 23.31" N Longitude: 119° 15' 38.66" W
Receiving Water Monitoring Station		
R-001	Receiving Water Sample Station	Located on the southeast shoreline of the Santa Clara River Estuary (Estuary). It may not be accessible when the Estuary is flooded. Latitude: 34° 13' 55.58" N Longitude: 119° 15' 27.59" W
R-002	Receiving Water Sample Station	Located at the south shoreline of the Santa Clara River Estuary. It may not be accessible when the Estuary is flooded. Latitude: 34° 13' 47.37" N Longitude: 119° 15' 43.15" W

Discharge Point Name	Monitoring Location Name	Monitoring Location Description
R-003	Receiving Water Sample Station	Located at the west shoreline of the Santa Clara River Estuary. (variable at the mouth of the outlet, where breaching has most recently occurred ¹)
R-004	Receiving Water Sample Station	Located at the northwest shoreline of the Santa Clara River Estuary, immediately downstream of the discharge point to the Estuary. Latitude: 34 ⁰ , 14', 04.15" N Longitude: 119 ⁰ , 15', 54.19" W
R-005	Receiving Water Sample Station	Located at the Harbor Boulevard Bridge crossing of the Santa Clara River at the Estuary boundary. R-005 was labeled as Station L-5 in the previous permit. This sampling location is for river water flowing into the Estuary, except when the Estuary is flooded. Latitude: 34 ⁰ , 14', 01.63" N Longitude: 119 ⁰ , 15', 23.79" W
M-001A	Receiving Water Sample Station	Located below the wildlife ponds at an existing structure with the approximate location of: Latitude: 34 ⁰ , 14', 11.9868" N Longitude: 119 ⁰ , 15', 32.421" W

The North latitude and West longitude information in Table E-1 are approximate for administrative purposes.

III. INFLUENT MONITORING REQUIREMENTS

Influent monitoring is required to determine compliance with NPDES permit conditions, assess treatment plant performance and assess effectiveness of the Pretreatment Program.

A. Monitoring Location Inf-001

1. The Discharger shall monitor influent to the facility at the Influent Pump Station as follows:

Table E-2. Influent Monitoring

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Flow	mgd	recorder/totalizer	continuous ²	²
pH	pH units	Grab	weekly	³
TSS	mg/L	24-hour composite	weekly	³
BOD ₅ 20°C ⁴	mg/L	24-hour composite	weekly	³
Nitrogen compounds ⁵	mg/L	Grab	monthly	³
Copper	µg/L	Grab	monthly	³
Selenium ⁶	µg/L	Grab	monthly	³
Lead	µg/L	Grab	monthly	³

¹Footnote deleted.

² Total daily flow and instantaneous peak daily flow (24-hr basis). Actual monitored flow shall be reported (not the maximum flow, i.e., design capacity).

³ Pollutants shall be analyzed using the analytical methods described in 40 CFR part 136; for priority pollutants the methods must meet the lowest MLs specified in Attachment 4 of the SIP, where no methods are specified for a given pollutant, by methods approved by this Regional Water Board or State Water Board.

⁴ Biological Oxygen Demand measured over 5 days at 20 degrees centigrade

⁵ Additional influent sampling for nitrogen species is required to identify causes for ammonia concentrations in effluent.

⁶ Additional influent testing is required to determine if a relationship exists between selenium, which may be present in groundwater, and effluent concentrations.

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Nickel	µg/L	Grab	monthly	³
Zinc	µg/L	Grab	quarterly	³
Remaining priority pollutants ⁷ excluding asbestos	µg/L	24-hour composite/grab for VOCs and Chromium VI	semiannually	³
Pesticides ⁸	µg/L	24-hour composite	semiannually	³

IV. EFFLUENT MONITORING REQUIREMENTS

Effluent monitoring is required to determine compliance with permit conditions and water quality standards, assess plant performance, identify operational problems, improve plant performance and provide information on wastewater characteristics and flows for use in interpreting water quality and biological data to inform any conclusions made concerning enhancement of the Estuary.

A. Monitoring Location Effluent Transfer Stations (M-001 and M-001F)

1. The Discharger shall monitor the discharge of tertiary-treated effluent at the Effluent Transfer Stations as compliance points as follows. If more than one analytical test method is listed for a given parameter, the Discharger must select from the listed methods:

Table E-3. Effluent Monitoring⁹

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Total waste flow	Mgd	recorder	continuous ¹⁰	²
Turbidity	NTU	recorder	continuous	³
Total residual chlorine (TRC)	mg/L	recorder	continuous ^{11, 12}	³
Total residual chlorine	mg/L	Grab ¹³	Daily ^{12, 14}	³

⁷ Priority pollutants as defined by the California Toxics Rule (CTR) in Attachment F.

⁸ Pesticides are, for purposes of this Order, the 6 constituents referred to in 40 CFR part 125.58 (p) (methoxychlor, demeton, guthion, malathion, mirex, and parathion).

⁹ Whole Effluent Toxicity monitoring is also required as described in Section V.

¹⁰ Where continuous monitoring of a constituent is required, the following shall be reported:

- a. Total waste flow – Total daily and peak daily flow (24-hr basis);
- b. Total residual chlorine (TRC) – Maximum daily value (24-hr basis);
- c. Turbidity – Maximum daily value, total amount of time each day the turbidity exceeded five turbidity units, flow-proportioned average daily value

¹¹ Total residual chlorine (TRC) shall be continuously recorded. The recorded charts shall be maintained by the Discharger for at least five years. The maximum daily peak, minimum daily peak, and daily average total residual chlorine shall be reported on the monthly monitoring reports.

¹² Continuous monitoring of TRC at the current location shall serve as an internal trigger for increased TRC end of pipe grab sampling if either of the following occur:

- a. TRC concentration excursions of up to 0.3 mg/L lasting greater than 15 minutes; or
- b. TRC concentration peaks in excess of 0.3 mg/L lasting greater than 1 minute and Additional end of pipe grab samples need not be taken if it can be demonstrated that a stoichiometrically appropriate amount of dechlorination chemical has been added to effectively dechlorinate the effluent to 0.1 mg/L or less for peaks in excess of 0.3 mg/L lasting more than 1 minute, but not for more than 5 minutes.

¹³ Grab samples shall be collected at end of pipe during peak flow.

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
(TRC)				
Total and fecal coliform	MPN/100 mL	Grab	Daily	3
enterococcus coliform	MPN/100 mL	Grab	5 times per 30 days	3
Settleable solids	ml/L	Grab	Daily	3
BOD ₅ 20°C	mg/L	24-hr comp.	Daily	3
Suspended solids	mg/L	24-hr comp.	Daily	3
Dissolved oxygen	mg/L	Grab	Daily	3
Temperature	°F	Grab	weekly	3
pH	pH units	Grab	weekly	3
Oil and grease	mg/L	Grab	weekly	3
Total dissolved solids	mg/L	24-hr comp.	monthly	3
Fluoride	mg/L	24-hr comp.	monthly	3
Phosphate as P	mg/L	24-hr comp.	monthly	3
Phosphorous	mg/L	24-hr comp.	monthly	3
Total Ammonia ¹⁵	mg/L	24-hr comp.	monthly	3
Nitrate nitrogen	mg/L	24-hr comp.	monthly	3
Nitrite nitrogen	mg/L	24-hr comp.	monthly	3
Organic nitrogen	mg/L	24-hr comp.	monthly	3
Total Kjeldahl nitrogen	mg/L	24-hr comp.	monthly	3
Detergents (as MBAS)	mg/L	24-hr comp.	monthly	3
Chlorophyll a	mg/L	Grab	monthly	3
Chronic toxicity	Tuc	24-hr comp.	monthly	See Section V.B.2.
Acute toxicity	TUa	24-hr comp.	annually	See Section V.A.2.
Barium	µg/L	24-hr comp.	semiannually	3
Cobalt	µg/L	24-hr comp.	semiannually	3
Copper	µg/L	24-hr comp.	monthly	3
Iron	µg/L	24-hr comp.	semiannually	3
Lead	µg/L	24-hr comp.	monthly	3
Mercury	µg/L	24-hr comp.	quarterly	3
Molybdenum	µg/L	24-hr comp.	semiannually	3
Nickel	µg/L	24-hr comp.	monthly	3
Selenium	µg/L	24-hr comp.	monthly	3
Silver	µg/L	24-hr comp.	quarterly	3
Vanadium	µg/L	24-hr comp.	semiannually	3
Zinc	µg/L	24-hr comp.	quarterly	3
Cyanide	µg/L	24-hr comp.	semiannually	3
Chlorodibromomethane	µg/L	Grab	semiannually	3
Chloroform	µg/L	24-hr comp.	semiannually	3

¹⁴ Daily grab samples shall be collected Monday through Friday only, except for holidays; and not on weekends.

¹⁵ Total ammonia means sum of unionized ammonia (NH₃) concentration and mass and ionized ammonia (NH₄⁺) concentration and mass.

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Dichlorobromomethane	µg/L	Grab	semiannually	3
Bis(2-ethylhexyl)phthalate	µg/L	Grab	semiannually	3
Bromoform	µg/L	Grab	semiannually	3
Acetone	µg/L	Grab	semiannually	3
Total xylene	µg/L	24-hr comp.	semiannually	3
TCDD	µg/L	24-hr comp.	semiannually	16
Remaining priority pollutants (excluding asbestos)	µg/L	grab/24-hr comp. for VOC	semiannually	3
Pesticides ⁸	µg/L	24-hr comp.	semiannually	3
Radioactivity (Including gross alpha, gross beta, combined radium-226 and radium-228, tritium, strontium-90 and uranium)	µg/L	24-hr comp.	semiannually	17

V. WHOLE EFFLUENT TOXICITY (WET) TESTING REQUIREMENTS

A. Acute Toxicity

1. Definition of Acute Toxicity

Acute toxicity is a measure of primarily lethal effects that occur over a 96-hour period. Acute toxicity shall be measured in percent survival measured in undiluted (100 percent) effluent.

- The average survival in the undiluted effluent for any 3 consecutive 96-hour static or continuous flow bioassay tests shall be at least 90 percent, and
- No single test shall produce less than 70 percent survival.

2. Acute Toxicity Effluent Monitoring Program

- Method. The Discharger shall conduct acute toxicity tests on 100 percent effluent and receiving water grab samples by methods specified in 40 CFR part 136, which

¹⁶ In accordance with the SIP, the Discharger shall conduct effluent monitoring for the seventeen 2,3,7,8-tetrachlorodibenzo-p-dioxin (2,3,7,8-TCDD or dioxin) congeners in the effluent and in the receiving water. Station RSW-004, located downstream of the discharge point. The Discharger shall use the appropriate Toxicity Equivalence Factor (TEF) to determine Toxic Equivalence (TEQ). Where TEQ equals the product between each of the 17 individual congeners' (i) concentration analytical result (C_i) and their corresponding Toxicity Equivalence Factor (TEF_i), (i.e., $TEQ_i = C_i \times TEF_i$). Compliance with the dioxin limitation shall be determined by the summation of the seventeen individual TEQs, or the following equation:

$$\text{Dioxin concentration in effluent} = \sum_{i=1}^{17} (TEQ_i) = \sum_{i=1}^{17} (C_i)(TEF_i)$$

¹⁷ Analyze these radiochemicals by the following USEPA methods: method 900.0 for gross alpha and gross beta, method 903.0 or 903.1 for radium-226, method 904.0 for radium-228, method 906.0 for tritium, method 905.0 for strontium-90, and method 908.0 for uranium. Analysis for combined radium-226 & 228 shall be conducted only if gross alpha results for the same sample exceed 15 pCi/L or beta greater than 50 pCi/L. If radium-226 & 228 exceeds the stipulated criteria, analyze for tritium, strontium-90 and uranium.

cites USEPA's *Methods for Measuring the Acute Toxicity of Effluents and Receiving Waters to Freshwater and Marine Organisms*, October, 2002 (EPA-821-R-02-012) or a more recent edition to ensure compliance.

- b. Test Species. The fathead minnow, *Pimephales promelas*, shall be used as the test species for fresh water discharges and the topsmelt, *Atherinops affinis*, shall be used as the test species for brackish discharges. However, if the salinity of the receiving water is between 1 to 32 parts per thousand (ppt), the Discharger may have the option of using the inland silverside, *Menidia beryllina*, instead of the topsmelt. The method for topsmelt is found in *USEPA's Methods for Measuring the Acute Toxicity of Effluents and Receiving Waters to Freshwater and Marine Organisms*, October 2002 (EPA-821-R-02-012).
- c. Alternate Reporting. In lieu of conducting the standard acute toxicity testing with the fathead minnow, the Discharger may elect to report the results or endpoint from the first 96 hours of the chronic toxicity test as the results of the acute toxicity test, but only if the Discharger uses USEPA's August 2002 Method (EPA-821-R-02-013) to conduct the chronic toxicity test.
- d. Acute Toxicity Accelerated Monitoring. If the effluent toxicity requirements in section IV.A. of this Order is not met, the Discharger shall conduct six additional tests, approximately every 2 weeks, over a 12-week period. The Discharger shall ensure that results of a failing acute toxicity test are received by the Discharger within 24 hours of completion of the test and the additional tests shall begin within 5 business days of receipt of the result. If the additional tests indicate compliance with acute toxicity limitation, the Discharger may resume regular testing.

However, if the extent of the acute toxicity of the receiving water upstream of the discharge is greater than the downstream and the results of the effluent acute toxicity test comply with acute toxicity limitation, the accelerated monitoring need not be implemented for the receiving water.

- e. Toxicity Identification Evaluation (TIE)
 - i. If the results of any two of the six accelerated tests are less than 90 percent survival, then the Discharger shall begin a TIE. The TIE shall include all reasonable steps to identify the sources of toxicity. Once the sources are identified, the Discharger shall take all reasonable steps to reduce toxicity to meet the objective.
 - ii. If the initial test and any of the additional six acute toxicity bioassay tests results are less than 70 percent survival, the Discharger shall immediately implement Initial Investigation Toxicity Reduction Evaluation (TRE) Work Plan. Once the sources are identified the Discharger shall take all reasonable steps to reduce toxicity to meet the requirements.

B. Chronic Toxicity Testing

1. Definition of Chronic Toxicity

Chronic toxicity is a measure of adverse sub-lethal effects in plants, animals, or invertebrates in a long-term test. The effects measured may include lethality or decreases in fertilization, growth, and reproduction.

2. Chronic Toxicity Effluent Monitoring Program

- a. Test Methods. The Discharger shall conduct critical life stage chronic toxicity tests on 24-hour composite 100 percent effluent samples or receiving water grab samples

in accordance with EPA's *Short Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms*, October 2002 (EPA-821-R-02-013) or USEPA's *Short Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Marine and Estuarine Organisms*, October 2002 (EPA-821-R-02-014), or current version.

b. Frequency

- i. Screening and Monitoring. The Discharger shall conduct the first chronic toxicity test screening for three consecutive months in 2014. The Discharger shall conduct short-term tests with the cladoceran, water flea (*Ceriodaphnia dubia* - survival and reproduction test), the fathead minnow (*Pimephales promelas* - larval survival and growth test), and the green alga (*Selenastrum capricornutum* - growth test) as an initial screening process for a minimum of three, but not to exceed, five suites of tests to account for potential variability of the effluent/receiving water. After this screening period, monitoring shall be conducted using the most sensitive species.
- ii. Re-screening is required every 24 months. The Discharger shall re-screen with the three species listed above and continue to monitor with the most sensitive species. If the first suite of re-screening tests demonstrates that the same species is the most sensitive then the re-screening does not need to include more than one suite of tests. If a different species is the most sensitive or if there is ambiguity, then the Discharger shall proceed with suites of screening tests for a minimum of 3, but not to exceed 5, suites.
- iii. Regular toxicity tests - After the screening period, monitoring shall be conducted monthly using the most sensitive species.

- c. Toxicity Units. The chronic toxicity of the effluent shall be expressed and reported in Chronic Toxic Units, TU_c , where,

$$TU_c = \frac{100}{NOEC}$$

The No Observable Effect Concentration (NOEC) is expressed as the maximum percent effluent concentration that causes no observable effect on test organisms, as determined by the results of a critical life stage toxicity test.

3. Accelerated Monitoring

If the chronic toxicity of the effluent or the receiving water downstream the discharge exceeds the monthly trigger median of 1.0 TU_c , the Discharger shall conduct 6 additional tests, approximately every 2 weeks, over a 12-week period. The samples shall be collected and the tests initiated no less than 7 days apart. The Discharger shall ensure that they receive results of a failing chronic toxicity test within 24 hours of the completion of the test and the additional tests shall begin within 5 business days of the receipt of the result. However, if the chronic toxicity of the receiving water upstream of the discharge is greater than the downstream and the TU_c of the effluent chronic toxicity test is less than or equal to a monthly median of 1 TU_c trigger, then accelerated monitoring need not be implemented for the receiving water.

- a. If any 3 out of the initial test and the six additional tests results exceed 1.0 TU_c the Discharger shall immediately implement the Initial Investigation of the TRE.

- b. If implementation of the Initial Investigation TRE Work Plan indicates the source of toxicity (e.g., a temporary plant upset, etc.), then the Discharger shall return to the normal sampling frequency required in Table 3 and Table 4 of this MRP.
 - c. If all of the 6 additional tests required above do not exceed 1 TU_c, then the Discharger may return to the normal sampling frequency.
 - d. If a TRE/TIE is initiated prior to completion of the accelerated testing schedule required, then the accelerated testing schedule may be terminated, or used as necessary in performing the TRE/TIE, as determined by the Executive Officer.
4. Quality Assurance
- a. Concurrent testing with a reference toxicant shall be conducted. Reference toxicant tests shall be conducted using the same test conditions as the effluent toxicity tests (e.g., same test duration, etc.).
 - b. If either the reference toxicant test or effluent test does not meet all test acceptability criteria (TAC) as specified in the test methods manual (EPA-821-R-02-012 and/or EPA-821-R-02-013), then the Discharger must re-sample and re-test within 14 days.
 - c. Control and dilution water should be receiving water or laboratory water, as appropriate, as described in the manual. If the dilution water used is different from the culture water, a second control using culture water shall be used.

C. TRE Work Plan

The Discharger shall prepare and submit a copy of the Discharger's initial investigation TRE Work Plan to the Executive Officer of the Regional Water Board for approval within 90 days of the effective date of this NPDES permit. If the Executive Officer does not disapprove the work plan within 60 days, the work plan shall become effective. The Discharger shall use USEPA manual EPA/833B-99/002 (municipal), or most current version, as guidance. At a minimum, the TRE Work Plan must contain the provisions in Attachment G. The TRE Work Plan shall describe the steps the Discharger intends to follow if toxicity is detected, and should include the following, at a minimum:

1. A description of the investigation and evaluation techniques that will be used to identify potential causes and sources of toxicity, effluent variability, and treatment system efficiency.
2. A description of the Facility's methods of maximizing in-house treatment efficiency and good housekeeping practices, and a list of all chemicals used in the operation of the Facility; and,
3. If a TIE is necessary, an indication of the person who would conduct the TIEs (i.e., an in-house expert or an outside contractor). See MRP section V for information.

D. Steps in TRE/TIE

1. If results of the implementation of the Facility's Initial Investigation TRE Work Plan indicate the need to continue the TRE/TIE, the Discharger shall expeditiously develop a more detailed TRE Work Plan for submittal to the Executive Officer within 15 days of completion of the initial investigation TRE. The detailed work plan shall include, but not be limited to the following:
 - a. Further actions to investigate and identify the cause of toxicity;
 - b. Actions the Discharger will take to mitigate the impact of the discharge and prevent the recurrence of toxicity; and,

- c. A schedule for these actions.
2. The following section summarizes the stepwise approach used in conducting the TRE:
 - a. Step 1 includes basic data collection.
 - b. Step 2 evaluates optimization of the treatment system operation, facility housekeeping, and selection and use of in-plant process chemicals.
 - c. If Steps 1 and 2 are unsuccessful, Step 3 implements a TIE and employment of all reasonable efforts using currently available TIE methodologies. The objective of the TIE shall be to identify the substance or combination of substances causing the observed toxicity.
 - d. Assuming successful identification or characterization of the toxicant(s), Step 4 evaluates final effluent treatment options.
 - e. Step 5 evaluates in-plant treatment options.
 - f. Step 6 consists of confirmation once a toxicity control method has been implemented.

Many recommended TRE elements parallel source control, pollution prevention, and storm water control program best management practices (BMPs). To prevent duplication of efforts, evidence of compliance with those requirements may be sufficient to comply with TRE requirements. By requiring the first steps of a TRE to be accelerated testing and review of the Facility's TRE Work Plan, a TRE may be ended in its early stages. All reasonable steps shall be taken to reduce toxicity to the required level. The TRE may be ended at any stage if monitoring indicates there are no longer toxicity violations.

3. The Discharger shall initiate a TIE as part of the TRE process to identify the cause(s) of toxicity. The Discharger shall use the USEPA acute manual, chronic manual, EPA/600/R-96-054 (Phase 1), EPA/600/R-92/080 (Phase 2), and EPA-600/R-92/081 (Phase 3), as guidance.
4. If a TRE/TIE is initiated prior to completion of the accelerated testing required in section V.D. of this program, then the accelerated testing schedule may be terminated, or used as necessary in performing the TRE/TIE, as determined by the Executive Officer.
5. Toxicity tests conducted as part of a TRE/TIE may also be used for compliance, if appropriate.
6. The Regional Water Board recognizes that toxicity may be episodic and identification of causes of and reduction of sources of toxicity may not be successful in all cases. Consideration of enforcement action by the Regional Water Board will be based, in part, on the Discharger's actions and efforts to identify and control or reduce sources of consistent toxicity.
 - a. If all the results of the 6 additional tests are in compliance with the chronic toxicity limitation, the Discharger may resume regular monthly testing.
 - b. If the results of any of the 6 accelerated tests exceeds the limitation, the Discharger shall continue to monitor weekly until 6 consecutive weekly tests are in compliance. At that time, the Discharger may resume regular monthly testing.
 - c. If the results of 2 of the 6 tests, or any 2 tests in a 6-week period, exceed the limitation, the Discharger shall initiate a TRE.

- d. If implementation of the initial investigation TRE Work Plan (see item D.3, above) indicates the source of toxicity (e.g., a temporary plant upset, etc.), then the Discharger shall return to the regular testing frequency.

E. Ammonia Removal

1. Except with prior approval from the Executive Officer of the Regional Water Board, ammonia shall not be removed from bioassay samples. The Discharger must demonstrate the effluent toxicity is caused by ammonia because of increasing test pH when conducting the toxicity test. It is important to distinguish the potential toxic effects of ammonia from other pH sensitive chemicals, such as certain heavy metals, sulfide, and cyanide. The following may be steps to demonstrate that the toxicity is caused by ammonia and no other toxicants before the Executive Officer would allow for control of pH in the test.
 - a. There is consistent toxicity in the effluent and the maximum pH in the toxicity test is in the range to cause toxicity due to increased pH.
 - b. Chronic ammonia concentrations in the effluent are greater than 4 mg/L total ammonia, based on guidance from USEPA.
 - c. Conduct graduated pH tests as specified in the toxicity identification evaluation methods. For example, mortality should be higher at pH 8 and lower at pH 6.
 - d. Treat the effluent with a zeolite column to remove ammonia. Mortality in the zeolite treated effluent should be lower than the non-zeolite treated effluent. Then add ammonia back to the zeolite-treated samples to confirm toxicity due to ammonia.
2. When it has been demonstrated that toxicity is due to ammonia because of increasing test pH, pH may be controlled using appropriate procedures which do not significantly alter the nature of the effluent, after submitting a written request to the Regional Water Board, and receiving written permission expressing approval from the Executive Officer of the Regional Water Board.

F. Reporting

The Discharger shall submit a full report of the toxicity test results, including any accelerated testing conducted during the month, as required by this NPDES permit. Test results shall be reported in Acute Toxicity Units (TU_a) or Chronic Toxicity Units (TU_c), as required, with the self-monitoring report (SMR) for the month in which the test is conducted. If an initial investigation indicates the source of toxicity and accelerated testing is unnecessary, pursuant to sections V.A. and V.B., then those results also shall be submitted with the SMR for the period in which the investigation occurred.

1. The full report shall be received by the Regional Water Board by the 15th day of the second month following sampling.
2. The full report shall consist of (1) the results; (2) the dates of sample collection and initiation of each toxicity test; (3) the toxicity limit; and, (4) printout of the toxicity program (ToxCalc or CETIS).
3. Test results for toxicity tests also shall be reported according to the appropriate manual chapter on Report Preparation and shall be attached to the SMR. Routine reporting shall include the following, at a minimum, as applicable, for each test, as appropriate:
 - a. Sample date(s)
 - b. Test initiation date
 - c. Test species

- d. End point value(s) for each dilution (e.g. number of young, growth rate, percent survival)
 - e. NOEC values in percent effluent
 - f. TU_c value(s), where $TU_c = \frac{100}{NOEC}$
 - g. Mean percent mortality (+standard deviation) after 96 hours in 100 percent effluent (if applicable)
 - h. NOEC and LOEC (Lowest Observable Effect Concentration) values for reference toxicant test(s)
 - i. Available water quality measurements for each test (e.g., pH, dissolved oxygen (dissolved oxygen), temperature, conductivity, hardness, salinity, ammonia).
4. The Discharger shall provide a compliance summary that includes a summary table of toxicity data from at least eleven of the most recent samples.
 5. The Discharger shall notify this Regional Water Board immediately of any toxicity exceedance and in writing 14 days after the receipt of the results of an effluent limit. The notification will describe actions the Discharger has taken or will take to investigate and correct the cause(s) of toxicity. It may also include a status report on any actions required by this Order, with a schedule for actions not yet completed. If no actions have been taken, the reasons shall be given.

VI. RECYCLING MONITORING REQUIREMENTS

Ventura Water Reclamation Facility is recycling wastewater under separate Waste Discharge Requirements (WDRs) and Water Recycling Requirements (WRRs) contained in Order No. 87-45, CI number 6190.

VII. RECEIVING WATER MONITORING REQUIREMENTS

A. Receiving Water Monitoring for Toxicity at Locations R-003, R-004, R-005, and M-001A

1. The Discharger shall monitor surface water for toxicity effects at R-003, R-004, and R-005. Toxicity will be assessed at the receiving water location at R-003, where the estuary discharges to the Ocean; at R-04, where the effluent channel through the tidal flat discharges to the estuary, at R-005, where the Santa Clara River enters the estuary above the outfall, and at M-001A, where the receiving water leaves the wildlife ponds, respectively, as follows:

Table E-4. Receiving Water Monitoring Requirements

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Copper	µg/L	Grab	monthly	2
Lead	µg/L	Grab	monthly	2
Mercury	µg/L	Grab	quarterly	2
Nickel	µg/L	Grab	monthly	2
Selenium	µg/L	Grab	monthly	2
Silver	µg/L	Grab	quarterly	2
Zinc	µg/L	Grab	quarterly	2
Arsenic	µg/L	Grab	quarterly	2
Chlorobromomethane	µg/L	Grab	semiannually	2

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Chloroform	µg/L	Grab	semiannually	2
Bromoform	µg/L	Grab	semiannually	2
Dichlorobromomethane	µg/L	Grab	semiannually	2
Bis(2-ethylhexyl)phthalate	µg/L	Grab	semiannually	2
Dioxin	µg/L	Grab	semiannually	16
Toulene	µg/L	Grab	quarterly	2
Remaining priority pollutants excluding as asbestos	µg/L	Grab	semiannually	2
Chronic toxicity	TUc	Grab	quarterly	See section V.D.
Acute toxicity ¹⁸	TUa	Grab	annually	See section V.C.

B. Receiving Water Monitoring for Nutrients at Locations R-001, R-002, R-003, R-004, R-005 and M-001A.

1. The Discharger shall monitor surface water for nutrient, dissolved oxygen, and toxicity effects at R-001, R-002, R-003, R-004, R-005, and M-001A as described below. The Regional Water Board may determine additional locations or constituents require monitoring within the Estuary based on the results of the Estuary Phase 3 special studies. See reopener requirements at VI.C.1.i.

Table E-5. Receiving Water Monitoring Requirements

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Total and fecal coliform	MPN/100 mL	Grab	Weekly	2
Enterococcus coliform	MPN/100 mL	Grab	5 times per month	2
TRC ¹⁹	mg/L	Grab	Weekly	2
Hardness	mg/L	Grab	Weekly	2
Salinity	ppt	Field	Weekly	2
pH	pH units	Grab	Weekly	2
Temperature	°F	Field	Weekly	2
DO	mg/L	Field	Weekly	2
Total phosphorous as P	mg/L	Grab	Monthly	2
Total Ammonia	mg/L	Grab	Monthly	2
Nitrate nitrogen	mg/L	Grab	Monthly	2
Nitrite nitrogen	mg/L	Grab	Monthly	2
Organic nitrogen	mg/L	Grab	Monthly	2
Total Kjeldahl nitrogen	mg/L	Grab	Monthly	2
Chlorophyll a	mg/L	Grab	Monthly	2

¹⁸ Acute toxicity test for the receiving water is conducted at Stations R-004 and R-003 only.

¹⁹ TRC test for the receiving water is conducted in Stations R-001 to R-004.

C. Observation Points for Surface Water Conditions at Locations R-001, R-002, R-003, R-004, R-005 and M-001A.

1. The Discharger shall observe water appearance at R-001, R-002, R-003, and R-004. R-005, and M-001A as follows:

Table E-6. Receiving Water Monitoring Requirements

Parameter	Minimum Sampling Frequency
Sludge banks or deposits	Weekly
Oil, grease, or slicks	Weekly
Foam	Weekly
Solids of waste origin	Weekly
Breaching (Flow from Estuary to Ocean)	Weekly

D. Sediment Monitoring at Locations R-003, R-004, and R-005.

1. The Discharger shall monitor sediment conditions at R-003, R-004, and R-005, as follows:

Table E-7. Sediment Monitoring Requirements

Location	Parameter	Minimum Sampling Frequency
Benthic Community	Where appropriate: identification of all organisms to lowest possible taxon; community structure analysis for each station; mean, range standard deviation, and 95 percent confidence limits, if appropriate, for value determined in the community analysis.	Annually
Upper 5 cm of Sediment	Arsenic, cadmium, chromium, copper, lead, mercury, nickel, silver, zinc, cyanide, phenolic compounds (chlorinated), phenolic compounds (non-chlorinated), total halogenated organic compounds, aldrin and dieldrin, endrin, HCH, chlordane, total DDT, DDT derivatives, total PCB, PCB derivatives, toxaphene, total PAH, PAH derivatives, detected priority pollutants, compounds on the local 303(d) list, dissolved sulfides (pore water), TOC and grain size (sufficiently detailed to calculate percent weight in relation to phi size)	Annually
Sediment Toxicity	Sediment toxicity monitoring following protocol described in the <i>Evaluation of Methods for Measuring Sediment Toxicity in California Bays and Estuaries</i> , March 2007.	Annually

a. Local Benthic Trends Survey:

This survey addresses the question whether benthic conditions under the influence of the discharge are changing over time. The data collected are used for regular assessment of trends in sediment contamination and biological response along a fixed grid of sites within the influence of the discharge.

- i. Until the Regional Water Board has determined the appropriate number and locations of sediment monitoring stations, based on the Estuary studies and input from the stakeholder group, at a minimum, 4 monitoring stations (R-003 to R-005 and M-001A, Attachment B) shall be sampled annually during the 3rd quarter (July, August, and September) for benthic conditions. The sampling method shall be approved by the Executive Officer.
- ii. Until the Regional Water Board has determined the appropriate number and locations of sediment monitoring stations, based on the Estuary studies and input from the stakeholder group, the following determinations shall be made for each station, where appropriate: identification of all organisms to lowest possible taxon; community structure analysis for each station; mean, range standard deviation, and 95 percent confidence limits, if appropriate, for value determined in the community analysis. The Discharger may be required to conduct additional "statistical analyses" to determine temporal and spatial trends..

b. Sediment/Chemical Monitoring

The Discharger shall collect grab samples for sediment and chemical monitoring. Until the Regional Water Board has determined the appropriate number and locations of sediment monitoring stations, based on the Estuary studies and input from the stakeholder group, one grab sample (upper 5 centimeters) shall be collected at 4 monitoring stations (R-003 to R-005 and M-001A) and analyzed for the following parameters:

Until the Regional Water Board has determined the appropriate number and locations of sediment monitoring stations, based on the Estuary studies and input from the stakeholder group, annual samples shall be taken during the 3rd quarter (July, August, and September) and monitored for the following constituents: arsenic, cadmium, chromium, copper, lead, mercury, nickel, silver, zinc, cyanide, phenolic compounds (chlorinated), phenolic compounds (non-chlorinated), total halogenated organic compounds, aldrin and dieldrin, endrin, HCH, chlordane, total DDT, DDT derivatives, total PCB, PCB derivatives, toxaphene, total PAH, PAH derivatives, detected priority pollutants, compounds on the local 303(d) list, dissolved sulfides (pore water), TOC and grain size (sufficiently detailed to calculate percent weight in relation to phi size).

c. Sediment Toxicity

Until the Regional Water Board has determined the appropriate number and locations of sediment monitoring stations, based on the Estuary studies and input from the stakeholder group, at a minimum, 4 monitoring stations (R-003 to R-005 and M-001A, Attachment B) shall be sampled annually during the 3rd quarter (July, August, and September) for sediment toxicity monitoring following protocol described in the *Evaluation of Methods for Measuring Sediment Toxicity in California Bays and Estuaries, March 2007*. The most sensitive species have to be used.

All sediment

samples must be collected on the same day.

VIII. OTHER MONITORING REQUIREMENTS

A. Additional Characterization of the Estuary

The Discharger shall complete Phase 3 Estuary Watershed Special Studies:

1. Phase 3 Estuary Studies – The Discharger shall perform additional estuary studies to provide sufficient information to allow the Regional Water Board to determine whether or not the continued discharge of effluent enhances the Estuary. See further description in Attachment F Section VI B.

B. Constituents of Emerging Concern (CECs) in Effluent

The Discharger shall conduct a special study to investigate the CECs in the effluent discharge as listed in the Table below. These constituents shall be monitored annually for at least 2 years. The Regional Water Board has determined that 2 years is an appropriate time period to determine those CECs that are present in wastewater treatment plant effluent. Monitoring results shall be reported as part of the annual report. Within six months of the effective date of this Order, the Discharger shall submit to the Executive Officer a CECs special study work plan for approval. Upon approval, the Discharger shall implement the work plan.

Table E-8. CECs to be measured in Effluent

Parameter	Units	Reporting Level
17 α -Ethinyl Estradiol	ng/L	0.5
17 β -Estradiol	ng/L	0.5
Estrone	ng/L	0.5
Bisphenol A	ng/L	10
Nonylphenol & Nonylphenol Polyethoxylates	ng/L	100
Octylphenol & Octylphenol Polyethoxylates	ng/L	100
Polybrominated Diphenyl Ethers (PBDE 28, 47, 99, 100, 153, 154, 183, 209)	ng/L	100 for PBDE 209 and 5 for all others
Amoxicillin	ng/L	10
Azithromycin	ng/L	10
Carbamazepine	ng/L	10
Caffeine	ng/L	10
N,N-Diethyl-m-toluamide (DEET)	ng/L	10
Dilantin(phenytoin)	ng/L	10
Gemfibrozil	ng/L	10
Ibuprofen (must be sampled if Acetaminophen is not)	ng/L	10
Iodinated Contrast Media (Iopromide)	ng/L	10
Sulfamethoxazole	ng/L	10
Trimethoprim	ng/L	10
Tris (2-chloroethyl phosphate)-TCEP, TCPP and TDCPP	ng/L	10
Triclosan	ng/L	10
Bifenthrin	ng/L	5
Permethrin	ng/L	10
Chlorpyrifos	ng/L	10
Galaxolide	ng/L	10
Diclofenac	ng/L	10
Perfluorooctane Sulfonate (PFOS)	ng/L	40
Fipronil	ng/L	2
Meprobamate	ng/L	10

1. The RL for 17 α -Ethinyl Estradiol, 17 β -Estradiol and Estrone is set at .5 ng/L, but ongoing research on analytical methods may result in improvements in the minimum

concentration of hormone which can be detected. Should improved methods become available, the reporting level for the compounds may be adjusted.

2. Reporting – By April 1st of each year (starting April 1, 2015), the Discharger shall submit to the Executive Officer, an annual summary report summarizing the monitoring results from the previous year. For example, the annual summary report due April 1, 2015, shall include CECs monitoring data from January to December 2014. Each annual summary report shall include a compilation of effluent monitoring data of CECs listed in the approved work plan, MLs, sample type, analytical methodology used, sampling date/time, QA/QC information, and an evaluation of cumulative CECs data collected to date as part of this special study. In addition, the first annual summary report due April 1, 2015, shall include a characterization of existing CECs data, i.e., all data collected outside of this special study.

C. Regional Watershed-Wide Monitoring Program for the Santa Clara River Watershed

Pursuant of 40 CFR parts 122.41(j) and 122.48(b), the monitoring program for a Discharger receiving an NPDES Permit must determine compliance with NPDES permit terms and conditions, and demonstrate that water quality standards are met.

Compliance monitoring focuses only on the quality of the discharge; it is not designed to assess the impact of the discharge on the receiving water in combination with other point source discharges or with any other sources of pollution (e.g., non-point source runoff, aerial fallout). Likewise, it is not designed to evaluate the current status of important ecological resources on a regional basis. The Santa Clarita Valley Sanitation District of Los Angeles County submitted the "Santa Clara River Watershed-wide Monitoring Program and Implementation Plan" (SCRWMP) to the Regional Water Board on December 15, 2011. This plan was developed in collaboration with stakeholders in the Santa Clara River Watershed and contains a design for an integrated regional monitoring program for this watershed.

Regional Water Board staff and interested stakeholders have yet to develop a funding mechanism to fully implement the SCRWMP, but hope to initiate the new program in 2014. The City of Ventura will be required to participate in implementation of the watershed-wide monitoring program. To achieve the goals of the SCRWMP, revisions to the Receiving Water Monitoring Requirements will be made, if necessary, under the direction of USEPA and the Regional Water Board.

IX. REPORTING REQUIREMENTS

A. General Monitoring and Reporting Requirements

The Discharger shall comply with all Standard Provisions (Attachment D) related to monitoring, reporting, and recordkeeping.

1. If there is no discharge during any reporting period, the report shall so state.
2. Each monitoring report shall contain a separate section titled "Summary of Non-Compliance" which discusses the compliance record and the corrective actions taken or planned that may be needed to bring the discharge into full compliance with waste discharge requirements. This section shall clearly list all non-compliance with discharge requirements, as well as all excursions of effluent limitations.
3. The Discharger shall inform the Regional Water Board well in advance of any proposed construction activity that could potentially affect compliance with applicable requirements.

B. Self-Monitoring Reports (SMRs)

1. At any time during the term of this Order, the State or Regional Water Board may notify the Discharger to submit SMRs ONLY through electronic means, using the State Water Board's California Integrated Water Quality System (CIWQS) Program website (<http://www.waterboards.ca.gov/ciwqs/index.html>). Until such notification is given, the Discharger shall submit hard copy SMRs in accordance with the requirements described in subsection B.5 below and submit information electronically. The CIWQS website will provide additional directions for SMR submittal in the event there will be service interruption for electronic submittal.
2. When requested, the Discharger shall electronically submit SMRs using the State Water Board's California Integrated Water Quality System (CIWQS) Program web site (<http://www.waterboards.ca.gov/ciwqs/index.html>). If for any reason hard copy SMRs will be required at the discretion of the Regional Water Board, the above requirement should be modified as appropriate. Additionally, the requirements may clarify what satisfies as submittal of hard copy SMRs. For example, "For this purpose, a hard copy signed penalty of perjury statement accompanying a CD with files in Portable Document Format (PDF) shall qualify as hard copy SMRs. Hard copy SMRs must be submitted to the Los Angeles Regional Water Board, signed and certified as required by the Standard Provisions (Attachment D), to the address listed below."
3. The Discharger shall report in the SMR the results for all monitoring specified in this MRP under sections III through IX. Additionally, the Discharger shall report in the SMR the results of any special studies, acute and chronic toxicity testing, TRE/TIE, the Pollutant Management Plan (PMP), and the Storm Water Pollution Prevention Plan (SWPPP) required by Special Provisions – section VI.C. of the Order. The Discharger shall submit quarterly and annual SMRs including the results of all required monitoring using USEPA-approved test methods or other test methods specified in this Order. SMRs are to include all new monitoring results obtained since the last SMR was submitted. If the Discharger monitors any pollutant more frequently than required by this Order, the results of this monitoring shall be included in the calculations and reporting of the data submitted in the SMR.
4. Monitoring periods and reporting for all required monitoring shall be completed according to the following schedule:

Table E-9. Monitoring Periods and Reporting Schedule

Sampling Frequency	Monitoring Period Begins On...	Monitoring Period	SMR Due Date
Continuous	Permit effective date	All	Submit with monthly SMR
Daily	Permit effective date	(Midnight through 11:59 PM) or any 24-hour period that reasonably represents a calendar day for purposes of sampling.	Submit with monthly SMR
Weekly	Sunday following permit effective date or on permit effective date if on a Sunday	Sunday through Saturday	Submit with monthly SMR
Monthly	First day of calendar month following permit effective date or on permit effective date if that date is first day of the month	1 st day of calendar month through last day of calendar month	By the 15 th day of the second month after the month of sampling
Quarterly	Closest of January 1, April 1, July 1, or October 1 following (or on) permit	January 1 through March 31 April 1 through June 30	May 15 August 15

Sampling Frequency	Monitoring Period Begins On...	Monitoring Period	SMR Due Date
	effective date	July 1 through September 30 October 1 through December 31	November 15 February 15
Semiannually	Closest of January 1 or July 1 following (or on) permit effective date	January 1 through June 30 July 1 through December 31	August 15 February 15
Annually	January 1 following (or on) permit effective date	January 1 through December 31	April 1
Annually for two years (CECs)	January 1 following (or on) permit effective date	July 1 through September 30	April 1

5. Reporting Protocols. The Discharger shall report with each sample result the applicable Reporting Level (RL) and the current Method Detection Limit (MDL), as determined by the procedure in 40 CFR part 136.

The Discharger shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:

- Sample results greater than or equal to the RL shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).
- Sample results less than the RL, but greater than or equal to the laboratory's MDL, shall be reported as "Detected, but Not Quantified," or DNQ. The estimated chemical concentration of the sample shall also be reported.

For the purposes of data collection, the laboratory shall write the estimated chemical concentration next to DNQ. The laboratory may, if such information is available, include numerical estimates of the data quality for the reported result. Numerical estimates of data quality may be percent accuracy ($\pm$ a percentage of the reported value), numerical ranges (low to high), or any other means considered appropriate by the laboratory.

- Sample results less than the laboratory's MDL shall be reported as "Not Detected," or ND.
 - Dischargers are to instruct laboratories to establish calibration standards so that the ML value (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Discharger to use analytical data derived from extrapolation beyond the lowest point of the calibration curve.
6. Compliance Determination. Compliance with effluent limitations for priority pollutants shall be determined using sample reporting protocols defined above and Attachment A. For purposes of reporting and administrative enforcement by the Los Angeles Regional Water Board and State Water Board, the Discharger shall be deemed out of compliance with effluent limitations if the concentration of the priority pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the reporting level (RL).
7. Multiple Sample Data. When determining compliance with an AMEL or MDEL for priority pollutants and more than one sample result is available, the Discharger shall compute the arithmetic mean unless the data set contains one or more reported determinations of "Detected, but Not Quantified" (DNQ) or "Not Detected" (ND). In those cases, the

Discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:

- a. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
 - b. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.
8. The Discharger shall submit SMRs in accordance with the following requirements:
- a. The Discharger shall arrange all reported data in a tabular format. The data shall be summarized to clearly illustrate whether the facility is operating in compliance with interim and/or final effluent limitations. The Discharger is not required to duplicate the submittal of data that is entered in a tabular format within CIWQS. When electronic submittal of data is required and CIWQS does not provide for entry into a tabular format within the system, the Discharger shall electronically submit the data in a tabular format as an attachment.
 - b. The Discharger shall attach a cover letter to the SMR. The information contained in the cover letter shall clearly identify violations of the WDRs; discuss corrective actions taken or planned; and the proposed time schedule for corrective actions. Identified violations must include a description of the requirement that was violated and a description of the violation.
 - c. SMRs must be submitted to the Regional Water Board, signed and certified as required by the Standard Provisions (Attachment D). Paper SMRs should be converted to a Portable Document Format (PDF). Documents that are less than 10 megabytes (MB) should be emailed to losangeles@waterboards.ca.gov. Documents that are 10 MB or larger should be transferred to a disk and mailed to the address listed below: (Reference the reports to Compliance File No. 1822 to facilitate routing to the appropriate staff and file.)

California Regional Water Quality Control Board

Los Angeles Region

320 West 4th Street, Suite 200

Los Angeles, CA 90013

Attention: Information Technology Unit

Dischargers who have been certified to only submit electronic SMRs to CIWQS should continue doing so, as previously required

C. Discharge Monitoring Reports (DMRs)

1. At any time during the term of this permit, the State Water Board or the Los Angeles Regional Water Board may notify the Discharger to electronically submit DMRs. Until such notification is given specifically for the submittal of DMRs, the Discharger shall submit DMRs in accordance with the requirements described below.

2. DMRs must be signed and certified as required by the standard provisions (Attachment D). The Discharger shall submit the original DMR and one copy of the DMR to the address listed below:

STANDARD MAIL	FEDEX/UPS/ OTHER PRIVATE CARRIERS
State Water Resources Control Board Division of Water Quality c/o DMR Processing Center PO Box 100 Sacramento, CA 95812-1000	State Water Resources Control Board Division of Water Quality c/o DMR Processing Center 1001 I Street, 15 th Floor Sacramento, CA 95814

3. All discharge monitoring results must be reported on the official USEPA pre-printed DMR forms (EPA Form 3320-1) or on self-generated forms that follow the exact same format of EPA Form 3320-1.

D. Other Reports

1. Annual Summary Report

By April 1 of each year, the Discharger shall submit an annual summary report containing a discussion of the previous year's influent/effluent analytical results and receiving water bacterial monitoring data. The annual summary report shall contain graphical and tabular summaries of the monitoring analytical data. The annual summary report shall also contain an overview of any plans for upgrades to the treatment plant's collection system, the treatment processes, or the outfall system. The Discharger shall submit an electronic copy to the Regional Water Board in accordance with the requirements described in subsection B.5 above.

- a. Each annual monitoring report shall contain a separate section titled "Reasonable Potential Analysis" which discusses whether or not reasonable potential was triggered for pollutants that do not have a final effluent limitation in the NPDES permit. This section shall contain the following statement: "The analytical results for this sampling period did/ did not trigger reasonable potential". If reasonable potential was triggered, then the following information should also be provided:
 - i. A list of the pollutant(s) that triggered reasonable potential;
 - ii. The Basin Plan or CTR criteria that was exceeded for each given pollutant;
 - iii. The concentration of the pollutant(s);
 - iv. The test method used to analyze the sample; and,
 - v. The date and time of sample collection.
2. The Discharger shall submit to the Regional Water Board, together with the first monitoring report required by this NPDES permit, a list of all chemicals and proprietary additives which could affect this waste discharge, including quantities of each. Any subsequent changes in types and/or quantities shall be reported promptly.
3. The Regional Water Board requires the Discharger to file with the Regional Water Board, within 90 days after the effective date of this Order, a technical report on his preventive (failsafe) and contingency (cleanup) plans for controlling accidental discharges, and for minimizing the effect of such events. The technical report should:

- a. Identify the possible sources of accidental loss, untreated waste bypass, and contaminated drainage. Loading and storage areas, power outage, waste treatment unit outage, and failure of process equipment, tanks, and pipes should be considered.
- b. Evaluate the effectiveness of present facilities and procedures and state when they become operational.
- c. Describe facilities and procedures needed for effective preventive and contingency plans.
- d. Predict the effectiveness of the proposed facilities and procedures and provide an implementation schedule contingent interim and final dates when they will be constructed, implemented, or operational.

The Discharger shall report the results of any special studies, acute and chronic toxicity testing, TRE/TIE, PMP, and Pollution Prevention Plan required by Special Provisions – VI.C. The Discharger shall submit reports in compliance with SMR reporting requirements described in subsection IX.B above.

ATTACHMENT F – FACT SHEET

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ATTACHMENT F – FACT SHEET

As described in section I, the Los Angeles Regional Water Quality Control Board (Regional Water Board) incorporates this Fact Sheet as findings of the Regional Water Board supporting the issuance of this Order. This Fact Sheet includes the legal requirements and technical rationale that serve as the basis for the requirements of this Order.

This Order has been prepared under a standardized format to accommodate a broad range of discharge requirements for Dischargers in California. Only those sections or subsections of this Order that are specifically identified as “not applicable” have been determined not to apply to this Discharger. Sections or subsections of this Order not specifically identified as “not applicable” are fully applicable to this Discharger.

I. PERMIT INFORMATION

The following table summarizes administrative information related to the facility.

Table F-1. Facility Information

WDID	4A560107001
Discharger	City of Ventura
Name of Facility	Ventura Water Reclamation Facility, Ventura
Facility Address	1400 Spinnaker Drive
	Ventura, CA 93002-0099
	Ventura County
Facility Contact, Title and Phone	Dan Pfeifer, Wastewater Utility Manager, (805) 677-4131
Authorized Person to Sign and Submit Reports	Dan Pfeifer, Wastewater Utility Manager, (805) 677-4131
Mailing Address	P.O. Box 99, Ventura, CA 93002-0099
Billing Address	P.O. Box 99, Ventura, CA 93002-0099
Type of Facility	Publicly-Owned Treatment Works (POTW)
Major or Minor Facility	Major
Threat to Water Quality	1
Complexity	A
Pretreatment Program	Y
Recycling Requirements	Producer
Facility Permitted Flow	9 million gallons per day (mgd)
Facility Design Flow	14 mgd
Watershed	Santa Clara River Watershed
Receiving Water	Santa Clara River Estuary via Wildlife Ponds
Receiving Water Type	Estuary

- A. The City of Ventura (hereinafter Discharger) is the owner and operator of Ventura Water Reclamation Facility (hereinafter Facility or Ventura WRF), a Publicly-Owned Treatment Works (POTW).

For the purposes of this Order, references to the “discharger” or “permittee” in applicable federal and state laws, regulations, plans, or policy are held to be equivalent to references to the Discharger herein.

- B. The Facility discharges wastewater to the Santa Clara River Estuary (Estuary), a water of the United States, tributary to the Pacific Ocean within the Santa Clara River Watershed. The Discharger was previously regulated by National Pollutant Discharge Elimination System (NPDES) Permit No. R4-2008-0011 (2008 Order) adopted on March 6, 2008 and expired on

March 6, 2013. Attachment B provides a map of the area around the Facility. Attachment C provides a flow schematic of the Facility.

Prior to making any change in the point of discharge, place of use, or purpose of use of treated wastewater that results in a decrease of flow in any portion of a watercourse, the Discharger must file a petition with the State Water Board, Division of Water Rights, and receive approval for such a change. The State Water Board retains the jurisdictional authority to enforce such requirements under California Water Code (CWC) section 1211.

- C. The Discharger filed a report of waste discharge (ROWD) and submitted an application for reissuance of its WDRs and NPDES permit on June 19, 2012. A site visit was conducted on July 19, 2012, to observe operations and collect additional data to develop permit limitations and requirements for waste discharge. The application was deemed complete on August 16, 2012.

II. FACILITY DESCRIPTION

The Discharger owns and operates the Ventura Water Reclamation Facility (VWRF), a tertiary wastewater treatment plant, with a design capacity of 14 million gallons per day (mgd), located at 1400 Spinnaker Road, Ventura, California. The design flow of 14 mgd is limited to 9 mgd discharge into the Santa Clara River Estuary, the 2008 average flow, at which time the Regional Water Board mandated studies to determine if the estuary water quality and beneficial uses were enhanced by the effluent¹. The Facility receives domestic, commercial, and industrial wastewater generated in the city of Ventura by an estimated population of 105,000.

A. Description of Wastewater and Biosolids Treatment and Controls

Treatment at the Facility consists of wastewater processing, biosolids processing, and a pasteurization demonstration project. Attachment C provides a flow schematic of the Facility.

1. Wastewater Processing

The treatment system consists of screening, grit removal, primary sedimentation, flow equalization, bio-augmentation re-aeration (BAR) with full nitrification and denitrification (NDN)², aeration with additional nitrification, activated sludge and mixed liquor recycling, secondary settling, pressurized tertiary filtration, chlorination with ammonia addition³, and dechlorination.

Concentrations of metals in the influent are generally controlled through the pretreatment program. However, because of high concentrations of copper, lead, nickel and zinc in the influent, the Facility improves metals removal by adding iron salt at the headworks.

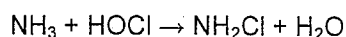
2. Biosolids Processing

The biosolids system consists of primary sludge thickening, dissolved air flotation (DAF) secondary sludge thickening, anaerobic digestion, and dewatering (using plate and frame filter presses).

¹ The prohibition against discharge of the entire design flow into the Estuary or additional outfalls may be lifted as discussed in the reopener requirements VI. C. 1.

² The Discharger completed the nitrification and partial denitrification system on October 3, 2012.

³ Ammonia (NH₃) has been added into chlorine contact basin since April 2004. Ammonia reacts with hypochlorous acid (HOCl) to form chloramine (NH₂Cl):



Chloramine is used as a disinfectant in municipal water systems as an alternative to chlorination and also exhibits less tendency to convert organic materials into chlorocarbons such as chloroform, dichlorobromomethane, dibromochloromethane, and carbon tetrachloride.

All of the Class B anaerobically digested sludge is dewatered, composted and disposed of at the Toland Road Landfill, permitted under Order R4-2011-0052, CI 6190.

3. Pasteurization Pilot Project

A small-scale pasteurization evaluation unit has been installed at the Facility and has been disinfecting 500,000 gallons, about 5 percent of the average daily flow, of wastewater per day since December 2011. The Discharger is considering disinfecting the full wastewater flow. A full-scale design of the system has the potential to generate enough electricity to power the entire Facility; saving the Discharger about half of its current energy costs.

Pasteurization uses natural gas combined with digester gas (a natural by-product of wastewater treatment) as fuel to drive a turbine (or turbines) to generate electricity. The hot exhaust air from the turbine (energy that is typically wasted) is then passed through a series of heat exchangers that increase the temperature of the wastewater to a level that disinfects the wastewater stream. The disinfected water is then cooled by transferring the heat of the disinfected water to incoming wastewater.

B. Discharge Points and Receiving Waters

1. Treated wastewater is discharged into the Estuary (Discharge Point 001) through a series of city-owned ponds. The ponds have soil bottoms and allow percolation.

Table F-2 Discharge Points

Discharge Point Number	001
Latitude	34° 14' 21.72" north
Longitude	119° 15' 31.33" west

2. Approximately 1 mgd of the treated wastewater percolates from the wildlife ponds into perched groundwater, part of the Oxnard Groundwater Basin. The groundwater can exhibit artesian conditions when the freshwater table is above sea level, as dictated by the tidal stage and the depth to water. The quality of groundwater in the Oxnard Plain Groundwater Basin has been diminished by seawater intrusion.
3. The mouth of the Santa Clara River is sometimes closed off by a sand bar so that a shallow lagoon, referred to as the Santa Clara River Estuary, is created. At times when the sand bar is breached, either by floodwaters or by mechanical means, the lagoon empties directly into the Pacific Ocean.

C. Summary of Existing Requirements and Self-Monitoring Report (SMR) Data

Effluent limitations contained in the existing Order for discharges from Discharge Point 001 (Monitoring Location M001) and representative monitoring data from the term of the previous Order are as follows:

Table F-3 Historic Effluent Limitations and Monitoring Data

Parameter	Units	Effluent Limitation			Monitoring Data (From March 2008 to April 2013)		
		Average Monthly	Average Weekly	Maximum Daily	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge
Biochemical oxygen demand (BOD ₅ 20°C)	mg/L	20	30	45	3.3	4	7
Total suspended solids (TSS)	mg/L	15	40	45	1.5	2.0	4.4
pH	--	--	--	--	--	--	6.7-7.3
Oil and Grease	mg/L	10	--	15	6.5	--	6.5
Settleable solids	mg/L	0.1	--	0.3	<0.1	--	<0.1
Total residual chlorine	mg/L	0.1	--	--	<0.1	--	2.7
MBAS	mg/l	0.5	--	--	0.16	--	0.16
Arsenic	µg/L	--	--	--	<2	--	<2
Cadmium	µg/L	--	--	--	<4	--	<4
Chromium (VI)	µg/L	--	--	--	<1	--	<1
Copper	µg/L	4.2	--	8.8	8.88	--	8.88
Lead	µg/L	--	--	--	9.44	--	9.44
Mercury	µg/L	0.05	--	0.10	<0.02	--	<0.02
Nickel	µg/L	--	--	--	35.4	--	35.4
Selenium	µg/L	--	--	--	13.7	--	13.7
Silver	µg/L	0.7	--	2.2	<0.2	--	<0.2
Thallium	µg/L	--	--	--	<1	--	<1
Zinc	µg/L	45	--	107	68.8	--	68.8
Summer Ammonia (May to October)	mg/l	0.045	--	0.30	0.98	--	1.2
Winter Ammonia (November to April)	mg/l	0.079	--	0.53	1.0	--	1.3
Nitrate + Nitrite as Nitrogen	mg/l	10	--	--	12.5	--	18.8
Nitrate as Nitrogen	mg/l	10	--	--	12.3	--	12.3
Nitrite as Nitrogen	mg/l	1	--	--	0.2	--	0.2
Cyanide	µg/L	--	--	--	<5	--	<5
Dioxin	µg/L	--	--	--	<5×10 ⁻⁶	--	<5×10 ⁻⁶
Benzene	µg/L	--	--	--	<0.2	--	<0.2

Parameter	Units	Effluent Limitation			Monitoring Data (From March 2008 to April 2013)		
		Average Monthly	Average Weekly	Maximum Daily	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge
Bromoform	µg/L	--	--	--	4.3	--	4.3
Carbon Tetrachloride	µg/L	--	--	--	<0.12	--	<0.12
Chlorodibromometh ane	µg/L	--	--	--	14.5	--	14.5
Chloroform	µg/L	--	--	--	36	--	36
Dichlorobromometha ne	µg/L	--	--	--	25.6	--	25.6
Ethylbenzene	µg/L	--	--	--	<0.34	--	<0.34
Methylene Chloride	µg/L	--	--	--	<0.25	--	<0.25
Tetrachloroethylene	µg/L	--	--	--	<0.03	--	<0.03
Toluene	µg/L	--	--	--	<0.2	--	<0.2
Pentachlorophenol	µg/L	--	--	--	<3.6	--	<3.6
Phenol	µg/L	--	--	--	<1.5	--	<1.5
2,4,6- Trichlorophenol	µg/L	--	--	--	<2.7	--	<2.7
Benzidine	µg/L	--	--	--	<4.4	--	<4.4
Benzo(a)Anthracene	µg/L	--	--	--	<7.8	--	<7.8
Benzo(a)Pyrene	µg/L	--	--	--	<2.5	--	<2.5
Benzo(b)Fluoranth ene	µg/L	--	--	--	<4.8	--	<4.8
Benzo(k)Fluoranth ene	µg/L	--	--	--	<2.5	--	<2.5
Bis(2- Chloroethyl)Ether	µg/L	--	--	--	<5.7	--	<5.7
Bis(2- Ethylhexyl)Phthalate	µg/L	--	--	--	<2.5	--	<2.5
Chrysene	µg/L	--	--	--	<2.0	--	<2.0
1,4-Dichlorobenzene	µg/L	--	--	--	<4.4	--	<4.4
3,3- Dichlorobenzidine	µg/L	--	--	--	<16.5	--	<16.5
Diethyl Phthalate	µg/L	--	--	--	<2.2	--	<2.2
Di-n-Butyl Phthalate	µg/L	--	--	--	<2.5	--	<2.5
1,2- Diphenylhydrazine	µg/L	--	--	--	<10	--	<10
Hexachlorobenzene	µg/L	--	--	--	<1.9	--	<1.9
Hexachloroethane	µg/L	--	--	--	<1.6	--	<1.6

Parameter	Units	Effluent Limitation			Monitoring Data (From March 2008 to April 2013)		
		Average Monthly	Average Weekly	Maximum Daily	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge
Indeno(1,2,3-cd)Pyrene	µg/L	--	--	--	<3.7	--	<3.7
N-Nitrosodi-n-Propylamine	µg/L	--	--	--	<10	--	<10
Aldrin	µg/L	--	--	--	<0.004	--	<0.004
Alpha-BHC	µg/L	--	--	--	<0.003	--	<0.003
Beta-BHC	µg/L	--	--	--	<0.006	--	<0.006
Gamma-BHC	µg/L	--	--	--	<0.004	--	<0.004
Chlordane	µg/L	--	--	--	<0.014	--	<0.014
p,p'-DDT	µg/L	--	--	--	<4.7	--	<4.7
p,p'-DDE	µg/L	--	--	--	<5.6	--	<5.6
p,p'-DDD	µg/L	--	--	--	<2.8	--	<2.8
Dieldrin	µg/L	--	--	--	<0.002	--	<0.002
Endosulfan I	µg/L	--	--	--	<0.014	--	<0.014
Endosulfan II	µg/L	--	--	--	<0.004	--	<0.004
Endrin	µg/L	--	--	--	<0.006	--	<0.006
Heptachlor	µg/L	--	--	--	<0.003	--	<0.003
Heptachlor Epoxide	µg/L	--	--	--	<0.083	--	<0.083
PCB 1016	µg/L	--	--	--	<0.065	--	<0.065
PCB 1221	µg/L	--	--	--	<0.065	--	<0.065
PCB 1232	µg/L	--	--	--	<0.065	--	<0.065
PCB 1242	µg/L	--	--	--	<0.065	--	<0.065
PCB 1248	µg/L	--	--	--	<0.065	--	<0.065
PCB 1254	µg/L	--	--	--	<0.065	--	<0.065
PCB 1260	µg/L	--	--	--	<0.065	--	<0.065
Toxaphene	µg/L	--	--	--	<10	--	<10

D. Compliance Summary

The following table summarizes the Facility's violations of subdivisions (h) and (i) of CWC section 13385, from March 2008 through December 2012. Compliance after December 2012 is currently under review by the Regional Water Board's Enforcement Unit.

On April 17, 2013, the Assistant Executive Officer issued Settlement Offer No. R4-2013-0053 for alleged violations of Regional Board Order Nos. R4-2008-0011 and R4-2008-0012, NPDES Permit No. CA0053651 for the Ventura WRF for \$591,000. The City of Ventura submitted a signed waiver of right to hearing on July 11, 2013.

Table F-4 Summary of Compliance History

Review Period	Exceeded Constituents	# of Times Exceeded	Regional Board Action
March 2008 through March 2010	Ammonia	3	MMP - Settlement
	Copper	1	
	Nitrogen Compound	94	
	pH	2	
	Total Coliform	56	
April 2010 through December 2012	Ammonia	151	Under Review
	Copper	12	
	Nitrate	2	
	Total Coliform	26	
	Turbidity	6	

1. Three Time Schedule Orders (TSOs) were issued between 2008 and 2013, to give the City time to make changes at the Facility to comply with the final effluent limitations for ammonia contained in the 2008 Order.
 - a. TSO No. R4-2008-0012 was issued concurrently with Order No. R4-2008-0011. It included interim requirements allowing the City three years to construct an ammonia removal process, such as NDN, at the Facility in order to comply with the final ammonia effluent limitations in the 2008 Order. The City upgraded the Facility, including modification of internal piping, aeration, and additional of tankage and pumping systems to assure complete biological nitrification and partial denitrification of effluent, by October 3, 2011. The Executive Officer extended the TSO by 90 days on June 16, 2011, to allow the City to complete maintenance of the aeration basins.
 - b. On September 17, 2012, the City emailed a letter to the Regional Water Board requesting another TSO with interim ammonia effluent limitations for ammonia. The NDN process, completed in 2011, effectively removes most ammonia from the effluent; however, ammonia must be added back into the effluent prior to the disinfection process to reduce total coliform and prevent the formation of trihalomethanes. TSO No. R4-2012-0166 was issued on December 6, 2012. It included interim ammonia limits and provided time for the Discharger to further explore process changes to reduce nitrogen concentrations. Because of the rules governing the issuance of TSOs, TSO No. R4-2012-0166 expired after only 3 months, on March 5, 2013.
 - c. TSO No. R4-2013-0047 was issued on March 7, 2013, in order to give the City additional time to consider strategies for ensuring long-term compliance with final

effluent limitations for ammonia. Since the issuance of the TSO, the City has been evaluating methods to reduce nitrogen loading to the Estuary. TSO No. R4-2013-0047 will expire once this Order becomes effective.

Table F-5 Ammonia Effluent Limitations in the 2008 Order

Constituent	Units	Effluent Limitations	
		Monthly Average	Daily Maximum
Summer Ammonia Nitrogen (April to October)	mg/L	0.045	0.30
Winter Ammonia Nitrogen (November to March)	mg/L	0.079	0.53

Table F-6 Interim Ammonia Effluent Limitations in Time Schedule Order

Constituent	Units	Effluent Limitations	
		Monthly Average	Daily Maximum
TSO Order No. R4-2008-012			
Summer Ammonia Nitrogen (April to October)	mg/L	2.1	2.5
Winter Ammonia Nitrogen (November to March)	mg/L	3.0	3.6
TSO Order Nos. R4-2013-0047 and R4-2012-0166			
Summer Ammonia Nitrogen (April to October)	mg/L	1.6	2.9
Winter Ammonia Nitrogen (November to March)	mg/L	1.3	1.4

E. Planned Changes

1. As discussed above, the Discharger is considering adding pasteurization as a disinfection alternative to chlorination, pending the outcome of the pasteurization disinfection pilot study and the special studies required in this Order.
2. Upon completion of the special studies, as discussed below, a preferred project or series of projects may be designed and constructed to optimize water quality compliance and beneficial uses in the Estuary.

III. APPLICABLE PLANS, POLICIES, AND REGULATIONS

The requirements contained in this Order are based on the requirements and authorities described in this section.

A. Legal Authorities

This Order serves as WDRs pursuant to article 4, chapter 4, division 7 of the CWC (commencing with section 13260). This Order is also issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the USEPA and chapter 5.5, division 7 of the CWC (commencing with section 13370). It shall serve as an NPDES permit for point source discharges from this facility to surface waters.

B. California Environmental Quality Act (CEQA)

Under CWC section 13389, this action to adopt an NPDES permit is exempt from the provisions of Chapter 3 of CEQA, (commencing with section 21100) of Division 13 of the Public Resources Code.

C. State and Federal Laws, Regulations, Policies, and Plans

1. **Water Quality Control Plan.** The Los Angeles Regional Water Board (Regional Water Board) adopted a Water Quality Control Plan for the Los Angeles Region (Basin Plan) on June 13, 1994 that designates beneficial uses, establishes water quality objectives (WQO), and contains implementation programs and policies to achieve those objectives for all waters addressed through the plan. Requirements in this Order implement the Basin Plan. In addition, the Basin Plan implements State Water Board Resolution 88-63, which established state policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal or domestic supply. Beneficial uses applicable to the Santa Clara River Estuary, the Pacific Ocean and the Oxnard Forebay groundwater basin are as follows:

Table F-7 Basin Plan Beneficial Uses

Discharge Point	Receiving Water Name	Beneficial Use(s)
001	Santa Clara River Estuary	<u>Existing:</u> Navigation (NAV); water contact recreation (REC-1); non-contact water recreation (REC-2); commercial and sport fishing (COMM); estuarine habitat (EST); marine habitat (MAR); wildlife habitat (WILD); rare, threatened, or endangered species ⁴ (RARE); migration of aquatic organisms, including fish ⁵ (MIGR); spawning, reproduction, and/or early development ⁶ (SPWN); and wetland ⁶ (WET).
001	Pacific Ocean, Nearshore ⁷	<u>Existing:</u> Industrial water supply (IND); NAV; REC-1; REC-2; COMM; MAR; WILD; preservation of biological habitats ⁸ (BIOL); RARE ⁹ ; MIGR-001 ¹⁰ ; SPWN ¹⁰ ; and, shellfish harvesting (SHELL).
001	Oxnard Plain Groundwater Basin (unconfined and perched)	<u>Existing</u> Municipal (MUN) and Agriculture (AGR) <u>Potential</u> Industrial Supply (IND)

⁴ One or more rare species utilize all ocean, bays, estuaries, and coastal wetlands for foraging and/or nesting.

⁵ Aquatic organisms utilize all bays, estuaries, lagoons and coastal wetlands, to a certain extent, for spawning and early development. This may include migration into areas, which are heavily influenced by freshwater inputs.

⁶ Waterbodies designated as WET may have wetlands habitat associated with only a portion of the waterbody. Any regulatory action would require a detailed analysis of the area.

⁷ Nearshore is defined as the zone bounded by the shoreline and a line 1,000 feet from the shoreline or the 30-foot depth contours, whichever is further from the shoreline.

⁸ Areas of Special Biological Significance (along coast from Latigo Point to Laguna Point) and Big Sycamore Canyon and Abalone Cove Ecological Reserves and Point Fermin Marine Life Refuge.

- a. Beneficial Uses of Surface Waters
 - i. The Facility discharges into an estuary that discharges into the Ocean when the sand bar is breached. The limits in this Order are based on the Basin Plan and the State Implementation Policy (SIP). However, consistent with the Ocean Plan, the effluent from the Facility may not negatively impact the beneficial uses of the Ocean.
 - ii. Under federal law, all waters are assumed to be "fishable" and "swimmable" unless a Use Attainability Analysis (UAA) has been done to justify the unattainability of these uses.
- b. Beneficial Uses of Groundwater
 - i. The wildlife ponds and the Estuary are known to be hydraulically connected to the underlying groundwater basin, the unconfined and perched Oxnard Plain Groundwater Basin. The Order specifies that effluent from the Facility may not negatively impact the beneficial uses of the groundwater basin.
 - ii. State Water Board Resolution No. 88-63 and Regional Water Board Resolution No. 89-03, known as the *Sources of Drinking Water Policy*, states that all surface waters and groundwaters of the state are considered to be suitable, or potentially suitable, for municipal or domestic water supply and should be so designated by the Regional Water Board with the following exceptions:
 - (a) The total dissolved solids (TDS) exceed 3,000 mg/L (5,000 μ S/cm, electrical conductivity) and it is not reasonably expected by Regional Water Boards to supply public water system;
 - (b) There is contamination, either by natural processes or by human activity (unrelated to the specific pollution incident), that cannot reasonably be treated for domestic use using either best management practices or best economically achievable treatment practices; or
 - (c) The water source does not provide sufficient water to supply a single well capable of producing an average, sustained yield of 200 gallons per day.
 - iii. The special studies required by this Order include an investigation of groundwater basin and the interaction between the groundwater, the Estuary and the wildlife ponds. The results of the special studies should be sufficient to allow the Regional Water Board to determine whether or not additional controls to protect groundwater are necessary and if the groundwater is a potential source of drinking water. See reopener requirements at VI.C.1.
- c. Requirements of this Order implement the Basin Plan and subsequent amendments including the following:
 - i. Ammonia - The 1994 Basin Plan contained water quality objectives (WQOs) for ammonia to protect aquatic life, in Tables 3-1 through Tables 3-4. Those ammonia objectives were revised, based on fresh and saltwater criteria according to the following Basin Plan amendments which are implemented in this Order.
 - (a) Freshwater Ammonia Water Objective – On April 25, 2002, the Regional Water Board adopted Resolution No. 2002-011, Amendment to the Water Quality Control Plan for the Los Angeles Region to Update the Ammonia Objectives for Inland Surface Waters (Including Enclosed Bays, Estuaries and Wetlands) with Beneficial Use Designations for Protection of Aquatic

Life (2002 Ammonia Amendment). The Basin Plan amendment was approved by the State Water Board, the Office of Administrative Law (OAL), and the USEPA on April 30, 2003, June 5, 2003, and June 19, 2003, respectively. The revised ammonia WQOs were adopted to be protective of aquatic life and are consistent with USEPA's 1999 ammonia criteria update.

- (b) Saltwater Ammonia Water Objective – On March 4, 2004, the Regional Water Board adopted the Resolution No. 2004-022, Amendment to the Water Quality Control Plan for the Los Angeles Region to Update the Ammonia Objectives for Inland Surface Waters Not Characteristic of Freshwater (Including Enclosed Bays, Estuaries and Wetlands) with Beneficial Use Designations for Protection of Aquatic Life (2004 Ammonia Amendment). The Basin Plan amendment is consistent with the USEPA Ambient Water Quality Criteria for Ammonia (Saltwater)-1989. The Basin Plan amendment was approved by the State Water Board, OAL, and USEPA on July 22, 2004, September 14, 2004, and May 19, 2005, respectively.

After the adoption of the 2008 Order, the Regional Water Board revised how the 2002 and 2004 Ammonia Amendments are incorporated into permits. USEPA completed additional toxicology testing and determined that the most sensitive species are long-lived fish such as salmonids (steelhead), not the short-lived species such as Ceriodaphnia (water flea), under some conditions. As a result, longer term variations in water quality conditions are now averaged in the calculation of the limit. Also, USEPA has directed permit writers to use the 90th and 50th percentiles of 30-day averages for pH and temperature in the calculation of the acute and chronic ammonia limit, respectively. Previously, the 99th and 90th percentile of 1-hour acute and 4-day chronic criteria had been used.

- ii. Chloride – On December 11, 2008, the Regional Water Board adopted the Resolution No. 2008-012, *Reconsideration of the Upper Santa Clara River Chloride TMDL Implementation Plan and Revise Chloride Water Quality Objectives*. The chloride Basin Plan amendment was approved by the State Water Board, OAL, and USEPA on October 20, 2009, January 26, 2010, and April 6, 2010, respectively. The TMDL does not set load allocations for Santa Clara River Reach 1, above Highway 101 and the Estuary. Page 3-13 of the Basin Plan indicates that there are no waterbody-specific objectives for TDS, chloride, sulfate or boron between the Highway 101 Bridge (receiving water station R-005) and the Estuary. Table 3-10 (page 3-19) of the Basin Plan indicates that the Oxnard Plain unconfined and perched aquifers have mineral objectives as follows: TDS is 3,000 mg/L, sulfate is 1,000 mg/L, and chloride is 500 mg/l. There is no boron objective.
- iii. Bacteria – The 1994 Basin Plan provided WQOs for bacteria to protect the beneficial uses of the receiving water body. On October 25, 2001, the Regional Water Board adopted the Resolution No. 2001-018, *Amendment to the Water Quality Control Plan for the Los Angeles Region to Update the Bacteria Objectives for Water Bodies Designated for Water Contact Recreation* revising the objectives based on freshwater and saltwater criteria. The bacteria Basin Plan amendment was approved by the State Water Board, OAL, and USEPA on July 18, 2002, September 19, 2002, and September 25, 2002, respectively.

- iv. **Copper** – Order No. R4-2008-00011, adopted by this Regional Water Board on March 6, 2008, calculated the copper limit based on a Water Effects Ratio (WER) study in the 2007 Updated Enhancement Study conducted by Nautilus Environmental and 2002 Metal Translator Factor (MTF) in the Metal Translator Study conducted by Entrix. A WER of 1.58, the lowest among 15 WERs calculated, and a MTF of .86 were applied. On December 11, 2008, the section IV.A.1.a. of the 2008 Order was amended to reflect a revised WER for copper, calculated using additional data provided by the Discharger. The amendment applied a revised WER of 2.08 based upon data collected during winter conditions between March 2004 and January 2005 when the toxicity effects were most pronounced. Another copper WER study was submitted to the Regional Water Board in February, 2010, proposing a value between 2.05 and 2.73 based on water samples collected on December 6, 2009, January 29, 2010, and February 2010. The results of the second WER study bracket the value used in the amendment 2008 Order, and the amended value of 2.08 is used here to calculate the copper limit..
 - v. **Metals Translator Study** – A metal translator study submitted June 27, 2002, and validated in 2007, quantifying the impact of the receiving water on metal toxicity. The metals translator was used to calculate metal limits in the 2008 Order and in this Order for copper (.86) and nickel (.81). Metal translators were not identified for other metals.
2. **Thermal Plan.** The State Water Board adopted the Water Quality Control Plan for Control of Temperature in the Coastal and Interstate Waters and Enclosed Bays and Estuaries of California (Thermal Plan) on January 7, 1971, and amended this plan on September 18, 1975. This plan contains temperature objectives for surface waters. Requirements of this Order implement the Thermal Plan.
 3. **Sediment Quality.** The State Water Board adopted the Water Quality Control Plan for Enclosed Bays and Estuaries – Part 1, Sediment Quality on September 16, 2008, and it became effective on August 25, 2009. This plan supersedes other narrative sediment quality objectives, and establishes new sediment quality objectives and related implementation provisions for specifically defined sediments in most bays and estuaries. Requirements of this Order implement sediment quality objectives of this Plan.
 4. **National Toxics Rule (NTR) and California Toxics Rule (CTR).** USEPA adopted the NTR on December 22, 1992, and later amended it on May 4, 1995 and November 9, 1999. About forty criteria in the NTR applied in California. On May 18, 2000, USEPA adopted the CTR. The CTR promulgated new toxics criteria for California and, in addition, incorporated the previously adopted NTR criteria that were applicable in the state. The CTR was amended on February 13, 2001. These rules contain federal water quality criteria for priority pollutants.

The CTR specifies that freshwater criteria apply at locations where the salinity is 1 part per thousand (ppt) or less 95 percent or more of the time, and marine water criteria apply at locations where the salinity is 10 ppt or more 95 percent or more of the time. At locations where salinities fall between 1 and 10 ppt, the more stringent of either fresh or marine waters apply.
 5. **State Implementation Policy (SIP).** On March 2, 2000, the State Water Board adopted the Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California (State Implementation Policy or SIP). The SIP became effective on April 28, 2000, with respect to the priority pollutant criteria promulgated for California by the USEPA through the NTR and to the priority pollutant objectives

established by the Los Angeles Regional Water Board in the Basin Plan. The SIP became effective on May 18, 2000, with respect to the priority pollutant criteria promulgated by the USEPA through the CTR. The State Water Board adopted amendments to the SIP on February 24, 2005, that became effective on July 13, 2005. The SIP establishes implementation provisions for priority pollutant criteria and objectives and provisions for chronic toxicity control. Requirements of this Order implement the SIP.

6. **Antidegradation Policy.** Federal regulation 40 CFR section 131.12 requires that the state water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State Water Board Resolution 68-16("Statement of Policy with Respect to Maintaining the Quality of the Waters of the State"). Resolution 68-16 is deemed to incorporate the federal antidegradation policy where the federal policy applies under federal law. Resolution 68-16 requires that existing water quality be maintained unless degradation is justified based on specific findings. The Los Angeles Regional Water Board's Basin Plan implements, and incorporates by reference, both the State and federal antidegradation policies. The discharge permitted in the Order is consistent with the antidegradation provisions of 40 CFR part 131.12 and State Water Board Resolution 68-16. See additional discussion at IV. D. 1.
7. **Anti-Backsliding Requirements.** Sections 402(o) and 303(d)(4) of the CWA and federal regulations at 40 CFR part 122.44(l) restrict backsliding in NPDES permits. These anti-backsliding provisions require that effluent limitations in a reissued permit be as stringent as those in the previous permit, with some exceptions in which limitations may be relaxed. See additional discussion at IV. D. 2.
8. **Endangered Species Act Requirements.** This Order does not authorize any act that results in adverse effects to state of federally listed species that utilize the Estuary or result in taking of a threatened or endangered species or any act that is now prohibited, or becomes prohibited in the future, under either the California Endangered Species Act (Fish and Game Code, §§ 2050 to 2097) or the Federal Endangered Species Act (16 USC §§ 1531 to 1544) nor does this Order require any such act if authorization cannot be obtained from the appropriate agency as specified by the Acts. This Order requires compliance with effluent limits, receiving water limits, and other requirements to protect the beneficial uses of waters of the state including protecting rare, threatened, or endangered species. The discharger is responsible for meeting all requirements of the applicable Endangered Species Act.
9. **State Parks.** The State of California Department of Parks and Recreation (State Parks) has declared McGrath State Beach and the surrounding 160 acres a natural preserve. The preserve includes the main channel of the Santa Clara River and adjacent natural lands of riparian shrub land and saltwater marsh. The purpose of the natural preserve is to protect and perpetuate the river ecosystem at the mouth of the Santa Clara River. Resource values of particular significance include estuarine waters, which are used extensively by a wide variety of waterfowl and other water-associated birds; nesting habitat of the endangered California least tern; and riparian shrub land and saltwater marsh communities. Flooding of the Estuary in 2012-2013 affected the use of McGrath State Beach as a campground. This Order does not address the authority to flood property not owned by the City of Ventura. Property right disputes are outside the responsibility of the Regional Water Board.
10. **Enclosed Bay and Estuaries Policy.** The Water Quality Control Policy for the Enclosed Bays and Estuaries of California (Enclosed Bays and Estuaries Policy), adopted by the

State Water Board as Resolution No. 95-84 on November 16, 1995, states that *"It is the policy of the State Board that the discharge of municipal wastewaters and industrial process waters (exclusive of cooling water discharges) to enclosed bays and estuaries, other than the San Francisco Bay-Delta system, shall be phased out at the earliest practicable date. Exceptions to this provision may be granted by a Regional Water Board only when the Regional Water Board finds that the wastewater in question would consistently be treated and discharged in such a manner that it would enhance the quality of receiving waters above that which would occur in the absence of the discharge."* The Enclosed Bay and Estuaries Policy does not define what is meant by "enhancement".

- a. While adopting Order No. 77-1008 in 1977, the Regional Water Board determined that the discharge from the Facility enhances the Estuary, based on a Facilities Plan⁹ submitted by the City in 1976.
- b. Order No. 00-143, adopted in 2000, also described effluent discharge as an enhancement, including extensive narrative limits and monitoring requirements designed to protect wetlands, the Estuary, and existing habitat.

Studies were completed between August 2002 and May 2007, as required in the 2000 Order, to provide the Regional Water Board with additional information to verify the enhancement finding. These studies considered (1) metal translators and impacts on endangered species including the Tidewater Goby, in 2004; (2) toxicology, ecology and hydrology in 2005 and 2007, (3) water budget, salinity, recycled water markets, resident species and WER in 2007. The studies documented how beneficial uses varied with season, water level and water quality. Table F-8 contains a summary of each study.

Table F-8 Discharger's Conclusions on Studies on Estuary Enhancement between 2002 and 2007

Study Title	Date	Conclusions
2002 Metals Translator Studies (Entrix, Inc)	Aug 2002	Determined a metals translator to protect the Estuary and its species.
VWRF Discharge Beneficial Uses on the Distribution and Utilization of Estuary Tidewater Goby (Entrix, Inc)	Sept 2004	Determined the distribution and habitat of the tidewater goby in the Estuary; the historical and current hydrology of the Santa Clara River and Estuary; and, the impacts and benefits of the VWRF discharge on the distribution within and use of the Estuary by tidewater goby.
2005 Toxicology, Ecology and Hydrology Final Report (Nautilus Environmental)	May 2005	Assessed the impacts and benefits of VWRF discharge on Estuary's sensitive species habitat and ecological conditions; Estuary hydrology and water balance; Estuary sediment and water quality; and, discharge water quality.
2007 Updated Toxicology, Ecology and Hydrology Final Report (Nautilus Environmental)	May 2007	Updated the 2005 Report.

⁹ The City filed a Facilities Plan for effluent utilization in 1976 that included a demonstration of enhancement of the Estuary due to the plant's freshwater discharge. The Facilities Plan indicated that some of the beneficial uses of the Estuary, such as fish and wildlife habitat and non-contact water recreation, were enhanced by the presence of the discharge. Consequently, Order No. 77-100, adopted by the Regional Water Board in May 1977, granted the City an exception to the discharge prohibition and allowed discharge of treated municipal wastewater into the Estuary. The demonstration of enhancement was based on an average discharge flow rate of 5.6 mgd.

Study Title	Date	Conclusions
Estuary Water Budget and Salinity Study (Kamman Hydrology & Engineering Inc.)	Apr 2007	Assessed the salinity and water balance information for the Estuary in a dry season.
Recycled Water Market Assessment (Kennedy/Jenks Consultants)	Apr 2007	Assessed the existing versus potential recycled water usage and customer locations for the City of Ventura.
Resident Species Study Santa Clara River Estuary (Entrix, Inc)	Sept 2007	Evaluated the most appropriate water quality criterion (freshwater, saltwater, or other) and discharge effluent limitations for metals and other constituents to protect the Estuary and its species.
2007 Metals Translator Study (Entrix, Inc)	Aug 2007	Evaluated the most appropriate discharge effluent limitations for metals to protect the Estuary and its species.
Water Effects Ratio Study (Nautilus Environmental)	May 2007	Evaluated the most appropriate discharge effluent limitations for metals to protect the Estuary and its species.

While adopting the 2008 Order, the Regional Water Board considered conflicting, yet credible, opinions from a variety of experts who predicted harm to endangered species, habitat, and recreation, both with and without the discharge. The Regional Water Board found that it had inadequate information with which to determine whether and to what extent the discharge that could be authorized by the permit continued to constitute an enhancement. Also, the Board lacked the information necessary to determine what, if any, negative impacts would occur to the Estuary if the discharge was prohibited, and therefore lacked the information necessary to dispute the previous enhancement finding.

The Regional Water Board chose to cap the allowable discharge at the 2000-2008 average annual level of 9 mgd and to require additional watershed-wide studies and a public process to determine the best approach for the Estuary. An Estuary Subwatershed Study, a Recycled Water Market Study, and a Treatment Wetlands Feasibility Study (collectively referred to as Phase 1 Studies) and a Phase 2 Estuary Subwatershed Study fulfilled requirements of the 2008 Order No. R4-2008-0011 and were completed between 2008 and 2013. The studies described specific infrastructure and recycled water projects and wetland treatment plans. Additional information was also gathered on groundwater quality and endangered species and habitat characteristics.

Table F-9 Discharger's Conclusions¹⁰ from Studies on Estuary Enhancement between 2008 and 2013

Study Title	Date	Conclusions
Recycled Water Market Study	Mar 2010	Addressed macro-level supply and demand issues and the ability to expand the local recycled water market for additional recycled water supply that could be generated to serve newly identified demand.
Treatment Wetlands Feasibility Study	Mar 2010	Identified the preferred wetland site locations, planning level design specifications, and available measures to maximize treatment for nutrients; modeled to predict projected discharge related

¹⁰ Stakeholders submitted comments regarding these studies. The conclusions given here represent the Discharger's interpretation of the studies. The Discharger's summary may not represent the conclusions made by other stakeholders or the staff of the Regional Water Board

Study Title	Date	Conclusions
		pollutant loads and concentrations entering and leaving wetlands and cost estimates for project level planning, design, permitting, and construction.
Phase 1 Estuary Subwatershed Study	Sept 2011	Determined discharge conditions, benefits and adverse impacts of the discharge on the Estuary's beneficial uses and sensitive, endangered and threatened species.
Phase 1 Estuary Subwatershed Study	Mar 2013	Identified and compared potential diversion and water reuse infrastructure alternatives, and addressed data gaps identified in the Phase 1 Final Amended Report (2011) to further facilitate a determination of discharge conditions, benefits and adverse impacts of the discharge on the Estuary's beneficial uses and sensitive, endangered and threatened species.

Since 2008, stakeholder workshops have been hosted by the City, consistent with the requirements of the 2008 Order. The Regional Water Board, the National Marine Fisheries Service, the United States Fish and Wildlife Service, the California Department of Parks and Recreation, the California Department of Fish and Wildlife, the United States Army Corp of Engineers, the United States Bureau of Reclamation, the Ventura County Farm Bureau, the United Water Conservation District, the County of Ventura Environmental Health, the Ventura County Watershed Protection District, the City of Oxnard, the Resource Conservation Partners, the Nature Conservancy, Heal the Bay, the Wishtoyo Foundation's Ventura Coastkeeper, the Ventura Audubon Society, the Surfrider Foundation, the Friends of the Santa Clara River and members of the public have attended these meetings.

- c. **Petitions.** Heal the Bay and the Discharger filed petitions to State Water Board challenging the 2008 Order. Heal the Bay requested a review of the discharge to the Estuary as a violation of the Enclosed Bay and Estuaries Policy. Heal the Bay withdrew the petition on February 6, 2012. The Discharger requested a review the final copper effluent limitations, stating that they were more stringent than required by federal law, technologically infeasible, and more stringent than required to protect the beneficial uses designated for the receiving waters of the Estuary. The Discharger withdrew the petition on April 13, 2011.
- d. **Legal Settlement.** Following adoption of the 2008 Order, the Wishtoyo Foundation's Ventura Coastkeeper (VCK) filed suit against the City of Ventura. On January 30, 2012, the City of Ventura, Heal the Bay, and Wishtoyo Foundation's Ventura Coastkeeper Program entered into a settlement agreement that provides a long-term plan for the City's reclamation facility and its highly treated water (Consent Decree). Entering into the settlement agreement followed 120 days of extensive public outreach, including a Town Hall meeting on November 10, 2011, which was attended by nearly 50 residents and other interested stakeholders.

The settlement's goal is to resolve a long-standing debate about the impact of the tertiary treated water on the sensitive ecology of the Estuary as well as associated legal and administrative actions against the City. The settlement outlines common goals and a collaborative process which relies on the best available science to decide how to use the reclaimed water produced by the wastewater facility in the future. The major points of the settlement include:

- i. Create opportunities to use between 50-100% of the treated water for landscaping, agricultural, or other reclamation uses to stretch water supplies and reduce or eliminate the amount of effluent released into the Estuary;
- ii. Build treatment wetlands to further improve water quality if treated water is released into the Estuary; and,
- iii. Work together with Ventura Water's customers to arrive at the most responsible and sustainable solution for the health of the Estuary and Ventura's water supply by 2018, and to fully implement this solution by 2025..

The settlement agreement requires the City to determine, through scientific analysis, the Maximum Ecologically Protective Diversion Volume (MEPDV). That MEPDV may be determined either by agreement from the City of Ventura, Heal the Bay and the Ventura Coastkeeper before January 1, 2018, or by a determination from a science review panel that is established in the Consent Decree if the City, Heal the Bay and Ventura Coastkeeper cannot agree to the MEPDV by or before January 1, 2018. After review and recommendation of the MEPDV by all resource agencies, the City of Ventura may propose to construct the necessary facilities to reach that volume.

The special studies described in this Order may provide the scientific analysis used to define the MEPDV, but the special studies must provide sufficient and meaningful information to determine if discharge enhances the Estuary. The MEPDV analysis may be used by the Regional Water Board staff in its evaluation of Estuary enhancement during the next revision of this Order, projected to take place in November of 2018.

The City of Ventura, Heal the Bay, and Wishtoyo's Ventura Coastkeeper program have made progress on technical studies toward the agreements described in their 2012 consent decree. The parties plan to complete additional technical work, but have not reached agreement on the conclusions completed."

11. **Recycling.** In accordance with statewide policies concerning water reclamation, this Regional Water Board strongly encourages, wherever practical, water recycling, water conservation, and use of storm water and dry-weather urban runoff.

D. Impaired Water Bodies on CWA Section 303(d) List

1. Integrated Report

The State Water Board proposed the California 2008-2010 Integrated Report from a compilation of the adopted Regional Water Boards' Integrated Reports containing 303(d) List of Impaired Waters and 305(b) Reports following recommendations from the Regional Water Boards and information solicited from the public and other interested parties. The Regional Water Boards' Integrated Reports were used to revise their 2006 CWA Section 303(d) List. On August 4, 2010, the State Water Board adopted the California 2008-2010 Integrated Report. On November 12, 2010, the USEPA approved California 2008-2010 Integrated Report Section 303(d) List of Impaired Waters requiring TMDLs for the Los Angeles Region.

The Santa Clara River Estuary is in the California 2008-2010 Integrated Report. The following pollutants were identified as impacting the receiving waters:

- a. Santa Clara River Estuary – Calwater Watershed 40311000 (Hydro. Unit No. 402.10 in Basin Plan)

Pollutants – ChemA, coliform bacteria, nitrate, toxaphene, and toxicity

- b. Santa Clara River Reach 1 (Estuary to Hwy 101 Bridge) – Calwater Watershed
40311000 (Hydro. Unit No. 402.10 in Basin Plan)

Pollutants – Toxicity

The cause for the toxicity is not described in the report, but toxicants controlled under the CTR and nutrients, which can be associated with eutrophication, dissolved oxygen depression and other effects toxic to aquatic life, are limited by this Order.

E. Other Plans, Policies and Regulations

1. **Secondary Treatment Regulations.** Part 133 of 40 CFR establishes the minimum levels of effluent quality to be achieved by secondary treatment. These limitations, established by USEPA, are incorporated into this Order, except where more stringent limitations are required by other applicable plans, policies, or regulations.
2. **Storm Water.** CWA section 402(p), as amended by the Water Quality Act of 1987, requires NPDES permits for storm water discharges. Pursuant to this requirement, in 1990, USEPA promulgated 40 CFR part 122.26 that established requirements for storm water discharges under an NPDES program. To facilitate compliance with federal regulations, on November 1991, the State Water Board issued a statewide general permit, *General NPDES Permit No. CAS000001 and Waste Discharge Requirements for Discharges of Storm Water Associated with Industrial Activities*. This permit was amended in September 1992 and reissued on April 17, 1997 in State Water Board Order No. 97-03-DWQ to regulate storm water discharges associated with industrial activity.

The Discharger developed and currently implements a Storm Water Pollution Prevention Plan (SWPPP), to comply with the State Water Board's (Order No. 97-03-DWQ).

3. **Sanitary Sewer Overflows (SSOs).** The CWA prohibits the discharge of pollutants from point sources to surface waters of the United States unless authorized under an NPDES permit. (33 United States Code sections 1311 and 1342). The State Water Board adopted the *Statewide General Waste Discharge Requirements for Sanitary Sewer Systems (SSS WDR)*, Water Quality Order No. 2006-0003 on May 2, 2006, to provide a consistent, statewide regulatory framework to address SSOs. The WDR requires public agencies that own or operate sanitary sewer systems to develop and implement sewer system management plans and report all SSOs to the State Water Board's online SSO database.

The requirements contained in this Order in sections VI.C.3.b, VI.C.4, and VI.C.5: are intended to be consistent with the requirements in the SSS WDR. The Regional Water Board recognizes that there are areas of overlapping interest between the NPDES permit conditions and the SSS WDR requirements. The requirements of the SSS WDR are considered the minimum thresholds (see finding 11 of Order NO. 2006-0003). The Regional Water Board will accept the documentation prepared by the Permittee under the SSS WDR for compliance purposes, as satisfying the requirements in sections VI.C.3.b, VI.C.4, and VI.C.5, provided for any more specific or stringent provisions enumerated in this Order, have also been addressed.

4. **Watershed Management.** This Regional Water Board has been implementing a Watershed Management Approach (WMA) to address water quality protection in the Los Angeles Region, as detailed in the Regional Water Board's Watershed Management Initiative (WMI). The WMI is designed to integrate various surface and ground water regulatory programs while promoting cooperative, collaborative efforts within a watershed. It is also designed to focus limited resources on key issues and use sound science. Information about the Santa Clara River Watershed and other watersheds in the

region can be obtained from the Regional Water Board's website at <http://www.waterboards.ca.gov/losangeles> and clicking on the word "Watersheds".

5. **Water Rights.** Prior to making any change in the point of discharge, place of use, or purpose of use of treated wastewater that results in a decrease of flow in any portion of a watercourse, the City must file a petition with the State Water Board, Division of Water Rights, and receive approval for such a change. The State Water Board retains the jurisdictional authority to enforce such requirements under CWC section 1211.

IV. RATIONALE FOR EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

The CWA requires point source dischargers to control the amount of conventional, non-conventional, and toxic pollutants that are discharged into the waters of the United States. The control of pollutants discharged is established through effluent limitations and other requirements in NPDES permits. There are two principal bases for effluent limitations in the Code of Federal Regulations: 40 CFR section 122.44(a) requires that permits include applicable technology-based limitations and standards; and 40 CFR section 122.44(d) requires that permits include water quality-based effluent limitations to attain and maintain applicable numeric and narrative water quality criteria to protect the beneficial uses of the receiving water.

A. Discharge Prohibitions

Effluent and receiving water limitations in this Order are based on the CWA, the Basin Plan, the State Water Board's plans and policies, USEPA guidance and regulations, and best practicable waste treatment technology. This order authorizes the discharge of tertiary-treated wastewater from Discharge Point 001 only. It does not authorize any other types of discharges.

B. Technology-Based Effluent Limitations (TBELs)

1. Scope and Authority

Section 301(b) of the CWA and implementing USEPA permit regulations at 40 CFR section 122.44 require that permits include conditions meeting applicable technology-based requirements at a minimum, and any more stringent effluent limitations necessary to meet applicable water quality standards. The discharge authorized by this Order must meet minimum federal technology-based requirements based on section 301 of the CWA established a required performance level--referred to as "secondary treatment"--that all POTWs were required to meet by July 1, 1977. Section 301(b)(1)(B) of the CWA required that EPA develop secondary treatment standards for POTWs, as defined in section 304(d)(1). Based on this statutory requirement, USEPA developed national secondary treatment regulations which are specified in 40 CFR part 133. These technology-based regulations apply to all POTWs and identify the minimum level of effluent quality to be attained by secondary treatment in terms of five-day biochemical oxygen demand (BOD), total suspended solids (TSS), and pH.

The CWA requires that TBELs be established based on several levels of controls:

- a. Best practicable treatment control technology (BPT) represents the average of the best existing performance by well-operated facilities within an industrial category or subcategory. BPT standards apply to toxic, conventional, and non-conventional pollutants.
- b. Best available technology economically achievable (BAT) represents the best existing performance of treatment technologies that are economically achievable within an industrial point source category. BAT standards apply to toxic and non-conventional pollutants.

- c. Best conventional pollutant control technology (BCT) represents the control from existing industrial point sources of conventional pollutants including BOD, TSS, fecal coliform, pH, and oil and grease. The BCT standard is established after considering a two-part reasonableness test. The first test compares the relationship between the costs of attaining a reduction in effluent discharge and the resulting benefits. The second test examines the cost and level of reduction of pollutants from the discharge from publicly owned treatment works to the cost and level of reduction of such pollutants from a class or category of industrial sources. Effluent limitations must be reasonable under both tests.
- d. New source performance standards (NSPS) represent the best available demonstrated control technology standards. The intent of NSPS guidelines is to set limitations that represent state-of-the-art treatment technology for new sources.

The CWA requires USEPA to develop effluent limitations, guidelines and standards (ELGs) representing application of BPT, BAT, BCT, and NSPS. Section 402(a)(1) of the CWA and 40 CFR section 125.3 authorize the use of best professional judgment (BPJ) to derive technology-based effluent limitations on a case-by-case basis where ELGs are not available for certain industrial categories and/or pollutants of concern. Where BPJ is used, the Regional Water Board must consider specific factors outlined in 40 CFR section 125.3.

2. Applicable TBELs

This Facility is subject to the technology-based regulations for the minimum level of effluent quality attainable by secondary treatment for BOD, TSS, and pH, as summarized in Table F-10. The 2008 Order established TBELs to meet applicable secondary treatment standards. These effluent limitations have been carried over from the previous Order to avoid backsliding. Mass-based effluent limitations, based on a flow rate capped at 9 mgd, are also included. The following table summarizes the TBELs for the discharge from the Facility:

Table F-10. Summary of TBELs

Parameter	Units	Effluent Limitations				
		Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum
BOD ₅ 20°C	mg/L	20	30	45	--	--
	lbs/day	2,340	3,500	5,260	--	--
TSS	mg/L	15	40	45	--	--
	lbs/day	1,750	4,670	5,250	--	--
pH	standard units	--	--	--	6.5	8.5
Removal Efficiency for BOD and TSS	percent	85	--	--	--	--

This Facility is also subject to TBELs contained in similar NPDES permits, for similar facilities, based on the treatment level achievable by tertiary-treated wastewater treatment systems. These effluent limitations are consistent with the State Water Board

precedential decision, State Water Board Order No. WQ 2004-0010 for the City of Woodland.

C. Water Quality-Based Effluent Limitations (WQBELs)

1. Scope and Authority

CWA Section 301(b) and 40 CFR section 122.44(d) require that permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards. This Order contains requirements, expressed as a technology equivalence requirement, more stringent than secondary treatment requirements that are necessary to meet applicable water quality standards. The rationale for these requirements, which consist of tertiary treatment or equivalent requirements or other provisions, is discussed in the section IV.C.2. of the Fact Sheet.

Section 122.44(d)(1)(i) of 40 CFR requires that permits include effluent limitations for all pollutants that are or may be discharged at levels that have the reasonable potential to cause or contribute to an exceedance of a water quality standard, including numeric and narrative objectives within a standard. Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, water quality-based effluent limitations (WQBELs) must be established using: (1) USEPA criteria guidance under CWA section 304(a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed state criterion or policy interpreting the state's narrative criterion, supplemented with other relevant information, as provided in section 122.44(d)(1)(vi).

The process for determining reasonable potential and calculating WQBELs when necessary is intended to protect the designated uses of the receiving water as specified in the Basin Plan, and achieve applicable water quality objectives and criteria that are contained in other state plans and policies, or any applicable water quality criteria contained in the CTR and NTR.

2. Applicable Beneficial Uses and Water Quality Criteria and Objectives

The Basin Plan designates beneficial uses, establishes WQOs, and contains implementation programs and policies to achieve those objectives for all waters addressed through the Basin Plan. The beneficial uses applicable to the Estuary are summarized in section III.C.1. of this Fact Sheet. The Basin Plan includes both narrative and numeric WQOs applicable to the receiving water as shown in the following discussions.

a. BOD and TSS

BOD is a measure of the quantity of the organic matter in the water and, therefore, the water's potential for becoming depleted in dissolved oxygen. As organic degradation takes place, bacteria and other decomposers use the oxygen in the water for respiration. Unless there is a steady supply of oxygen to the system, the water will quickly become depleted of oxygen. Adequate dissolved oxygen levels are required to support aquatic life. Depressions of dissolved oxygen can lead to anaerobic conditions resulting in odors, or, in extended durations of anaerobic conditions or higher magnitude cases, in fish kills.

The Ventura WRP permit provides tertiary treatment requirements, such as, the BOD₅20°C and TSS limits that are more stringent than secondary treatment

requirements, based on best professional judgment (BPJ). The Facility achieves solids removal that is better than secondary-treated wastewater by adding chemical coagulants to enhance the precipitation of solids, and by filtering the effluent.

Total Suspended Solids (TSS) is another measure of the quantity of organic and mineral matter in the water. High TSS usually accompanies high BOD, and together they quantify treatment plant performance in removing organic and inorganic solid waste.

40 CFR part 133 describes the minimum level of effluent quality attainable by secondary treatment for BOD and TSS as:

- i. The 30-day average shall not exceed 30 mg/L, and
- ii. The 7-day average shall not exceed 45 mg/L.

The limits in this Order are based on tertiary treatment and may be more stringent than secondary treatment requirements. The Facility achieves solids removal that is better than secondary-treated wastewater by adding chemical coagulants to enhance the precipitation of solids and by filtering the secondary effluent.

The monthly average, weekly average, and daily maximum limits cannot be removed because none of the anti-backsliding exceptions apply. Those limits were all included in the previous Order and the Discharger has been able to meet all three limits (monthly average, the weekly average, and the daily maximum) for both BOD and TSS.

In addition to having mass-based and concentration-based effluent limitations for BOD and TSS, the Facility also has a percent removal requirement for these two constituents. In accordance with 40 CFR parts 133.102(a)(3) and 133.102(b)(3), the 30-day (monthly) average percent removal shall not be less than 85 percent. Percent removal is defined as a percentage expression of the removal efficiency across a treatment plant for a given pollutant parameter, as determined from the 30-day (monthly) average values of the raw wastewater influent pollutant concentrations to the Facility and the 30-day (monthly) average values of the effluent pollutant concentrations for a given time period.

b. Settleable solids

Excessive deposition of sediments can destroy spawning habitat, blanket benthic (bottom dwelling) organisms and abrade the gills of larval fish. The limits for settleable solids are based on the Basin Plan (page 3-16) narrative, "*Waters shall not contain suspended or settleable material in concentrations that cause nuisance or adversely affect beneficial uses.*" The numeric limits are empirically based on results obtained from the settleable solids 1-hour test, using an Imhoff cone.

It is impracticable to use a 7-day average limitation, because short term spikes of settleable solid levels that would be permissible under a 7-day average scheme would not be adequately protective of all beneficial uses. The monthly average and the daily maximum limits cannot be removed because none of the anti-backsliding exceptions apply. The monthly average and daily maximum limits were both included in the previous Order and the Discharger has been able to meet both limits.

c. Oil and grease

Oil and grease are not readily soluble in water and form a film on the water surface. Oily films can coat birds and aquatic organisms, impacting respiration and thermal regulation, and causing death. Oil and grease can also cause nuisance conditions

(odors and taste), are aesthetically unpleasant, and can restrict a wide variety of beneficial uses. The limits for oil and grease are based on the Basin Plan (page 3-11) narrative, *"Waters shall not contain oils, greases, waxes, or other materials in concentrations that result in a visible film or coating on the surface of the water or on objects in the water, that cause nuisance, or that otherwise adversely affect beneficial uses."*

The numeric limits are empirically based on concentrations at which an oily sheen becomes visible in water. It is impracticable to use a 7-day average limitation, because spikes that occur under a 7-day average scheme could cause visible oil sheen. A 7-day average scheme would not be sufficiently protective of beneficial uses. The monthly average and the daily maximum limits cannot be removed because none of the anti-backsliding exceptions apply. Both limits were included in the previous Order and the Discharger has been able to meet both limits.

d. Residual chlorine

Disinfection of wastewaters with chlorine produces chlorine residual. Chlorine and its reaction products are toxic to aquatic life. The limit for residual chlorine is based on the Basin Plan (page 3-9) narrative, *"Chlorine residual shall not be present in surface water discharges at concentrations that exceed 0.1 mg/L and shall not persist in receiving waters at any concentration that causes impairment of beneficial uses"*.

It is impracticable to use a 7-day average or a 30-day (monthly) average limitation, because it is not as protective as of beneficial uses as a daily maximum limitation is. Chlorine is very toxic to aquatic life and short-term exposures of chlorine may cause fish kills. The Discharger has been able to maintain a chlorine residual at or below 0.1 mg/L.

e. TDS, Chloride, Sulfate, and Boron

Page 3-12 of the Basin Plan indicates that there are no water body specific objectives in surface water for Total Dissolved Solids (TDS), chloride, sulfate and boron.

The perched and unconfined Oxnard Plain Forebay Groundwater Basin, which is continuous with freshwater underlying the discharge location, has limits on page 3-19 for TDS (3,000 mg/L), sulfate (1,000 mg/L), and chloride (500 mg/L).

Upon completion of the special study, groundwater limits may be reconsidered as described in the reopener provision at VI. C. I..

f. Methylene Blue Activated Substances (MBAS)

The MBAS procedure tests for the presence of anionic surfactants (detergents) in surface and ground waters. Surfactants disturb the water surface tension, which affects insects and can affect gills in aquatic life. The MBAS can also impart an unpleasant soapy taste to water, as well as cause scum and foaming in waters, which impact the aesthetic quality of both surface and ground waters.

Given the nature of the Facility (a POTW) which accepts domestic wash water into the sewer system and treatment plant and the characteristics of the wastes discharged, the discharge has reasonable potential to exceed both the numeric MBAS WQO and the narrative WQO for prohibition of floating material such as foams and scums. Therefore, an effluent limitation is required based upon BPJ.

g. Nitrogen Compounds/Nutrient Compounds

- i. **Total inorganic nitrogen ($\text{NO}_2 + \text{NO}_3$ as N).** Total inorganic nitrogen is the sum of nitrate-nitrogen and nitrite-nitrogen. High nitrate levels in drinking water can cause health problems in humans. Infants are particularly sensitive and can develop methemoglobinemia (blue-baby syndrome). Nitrogen is also considered a nutrient. Excessive amounts of nutrients can lead to other water quality impairments, ex. algae¹¹. Although there are no water body specific objectives for nitrogen in the Estuary, based on the Discharger's effluent data from their SMRs, the discharge has reasonable potential to exceed the numeric nitrate plus nitrite as nitrogen 10 mg/L WQO, specified in the Basin Plan page 3-11. An effluent limitation is required based upon BPJ. The Discharger will have to meet the 10 mg/L WQO at the end-of-pipe, since no dilution credit is appropriate at this time. This effluent limitation in concentration shall not apply when the concentration limitation for nitrate as nitrogen or nitrite as nitrogen is exceeded. This effluent limitation in mass load shall not apply when the mass load limitation for nitrate as nitrogen or nitrite as nitrogen is exceeded.
- ii. **Nitrite as Nitrogen.** Chapter 3 of the Basin Plan (page 3-11) contains the following WQO, "*Waters shall not exceed the 10 mg/L nitrogen as nitrate-nitrogen plus nitrite-nitrogen ($\text{NO}_3\text{-N} + \text{NO}_2\text{-N}$), 45 mg/L as nitrate (NO_3), 10 mg/L as nitrate-nitrogen ($\text{NO}_3\text{-N}$), or 1 mg/L as nitrite-nitrogen ($\text{NO}_2\text{-N}$) or as otherwise designated in Table 3-8.*" The nitrite effluent concentrations may exceed the 1 mg/L WQO specified in the Basin Plan based on the BPJ, if the oxidation process of the nitrification is not complete. A final nitrite limitation of 1 mg/L has been added to this Order.
- iii. **Ammonia.** Ammonia is a pollutant routinely found in the wastewater effluent of POTWs, in landfill-leachate, as well as in run-off from agricultural fields where commercial fertilizers and animal manure are applied. Ammonia exists in two forms – un-ionized ammonia (NH_3) and the ammonium ion (NH_4^+). They are both toxic, but the neutral, un-ionized ammonia species is much more toxic, because it is able to diffuse across the epithelial membranes of aquatic organisms much more readily than the charged ammonium ion. The form of ammonia is primarily a function of pH, but it is also affected by temperature and other factors. Additional impacts can also occur as the oxidation of ammonia lowers the dissolved oxygen content of the water, further stressing aquatic organisms. Oxidation of ammonia to nitrate may lead to groundwater impacts in areas of recharge. Ammonia also combines with chlorine (often both are present in POTW treated effluent discharges) to form chloramines – persistent toxic compounds that extend the effects of ammonia and chlorine downstream.

The Facility began operating a NDN system on October 3, 2011. The Facility's performance related to removing ammonia has improved since 2008. TSOs were issued in 2012 and 2013 because ammonia add-back after chlorination elevated ammonia in the effluent and the City needs additional time to consider and implement a long-term solution.

An RPA was conducted for total ammonia, as NH_3 plus NH_4^+ , using the Discharger's effluent data from their SMRs since October 2011. The RPA

¹¹ Excessive growth of algae and/or other aquatic plants can degrade water quality. Algal blooms sometimes occur naturally, but they are often the result of excess nutrients (i.e., nitrogen, phosphorus) from waste discharges or nonpoint sources. These algal blooms can lead to problems with tastes, odors, color, and increased turbidity and can depress the dissolved oxygen content of the water, leading to fish kills. Floating algal scum and algal mats are also an aesthetically unpleasant nuisance.

compares the effluent data with the Basin Plan WQOs. Because the Discharger's effluent has exceeded the WQOs for total ammonia between 2011 and 2013, there is reasonable potential that the discharge will cause or contribute to an exceedance of the Basin Plan WQOs. Consistent with 40 CFR part 122.44(d), this Order contains numeric effluent limitations for total ammonia.

Ammonia saltwater criteria are salinity, pH, and temperature dependent. Because there are such wide salinity, pH, and temperature fluctuations in the Estuary during summer (from May to October) and winter (from November to April) months, seasonal total ammonia effluent limitations provide better protection of living organisms in the Estuary and have been included in this Order.

- (1) The Regional Water Board has adopted NPDES permits recently using an approach for calculating both the end-of-pipe limitations for ammonia, as well as receiving water limitations that address site-specific characteristics of effluent, as well as the receiving water. The procedures for calculating the total ammonia effluent limitations are based on the 2002 and 2004 Ammonia Amendments are as follows. For complete calculations see Attachment M.

One-Hour Average Total Ammonia Saltwater Quality Objective as Criteria Maximum Concentration (CMC)

Total Ammonia =

$$0.233 + 0.233 \times 10^{(pK_a^S + 0.0324 \times (298 - T) + 0.0415 \times P/T - pH)}$$

Where: P = 1 atm, T = temperature ($^{\circ}$ K) and $pK_a^S = 0.116 * I + 9.245$, the stoichiometric acid hydrolysis constant of ammonium ions in saltwater on I, I = $19.9273 * S(1000 - 1.005109S)^{-1}$, the molar ionic strength of saltwater based on S where S = salinity

The following represents the 90th percentile of receiving water data after implementation of NDN treatment at R-004 for summer and winter :

- (i) pH: 7.64 (summer) and 8.57 (winter)
- (ii) temperature: 22.95 $^{\circ}$ C (summer) and 18.597 $^{\circ}$ C (winter)
- (iii) Salinity: 1.54 (summer) and 2.09 (winter)
- (iv) I: 0.03 (summer) and .0416 (winter)
- (v) pK_a^S : 9.25 (summer) and 9.249 (winter)

Use of the 90th percentile pH, temperature, and salinity to set effluent limitations is appropriate because of the shorter time scale of the one-hour average. It is conservative, because it is overprotective 90 percent of the time. Using the values in the formula above, the resulting One-hour

Average Objective is equal to 11.41 mg/L for the summer months and 2.05 mg/L for the winter months.

Four-Day Average Total Ammonia Saltwater Quality Objective as Criteria Continuous Concentration (CCC)

Total Ammonia =

$$0.035 + 0.035 \times 10^{(pK_a^S + 0.0324 \times (298 - T) + 0.0415 \times P/T - pH)}$$

Where: P = 1 atm, T = temperature ($^{\circ}$ K) and $pK_a^S = 0.116 \cdot I + 9.245$, the stoichiometric acid hydrolysis constant of ammonium ions in saltwater on I, $I = 19.9273 \cdot S(1000 - 1.005109S)^{-1}$, the molar ionic strength of saltwater based on S, where S = salinity.

The following represents the 50th percentile of receiving water data collected after implementation of NDN at R-004:

- (i) pH: 7.49 (summer) and 7.91 (winter)
- (ii) temperature: 21.21 $^{\circ}$ C (summer) and 14.97 $^{\circ}$ C (winter)
- (iii) Salinity: 1.3 (summer) and 1.65 (winter)
- (iv) I: .0259 (summer) and .033 (winter)
- (v) pK_a^S : 9.248 (summer) and 9.25 (winter)

Use of the 50th percentile pH, temperature, and salinity is appropriate to set the 4-day average objective, because the 4-day average represents more long-term conditions. Using the values in the formula above, the resulting 4-day Average Objective is equal to 2.7 mg/L for the summer period and 1.649 mg/L for the winter period.

Translation of Total Ammonia Saltwater WQOs into Effluent Limitations

In order to translate the WQOs for total ammonia as described in the preceding discussions into effluent limitations, the Implementation Provisions of the 2002 Basin Plan Amendment, section 5 – *Translation of Objectives into Effluent Limits*, was followed and is discussed below. This method is similar to the method contained the SIP. The method is also consistent with that outlined in the USEPA *Technical Support Document for Water Quality-based Toxics Control* (1991) (TSD).

- (i) Identify applicable water quality criteria: Four-day Average = 2.7 mg/L (summer) and 1.65 mg/L (winter) and One-hour Average = 11.41 mg/L (summer) and 2.05 mg/L (winter)
- (ii) For each WQO, calculate the effluent concentration allowance (ECA) using the steady-state mass balance model. Since mixing has not

been allowed by the Regional Water Board, this equation applies:
 $ECA = WQO$

(iii) Determine the Long-Term Average discharge condition (LTA) by multiplying each ECA with a factor (multiplier) that adjusts for variability. The coefficient of variation (CV) is calculated from the weekly averages of the effluent ammonia at all monitoring stations by summer and winter as documented in Attachment K-6.

(iv) Find ECA multiplier, when CV = .2597 (summer) and .23989 (winter):
 $ECA \text{ multiplier}_{4\text{-day}99} = 0.74646$ (summer) and 0.762728 (winter) and
 $ECA \text{ multiplier}_{1\text{-hour}99} = 0.570296$ (summer) and 0.593201 (winter)

(v) Use the LTA equations:

$$LTA_{4\text{-day}99} = ECA_{4\text{-day}} \times ECA \text{ multiplier}_{4\text{-day}99} = 2.7 \text{ mg/L} \times 0.747 = 2.01 \text{ mg/L (summer)}$$

$$LTA_{1\text{-hour}99} = ECA_{1\text{-hour}} \times ECA \text{ multiplier}_{1\text{-hour}99} = 11.41 \text{ mg/L} \times 0.570 = 6.50 \text{ mg/L (summer)}$$

$$LTA_{4\text{-day}99} = ECA_{4\text{-day}} \times ECA \text{ multiplier}_{4\text{-day}99} = 1.65 \text{ mg/L} \times 0.763 = 1.26 \text{ mg/L (winter)}$$

$$LTA_{1\text{-hour}99} = ECA_{1\text{-hour}} \times ECA \text{ multiplier}_{1\text{-hour}99} = 2.05 \text{ mg/L} \times 0.593 = 1.216 \text{ mg/L (winter)}$$

(vi) Select the (most limiting) of the LTAs derived in Step 3:
 $(LTA_{\min})LTA_{\min} = 2.012323 \text{ mg/L (summer)}$ and $LTA_{\min} = 1.21589 \text{ mg/L (winter)}$

(vii) Calculate maximum daily effluent limit (MDEL) and average monthly effluent limit (AMEL) by multiplying $LTA_{\min}$ as selected in Step 4 with a factor (multiplier).

The CV is calculated from the effluent, as above = .2597 (summer) and .23989 (winter).

$AMEL \text{ multiplier}_{95} = 1.22672$ (summer) and 1.208601 (winter)
 $MDEL \text{ multiplier}_{99} = 1.753475$ (summer) and 1.685768 (winter)

$$AMEL = LTA_{\min} \times AMEL \text{ multiplier}_{95} = 2.012323 \text{ mg/L} \times 1.226728 = 2.46857 \approx 2.47 \text{ mg/L (summer)}$$

$$MDEL = LTA_{\min} \times MDEL \text{ multiplier}_{99} = 2.012323 \text{ mg/L} \times 1.753475 = 3.528559 \approx 3.53 \text{ mg/L (summer)}$$

$$AMEL = LTA_{\min} \times AMEL \text{ multiplier}_{95} = 1.21589 \text{ mg/L} \times 1.2080601 = 1.469525 \approx 1.47 \text{ mg/L (winter)}$$

$$MDEL = LTA_{\min} \times MDEL \text{ multiplier}_{99} = 1.21589 \text{ mg/L} \times 1.685768 = 2.049708 \approx 2.05 \text{ mg/L (winter)}$$

Table F-11. WQBEL Total Ammonia Effluent Limitations

Constituent	AMEL (mg/L)	MDEL (mg/L)
Total Ammonia (summer months)	2.47	3.53
Total Ammonia (winter months)	1.47	2.05

(2) Performance Total Ammonia Effluent Limitations

The 2008 Order had more restrictive total ammonia limits (Summer AMEL 0.045 and Winter AMEL .079 mg/L) than those calculated above (Summer AMEL 2.47 and Winter AMEL 1.47 mg/L) fusing 2011-2013 receiving water data and current procedures for calculating ammonia effluent limits. Because this Order includes Estuary and Nutrient, Dissolved Oxygen and Toxicity studies, effluent limits for total ammonia in the Order are based on performance. The procedures for calculating the ammonia nitrogen effluent limitation based on the Basin Plan amendments is determined by calculating the 95% and 99% total ammonia effluent concentration for winter and summer conditions using Minitab software.

Table F-12a. Total Ammonia Effluent Performance

Constituent	AMEL (mg/L) (95% probability)	MDEL (mg/L) (99% probability)
Total Ammonia (summer months)	1.07	1.17
Total Ammonia (winter months)	1.3	1.4

(3) Combined WQBEL and Performance Total Ammonia Effluent Limits

The Performance Based Effluent Limits are more stringent than the WQBELs for every calculated limit. For example, the AMEL for total ammonia in the winter (1.3 mg/L) in the performance ammonia limit is lower than the WQBEL calculated for winter (1.47 mg/L). The most protective of the limits is applied to create the final ammonia effluent limits, with an AMEL in the winter of 1.47 as shown in Table F-12b.

Table F-12b. Final Effluent Ammonia Limits

Constituent	AMEL (mg/L) (95% probability)	MDEL (mg/L) (99% probability)
Total Ammonia (summer months)	1.07	1.17
Total Ammonia (winter months)	1.3	1.4

A reopener provision, described at VI.C.I., allows the Order and the ammonia limits to be reconsidered upon completion of the Nutrient, Dissolved Oxygen and Toxicity Study.

h. Coliform/Bacteria

Total coliform, fecal coliform, and enterococcus bacteria are used to indicate the likelihood of pathogenic bacteria in surface waters. Given the nature of the Facility, a wastewater treatment plant, pathogens are likely to be present in the effluent in cases where the disinfection process is not operating adequately. As such, this Order contains the following limitations:

i. Effluent Limitations:

- (a) The 7-day median number of total coliform at some point in the treatment process must not exceed 2.2 Most Probable Number (MPN) per 100 milliliters;
- (b) The number of total coliform must not exceed 23 MPN per 100 milliliters in more than one sample within any 30-day period; and,
- (c) No sample shall exceed an MPN of 240 total coliform bacteria per 100 milliliters.

These disinfection-based effluent limitations for coliform are for human health protection and are consistent with disinfected tertiary recycled water requirements (CCRs title 22, chapter 3, section 60301.230) established by the CDPH. On July 8, 2010, the Regional Water Board adopted Resolution No. R-0010-006, the Santa Clara River Bacteria TMDL, also enacting that the 7-day median number of total coliform at some point in the treatment process must not exceed 2.2 MPN per 100 milliliters.

These limits for total coliform must be met at the point of the treatment train immediately following disinfection, as a measure of the effectiveness of the disinfection process.

ii. Receiving Water Limitation

- (a) Geometric Mean Limits: Total coliform density shall not exceed 1,000/100 mL; Fecal coliform density shall not exceed 200 per 100 mL; Enterococcus density shall not exceed 35 per 100 mL.
- (b) Single Sample Limits: Total coliform density shall not exceed 10,000 per 100 mL; Fecal coliform density shall not exceed 400 per 100 mL; Enterococcus density shall not exceed 104 per 100 mL; and, Total coliform density shall not exceed 1,000 per 100 mL, if the ratio of fecal-to-total coliform exceeds 0.1.

i. pH

The hydrogen ion activity of water (pH) is measured on a logarithmic scale, ranging from 0 to 14. While the pH of "pure" water at 25°C is 7.0, the pH of natural waters is usually slightly basic due to the solubility of carbon dioxide from the atmosphere. Minor changes from natural conditions can harm aquatic life. In accordance with 40 CFR part 133.102(c), the effluent values for pH shall be maintained within the limits of 6.0 to 9.0 unless the POTW demonstrates that (1) inorganic chemicals are not added to the waste stream as part of the treatment process; and (2) contributions from industrial sources do not cause the pH of the effluent to be less than 6.0 or greater than 9.0. The effluent limitation for pH in this Order requiring that the wastes

discharged shall at all times be within the range of 6.5 to 8.5 is taken from the Basin Plan (page 3-15) which reads *"the pH of inland surface waters shall not be depressed below 6.5 or raised above 8.5 as a result of waste discharge."*

j. Temperature

USEPA document, Quality Criteria for Water 1986 [EPA 440/5-86-001, May 1, 1986], also referred to as the Gold Book, discusses temperature and its effects on beneficial uses, such as recreation and aquatic life.

- i. The Federal Water Pollution Control Administration in 1967 called temperature "a catalyst, a depressant, an activator, a restrictor, a stimulator, a controller, a killer, and one of the most important water quality characteristics to life in water." The suitability of water for total body immersion is greatly affected by temperature. Depending on the amount of activity by the swimmer, comfortable temperatures range from 20°C to 30°C (68 °F to 86 °F).
- ii. Temperature also affects the self-purification phenomenon in water bodies and therefore the aesthetic and sanitary qualities that exist. Increased temperatures accelerate the biodegradation of organic material both in the overlying water and in bottom deposits which makes increased demands on the dissolved oxygen resources of a given system. The typical situation is exacerbated by the fact that oxygen becomes less soluble as water temperature increases. Thus, greater demands are exerted on an increasingly scarce resource which may lead to total oxygen depletion and obnoxious septic conditions. Increased temperature may increase the odor of water because of the increased volatility of odor-causing compounds. Odor problems associated with plankton may also be aggravated.
- iii. Temperature changes in water bodies can alter the existing aquatic community. Coutant (1972) has reviewed the effects of temperature on aquatic life reproduction and development. Reproductive elements are noted as perhaps the most thermally restricted of all life phases, assuming other factors are at or near optimum levels. Natural short-term temperature fluctuations appear to cause reduced reproduction of fish and invertebrates.

The Basin Plan lists temperature requirements for the receiving waters. Based on the requirements of the Basin Plan and a white paper developed by Regional Water Board staff entitled *Temperature and Dissolved Oxygen Impacts on Biota in Tidal Estuaries and Enclosed Bays in the Los Angeles Region*, a maximum effluent temperature limitation of 86 °F is included in the Order. The white paper evaluated the optimum temperatures for steelhead, topmelt, ghost shrimp, brown rock crab, jackknife clam, and blue mussel. The new temperature effluent limitation is reflective of new information available that indicates that the 100°F temperature is not protective of aquatic organisms. A survey was completed for several kinds of fish and the 86°F temperature was found to be protective. It is impracticable to use a 7-day average or a 30-day average limitation for temperature, because it is not as protective as of beneficial uses as a daily maximum limitation is. A daily maximum limitation is necessary to protect aquatic life and is consistent with the fishable/swimmable goals of the CWA. Section IV.A.2.b. of the Order contains the following effluent limitation for temperature: *"The temperature of wastes discharged shall not exceed 86°F, except as a result of external ambient temperature."*

The above effluent limitation for temperature has been quoted in all recent NPDES permits adopted by this Regional Water Board.

Section V.A.1. of the Order explains how compliance with the receiving water temperature limitation will be determined.

k. Turbidity

Turbidity is an expression of the optical property that causes light to be scattered in water due to particulate matter such as clay, silt, organic matter, and microscopic organisms. Turbidity can result in a variety of water quality impairments. The effluent limitation for turbidity which reads, *"For the protection of the water contact recreation beneficial use, the wastes discharged to water courses shall have received adequate treatment, so that the turbidity of the wastewater does not exceed: (a) a daily average of 2 Nephelometric turbidity units (NTUs); and (b) 5 NTUs more than 5 percent of the time (72 minutes) during any 24 hour period,"* is based on the Basin Plan (page 3-17) and Section 60301.320 of title 22, chapter 3, "Filtered Wastewater" of the CCRs.

l. Radioactivity

Radioactive substances are generally present in natural waters in extremely low concentrations. Mining or industrial activities increase the amount of radioactive substances in waters to levels that are harmful to aquatic life, wildlife, or humans. The existing effluent limitation for radioactivity which reads, *"Radionuclides shall not be present in concentrations that are deleterious to human, plant, animal, or aquatic life or that result in accumulation of radionuclides in the food web to an extent that presents a hazard to human, plant, animal, or aquatic life"* is based on the radioactive substances specified in the Basin Plan in order to protect the surface water beneficial use and human health.

m. Biostimulatory Substances

Biostimulatory substances include excess nutrients (nitrogen and phosphorus) and other compounds that stimulate aquatic growth. In addition to being aesthetical unpleasant (causing taste, odor, or color problems), this excessive growth can also cause other water quality problems. The limits for biostimulatory substances are based on the Basin Plan (page 3-8) narrative, *"Waters shall not contain biostimulatory substances in concentrations that promote aquatic growth to the extent that such growth causes nuisance or adversely affects beneficial uses."*

n. CTR and SIP

The CTR and the SIP specify numeric objectives for toxic substances and the procedures whereby these objectives are to be implemented. The procedures include those used to conduct an RPA to determine the need for effluent limitations for priority and non-priority pollutants.

3. Determining the Need for WQBELs

Pursuant to 40 CFR part 122.45(d)(2), for POTW continuous discharges, all permit effluent limitations, standards, and prohibitions, including those necessary to achieve water quality standards, shall, unless impracticable, be stated as average weekly and average monthly discharge limitations (AWEL and AMEL, respectively). It is impracticable to include only average weekly and average monthly effluent limitations in the permit, because a single daily discharge of certain pollutants, in excess amounts, can cause violations of WQOs. The effects of pollutants on aquatic organisms are often rapid. For many pollutants, an average weekly or average monthly effluent limitation

alone is not sufficiently protective of beneficial uses. As a result, Maximum Daily Effluent Limits (MDELs, as referenced in 40 CFR part 122.45(d)(1)), are included in the permit for certain constituents.

Priority pollutant water quality criteria in the CTR are applicable to the Estuary. The CTR contains both saltwater and freshwater criteria. Because a distinct separation generally does not exist between freshwater and saltwater aquatic communities, the following apply in accordance with 40 CFR part 131.38(c)(3): freshwater criteria apply at salinities of 1 part per thousand (ppt) and below at locations where this condition occurs 95 percent or more of the time; saltwater criteria apply at salinities of 10 ppt and above at locations where this occurs 95 percent or more of the time; and at salinities between 1 and 10 ppt the more stringent of the two apply. The CTR criteria for freshwater or human health for consumption of organisms, whichever is most stringent, are used to prescribe the effluent limitations in the tentative Order to protect the beneficial uses of the Estuary.

Some water quality criteria are hardness dependent. The Discharger provided hardness data collected from the Estuary. The receiving water hardness values ranged from 229 to 5270 mg/L after 2009. Since most of the receiving water hardness data (776 of 780 data points) and their average hardness as CaCO₃ (920 mg/L) are greater than 400 mg/L as CaCO₃, in accordance with the SIP/CTR procedures, the 400 mg/L hardness cap will be used in calculating metals criteria for evaluation of reasonable potential.

In accordance with Section 1.3 of the SIP, the Regional Water Board conducted an RPA for each priority pollutant with an applicable criterion or objective to determine if a WQBEL is required in the permit. The Regional Water Board analyzed effluent data to determine if a pollutant in a discharge has a reasonable potential to cause or contribute to an excursion above a state water quality standard. For all parameters that demonstrate reasonable potential, numeric WQBELs are required. The RPA considers water quality criteria from the CTR and NTR, and when applicable, WQOs specified in the Basin Plan. To conduct the RPA, the Regional Water Board staff identified the maximum effluent concentration (MEC) and maximum background concentration in the receiving water for each constituent, based on data provided by the Discharger. The monitoring data used are the available data from the 2008 Order.

Section 1.3 of the SIP provides the procedures for determining reasonable potential to exceed applicable water quality criteria and objectives. The SIP specifies three triggers to complete a RPA:

- a. Trigger 1 – If the MEC is greater than or equal to the CTR water quality criteria or applicable objective (C), a limitation is needed.
- b. Trigger 2 – If background water quality (B) > C and the pollutant is detected in the effluent, a limitation is needed.
- c. Trigger 3 – If other related information such as CWA 303(d) listing for a pollutant, discharge type, or compliance history, then BPJ is used to determine that a limit is needed.

Sufficient effluent and ambient data are needed to conduct a complete RPA. If data are not sufficient, the Discharger will be required to gather the appropriate data for the Regional Water Board to conduct the RPA. Upon review of the data, and if the Regional Water Board determines that WQBELs are needed to protect the beneficial uses, the NPDES permit will be reopened for appropriate modification.

At least 10 data sets for effluent and receiving water monitoring for the period from March 2008 through April 2013 were available. In addition, samples for certain priority

pollutants were collected as required by the existing Monitoring and Reporting Program (MRP; Attachment E); these data were also used to complete the RPA.

The RPA was performed for the priority pollutants regulated in the CTR for which data are available. Based on the RPA, pollutants that demonstrate reasonable potential are copper, selenium, lead and nickel. Refer to Attachments K, M and N for a summary of the RPA and associated effluent limitation calculations. The following table summarizes the RPA.

Table F-13. Summary of RPA

CTR No.	Constituent	Water Quality Criteria (C) µg/L	Max Effluent Conc. (MEC) µg/L	Max Receiving Water Conc.(B) µg/L	RPA Result - Need Limit?	Reason
1	Antimony	4300	<1	<1	No	B<C, MEC<C
2	Arsenic	36	<2	7.7	No	B<C, MEC<C
3	Beryllium	Narrative	<2	<.2	No	No criteria
4	Cadmium	9.4	<4	<4.0	No	B<C, MEC<C
5a	Chromium III	N.A.	<7	<9.05	No	No criteria
5b	Chromium VI	11.4	<1	-	No	MEC<C
6	Copper	6.86	8.52	4.97	Yes	B<C, MEC>C
7	Lead	8.5	9.4	7.63	Yes	MEC>C
8	Mercury	0.051	<0.02	<2	No	MEC<C
9	Nickel	10.2	35.4	50.2	Yes	B>C, MEC>C
10	Selenium	5	13.7	8.07	Yes	B>C, MEC>C
11	Silver	2.2	<0.2	<.2	No	MEC<C, B<C
12	Thallium	6.3	<1	<1.0	No	MEC<C, B<C
13	Zinc	96	68.8	32.5	No	C>MEC, B<C
14	Cyanide	1	<5	<5	No	B<C, MEC<C
15	Asbestos	NA	NA	-	No	NA
16	2,3,7,8-TCDD (Dioxin)	1.4×10^{-08}	$<1 \times 10^{-05}$	-	No	All ND in Effluent
17	Acrolein	780	<0.6	<0.6	No	All ND in Effluent
18	Acrylonitrile	0.66	<0.5	<0.5	No	All ND in Effluent
19	Benzene	71	<0.2	<0.20	No	All ND in Effluent
20	Bromoform	3600	4.3	2.96	No	MEC<C, B<C
21	Carbon Tetrachloride	4.4	<0.12	<0.12	No	All ND in Effluent
22	Chlorobenzene	21,000	<0.2	<0.20	No	All ND in Effluent
23	Dibromochloromethane ¹⁸	34	14.5	3.63	No	MEC<C, B<C
24	Chloroethane	No criteria	<0.52	<0.52	No	No criteria
25	2-chloroethyl vinyl ether	No criteria	<0.13	<0.13	No	No criteria
26	Chloroform	No criteria	36	10.9	No	No criteria
27	Dichlorobromomethane ¹⁸	46	25.6	4.89	No	MEC<C, B<C
28	1,1-dichloroethane	5	<0.07	<0.07	No	MEC<C, B<C
29	1,2-dichloroethane	99	<0.03	<0.03	No	All ND in Effluent
30	1,1-dichloroethylene	3.2	<0.13	<0.13	No	All ND in Effluent
31	1,2-dichloropropane	39	<0.04	<0.34	No	All ND in Effluent
32	1,3-dichloropropylene	1,700	<0.68	<0.34	No	All ND in Effluent
33	Ethylbenzene	29,000	<0.34	<0.34	No	All ND in Effluent

CTR No.	Constituent	Water Quality Criteria (C) µg/L	Max Effluent Conc. (MEC) µg/L	Max Receiving Water Conc.(B) µg/L	RPA Result - Need Limit?	Reason
34	Methyl bromide	4,000	<0.5	<0.50	No	All ND in Effluent
35	Methyl chloride	No criteria	<0.08	<0.08	No	No criteria
36	Methylene chloride	1,600	<0.25	<0.25	No	All ND in Effluent
37	1,1,2,2-tetrachloroethane	11	<0.03	<0.03	No	All ND in Effluent
38	Tetrachloroethylene	8.85	<0.03	<0.03	No	All ND in Effluent
39	Toluene	200,000	<0.2	1.3	No	All ND in Effluent
40	Trans 1,2-Dichloroethylene	140,000	<0.1	<0.1	No	All ND in Effluent
41	1,1,1-Trichloroethane	No criteria	<0.03	<0.03	No	All ND in Effluent
42	1,1,2-Trichloroethane	42	<0.02	<0.02	No	All ND in Effluent
43	Trichloroethylene	81	<0.12	<0.12	No	All ND in Effluent
44	Vinyl Chloride	525	<0.18	<0.18	No	All ND in Effluent
45	2-chlorophenol	400	<3.3	<3.3	No	All ND in Effluent
46	2,4-dichlorophenol	790	<2.7	<2.7	No	All ND in Effluent
47	2,4-dimethylphenol	2,300	<2.7	<2.7	No	All ND in Effluent
48	2-methyl-4,6-dinitrophenol	765	<10	<10	No	All ND in Effluent
49	2,4-dinitrophenol	14,000	<42	<42	No	All ND in Effluent
50	2-nitrophenol	No criteria	<3.6	<3.6	No	No criteria
51	4-nitrophenol	No criteria	<2.4	<2.4	No	No criteria
52	3-Methyl-4-Chlorophenol	No criteria	<3	<3.0	No	No criteria
53	Pentachlorophenol	7.9	<3.6	<3.6	No	All ND in Effluent
54	Phenol	4,600,000	<1.5	<1.5	No	All ND in Effluent
55	2,4,6-trichlorophenol	6.5	<2.7	<2.7	No	All ND in Effluent
56	Acenaphthene	2,700	<2	<2.0	No	All ND in Effluent
57	Acenaphthylene	No criteria	<1.9	<1.9	No	No criteria
58	Anthracene	110,000	<1.9	<1.9	No	All ND in Effluent
59	Benidine	0.00054	<4.4	<4.4	No	All ND in Effluent
60	Benzo(a)Anthracene	0.049	<7.8	<7.8	No	All ND in Effluent
61	Benzo(a)Pyrene	0.049	<2.5	<2.5	No	All ND in Effluent
62	Benzo(b)Fluoranthene	0.049	<4.8	<4.8	No	All ND in Effluent
63	Benzo(ghi)Perylene	No criteria	<4.1	<4.1	No	No criteria
64	Benzo(k)Fluoranthene	0.049	<2.5	<2.5	No	All ND in Effluent
65	Bis(2-Chloroethoxy) methane	No criteria	<5.3	<5.3	No	No criteria
66	Bis(2-Chloroethyl)Ether	1.4	<5.7	<5.7	No	All ND in Effluent
67	Bis(2-Chloroisopropyl) Ether	170,000	<5.7	<5.7	No	All ND in Effluent
68	Bis(2-Ethylhexyl)Phthalate ¹⁸	5.9	<2.5	7.1	No	All ND in Effluent
69	4-Bromophenyl Phenyl Ether	No criteria	<1.9	<1.9	No	No criteria
70	Butylbenzyl Phthalate	5,200	<2.5	<2.5	No	All ND in Effluent
71	2-Chloronaphthalene	4,300	<1.9	<1.9	No	All ND in Effluent
72	4-Chlorophenyl Phenyl	No criteria	<4.2	<4.2	No	No criteria

CTR No.	Constituent	Water Quality Criteria (C) µg/L	Max Effluent Conc. (MEC) µg/L	Max Receiving Water Conc.(B) µg/L	RPA Result - Need Limit?	Reason
	Ether					
73	Chrysene	0.049	<2.0	<2.0	No	All ND in Effluent
74	Dibenzo(a,h) Anthracene	0.049	<2.5	<2.5	No	All ND in Effluent
75	1,2-Dichlorobenzene	17,000	<1.9	<1.9	No	All ND in Effluent
76	1,3-Dichlorobenzene	2,600	<2.5	<1.9	No	All ND in Effluent
77	1,4-Dichlorobenzene	2,600	<4.4	<4.4	No	All ND in Effluent
78	3-3'-Dichlorobenzidine	0.077	<16.5	<16.5	No	All ND in Effluent
79	Diethyl Phthalate	120,000	<2.2	<2.2	No	All ND in Effluent
80	Dimethyl Phthalate	2,900,000	<1.6	<1.6	No	All ND in Effluent
81	Di-n-Butyl Phthalate	12,000	<2.5	<2.5	No	All ND in Effluent
82	2-4-Dinitrotoluene	9.1	<5.7	<5.7	No	All ND in Effluent
83	2-6-Dinitrotoluene	No criteria	<1.9	<1.9	No	No criteria
84	Di-n-Octyl Phthalate	No criteria	<2.5	<2.5	No	No criteria
85	1,2-Diphenylhydrazine	0.54	<10	<10	No	All ND in Effluent
86	Fluoranthene	370	<2.2	<2.2	No	All ND in Effluent
87	Fluorene	14,000	<1.9	<1.9	No	All ND in Effluent
88	Hexachlorobenzene	0.00077	<1.9	<1.9	No	All ND in Effluent
89	Hexachlorobutadiene	50	<0.9	<0.9	No	All ND in Effluent
90	Hexachlorocyclopentadiene	17,000	<1.9	<1.9	No	All ND in Effluent
91	Hexachloroethane	8.9	<1.6	<1.6	No	All ND in Effluent
92	Indeno(1,2,3-cd)Pyrene	0.049	<3.7	<3.7	No	All ND in Effluent
93	Isophorone	600	<2.2	<2.2	No	All ND in Effluent
94	Naphthalene	No criteria	<10	<10	No	No criteria
95	Nitrobenzene	1,900	<1.9	<1.9	No	All ND in Effluent
96	N-Nitrosodimethylamine	8.1	<0.15	<1.5	No	All ND in Effluent
97	N-Nitrosodi-n-Propylamine	1.4	<10	<10	No	All ND in Effluent
98	N-Nitrosodiphenylamine	16	<10	<10	No	All ND in Effluent
99	Phenanthrene	No criteria	<5.4	<5.4	No	No criteria
100	Pyrene	11,000	<1.9	<1.9	No	All ND in Effluent
101	1,2,4-Trichlorobenzene	No criteria	<1.9	<1.9	No	No criteria
102	Aldrin	0.00014	<0.004	<0.004	No	All ND in Effluent
103	Alpha-BHC	0.013	<0.003	<0.003	No	All ND in Effluent
104	Beta-BHC	0.046	<0.006	<0.006	No	All ND in Effluent
105	Gamma-BHC (Lindane)	0.063	<0.004	<0.004	No	All ND in Effluent
106	Delta-BHC	No criteria	<0.009	<0.009	No	No criteria
107	Chlordane	0.00059	<0.014	<0.014	No	All ND in Effluent
108	4,4'-DDT	0.00059	<4.7	<4.7	No	All ND in Effluent
109	4,4'-DDE	0.00059	<5.6	<5.6	No	All ND in Effluent
110	4,4'-DDD	0.00084	<2.8	<2.8	No	All ND in Effluent
111	Dieldrin	0.00014	<0.002	<0.002	No	All ND in Effluent
112	Alpha-Endosulfan	No criteria	<0.014	<0.014	No	No criteria
113	Beta-Endosulfan	No criteria	<0.004	<0.014	No	No criteria

CTR No.	Constituent	Water Quality Criteria (C) µg/L	Max Effluent Conc. (MEC) µg/L	Max Receiving Water Conc.(B) µg/L	RPA Result - Need Limit?	Reason
114	Endosulfan Sulfate	240	<0.066	<0.066	No	All ND in Effluent
115	Endrin	0.0023	<0.006	<0.006	No	All ND in Effluent
116	Endrin Aldehyde	0.81	<0.023	<0.023	No	All ND in Effluent
117	Heptachlor	0.00021	<0.003	<0.003	No	All ND in Effluent
118	Heptachlor Epoxide	0.00011	<0.083	<0.083	No	All ND in Effluent
119	PCB 1016	0.00017	<0.39	<0.39	No	All ND in Effluent
120	PCB 1221	0.00017	<0.39	<0.65	No	All ND in Effluent
121	PCB 1232	0.00017	<0.39	<0.65	No	All ND in Effluent
122	PCB 1242	0.00017	<0.39	<0.65	No	All ND in Effluent
123	PCB 1248	0.00017	<0.39	<0.65	No	All ND in Effluent
124	PCB 1254	0.00017	<0.39	<0.65	No	All ND in Effluent
125	PCB 1260	0.00017	<0.39	<0.65	No	All ND in Effluent
126	Toxaphene	0.00075	<10	<10	No	All ND in Effluent

4. WQBEL Calculations

Final WQBELs are based on monitoring results and following the calculation process outlined in section 1.4 of the CTR and the SIP. A table providing the calculation for all applicable WQBELs for this Order is provided in Attachment P of this Order.

a. Calculation Options

Once the RPA has been conducted using either the Technical Support Document (TSD) or the SIP methodologies, WQBELs are calculated. Alternative procedures for calculating WQBELs include:

- Use wasteload allocation (WLA) from applicable TMDL;
- Use a steady-state model to derive MDELs and AMELs; and,
- Where sufficient data exist, use a dynamic model which has been approved by the State Water Board.

b. WQBELs Calculations

WQBELs based on human health and aquatic life criterion were established for this Order. The process for developing these limits is in accordance with section 1.4 of the SIP. Attachments K, M and N summarize the development and calculation of all WQBELs for this Order using the process described below.

- Lead: Tiers 1 and 2 of the SIP triggered reasonable potential for lead because the MEC exceeded the criteria ($MEC > C$), there was an exceedance of WQOs in the receiving water ($B > C$) and the pollutant is detected in the effluent. Therefore, final effluent limitations have been prescribed for lead.

- For each constituent requiring an effluent limit, identify the applicable water quality criteria or objective. For each criterion determine the ECA using the following steady state equation:

$$ECA = C + D(C-B) \text{ when } C > B, \text{ and}$$

$$ECA = C \text{ when } C \leq B$$

where

C = the priority pollutant criterion/objective, adjusted if necessary for hardness, pH and translators. In this Order a hardness value of 400 mg/L (as CaCO_3) was used for development of hardness-dependent criteria.

D = the dilution credit

B = the ambient background concentration

For this Order, dilution was not allowed due to nature of the receiving water and quantity of the effluent.

For lead, the applicable saltwater quality criteria are:

$$\text{ECA}_{\text{acute}} = 220.82 \text{ } \mu\text{g/L}$$

$$\text{ECA}_{\text{chronic}} = 8.52 \text{ } \mu\text{g/L}$$

No metal translator or water effects ratio were identified to modify the criterion.

- (b) For each ECA based on aquatic life criterion/objective, determine the LTA by multiplying the ECA by a factor (multiplier). The multiplier is statistically based factor that adjusts the ECA to account for effluent variability. The value of the multiplier varies depending on the coefficient of variation (CV) of the data set and whether it is an acute or chronic criterion/objective. Table 1 of the SIP provides pre-calculated values for the multipliers based on the value of the CV. Equations to develop the multipliers in place of using values in the tables are provided in section 1.4, step 3 of the SIP and will not be repeated here.

$$\text{LTA}_{\text{acute}} = \text{ECA}_{\text{acute}} \times \text{Multiplier}_{\text{acute}}$$

$$\text{LTA}_{\text{chronic}} = \text{ECA}_{\text{chronic}} \times \text{Multiplier}_{\text{chronic}}$$

The CV for the data set must be determined before the multipliers can be selected and will vary depending on the number of samples and the standard deviation of a data set. If the data set is less than 10 samples, or at least 80 percent of the samples in the data set are reported non-detect, the CV is set equal to 0.6.

- (c) For lead, the following data was used to develop the acute and chronic LTA using Table 1 of the SIP: Because there are 11 samples, of which 10 are nondetect, the CV is equal to 0.6 and

$$\text{Multiplier}_{\text{acute}} = 0.321$$

$$\text{Multiplier}_{\text{chronic}} = 0.527$$

$$\text{LTA}_{\text{acute}} = 220.82 \text{ } \mu\text{g/L} \times 0.3211 = 70.9 \text{ } \mu\text{g/L}$$

$$\text{LTA}_{\text{chronic}} = 8.52 \text{ } \mu\text{g/L} \times 0.5274 = 4.483 \text{ } \mu\text{g/L}$$

- (d) Select the most limiting (lowest) of the LTA.

$$\text{LTA} = \text{most limiting of } \text{LTA}_{\text{acute}} \text{ or } \text{LTA}_{\text{chronic}}$$

For lead, the most limiting LTA was the $\text{LTA}_{\text{acute}}$

$$LTA = 4.483 \mu\text{g/L}$$

- (e) Calculate the WQBELs by multiplying the LTA by a factor (multiplier). WQBELs are expressed as AMELs and MDELs. The multiplier is statistically based factor that adjusts the LTA for the averaging periods and exceedance frequencies of the criteria/objectives and the effluent limitations. The value of the multiplier varies depending on the probability basis, the CV of the data set, the number of samples (for AMEL) and whether it is monthly or daily limit. Table 2 of the SIP provides the pre-calculated values for the multipliers based on the value of the CV and the number of samples. Equations to develop the multipliers in place of using values in the tables are provided in section 1.4, step 5 of the SIP and will not be repeated here.

$$AMEL_{\text{aquatic life}} = LTA \times AMEL_{\text{multiplier}}$$

$$MDEL_{\text{aquatic life}} = LTA \times MDEL_{\text{multiplier}}$$

AMEL multipliers are based on 95th percentile occurrence probability, and the MDEL multipliers are based on the 99th percentile occurrence probability. If the number of samples is less than 4, the default number of samples to be used is 4.

For lead, the following data was used to develop the AMEL and MDEL for aquatic life using table 2 of the SIP. The number of samples is 11 and the CV is 0.6.

$$\text{Multiplier}_{MDEL} = 3.1145$$

$$\text{Multiplier}_{AMEL} = 1.5524$$

$$AMEL_{\text{aquatic life}} = 4.483 \times 1.55 = 6.96 \mu\text{g/L}$$

$$MDEL_{\text{aquatic life}} = 4.49 \times 3.11 = 13.96 \mu\text{g/L}$$

- (f) For the ECA based on human health, set the AMEL equal to the $ECA_{\text{human health}}$. An $AMEL_{\text{human health}}$ for lead has not been established, so the discharge limits are based on potential impacts to aquatic life.
- (g) Select the lower of the AMEL and MDEL based on aquatic life and human health water quality based effluent limit for this Order.

Table F-14 Lead WQBEL

Constituent	AMEL _{aquatic life} (mg/L) (95% probability)	MDEL _{aquatic life} (mg/L) (99% probability)	AMEL _{human health} (mg/L) (95% probability)	MDEL _{human health} (mg/L) (99% probability)
Lead	7.0	14	No criteria	No criteria

- ii. Copper: Tiers 1 of the SIP triggered reasonable potential for copper because the MEC exceeded the criteria ($MEC > C$) and the pollutant is detected in the effluent..

The salt water continuous critical concentration (CCC) from the SIP, as total recoverable copper, is 3.7349 $\mu\text{g/L}$ and the most restrictive SIP criterion. Applying the metal translator of .86 from the Entrix study of June 27, 2002, the

CCC is 4.3430 µg/L, as described in Attachment N part 2. The WER applied for copper was 2.08, based on Order R4-2008-0011, which amended the 2008 NPDES permit. The resulting CCC criteria is 9.0333 µg/L. This value is used with a coefficient of variation of .79 and the maximum effluent concentration of 8.88 mg/L in the calculation of the limits in Attachment N part 3.

- iii. Nickel: Tier 1 and Tier 2 of the SIP RPA procedures were triggered for nickel because the MEC exceeded the criteria ($MEC > C$), there was an exceedance of WQOs in the receiving water ($B > C$) and the pollutant is detected in the effluent. A metal translator of .81, based on the Entrix study of June 27, 2002, was applied to calculate the final effluent limitations for nickel.
- iv. Selenium: Tier 1 and Tier 2 of the SIP RPA procedures were triggered for selenium because the MEC exceeded the criteria ($MEC > C$), there was an exceedance of WQOs in the receiving water ($B > C$) and the pollutant is detected in the effluent. When the freshwater quality objective of 5 mg/L, the most restrictive WQO was considered, a final effluent limitations could be established.

c. Impracticability Analysis

Federal NPDES regulations contained in 40 CFR part 122.45 continuous dischargers, states that all permit limitations, standards, and prohibitions, including those to achieve water quality standards, shall unless impracticable be stated as maximum daily and average monthly discharge limitations for all dischargers other than POTWs.

As stated by USEPA in its long standing guidance for developing WQBELs, average limitations, alone, are not practical for limiting acute, chronic, and human health toxic effects.

For example, a POTW sampling for a toxicant to evaluate compliance with a 7-day average limitation could fully comply with this average limit, but still be discharging toxic effluent on one, two, three, or up to four of these seven days and not be meeting 1-hour average acute criteria or 4-day average chronic criteria. For these reason, USEPA recommends daily maximum and 30-day average limits for regulating toxics in all NPDES discharges. For the purposes of protecting the acute effects of discharges containing toxicants (CTR human health for the ingestion of fish), daily maximum limitations have been established in this NPDES permit for mercury because it is considered to be a carcinogen, endocrine disruptor, and is bioaccumulative.

A 7-day average alone would not protect one, two, three, or four days of discharging pollutants in excess of the acute and chronic criteria. Fish exposed to these endocrine disrupting chemicals will be passed on to the human consumer. Endocrine disrupters alter hormonal functions by several means. These substances can:

- i. Mimic or partly mimic the sex steroid hormones estrogens and androgens (the male sex hormone) by binding to hormone receptors or influencing cell signaling pathways.
- ii. Block, prevent and alter hormonal binding to hormone receptors or influencing cell signaling pathways.
- iii. Alter production and breakdown of natural hormones.

iv. Modify the making and function of hormone receptors.

d. Mass-based limits

40 CFR part 122.45(f)(1) requires that, except under certain conditions, all NPDES permit limits, standards, or prohibitions be expressed in terms of mass units. 40 CFR part 122.45(f)(2) allows the permit writer, at his/her discretion, to express limits in additional units (e.g., concentration units). The regulations mandate that, where limits are expressed in more than one unit, the permittee must comply with both.

Generally, mass-based limits ensure that proper treatment, and not dilution, is employed to comply with the final effluent concentration limits. Concentration-based effluent limits, on the other hand, discourage the reduction in treatment efficiency during low-flow periods and require proper operation of the treatment units at all times. In the absence of concentration-based effluent limits, a permittee would be able to increase the effluent concentration (i.e., reduce its level of treatment) during low-flow periods and still meet its mass-based limits. To account for this, this Order includes mass and concentration limits for some constituents.

The Facility design flow is 14 mgd. Because the Regional Water Board capped the flow at 9 mgd in the 2008 Order, the flow of 9 mgd is used to calculate mass based limits in this Order.

e. Final WQBELs

Summaries of the WQBELs required by this Order are described below.

Table F-15. Summary of WQBEL

Constituent	Units	Effluent Limitations	
		Average Monthly	Maximum Daily
Summer Total Ammonia (May – October)	mg/L	1.07	1.17
	lbs/day	80	88
Winter Total Ammonia (November to April)	mg/L	1.3	1.4
	lbs/day	98	105
Nitrite + Nitrate as Nitrogen	mg/L	10	--
	lbs/day	750	--
Nitrite as Nitrogen	mg/L	1	--
	lbs/day	75	--
Nitrate as Nitrogen	mg/L	10	--
	lbs/day	750	--
Copper	µg/L	6.1	14
	lbs/day	.45	1.1
Selenium	µg/L	2.9	8.2
	lbs/day	.22	.62
Lead	µg/L	7.0	14
	lbs/day	0.52	1.1
Nickel	µg/L	7.2	18.8
	lbs/day	.54	1.4

5. Whole Effluent Toxicity (WET)

Because of the nature of industrial and domestic discharges into the POTW sewershed, it is possible that other toxic constituents could be present in the Facility's effluent, or could have synergistic or additive effects. Also, because numeric limits for certain toxic constituents that did not show reasonable potential have been removed, the acute toxicity limit may provide a backstop to preventing the discharge of toxic pollutants in toxic amounts. As such, the permit contains effluent limitations for toxicity.

The toxicity numeric effluent limitations are based on:

- a. 40 CFR part 122.44(d)(v) – limits on WET are necessary when chemical-specific limits are not sufficient to attain and maintain applicable numeric or narrative water quality standards;
- b. 40 CFR part 122.44(d)(vi)(A) – where a state has not developed a water quality criterion for a specific pollutant that is present in the effluent and has reasonable potential, the permitting authority can establish effluent limits using numeric water quality criterion;
- c. Basin Plan objectives and implementation provisions for toxicity;
- d. USEPA Regions 9 & 10 Guidance for Implementing WET Programs, May 31, 1996;
- e. Whole Effluent Toxicity (WET) Control Policy, July 1994; and,
- f. Technical Support Document (TSD) (several chapters and Appendix B).

The circumstances warranting a numeric chronic toxicity effluent limitation when there is reasonable potential were under review by the State Water Board in SWRCB/OCC Files A-1496 & A-1496(a) [Los Coyotes/Long Beach Petitions]. On September 16, 2003, at a public hearing, the State Water Board adopted Order No. 2003-0012, deferring the issue of numeric chronic toxicity effluent limitations until a subsequent Phase of the SIP is adopted. In the meantime, the State Water Board replaced the numeric chronic toxicity limit with a narrative effluent limitation and a 1 TUc trigger, in the Long Beach and Los Coyotes WRP NPDES permits. This permit contains a similar narrative chronic toxicity effluent limitation, with a numeric trigger for accelerated monitoring. Phase 2 of the SIP has been adopted; however, the toxicity control provisions were not revised.

On January 17, 2006, the State Water Board Division of Water Quality held a CEQA scoping meeting to seek input on the scope and content of the environmental information that should be considered in the planned revisions of the SI). However, the Toxicity Control Provisions of the SIP continue unchanged.

This Order contains a reopener to allow the Regional Water Board to modify the permit, if necessary, consistent with any new policy, law, or regulation. Until such time, this Order will have toxicity limitations that are consistent with the State Water Board's precedential decision.

- g. Acute Toxicity Limitation:

The Dischargers shall test for acute toxicity by using USEPA's *Methods for Measuring the Acute Toxicity of Effluents and Receiving Waters to Freshwater and Marine Organisms*, October 2002 (EPA-821-R-02-012). Acute toxicity provisions are derived from the Basin Plan's toxicity standards (Basin Plan 3-16 and 3-17). The

provisions require the Discharger to accelerate acute toxicity monitoring and take further actions to identify the source of toxicity and to reduce acute toxicity.

h. Chronic Toxicity Limitation and Requirements:

Chronic toxicity provisions are derived from the Basin Plan's toxicity standards (Basin Plan 3-16 and 3-17). The provisions require the Discharger to accelerate chronic toxicity monitoring and take further actions to identify the source of toxicity and to reduce chronic toxicity. The monthly median trigger of 1.0 TU_c for chronic toxicity is based on *USEPA Regions 9 & 10 Guidance for Implementing Whole Effluent Toxicity (WET) Programs*, Final May 31, 1996 (chapter 2 – Developing WET Permitting Conditions, page 2-8). In cases where effluent receives no dilution or where mixing zones are not allowed, the 1.0 TU_c chronic criterion should be expressed as a monthly median. The “median” is defined as the middle value in a distribution, above which and below which lays an equal number of values. For example, if the results of the WET testing for a month were 1.5, 1.0, and 1.0 TU_c, the median would be 1.0 TU_c.

The *USEPA Regions 9 & 10 Guidance for Implementing Whole Effluent Toxicity (WET) Programs*, Final May 31, 1996 (chapter 2 – Developing WET Permitting Conditions, page 2-8) recommends two alternatives for setting up maximum daily limit: using 2.0 TU_c as the maximum daily limit; or using a statistical approach outlined in the Technical Guidance Document to develop a MDEL. In this Order, neither a MDEL nor a trigger for chronic toxicity is prescribed.

D. Final Effluent Limitation Considerations

1. Anti-Backsliding Requirements

CWA section 402(o) contains a general provision that prohibits a renewed permit from containing effluent limitations that are less stringent than the comparable effluent limitations in the previous permit, with certain exceptions. Except as discussed further below, the effluent limitations in this Order are at least as stringent as the effluent limitations in the previous permit, Order No. R4-2008-0208. CWA section 402(o)(2)(B)(i) allows a permit to contain less stringent effluent limitations where “information is available which was not available at the time of permit issuance (other than revised regulations, guidance, or test methods) and which would have justified the application of a less stringent effluent limitation at the time of permit issuance”.

CWA section 402(o)(2)(E) allows a permit to contain less stringent effluent limitations where “the permittee has installed the treatment facilities required to meet the effluent limitations in the previous permit and has properly operated and maintained the facilities but has nevertheless been unable to achieve the previous effluent limitations, in which case the limitations in the [renewed] permit may reflect the level of pollutant control actually achieved (but shall not be less stringent than required by effluent guidelines in effect at the time of permit renewal, reissuance, or modification).”

In this Order, effluent limitations for mercury, silver and zinc have been removed because analysis of new information indicates that the pollutants are not being discharged at concentrations that have the reasonable potential to cause or contribute to exceedances of water quality standards.

The removal of effluent limitations for these constituents is consistent with the anti-backsliding requirements of CWA section 402(o)(2)(B)(i) because new information indicates that no effluent limitations are required. In this Order the new MDELs and AMELs for copper are less stringent than those in the previous Order. The effluent limitations for copper in the previous permit were based on a May 2005 WER Study and

applied a WER of 1.58 for dissolved copper. Since adoption of the previous permit, the discharger completed a new WER study, submitted in February 2010. The effluent limitations for copper in this Order were derived using the calculation in the SIP and application of the 2010 WER Study, which relies on the geometric mean of mean WERs. The calculated values are 2.05 and 2.73 for dissolved and total copper, respectively. Staff review of this information concluded that the total copper WER, for the most critical conditions in the winter when the berm was open, should be used to calculate the WER. As a result, Order R4-2008-0208, was amended to the 2008 Order to establish a WER of 2.08, the value used here.

The new effluent limitations for copper are consistent with the anti-backsliding requirements of CWA section 402(o)(2)(B)(i) because the new effluent limitations were derived based on an evaluation of new information not available at the time of adoption of the previous permit.

In this Order, the new AMELs and MDELs for ammonia are less stringent than the effluent limitations for ammonia contained in the previous permit. Since adoption of the previous permit, the Discharger has installed treatment facilities to meet the effluent limitations of the previous permit and has properly operated and maintained the facilities, but has not been able to achieve the effluent limitations. In addition, USEPA has updated guidance regarding implementation of its 1999 ammonia criteria. The Basin Plan also provides direction on application of WQOs for ammonia and reads "*identify the WQOs for ammonia for the receiving water immediately downstream of the discharge.*"¹² Receiving water stations R-001, R-002, R-003, R-004 and R-005 are located at the margins of the Estuary. Receiving water station R-004 is located immediately downstream of the discharge. The ammonia effluent limitations in the previous permit were based on the most restrictive conditions at receiving water station R-001 (rather than on R-004) the 2004 ammonia saltwater Basin Plan Amendment, the USEPA's updated 1999 ammonia criteria, ammonia data, and the 2000 CTR. The new effluent limitations for ammonia are calculated using receiving water conditions at R-004, the point closest to the outfall as required by the Basin Plan, on Estuary studies completed between 2008 and 2013, and on the performance of the treatment facilities, and taking into account USEPA's updated 1999 ammonia criteria. Calculating the effluent limitations based only on the receiving water conditions at R-004 and on USEPA's updated ammonia criteria would result in significantly less stringent effluent limitations than contained in the previous permit. The new effluent limitations are, therefore, more stringent than the calculated limitations and are based on the performance of the facilities, but are less stringent than the previous permit. This relaxation of effluent limitations is consistent with the anti-backsliding requirements of CWA sections 402(o)(2)(B)(i) and 402(o)(2)(E) because (1) the new effluent limitations were derived based on an evaluation of new information not available at the time of adoption of the previous permit, in particular new Estuary studies, (2) the discharger has installed and is properly operating and maintaining treatment facilities since adoption of the previous permit, (3) the discharger has not been able to achieve the limitations in the previous permit, and (4) the new effluent limitations at least reflect the level of pollutant control actually achieved (i.e., the performance based levels). In addition, the new effluent limitations will not result in degradation of the receiving water because they are derived based on criteria sufficient to be protective of the beneficial uses of the receiving waters and will not cause degradation as compared to the previous permit. These effluent limitations will be reconsidered and the permit may be reopened to address new information gathered regarding the Estuary.

¹² Basin Plan amendment for Inland Surface Water Ammonia Objectives for Saltwater March 4, 2004.

2. Antidegradation Policies

The Regional Water Board is required under CWA section 301(b)(1)(C) and its implementing regulations (40 CFR part 122.4(a); 40 CFR part 122.4(d); 40 CFR part 122.44(d)) to establish conditions in NPDES permits that ensure compliance with state water quality standards, including antidegradation requirements. The federal antidegradation policy (40 CFR part 131.12(a)(1)) requires that *"existing instream water uses and the level of water quality necessary to protect the existing uses shall be maintained and protected."* As defined in 40 CFR part 131.3(e), *"[e]xisting uses are those uses actually attained in the water body on or after November 28, 1975, whether or not they are included in the water quality standards"* (emphasis added).

40 CFR part 131.12 requires that the state water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State Water Board Resolution No. 68-16. Resolution No. 68-16 incorporates the federal antidegradation policy where the federal policy applies under federal law. Resolution No. 68-16 requires that existing water quality be maintained unless degradation is justified based on specific findings. The Regional Water Board's Basin Plan implements, and incorporates by reference, both state and federal antidegradation policies.

The effluent limit for copper is higher in this permit than in the previous permit. The process for developing the limit was consistent with the previous permit, but the effluent data used to calculate the limit was updated. Because plant performance with respect to metals has not changed, the level of copper in effluent is not expected to increase. The effluent limits for ammonia are also higher in this permit than in the previous permit. However, the limits in this permit are performance-based and were calculated consistently with the interim effluent limits for ammonia contained in the TSOs issued between 2008 and 2013 (see Fact Sheet Section II.D.1). This permit requires consistent performance by the facility and does not allow degradation of the waterbody.

3. Stringency of Requirements for Individual Pollutants

This Order contains both technology-based and water quality-based effluent limitations for individual pollutants. The TBELs consist of restrictions on BOD, TSS, pH, and percent removal of BOD and TSS. Restrictions on BOD, TSS, and pH are discussed in section IV.B. of this Fact Sheet. This Order's technology-based pollutant restrictions implement the minimum, applicable federal technology-based requirements.

Water quality-based effluent limitations have been derived to implement water quality objectives that protect beneficial uses. Both the beneficial uses and the water quality objectives have been approved pursuant to federal law and are the applicable federal water quality standards. To the extent that toxic pollutants WQBELs were derived from the CTR, the CTR is the applicable standard pursuant to 40 CFR section 131.38. The procedures for calculating the individual water quality-based effluent limitations for priority pollutants are based on the CTR implemented by the SIP, which was approved by USEPA on May 18, 2000. All beneficial uses and water quality objectives contained in the Basin Plan were approved under state law and submitted to and approved by USEPA prior to May 30, 2000. Any water quality objectives and beneficial uses submitted to USEPA prior to May 30, 2000, but not approved by USEPA before that date, are nonetheless "applicable water quality standards for purposes of the CWA"

pursuant to 40 CFR section 131.21(c)(1). Collectively, this Order's restrictions on individual pollutants are no more stringent than required to implement the requirements of the CWA.

Table F-16. Summary of Final Effluent Limitations for Discharge Point 001

Parameter	Units	Effluent Limitations					Basis
		Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum	
BOD ₅ 20°C	mg/L	20	30	45	--	--	Existing
	lbs/day	2,340	3,500	5,260	--	--	
TSS	mg/L	15	40	45	--	--	Existing
	lbs/day	1,750	4,670	5,250	--	--	
pH	pH unit	--	--	--	6.5	8.5	Existing
Oil and Grease	mg/L	10	--	15	--	--	Existing
	lbs/day	750	--	1,100	--	--	
Settleable Solids	ml/L	0.1	--	0.3	--	--	Existing
Total Residual Chlorine	mg/L	0.1	--	--	--	--	Existing
	lbs/day	7.5	--	--	--	--	
MBAS	mg/L	0.5	--	--	--	--	Basin Plan, BPJ
	lbs/day	38	--	--	--	--	
Summer Total Ammonia	mg/L	1.07		1.17	--	--	Performance
	lbs/day	80		88	--	--	
Winter Total Ammonia	mg/L	1.3		1.4	--	--	Performance, Basin Plan
	lbs/day	98		105	--	--	
Nitrate + Nitrite as Nitrogen ¹³	mg/L	10	--	--	--	--	Basin Plan
	lbs/day	750	--	--	--	--	
Nitrite as Nitrogen	mg/L	10	--	--	--	--	Basin Plan, BPJ
	lbs/day	750	--	--	--	--	
Nitrate as Nitrogen	mg/L	10	--	--	--	--	Basin Plan
	lbs/day	750	--	--	--	--	
Copper	µg/L	6.1	--	14	--	--	CTR, SIP, WER, Metal Translator
	lbs/day	.46	--	1.1	--	--	
Selenium	µg/L	2.9	--	8.2	--	--	CTR, SIP
	lbs/day	.22	--	.62	--	--	

¹³ This effluent limitation in concentration shall not apply when the concentration limitation for nitrate as nitrogen or nitrite as nitrogen is exceeded. This effluent limitation in mass load shall not apply when the mass load limitation for nitrate as nitrogen or nitrite as nitrogen is exceeded

Parameter	Units	Effluent Limitations					Basis
		Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum	
Lead	µg/L	7.0	--	14	--	--	CTR, SIP
	lbs/day	0.52	--	1.1	--	--	
Nickel	µg/L	7.2	--	18.8	--	--	CTR, SIP. Metal Translator
	lbs/day	.54	--	1.4	--	--	

E. Recycling Specifications

The reuse of the reclaimed water is regulated under a separate WDRs and Water Recycling Requirements (WRRs) for the City of Ventura, Order No. 87-45, CI No. 6190. The Discharger is in the process of planning an expansion of the recycled water system. Once the plans are completed and the California Department of Public Health has approved the expansion, the WDR/WRR will be renewed.

V. RATIONALE FOR RECEIVING WATER LIMITATIONS

A. Surface Water

The Basin Plan contains numeric and narrative WQOs applicable to all surface waters within the Los Angeles Region. WQOs include an objective to maintain the high quality waters pursuant to 40 CFR part 131.12 and State Water Board Resolution No. 68-16. Receiving water limitations in this Order are included to ensure protection of beneficial uses of the receiving water.

B. Groundwater

The fresh groundwater beneath the Facility is in the Oxnard Plain Groundwater Basin. Groundwater infiltration and exfiltration has been documented at the Facility. The wildlife ponds do not have lined bottoms, and a loss of about 1 mgd is known to occur at the ponds, either through evaporation or seepage into the groundwater.

The loss of treated effluent from the ponds into the underlying groundwater is evidence of a hydraulic connection between the surface water and the groundwater. The water table, and hence, the location of the saline/freshwater interface, is known to vary between closed-flooded estuary conditions and open-breaching estuary conditions. The boundary between the groundwater basin (fresh) and the ocean (saline) is not a fixed point. Even though groundwater limits are not included in this Order, impacts on groundwater are prohibited.

VI. RATIONALE FOR PROVISIONS

A. Standard Provisions

Standard Provisions, which apply to all NPDES permits in accordance with 40 CFR section 122.41, and additional conditions applicable to specified categories of permits in accordance with 40 CFR section 122.42, are provided in Attachment D. The discharger must comply with all standard provisions and with those additional conditions that are applicable under section 122.42.

Sections 122.41(a)(1) and (b) through (n) of 40 CFR establish conditions that apply to all state-issued NPDES permits. These conditions must be incorporated into the permits either expressly or by reference. If incorporated by reference, a specific citation to the regulations must be included in the Order. Section 123.25(a)(12) of 40 CFR allows the state to omit or modify conditions to impose more stringent requirements. In accordance with 40 CFR section 123.25, this Order omits federal conditions that address enforcement authority specified in 40

CFR sections 122.41(j)(5) and (k)(2) because the enforcement authority under the Water Code is more stringent. In lieu of these conditions, this Order incorporates by reference CWC section 13387(e).

B. Special Provisions

1. Reopener Provisions

This provision is based on 40 CFR part 123. The Regional Water Board may reopen the NPDES permit to modify permit conditions and requirements. Causes for modifications include the promulgation of new regulations, modification in sludge use or disposal practices, or adoption of new regulations by the State Water Board or Regional Water Board, including revisions to the Basin Plan.

2. Special Studies and Additional Monitoring Requirements

a. Phase 3 Estuary Studies

The Discharger shall perform additional estuary studies to provide sufficient information to allow the Regional Water Board to determine whether or not the continued discharge of effluent enhances the Estuary. The study will clarify the City's water budget analysis for the Santa Clara River Estuary, to determine whether any effluent discharge is needed to sustain the Estuary's native species, and if so how much.

- i. Nutrient, Dissolved Oxygen and Toxicity Special Study - The Discharger must perform a special study to identify the cause of nutrient, dissolved oxygen and toxicity impairments in the Estuary. If it is determined that the effluent from the Facility is causing the impairments, the Facility must propose a plan for reducing nutrient loading, including ammonia, nitrogen and phosphorus loading and toxicity impairments.
- ii. Groundwater Special Study – the discharger must perform a special study to document the interaction between the estuary, discharge and groundwater and determine if the beneficial use of MUN applies to the water impacted by the discharge.
- iii. The work plans for the Phase 3 Estuary, Nutrient, Dissolved Oxygen and Toxicity and Groundwater special studies shall be submitted 180 days following permit adoption for approval by the Executive Officer. The scope of the work of the Phase 3 Estuary study must be sufficient to support a staff finding concerning enhancement of the estuary in conformance with the Enclosed Bays and Estuary Policy.
- iv. The Phase 3 Estuary Special Studies must be completed with final reports submitted no later than January 1, 2018. The Nutrient, Dissolved Oxygen and Toxicity, and Groundwater Special Studies must be completed with final reports submitted no later 180 days before the expiration of this order.

b. CECs Special Study

Advancements in analytical technology over the last decade have dramatically increased the number of chemicals that can be detected and greatly decreased the concentrations at which chemicals can be detected. This new ability to detect trace levels of chemical concentrations has expanded the understanding of the kinds of contaminants present in the water and wastewater. Many man-made chemicals, particularly pesticides, pharmaceuticals and personal care products, have been found in waters across the United States.

Collectively, these compounds are referred to as Emerging Constituents (ECs) or CECs because their presence is starting to be revealed by rapid advances in analytical technology. Despite recent improvements in analytical science, the extent of contamination by CEC's in the environment and a lack of experience in CEC measurement makes interpretation of CEC data difficult. For these reasons, CECs are unregulated chemicals, for which no water quality standards have been established.

Recent publications and media reports on CECs have increased public awareness of the issue, providing an impetus for CEC investigations around the country, including local efforts by the City of Los Angeles and Southern California Coastal Water Research Project (SCCWRP). For instance, starting 2005, the City of Los Angeles has been conducting a special study as part of the Order No. 2005-0020, whose results suggest that the presence of natural and synthetic estrogen hormones has caused feminization of male fish (hornyhead turbot) in Santa Monica Bay, especially near the Hyperion Treatment Plant outfall. In January 2010, SCCWRP convened a workshop where 50 scientists, water quality managers, and stakeholders discussed and collaborated on developing an effective CEC monitoring and management strategy that is protective of water quality. Outcomes of this workshop include recommended lists of CECs for monitoring in recycled water (for groundwater concerns) and for monitoring in ambient waters, including ocean waters. The final report of Monitoring Strategies for CECs in Recycled Water was published on June 25, 2010.

The Discharger shall initiate an investigation of CECs in the Discharger's effluent by conducting a special study. Specifically, within 90 days of the effective date of this Order, the Discharger shall develop a CEC Special Study Work Plan and submit for approval by the Executive Officer of this Regional Water Board. Immediately upon approval of the work plan, the Discharger shall fully implement the special study.

c. Toxicity Reduction Requirements

If the discharge consistently exceeds an effluent limitation for toxicity, the Discharger needs to conduct Toxicity Identification Evaluation(TIE)/Toxicity Reduction Evaluation (TRE) detailed in section V of the MRP. The TRE will help the Discharger identify the possible source(s) of toxicity. Once the source(s) of toxicity is identified, the Discharger shall take all reasonable steps to reduce toxicity to the required level.

d. Treatment Plant Capacity

The treatment plant capacity study required by this Order shall serve as an indicator for the Regional Water Board regarding Facility's increasing hydraulic capacity and growth in the service area.

3. Best Management Practices and Pollution Prevention

a. Storm Water Pollution Prevention Plan (SWPPP)

CWA section 402(p), as amended by the Water Quality Act of 1987, requires NPDES permits for storm water discharges. Pursuant to this requirement, in 1990, USEPA promulgated 40 CFR part 122.26 that established requirements for storm water discharges under an NPDES permit. To facilitate compliance with federal regulations, on November 1991, the State Board issued a statewide general permit, *General NPDES Permit No. CAS000001 and Waste Discharge Requirements for Discharges of Storm Water Associated with Industrial Activities*. This permit was amended in September 1992 and reissued on April 17, 1997 as State Board Order

No. 97-03-DWQ. The Facility is covered under this general permit and an updated SWPPP is required.

b. **Spill Contingency Plan (SCCP)**

Since spill or overflow is a common event in the treatment plant service areas, this Order requires the Discharger to review and update, if necessary, SCCP after each incident. The Discharger shall ensure that the up-to-date SCCP is readily available to the sewage system personnel at all times and that the sewage personnel are familiar with it.

c. **Pollutant Minimization Program (PMP)**

This provision is based on the requirements of section 2.4.5 of the SIP.

4. Construction, Operation, and Maintenance Specifications

This provision is based on the requirements of 40 CFR part 122.41(e) and the previous Order.

5. Special Provisions for Municipal Facilities (POTWs Only)

a. **Biosolids Requirements.** To implement CWA section 405(d), on February 19, 1993, USEPA promulgated 40 CFR part 503 to regulate the use and disposal of municipal sewage sludge. This regulation was amended on September 3, 1999. The regulation requires that producers of sewage sludge meet certain reporting, handling, and disposal requirements. It is the responsibility of the Discharger to comply with said regulations that are enforceable by USEPA, because California has not been delegated the authority to implement this program. The Discharger is also responsible for compliance with WDRs and NPDES permits for the generation, transport and application of biosolids issued by the State Water Board, other Regional Water Boards, the Arizona Department of Environmental Quality or USEPA, to whose jurisdiction the Facility's biosolids will be transported and applied.

b. **Pretreatment Requirements.** This Order contains pretreatment requirements consistent with applicable effluent limitations, national standards of performance, and toxic and performance effluent standards established pursuant to sections 208(b), 301, 302, 303(d), 304, 306, 307, 403, 404, 405, and 501 of the CWA, and amendments thereto. This Order contains requirements for the implementation of an effective pretreatment program pursuant to section 307 of the CWA; 40 CFR parts 35 and 403; and/or section 2233, title 23 of the CCR.

Applications for renewal or modification of this Order must contain information about industrial discharges to the POTW pursuant to 40 CFR 122.21(j)(6). Pursuant to 40 CFR 122.42(b) and Provision VII.A of Attachment D, Standard Provisions, of this Order, the Discharger shall provide adequate notice of any new introduction of pollutants or substantial change in the volume or character of pollutants from industrial discharges which were not included in the permit application." and "Pursuant to 40 CFR 122.44(j)(1), the Discharger shall annually identify and report, in terms of character and volume of pollutants, any Significant Industrial Users discharging to the POTW subject to Pretreatment Standards under section 307(b) of the CWA and 40 CFR 403.

c. **Spill Reporting Requirements.** This Order established a reporting protocol for how different types of spills, overflow or bypasses of raw or partially treated sewage from its collection system or treatment plant covered by this Order shall be reported to regulatory agencies.

The State Water Board issued General Waste Discharge Requirements for Sanitary Sewer Systems, Water Quality Order 2006-0003-DWQ (General Order) on May 2, 2006. The Monitoring and Reporting Requirements for the General Order were amended by Water Quality Order WQ 2008-0002-EXEC on February 20, 2008. The General Order requires public agencies that own or operate sanitary sewer systems with greater than one mile of pipes or sewer lines to enroll for coverage under the General Order. The General Order requires agencies to develop sanitary sewer management plans (SSMPs) and report all sanitary sewer overflows (SSOs), among other requirements and prohibitions.

Furthermore, the General Order contains requirements for operation and maintenance of collection systems and for reporting and mitigating SSOs. Inasmuch that the Discharger's collection system is part of the system that is subject to this Order, certain standard provisions are applicable as specified in Provisions, section VI.C.5. For instance, the 24-hour reporting requirements in this Order are not included in the General Order. The Discharger must comply with both the General Order and this Order. The Discharger and public agencies that are discharging wastewater into the facility were required to obtain enrollment for regulation under the General Order by December 1, 2006.

VII. RATIONALE FOR MONITORING AND REPORTING REQUIREMENTS

Section 308(a) of the CWA and sections 122.41(h), (j)-(l), 122.44(i), and 122.48 of 40 CFR require that all NPDES permits specify monitoring and reporting requirements. CWC sections 13267 and 13383 also authorize the Regional Water Board to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. The MRP, Attachment E of this Order, establishes monitoring, reporting, and recordkeeping requirements that implement federal and state requirements. The following provides the rationale for the monitoring and reporting requirements contained in the MRP for this facility.

A. Influent Monitoring

Influent monitoring is required to determine compliance with permit conditions, assess treatment plant performance, and assess effectiveness of the Pretreatment Program.

Influent monitoring in this Order follows the influent monitoring requirements in the previous Order with minor modification. The monitoring frequencies for selenium, lead and nickel have been increased to monthly.

B. Effluent Monitoring

The Discharger is required to conduct monitoring of the permitted discharges in order to evaluate compliance with permit conditions. Monitoring requirements are given in the MRP (Attachment E). This provision requires compliance with the MRP, and is based on 40 CFR parts 122.44(i), 122.62, 122.63, and 124.5. The MRP is a standard requirement in almost all NPDES permits (including this Order) issued by the Regional Water Board. In addition to containing definition of terms, it specifies general sampling/analytical protocols and the requirements of reporting spills, violation, and routine monitoring data in accordance with NPDES regulations, the CWC, and Regional Water Board policies. The MRP also contains sampling program specific for the Discharger's wastewater treatment plant. It defines the sampling stations and frequency, pollutants to be monitored, and additional reporting requirements. Pollutants to be monitored include all pollutants for which effluent limitations are specified. Further, in accordance with section 1.3 of the SIP, periodic monitoring is required for all priority pollutants defined by the CTR for which criteria apply and for which no effluent limitations have been established to evaluate reasonable potential to cause or contribute to an excursion above a water quality standard.

Monitoring for those pollutants expected to be present in the discharge from the Facility will be required as shown on the MRP and as required in the SIP.

Table F-17 compares the effluent monitoring frequency of the previous Order and this Order. On May 19, 2010, Regional Water Board staff wrote a memo to the Board members regarding the determination of POTW's effluent monitoring frequency for priority pollutants, pesticides, and radioactive pollutants. Regional Water Board staff used a matrix of criteria, based on BPJ, to set the effluent and receiving monitoring frequencies. The monitoring frequencies for these pollutants, which vary from monthly, to quarterly, to semiannually, are generally set based on the following three criteria:

1. Criterion 1: Monitoring frequency will be monthly, for those pollutants with reasonable potential to exceed or contribute to exceedances of WQOs (i.e. monitoring has shown exceedances of the objectives, etc.);
2. Criterion 2: Monitoring frequency will be quarterly, for those pollutants in which some or all of the historic effluent monitoring data detected the pollutants, but without reasonable potential to exceed or contribute to exceedances of WQOs; or,

3. Criterion 3: Monitoring frequency will be semiannually, for those pollutants in which all of the historic effluent monitoring data have had non-detected concentrations of the pollutants and without current reasonable potential to exceed or contribute to exceedances of WQOs.

Table F-17. Effluent Monitoring Program Comparison Table

Parameter	Monitoring Frequency (Order No. R4-2008-0011)	Monitoring Frequency (Order No. R4-2013-0174)	Change of Monitoring Frequency
Total residual chlorine	Continuous	Continuous	No
Total residual chlorine	5 days/week	5 days/week	No
Total waste flow	Continuous	Continuous	No
Turbidity	Continuous	Continuous	No
Total and fecal coliform	Daily	Daily	No
Settleable solids	Daily	Daily	No
BOD ₅ 20°C	Daily	Daily	No
TSS	Daily	Daily	No
Dissolved oxygen	Daily	Daily	No
Temperature	Weekly	Weekly	No
pH	Weekly	Weekly	No
Oil and grease	Weekly	Weekly	No
TDS	Monthly	Monthly	No
Fluoride	Monthly	Monthly	No
Phosphate as P	Monthly	Monthly	No
Phosphorous	Monthly	Monthly	No
Total Ammonia	Monthly	Monthly	No
Nitrate nitrogen	Monthly	Monthly	No
Nitrite nitrogen	Monthly	Monthly	No
Organic nitrogen	Monthly	Monthly	No
Total Kjeldahl nitrogen	Monthly	Monthly	No
Detergents (as MBAS)	Monthly	Monthly	No
Chlorophyll a	Monthly	Monthly	No
Cyanide	Monthly	Semiannually	Decrease
Chronic toxicity	Monthly	Monthly	No
Acute toxicity	Annually	Annually	No
Aluminum	Semiannually	Semiannually	No
Antimony	Semiannually	Semiannually	No
Arsenic	Semiannually	Semiannually	No
Barium	Semiannually	Semiannually	No
Beryllium	Semiannually	Semiannually	No
Cadmium	Semiannually	Semiannually	No
Chromium VI	Semiannually	Semiannually	No
Cobalt	Semiannually	Semiannually	No
Copper	Monthly	Monthly	No

Parameter	Monitoring Frequency (Order No. R4-2008-0011)	Monitoring Frequency (Order No. R4-2013-0174)	Change of Monitoring Frequency
Iron	Semiannually	Semiannually	No
Lead	Semiannually	Monthly	Increase
Mercury	Monthly	Semiannually	Decrease
Molybdenum	Semiannually	Semiannually	No
Nickel	Semiannually	Monthly	Increase
Selenium	Semiannually	Monthly	Increase
Silver	Monthly	Semiannually	Decrease
Thallium	Semiannually	Semiannually	No
Vanadium	Semiannually	Semiannually	No
Zinc	Monthly	Quarterly	Decrease
Chlorodibromomethane	Monthly	Semiannually	Decrease
Dichlorobromomethane	Monthly	Semiannually	Decrease
Bis(2-ethylhexyl)phthalate	Monthly	Semiannually	Decrease
Acetone	Semiannually	Semiannually	No
Total xylene	Semiannually	Semiannually	No
Toulene	Semiannually	Semiannually	No
Chloroform	Semiannually	Semiannually	No
Bromoform	Semiannually	Semiannually	No
Remaining priority pollutants (excluding asbestos)	Semiannually	Semiannually	No
Pesticides	Semiannually	Semiannually	No
Radioactivity (Including gross alpha, gross beta, combined radium-226 and radium- 228, tritium, strontium- 90 and uranium)	Semiannually	Semiannually	No
CECs	Biannually	Annually for 2 years	Increase

The reduction of monitoring frequencies for priority pollutants listed in the above table is based on the fact that previous monitoring data for these pollutants indicate that the discharge did not demonstrate reasonable potential to exceed water quality standards.

However, monitoring frequency for priority pollutants with reasonable potential to exceed the WQOs is increased or maintained.

C. Whole Effluent Toxicity Testing Requirements

As discussed on page F-45, WET testing protects the receiving water quality from the aggregate toxic effect of a mixture of pollutants in the effluent. An acute toxicity test is conducted over a short time period and measures mortality. A chronic toxicity test is conducted over a longer period of time and may measure mortality, reproduction, and growth.

This requirement establishes conditions and protocol by which compliance with the Basin Plan narrative WQO for toxicity will be demonstrated and in accordance with section 4.0 of

the SIP. Conditions include required monitoring and evaluation of the effluent for acute and chronic toxicity and numerical values for chronic toxicity evaluation to be used as "triggers" for initiating accelerated monitoring and toxicity reduction evaluation(s).

D. Receiving Water Monitoring

1. Surface Water

Receiving water monitoring is required to determine compliance with receiving water limitations and to characterize the water quality of the receiving water. Requirements are based on the Basin Plan. The receiving water monitoring program in this Order includes the following modifications to the existing receiving water monitoring program:

- a. Monitoring for priority pollutants and chronic toxicity is required at station R-004, R-003 and R-005.
- b. The reduction of monitoring frequencies for priority pollutants listed in table F-18 is based on the fact that previous effluent monitoring data for these pollutants indicate that the discharge did not demonstrate reasonable potential to exceed or contribute to exceedances of water quality standards. However, monitoring frequency for priority pollutants and chronic toxicity with reasonable potential to exceed or contribute to exceedances of the WQOs has been increased.

Table F-18: Receiving Water Monitoring Program Comparison Table

Parameter	Monitoring Frequency (2008 Order)	Monitoring Frequency (2013 Order)
Lead	Semiannually	Monthly
Nickel	Semiannually	Monthly
Copper	Monthly	Monthly
Mercury	Monthly	Semiannually
Silver	Monthly	Quarterly
Zinc	Monthly	Quarterly
Cyanide	Monthly	Semiannually
Selenium	--	Monthly
Bromoform	Monthly	Semiannually
Chlorodibromomethane	Monthly	Semiannually
Chloroform	Monthly	Semiannually
Dichlorobromomethane	Monthly	Semiannually
Bis(2-ethylhexyl)phthalate	Monthly	Semiannually
Remaining priority pollutants (excluding asbestos)	Semiannually	Semiannually
Chronic toxicity	Quarterly	Quarterly

2. Groundwater

None required at this time.

E. Other Monitoring Requirements

1. Biosolids/Sludge Monitoring

This section establishes monitoring and reporting requirements for the storage, handling and disposal practices of sludge generated from the operation of this Facility, as described in Attachment I.

VIII. PUBLIC PARTICIPATION

The Regional Water Board considered the issuance of WDRs that will serve as an NPDES permit for the Ventura Water Reclamation Facility. As a step in the WDR adoption process, the Regional Water Board staff developed tentative WDRs and encouraged public participation in the WDR adoption process.

A. Notification of Interested Parties

The Regional Water Board notified the Discharger and interested agencies and persons of its intent to prescribe WDRs for the discharge and provided them with an opportunity to submit their written comments and recommendations. Notification was provided by posting a notice in a local newspaper and by posting a notice at the Facility in Ventura, California.

The public had access to the agenda and any changes in dates and locations through the Regional Water Board's website: www.waterboards.ca.gov/losangeles. The public was provided access to the agenda including any changes in dates and locations.

B. Written Comments

Interested persons were invited to submit written comments concerning tentative WDRs as provided through the notification process. Comments were due either in person or by mail to the Executive Office at the Regional Water Board at 320 West Fourth Street Suite 200 Los Angeles California, 90013-2343.

To be fully responded to by staff and considered by the Regional Water Board, the written comments were due at the Regional Water Board office by 12:00 pm (noon) on October 7, 2013. Comments or evidence submitted after that date may not be included in the record for this matter.

C. Public Hearing

The Regional Water Board will hold a public hearing on the tentative WDRs during its regular Regional Water Board meeting on the following date and time and at the following location:

Date: November 7, 2013
Time: 9:00 a.m.
Location: City of Simi Valley, Council Chambers
2929 Tapo Canyon Road
Simi Valley, California

Interested persons were invited to attend. At the public hearing, the Regional Water Board heard testimony pertinent to the discharge, WDRs, and NPDES permit. For accuracy of record, important testimony was requested in writing.

D. Reconsideration of Waste Discharge Requirements

Any aggrieved person may petition the State Water Board to review the decision of the Regional Water Board regarding the final WDRs. The petition must be received by the State Water Board at the following address within 30 calendar days of the Regional Water Board's action:

State Water Resources Control Board
Office of Chief Counsel
P.O. Box 100, 1001 I Street
Sacramento, CA 95812-0100

For instructions on how to file a petition for review, see
http://www.waterboards.ca.gov/public_notices/petitions/water_quality/wqpetition_instr.shtml

E. Information and Copying

The Report of Waste Discharge, other supporting documents, and comments received are on file and may be inspected at the address above at any time between 8:30 a.m. and 4:45 p.m., Monday through Friday. Copying of documents may be arranged through the Regional Water Board by calling (213) 576-6600.

F. Register of Interested Persons

Any person interested in being placed on the mailing list for information regarding the WDRs and NPDES permit should contact the Regional Water Board, reference this facility, and provide a name, address, and phone number.

G. Additional Information

Requests for additional information or questions regarding this order should be directed to Elizabeth Erickson at (213) 576 6665.

ATTACHMENT G – GENERIC TOXICITY REDUCTION EVALUATION (TRE) WORKPLAN (POTW)

1. Information and Data Acquisition

a. Operations and performance review

- i. NPDES permit requirements
 - (1) Effluent limitations
 - (2) Special conditions
 - (3) Monitoring data and compliance history
- ii. POTW design criteria
 - (1) Hydraulic loading capacities
 - (2) Pollutant loading capacities
 - (3) Biodegradation kinetics calculations/assumptions
- iii. Influent and effluent conventional pollutant data
 - (1) Biochemical oxygen demand (BOD₅)
 - (2) Chemical oxygen demand (COD)
 - (3) Suspended solids (SS)
 - (4) Ammonia
 - (5) Residual chlorine
 - (6) pH
- iv. Process control data
 - (1) Primary sedimentation - hydraulic loading capacity and BOD and SS removal
 - (2) Activated sludge - Food-to-microorganism (F/M) ratio, mean cell residence time (MCRT), mixed liquor suspended solids (MLSS), sludge yield, and BOD and COD removal
 - (3) Secondary clarification - hydraulic and solids loading capacity, sludge volume index and sludge blanket depth
- v. Operations information
 - (1) Operating logs
 - (2) Standard operating procedures
 - (3) Operations and maintenance practices
- vi. Process sidestream characterization data
 - (1) Sludge processing sidestreams
 - (2) Tertiary filter backwash
 - (3) Cooling water
- vii. Combined sewer overflow (CSO) bypass data
 - (1) Frequency
 - (2) Volume
- viii. Chemical coagulant usage for wastewater treatment and sludge processing
 - (1) Polymer
 - (2) Ferric chloride
 - (3) Alum

- b. POTW influent and effluent characterization data**
 - i. Toxicity
 - ii. Priority pollutants
 - iii. Hazardous pollutants
 - iv. SARA 313 pollutants,
 - v. Other chemical-specific monitoring results
- c. Sewage residuals (raw, digested, thickened and dewatered sludge and incinerator ash) characterization data**
 - i. EP toxicity
 - ii. Toxicity Characteristic Leaching Procedure (TCLP)
 - iii. Chemical analysis
- d. Industrial waste survey (IWS)**
 - i. Information on IUs with categorical standards or local limits and other significant non-categorical IUs
 - ii. Number of IUs
 - iii. Discharge flow
 - iv. Standard Industrial Classification (SIC) code
 - v. Wastewater flow
 - (1) Types and concentrations of pollutants in the discharge
 - (2) Products manufactured
 - vi. Description of pretreatment facilities and operating practices
 - vii. Annual pretreatment report
 - viii. Schematic of sewer collection system
 - ix. POTW monitoring data
 - (1) Discharge characterization data
 - (2) Spill prevention and control procedures
 - (3) Hazardous waste generation
 - x. IU self-monitoring data
 - (1) Description of operations
 - (2) Flow measurements
 - (3) Discharge characterization data
 - (4) Notice of sludge loading
 - (5) Compliance schedule (if out of compliance)
 - xi. Technically based local limits compliance reports
 - xii. Waste hauler monitoring data manifests
 - xiii. Evidence of POTW treatment interferences (i.e., biological process inhibition)

ATTACHMENT H – STORM WATER POLLUTION PREVENTION PLAN REQUIREMENTS

SECTION A: STORM WATER POLLUTION PREVENTION PLAN REQUIREMENTS¹

1. Implementation Schedule

A storm water pollution prevention plan (SWPPP) shall be developed and implemented for each facility covered by this General Permit in accordance with the following schedule.

- a. Facility operators beginning industrial activities before October 1, 1992 shall develop and implement the SWPPP no later than October 1, 1992. Facility operators beginning industrial activities after October 1, 1992 shall develop and implement the SWPPP when industrial activities begin.
- b. Existing facility operators that submitted a Notice of Intent (NOI), pursuant to State Water Resources Control Board (State Water Board) Order No. 91-013-DWQ (as amended by Order No. 92-12) or San Francisco Bay Regional Water Quality Control Board (Regional Water Board) Order No. 92-11 (as amended by Order No. 92-116), shall continue to implement their existing SWPPP and shall implement any necessary revisions to their SWPPP in a timely manner, but in no case later than August 1, 1997.

2. Objectives

The SWPPP has two major objectives: (a) to identify and evaluate sources of pollutants associated with industrial activities that may affect the quality of storm water discharges and authorized non-storm water discharges from the facility; and (b) to identify and implement site-specific best management practices (BMPs) to reduce or prevent pollutants associated with industrial activities in storm water discharges and authorized non-storm water discharges. BMPs may include a variety of pollution prevention measures or other low-cost and pollution control measures. They are generally categorized as non-structural BMPs (activity schedules, prohibitions of practices, maintenance procedures, and other low-cost measures) and as structural BMPs (treatment measures, run-off controls, over-head coverage.) To achieve these objectives, facility operators should consider the five phase process for SWPPP development and implementation as shown in Table A.

The SWPPP requirements are designed to be sufficiently flexible to meet the needs of various facilities. SWPPP requirements that are not applicable to a facility should not be included in the SWPPP.

A facility's SWPPP is a written document that shall contain a compliance activity schedule, a description of industrial activities and pollutant sources, descriptions of BMPs, drawings, maps, and relevant copies or references of parts of other plans. The SWPPP

¹ From State Water Board's Water Quality Order No. 97-03-DWQ (NPDES General Permit No. CAS000001), Waste Discharge Requirements for Discharges of Storm Water Associated with Industrial Activities Excluding Construction Activities.

shall be revised whenever appropriate and shall be readily available for review by facility employees or Regional Water Board inspectors.

3. Planning and Organization

a. Pollution Prevention Team

The SWPPP shall identify a specific individual or individuals and their positions within the facility organization as members of a storm water pollution prevention team responsible for developing the SWPPP, assisting the facility manager in SWPPP implementation and revision, and conducting all monitoring program activities required in Section B of this General Permit. The SWPPP shall clearly identify the General Permit related responsibilities, duties, and activities of each team member. For small facilities, storm water pollution prevention teams may consist of one individual where appropriate.

b. Review Other Requirements and Existing Facility Plans

The SWPPP may incorporate or reference the appropriate elements of other regulatory requirements. Facility operators should review all local, State, and Federal requirements that impact, complement, or are consistent with the requirements of this General Permit. Facility operators should identify any existing facility plans that contain storm water pollutant control measures or relate to the requirements of this General Permit. As examples, facility operators whose facilities are subject to Federal Spill Prevention Control and Countermeasures' requirements should already have instituted a plan to control spills of certain hazardous materials. Similarly, facility operators whose facilities are subject to air quality related permits and regulations may already have evaluated industrial activities that generate dust or particulates.

4. Site Map

The SWPPP shall include a site map. The site map shall be provided on an 8-½ x 11 inch or larger sheet and include notes, legends, and other data as appropriate to ensure that the site map is clear and understandable. If necessary, facility operators may provide the required information on multiple site maps.

**TABLE A
FIVE PHASES FOR DEVELOPING AND IMPLEMENTING INDUSTRIAL
STORM WATER POLLUTION PREVENTION PLANS**

PLANNING AND ORGANIZATION

Form Pollution Prevention Team
Review other plans

ASSESSMENT PHASE

Develop a site map
Identify potential pollutant sources
Inventory of materials and chemicals
List significant spills and leaks
Identify non-storm water discharges
Assess pollutant Risks

BEST MANAGEMENT PRACTICES IDENTIFICATION PHASE

Non-structural BMPs
Structural BMPs
Select activity and site-specific BMPs

IMPLEMENTATION PHASE

Train employees
Implement BMPs
Conduct recordkeeping and reporting

EVALUATION / MONITORING

Conduct annual site evaluation
Review monitoring information
Evaluate BMPs
Review and revise SWPPP

The following information shall be included on the site map:

- a. The facility boundaries; the outline of all storm water drainage areas within the facility boundaries; portions of the drainage area impacted by run-on from surrounding areas; and direction of flow of each drainage area, on-site surface water bodies, and areas of soil erosion. The map shall also identify nearby water bodies (such as rivers, lakes, and ponds) and municipal storm drain inlets where the facility's storm water discharges and authorized non-storm water discharges may be received.
- b. The location of the storm water collection and conveyance system, associated points of discharge, and direction of flow. Include any structural control measures that affect storm water discharges, authorized non-storm water discharges, and run-on. Examples of structural control measures are catch basins, berms, detention ponds, secondary containment, oil/water separators, diversion barriers, etc.
- c. An outline of all impervious areas of the facility, including paved areas, buildings, covered storage areas, or other roofed structures.
- d. Locations where materials are directly exposed to precipitation and the locations where significant spills or leaks identified in Section A.6.a.iv. below have occurred.
- e. Areas of industrial activity. This shall include the locations of all storage areas and storage tanks, shipping and receiving areas, fueling areas, vehicle and equipment storage/maintenance areas, material handling and processing areas, waste treatment and disposal areas, dust or particulate generating areas, cleaning and rinsing areas, and other areas of industrial activity which are potential pollutant sources.

5. List of Significant Materials

The SWPPP shall include a list of significant materials handled and stored at the site. For each material on the list, describe the locations where the material is being stored, received, shipped, and handled, as well as the typical quantities and frequency. Materials shall include raw materials, intermediate products, final or finished products, recycled materials, and waste or disposed materials.

6. Description of Potential Pollutant Sources

- a. The SWPPP shall include a narrative description of the facility's industrial activities, as identified in Section A.4.e above, associated potential pollutant sources, and potential pollutants that could be discharged in storm water discharges or authorized non-storm water discharges. At a minimum, the following items related to a facility's industrial activities shall be considered:
 - i. Industrial Processes

Describe each industrial process, the type, characteristics, and quantity of significant materials used in or resulting from the process, and a description of the manufacturing, cleaning, rinsing, recycling, disposal, or other activities related to the process. Where applicable, areas protected by containment structures and the corresponding containment capacity shall be described.

ii. Material Handling and Storage Areas

Describe each handling and storage area, type, characteristics, and quantity of significant materials handled or stored, description of the shipping, receiving, and loading procedures, and the spill or leak prevention and response procedures. Where applicable, areas protected by containment structures and the corresponding containment capacity shall be described.

iii. Dust and Particulate Generating Activities

Describe all industrial activities that generate dust or particulates that may be deposited within the facility's boundaries and identify their discharge locations; the characteristics of dust and particulate pollutants; the approximate quantity of dust and particulate pollutants that may be deposited within the facility boundaries; and a description of the primary areas of the facility where dust and particulate pollutants would settle.

iv. Significant Spills and Leaks

Describe materials that have spilled or leaked in significant quantities in storm water discharges or non-storm water discharges since April 17, 1994. Include toxic chemicals (listed in 40 CFR, Part 302) that have been discharged to storm water as reported on U.S. Environmental Protection Agency (U.S. EPA) Form R, and oil and hazardous substances in excess of reportable quantities (see 40 Code of Federal Regulations [CFR], Parts 110, 117, and 302).

The description shall include the type, characteristics, and approximate quantity of the material spilled or leaked, the cleanup or remedial actions that have occurred or are planned, the approximate remaining quantity of materials that may be exposed to storm water or non-storm water discharges, and the preventative measures taken to ensure spill or leaks do not reoccur. Such list shall be updated as appropriate during the term of this General Permit.

v. Non-Storm Water Discharges

Facility operators shall investigate the facility to identify all non-storm water discharges and their sources. As part of this investigation, all drains (inlets and outlets) shall be evaluated to identify whether they connect to the storm drain system.

All non-storm water discharges shall be described. This shall include the source, quantity, frequency, and characteristics of the non-storm water discharges and associated drainage area.

Non-storm water discharges that contain significant quantities of pollutants or that do not meet the conditions provided in Special Conditions D. are prohibited by this General Permit (Examples of prohibited non-storm water discharges are contact and non-contact cooling water, boiler blowdown, rinse water, wash water, etc.). Non-storm water discharges that meet the conditions provided in Special Condition D. are authorized by this General Permit. The SWPPP must include BMPs to prevent or reduce contact of non-storm water discharges with significant materials or equipment.

vi. Soil Erosion

Describe the facility locations where soil erosion may occur as a result of industrial activity, storm water discharges associated with industrial activity, or authorized non-storm water discharges.

- b. The SWPPP shall include a summary of all areas of industrial activities, potential pollutant sources, and potential pollutants. This information should be summarized similar to Table B. The last column of Table B, "Control Practices", should be completed in accordance with Section A.8. below.

7. Assessment of Potential Pollutant Sources

- a. The SWPPP shall include a narrative assessment of all industrial activities and potential pollutant sources as described in A.6. above to determine:
- i. Which areas of the facility are likely sources of pollutants in storm water discharges and authorized non-storm water discharges, and
 - ii. Which pollutants are likely to be present in storm water discharges and authorized non-storm water discharges. Facility operators shall consider and evaluate various factors when performing this assessment such as current storm water BMPs; quantities of significant materials handled, produced, stored, or disposed of; likelihood of exposure to storm water or authorized non-storm water discharges; history of spill or leaks; and run-on from outside sources.
- b. Facility operators shall summarize the areas of the facility that are likely sources of pollutants and the corresponding pollutants that are likely to be present in storm water discharges and authorized non-storm water discharges.

Facility operators are required to develop and implement additional BMPs as appropriate and necessary to prevent or reduce pollutants associated with each pollutant source. The BMPs will be narratively described in Section 8 below.

8. Storm Water Best Management Practices

The SWPPP shall include a narrative description of the storm water BMPs to be implemented at the facility for each potential pollutant and its source identified in the site assessment phase (Sections A.6. and 7. above). The BMPs shall be developed and implemented to reduce or prevent pollutants in storm water discharges and authorized non-storm water discharges. Each pollutant and its source may require one or more BMPs. Some BMPs may be implemented for multiple pollutants and their sources, while other BMPs will be implemented for a very specific pollutant and its source:

TABLE B
EXAMPLE

ASSESSMENT OF POTENTIAL POLLUTION SOURCES AND
CORRESPONDING BEST MANAGEMENT PRACTICES
SUMMARY

Area	Activity	Pollutant Source	Pollutant	Best Management Practices
Vehicle & Equipment Fueling	Fueling	Spills and leaks during delivery. Spills caused by topping off fuel tanks. Hosing or washing down fuel oil fuel area. Leaking storage tanks. Rainfall running off fuel oil, and rainfall running onto and off fueling area.	fuel oil	Use spill and overflow protection. Minimize run-on of storm water into the fueling area. Cover fueling area. Use dry cleanup methods rather than hosing down area. Implement proper spill prevention control program. Implement adequate preventative maintenance program to preventive tank and line leaks. Inspect fueling areas regularly to detect problems before they occur. Train employees on proper fueling, cleanup, and spill response techniques.

The description of the BMPs shall identify the BMPs as (1) existing BMPs, (2) existing BMPs to be revised and implemented, or (3) new BMPs to be implemented. The description shall also include a discussion on the effectiveness of each BMP to reduce or

prevent pollutants in storm water discharges and authorized non-storm water discharges. The SWPPP shall provide a summary of all BMPs implemented for each pollutant source. This information should be summarized similar to Table B.

Facility operators shall consider the following BMPs for implementation at the facility:

a. Non-Structural BMPs

Non-structural BMPs generally consist of processes, prohibitions, procedures, schedule of activities, etc., that prevent pollutants associated with industrial activity from contacting with storm water discharges and authorized non-storm water discharges. They are considered low technology, cost-effective measures. Facility operators should consider all possible non-structural BMPs options before considering additional structural BMPs (see Section A.8.b. below). Below is a list of non-structural BMPs that should be considered:

i. Good Housekeeping

Good housekeeping generally consist of practical procedures to maintain a clean and orderly facility.

ii. Preventive Maintenance

Preventive maintenance includes the regular inspection and maintenance of structural storm water controls (catch basins, oil/water separators, etc.) as well as other facility equipment and systems.

iii. Spill Response

This includes spill clean-up procedures and necessary clean-up equipment based upon the quantities and locations of significant materials that may spill or leak.

iv. Material Handling and Storage

This includes all procedures to minimize the potential for spills and leaks and to minimize exposure of significant materials to storm water and authorized non-storm water discharges.

v. Employee Training

This includes training of personnel who are responsible for (1) implementing activities identified in the SWPPP, (2) conducting inspections, sampling, and visual observations, and (3) managing storm water. Training should address topics such as spill response, good housekeeping, and material handling procedures, and actions necessary to implement all BMPs identified in the SWPPP. The SWPPP shall identify periodic dates for such training. Records shall be maintained of all training sessions held.

vi. Waste Handling/Recycling

This includes the procedures or processes to handle, store, or dispose of waste materials or recyclable materials.

vii. Recordkeeping and Internal Reporting

This includes the procedures to ensure that all records of inspections, spills, maintenance activities, corrective actions, visual observations, etc., are developed, retained, and provided, as necessary, to the appropriate facility personnel.

viii. Erosion Control and Site Stabilization

This includes a description of all sediment and erosion control activities. This may include the planting and maintenance of vegetation, diversion of run-on and runoff, placement of sandbags, silt screens, or other sediment control devices, etc.

ix. Inspections

This includes, in addition to the preventative maintenance inspections identified above, an inspection schedule of all potential pollutant sources. Tracking and follow-up procedures shall be described to ensure adequate corrective actions are taken and SWPPPs are made.

x. Quality Assurance

This includes the procedures to ensure that all elements of the SWPPP and Monitoring Program are adequately conducted.

b. Structural BMPs

Where non-structural BMPs as identified in Section A.8.a. above are not effective, structural BMPs shall be considered. Structural BMPs generally consist of structural devices that reduce or prevent pollutants in storm water discharges and authorized non-storm water discharges. Below is a list of structural BMPs that should be considered:

i. Overhead Coverage

This includes structures that provide horizontal coverage of materials, chemicals, and pollutant sources from contact with storm water and authorized non-storm water discharges.

ii. Retention Ponds

This includes basins, ponds, surface impoundments, bermed areas, etc. that do not allow storm water to discharge from the facility.

iii. Control Devices

This includes berms or other devices that channel or route run-on and runoff away from pollutant sources.

iv. Secondary Containment Structures

This generally includes containment structures around storage tanks and other areas for the purpose of collecting any leaks or spills.

v. Treatment

This includes inlet controls, infiltration devices, oil/water separators, detention ponds, vegetative swales, etc. that reduce the pollutants in storm water discharges and authorized non-storm water discharges.

9. Annual Comprehensive Site Compliance Evaluation

The facility operator shall conduct one comprehensive site compliance evaluation (evaluation) in each reporting period (July 1-June 30). Evaluations shall be conducted within 8-16 months of each other. The SWPPP shall be revised, as appropriate, and the revisions implemented within 90 days of the evaluation. Evaluations shall include the following:

- a. A review of all visual observation records, inspection records, and sampling and analysis results.
- b. A visual inspection of all potential pollutant sources for evidence of, or the potential for, pollutants entering the drainage system.
- c. A review and evaluation of all BMPs (both structural and non-structural) to determine whether the BMPs are adequate, properly implemented and maintained, or whether additional BMPs are needed. A visual inspection of equipment needed to implement the SWPPP, such as spill response equipment, shall be included.
- d. An evaluation report that includes, (i) identification of personnel performing the evaluation, (ii) the date(s) of the evaluation, (iii) necessary SWPPP revisions, (iv) schedule, as required in Section A.10.e, for implementing SWPPP revisions, (v) any incidents of non-compliance and the corrective actions taken, and (vi) a certification that the facility operator is in compliance with this General Permit. If the above certification cannot be provided, explain in the evaluation report why the facility operator is not in compliance with this General Permit. The evaluation report shall be submitted as part of the annual report, retained for at least five years, and signed and certified in accordance with Standard Provisions 9. and 10. of Section C. of this General Permit.

10. SWPPP General Requirements

- a. The SWPPP shall be retained on site and made available upon request of a representative of the Regional Water Board and/or local storm water management agency (local agency) which receives the storm water discharges.
- b. The Regional Water Board and/or local agency may notify the facility operator when the SWPPP does not meet one or more of the minimum requirements of this Section. As requested by the Regional Water Board and/or local agency, the facility operator shall submit an SWPPP revision and implementation schedule that meets the minimum requirements of this section to the Regional Water Board and/or local agency that requested the SWPPP revisions. Within 14 days after implementing the required SWPPP revisions, the facility operator shall provide written certification to the Regional Water Board and/or local agency that the revisions have been implemented.
- c. The SWPPP shall be revised, as appropriate, and implemented prior to changes in industrial activities which (i) may significantly increase the quantities of pollutants in storm water discharge, (ii) cause a new area of industrial activity at the facility to be exposed to storm water, or (iii) begin an industrial activity which would introduce a new pollutant source at the facility.
- d. Other than as provided in Provisions B.11, B.12, and E.2 of the General Permit, the SWPPP shall be revised and implemented in a timely manner, but in no case more than 90 days after a facility operator determines that the SWPPP is in violation of any requirement(s) of this General Permit.
- e. When any part of the SWPPP is infeasible to implement by the deadlines specified in Provision E.2 or Sections A.1, A.9, A.10.c, and A.10.d of this General Permit due to proposed significant structural changes, the facility operator shall submit a report to the Regional Water Board prior to the applicable deadline that (i) describes the portion of the SWPPP that is infeasible to implement by the deadline, (ii) provides justification for a time extension, (iii) provides a schedule for completing and implementing that portion of the SWPPP, and (iv) describes the BMPs that will be implemented in the interim period to reduce or prevent pollutants in storm water discharges and authorized non-storm water discharges. Such reports are subject to Regional Water Board approval and/or modifications. Facility operators shall provide written notification to the Regional Water Board within 14 days after the SWPPP revisions are implemented.
- f. The SWPPP shall be provided, upon request, to the Regional Water Board. The SWPPP is considered a report that shall be available to the public by the Regional Water Board under Section 308(b) of the Clean Water Act.

ATTACHMENT I

BIOSOLIDS USE AND DISPOSAL REQUIREMENTS

1. All biosolids generated by the Discharger shall be reused or disposed of in compliance with the applicable portions of:
 - a. 40 CFR 503: for biosolids that are land applied, placed in surface disposal sites (dedicated land disposal sites or monofills), or incinerated; 40 CFR 503 Subpart B (land application) applies to biosolids placed on the land for the purpose of providing nutrients or conditioning the soil for crops or vegetation. 40 CFR 503 Subpart C (surface disposal) applies to biosolids placed on the land for the purpose of disposal.
 - b. 40 CFR 258: for biosolids disposed of in Municipal Solid Waste landfills.
 - c. 40 CFR 257: for all biosolids disposal practices not covered under 40 CFR 258 or 503.
2. The Discharger is responsible for assuring that all biosolids from its facility are used or disposed of in accordance with 40 CFR 503, whether the Discharger reuses or disposes of the biosolids itself or transfers them to another party for further treatment, reuse, or disposal. The Discharger is responsible for informing subsequent preparers, applicers, or disposers of the requirements they must meet under 40 CFR 503.
3. Duty to mitigate: The Discharger shall take all reasonable steps to prevent or minimize any biosolids use or disposal which may adversely impact human health or the environment.
4. No biosolids shall be allowed to enter wetland or other waters of the United States.
5. Biosolids treatment, storage, and use or disposal shall not contaminate groundwater.
6. Biosolids treatment, storage, and use or disposal shall not create a nuisance such as objectionable odors or flies.
7. The Discharger shall assure that haulers who transport biosolids off site for further treatment, storage, reuse, or disposal take all necessary measures to keep the biosolids contained.
8. If biosolids are stored for over two years from the time they are generated, the Discharger must ensure compliance with all the requirements for surface disposal under 40 CFR 503 Subpart C, or must submit a written request to EPA with the information in 503.20 (b), requesting permission for longer temporary storage.

9. Sewage sludge containing more than 50 mg/kg PCB's shall be disposed of in accordance with 40 CFR 761.
10. Any off-site biosolids treatment, storage, use or disposal site operated by the Discharger within Region 4 (Los Angeles Region of RWQCB) that is not subject to its own Waste Discharge Requirements shall have facilities adequate to divert surface runoff from the adjacent area, to protect the site boundaries from erosion, and to prevent any conditions that would cause drainage from the materials in the disposal site to escape from the site. Adequate protection is defined as protected from at least a 100-year storm and from the highest tidal stage that may occur.
11. Inspection and Entry: The Regional Board, USEPA or an authorized representative thereof, upon the presentation of credentials, shall be allowed by the Discharger, directly or through contractual arrangements with their biosolids management contractors, to:
 - a. enter upon all premises where biosolids are produced by the Discharger and all premises where Discharger biosolids are further treated, stored, used, or disposed, either by the Discharger or by another party to whom the Discharger transfers the biosolids for further treatment, storage, use, or disposal;
 - b. have access to and copy any records that must be kept under the conditions of this permit or of 40 CFR 503, by the Discharger or by another party to whom the Discharger transfers the biosolids for further treatment, storage, use, or disposal; and
 - c. inspect any facilities, equipment (including monitoring and control equipment), practices, or operations used in the production of biosolids and further treatment, storage, use, or disposal by the Discharger or by another party to whom the Discharger transfers the biosolids for further treatment, storage, use, or disposal.
12. Monitoring shall be conducted as follows:
 - a. Biosolids shall be tested for the metals required in section 503.16 (for land application) or 503.26 (for surface disposal), using the methods in "Test Methods for Evaluating Solids Waste, Physical/Chemical Methods" (SW-846), as required in 503.8(b)(4), at the following minimum frequencies:

<u>Volume (dry metric tons/year)</u>	<u>Frequency</u>
0 – 290	once per year
290 – 1500	once per quarter
1500 – 15000	once per 60 days
> 15000	once per month

For accumulated, previously untested biosolids, the Discharge~ shall develop a

representative sampling plan, which addresses the number and location of sampling points, and collect representative samples.

Test results shall be expressed in mg pollutant per kg biosolids on a 100% dry weight basis.

Biosolids to be land applied shall be tested for Organic-N, ammonium-N, and nitrate-N at the frequencies required above.

- b. Prior to land application, the Discharger shall demonstrate that the biosolids meet Class A or Class B pathogen reduction levels by one of the methods listed in 503.32. Prior to disposal in a surface disposal site, the Discharger shall demonstrate that the biosolids meet Class B levels or shall ensure that the site is covered at the end of each operating day.
 - c. For biosolids that are land applied or placed in a surface disposal site, the Discharger shall track and keep records of the operational parameters used to achieve Vector Attraction Reduction requirements in 503.33 (b).
 - d. Class 1 facilities (facilities with pretreatment programs or others designated as Class 1 by the Regional Administrator) and Federal facilities with > 5 mgd influent flow shall sample biosolids for pollutants listed under Section 307 (a) of the Act (as required in the pretreatment section of the permit for POTWs with pretreatment programs.) Class 1 facilities and Federal Facilities with > 5 mgd influent flow shall test dioxins/dibenzofurans using a detection limit of < 1 pg/g during their next sampling period if they have not done so within the past 5 years and once per 5 years thereafter.
 - e. The biosolids shall be tested annually, or more frequently if necessary to determine hazardousness in accordance with California Law.
 - f. If biosolids are placed in a surface disposal site (dedicated land disposal site or monofill), a qualified groundwater scientist shall develop a groundwater monitoring program for the site, or shall certify that the placement of biosolids on the site will not contaminate an aquifer.
 - g. Biosolids placed in a municipal landfill shall be tested semi-annually by the Paint Filter Test (SW-846, Method 9095) to demonstrate that there are no free liquids.
13. The Discharger either directly or through contractual arrangements with their biosolids management contractors shall comply with the following 40 CFR 503 notification requirements:
- a. A reuse/disposal plan shall be submitted to EPA Region IX Coordinator and, in the absence of other state or regional reporting requirements, to the state permitting

agency, prior to the use or disposal of any biosolids from this facility to a new or previously unreported site. The plan shall be submitted by the land applier of the biosolids and shall include, a description and a topographic map of the proposed site(s) for reuse or disposal, names and addresses of the applier(s) and site owner(s), and a list of any state or local permits which must be obtained. For land application sites, the plan shall include a description of the crops or vegetation to be grown, proposed nitrogen loadings to be used for the crops, and a groundwater monitoring plan if one exists.

- b. If the Discharger biosolids do not meet 503.13 Table 3 metals concentration limits, the Discharger must require their land applier to contact the state permitting authority to determine whether bulk biosolids subject to the cumulative pollutant loading rates in 503.12(b)(2) have been applied to the site since July 20, 1993, and, if so, the cumulative amount of pollutants applied to date, and background concentration, if known. The Discharger shall then notify EPA Region IX Coordinator of this information.
 - c. For biosolids that are land applied, the Discharger shall notify the applier in writing of the nitrogen content of the biosolids, and the applier's requirements under 503, including the requirements that the applier certify that the requirement to obtain information in Subpart A, and that the management practices, site restrictions, and any applicable vector attraction reduction requirements Subpart D have been met. The Discharger shall require the applier to certify at the end of 38 months following application of Class B biosolids that those harvesting restrictions in effect for up to 38 months have been met.
 - d. If bulk biosolids are shipped to another State or to Indian Lands, the Discharger must send written notice prior to the initial application of bulk biosolids to the permitting authorities in the receiving State or Indian Land (the EPA Regional Office for the area and the State/Indian authorities).
 - e. Notification of 503 non-compliance: The Discharger shall require appliers of their biosolids to notify EPA Region 9 and their state permitting agency of any noncompliance within 24 hours if the non-compliance may seriously endanger health or the environment. For other instances of non-compliance, the Discharger shall require appliers of their biosolids to notify EPA Region 9 and their state permitting agency of the non-compliance in writing within 10 working days of becoming aware of the non-compliance.
14. The Discharger shall submit an annual biosolids report to EPA Region IX Biosolids Coordinator and the Los Angeles Regional Water Quality Control Board by February 19 of each year for the period covering the previous calendar year. The report shall include:
- a. The amount of biosolids generated that year, in dry metric tons, and the amount

accumulated from previous years.

- b. Results of all pollutant monitoring required in the Monitoring Section above.
 - c. Descriptions of pathogen reduction methods, and vector attraction reduction methods, as required in 503.17 and 503.27.
 - d. Results of any groundwater monitoring or certification by groundwater scientist that the placement of biosolids in a surface disposal site will not contaminate an aquifer.
 - e. Names and addresses of land appliers and surface disposal site operators, and volumes applied (dry metric tons).
 - f. Names and addresses of persons who received biosolids for storage, further treatment, disposal in a municipal waste landfill, or for other reuse/disposal methods not covered in 14.c, above, and volumes delivered to each.
15. The Discharger shall require all parties contracted to manage their biosolids to submit an annual biosolids report to EPA Region IX Biosolids Coordinator by February 19 of each year for the period covering the previous calendar year. The report shall include:
- a. Names and addresses of land appliers and surface disposal site operators, name, location (latitude/longitude), and size (hectares) of site(s), volumes applied/disposed (dry metric tons) and for land application, biosolids loading rates (metric tons per hectare), nitrogen loading rates (kg/ha), dates of applications, crops grown, dates of seeding and harvesting and certifications that the requirement to obtain information in 503.12(e)(2), management practices in 503.14 and site restrictions in 503.32(b)(5) have been met.

Attachment K-1 Metals Effluent Data															
Constituent	Copper	1/2 copper	Lead	1/2 Lead	Nickel		Selenium				1/2 nickel	Zinc	1/2 zinc	Selenium	1/2 Selenium
Mar-08	8.52	8.52	9.4	9.4	3.44	3.44					3.44	38.5	38.5		
Apr-08	3.72	3.72										35.6	35.6		
May-08	3.26	3.26										26.6	26.6		
Jun-08	<2	1.00										18.9	18.9		
Jul-08	<2	1.00										22.9	22.9		
Aug-08	<2	1.00	<5	0.3	<10	5.00	8.78				5.00	21.8	21.8	8.78	8.78
Sep-08	3.29	3.29										19.4	19.4		
Oct-08	<2	1.00										14.9	14.9		
Nov-08	4.04	4.04										33.3	33.3		
Dec-08	<2	1.00										21.1	21.1		
Jan-09	<2	1.00										37.1	37.1		
Feb-09	3.24	3.24	<5	0.3	4.66	4.66	2.0				4.66	22.6	22.6	<2	1
Mar-09	<2	1.00										23.3	23.3		
Apr-09	<2	1.00										22.7	22.7		
May-09	<2	1.00										24.7	24.7		
Jun-09	<2	1.00										17.6	17.6		
Jul-09	<2	1.00										19.6	19.6		
Aug-09	2.09	2.09	<5	0.3	4.14	4.14	2.0				4.14	23.8	23.8	<2	1
Sep-09	<2	1.00										22.2	22.2		
Oct-09	<2	1.00										23.2	23.2		
Nov-09	<2	1.00										18.7	18.7		
Dec-09	<2	1.00										30.2	30.2		
Jan-10	3.74	3.74										35.3	35.3		
Feb-10	4.75	4.75	<5	0.3	5.03	5.03	2.0				5.03	37.6	37.6	<2	1
Mar-10	4.36	4.36										34.4	34.4		
Apr-10	2.36	2.36										41.1	41.1		
May-10	<2	1.00										32.2	32.2		
Jun-10	2.95	2.95										42.0	42.0		
Jul-10	<2	1.00										34.6	34.6		
Aug-10	2.92	2.92	<5	0.3	5.57	5.57	2.0				5.57	44.7	44.7	<2	1
Sep-10	<2	1.00										33.5	33.5		
Oct-10	4.95	4.95										32.0	32.0		
Nov-10	3.62	3.62										28.6	28.6		
Dec-10	4.16	4.16										33.8	33.8		
Jan-11	2.86	2.86										30.5	30.5		
Feb-11	4.09	4.09	<5	0.3	11.40	11.40	13.7				11.40	26.1	26.1	13.7	13.7
Mar-11	2.61	2.61										33.0	33.0		
Apr-11	2.87	2.87										35.0	35.0		
May-11	<2	1.00										<2	1.0		
Jun-11	4.94	4.94										34.8	34.8		
Jul-11	<2	1.00										34.2	34.2		
Aug-11	<2	1.00	<5	0.3	<1	0.50	2.0				0.50	35.4	35.4	<2	1
Sep-11	<2	1.00										42.1	42.1		
Oct-11	5.81	5.81										36.5	36.5		
Nov-11	4.35	4.35										18.5	18.5		
Dec-11	7.63	7.63										68.8	68.8		
Jan-12	4.48	4.48										34.7	34.7		
Feb-12	5.00	5.00	<5	0.3	23.30	23.30	2.0				23.30	31.3	31.3	<2	1
Mar-12	5.07	5.07										34.5	34.5		
Apr-12	<2	1.00										35.7	35.7		
May-12	<2	1.00										35.7	35.7		
Jun-12	<2	1.00										27.8	27.8		
Jul-12	<2	1.00										12.8	12.8		
Aug-12	<2	1.00	<5	0.3	35.40	35.40	2.0				35.40	31.9	31.9	<2	1
Sep-12	<2	1.00										31.4	31.4		
Oct-12	<2	1.00													
Nov-12	8.88	8.88										36.9	36.9		
Dec-12	<2	1.00										11.1	11.1		
Jan-13	<2	1.00										39.3	39.3		
Feb-13	2.89	2.89	<5	0.3	3.69						3.69	40.0	40.0	<2	1
Mar-13	<2	1.00										39.3	39.3		

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Data	Ethylbenzene		Methyl Bromide		Methyl Chloride		Methylene Chloride		1,1,2,2-Tetrachloroethane		Tetrachloroethylene		Toluene		1,2-Trans-Dichloroethylene		1,1,1-Trichloroethane		1,1,2-Trichloroethane		Trichloroethylene		Vinyl Chloride		2-Chlorophenol		2,4-Dichlorophenol		2,4-Dimethylphenol		2-Methyl-4,6-Dinitrophenol
Source	33		34		35		36		37		38		39		40		41		42		43		44		45		46		47		48
ata points	10		10		10		10		10		10		10		8		10		10		10		10		10		10		10		10
SMR																															
SMR																															
SMR																															
SMR																															
SMR	0.34	<	0.50	<	0.08	<	0.25	<	0.03	<	0.03	<	0.20	<	0.10	<	0.03	<	0.02	<	0.12	<	0.18	<	3.3	<	2.7	<	2.7	<	10
SMR																															
SMR																															
SMR																															
SMR	0.34	<	0.50	<	0.08	<	0.25	<	0.03	<	0.03	<	0.20	<	0.10	<	0.03	<	0.02	<	0.12	<	0.18	<	3.3	<	2.7	<	2.7	<	10
SMR																															
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SMR																															
SMR	0.34	<	0.50	<	0.08	<	0.25	<	0.03	<	0.03	<	0.20	<	0.10	<	0.03	<	0.02	<	0.12	<	0.18	<	3.3	<	2.7	<	2.7	<	10
SMR																															
SMR																															
SMR																															
SMR	0.34	<	0.50	<	0.08	<	0.25	<	0.03	<	0.03	<	0.20	<	0.10	<	0.03	<	0.02	<	0.12	<	0.18	<	3.3	<	2.7	<	2.7	<	10
SMR																															
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SMR	0.34	<	0.50	<	0.08	<	0.25	<	0.03	<	0.03	<	0.20	<	0.10	<	0.03	<	0.02	<	0.12	<	0.18	<	3.3	<	2.7	<	2.7	<	10
SMR																															
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Data		2,4-Dinitrophenol		2-Nitrophenol		4-Nitrophenol		3-Methyl-4-Chlorophenol		Pentachlorophenol		Phenol		2,4,6-Trichlorophenol		Acenaphthene		Acenaphthylene		Anthracene		Benzidine		Benzo(a)Anthracene		Benzo(a)Pyrene		Benzo(b)Fluoranthene		Benzo(ghi)Perylene	
Source		49		50		51		52		53		54		55		56		57.00		58		59		60		61		62		63	
ata points		9		9		9		9		10		9		9		10		10		10		10		10		10		10		10	
SMR																															
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SMR	<	42	<	3.6	<	2.4	<	3	<	3.60	<	1.5	<	2.7	<	2	<	1.90	<	1.9	<	4.4	<	7.8	<	2.5	<	4.8	<	4.1	<
SMR																															
SMR																															
SMR																															
SMR																															
SMR																															
SMR	<	42	<	3.6	<	2.4	<	3	<	3.60	<	1.5	<	2.7	<	2	<	1.90	<	1.9	<	4.4	<	7.8	<	2.5	<	4.8	<	4.1	<
SMR																															
SMR																															
SMR																															
SMR																															
SMR	<	42	<	3.6	<	2.4	<	3	<	3.60	<	1.5	<	2.7	<	2	<	1.90	<	1.9	<	4.4	<	7.8	<	2.5	<	4.8	<	4.1	<
SMR																															
SMR																															
SMR																															
SMR	<	42	<	3.6	<	2.4	<	3	<	3.60	<	1.5	<	2.7	<	2	<	1.90	<	1.9	<	4.4	<	7.8	<	2.5	<	4.8	<	4.1	<
SMR																															
SMR																															
SMR																															
SMR	<	42	<	1.9	<	3.6	<	3	<	3.60	<	1.5	<																		

Data		Benzok(j)Fluoranthene		Bis(2-Chloroethoxy)Methane		Bis(2-Chloroethyl)Ether		Bis(2-Chloroisopropyl)Ether		Bis(2-Ethylhexyl)Phthalate		4-Bromophenyl Phenyl Ether		Butylbenzyl Phthalate		2-Chloronaphthalene		4-Chlorophenyl Phenyl Ether		Chrysene		Dibenzo(a,h)Anthracene		1,2-Dichlorobenzene		1,3-Dichlorobenzene		1,4-Dichlorobenzene		3,3'-Dichlorobenzidine		Diethyl Phthalate
Source	64		65		66		67		68		69		70		71		72		73		74		75		76		77		78		79	
ata points	10		10		10		10		20		10		10		10		10		10		10		10		10		10		10		10	
	SMR						<	2.5																								
	SMR						<	2.5																								
	SMR						<	2.5																								
	SMR						<	2.5																								
	SMR	2.5	<	5.3	<	5.7	<	5.7	<	2.5	<	1.9	<	2.5	<	1.9	<	4.2	<	2	<	2.5	<	1.9	<	1.9	<	4.4	<	16.5	<	2.2
	SMR						<	2.5																								
	SMR						<	2.5																								
	SMR						<	2.5																								
	SMR						<	2.5																								
	SMR	2.5	<	5.3	<	5.7	<	5.7	<	2.5	<	1.9	<	2.5	<	1.9	<	4.2	<	2	<	2.5	<	1.9	<	1.9	<	4.4	<	16.5	<	2.2
	SMR						<	2.5																								
	SMR																															
	SMR																															
	SMR	2.5	<	5.3	<	5.7	<	5.7	<	2.5	<	1.9	<	2.5	<	1.9	<	4.2	<	2	<	2.5	<	1.9	<	1.9	<	4.4	<	16.5	<	2.2
	SMR																															
	SMR																															
	SMR																															
	SMR	2.5	<	5.3	<	5.7	<	5.7	<	2.5	<	1.9	<	2.5	<	1.9	<	4.2	<	2	<	2.5	<	1.9	<	1.9	<	4.4	<	16.5	<	2.2
	SMR																															
	SMR																															
	SMR																															
	SMR	2.5	<	5.3	<	5.7	<	5.7	<	2.5	<	1.9	<	2.5	<	1.9	<	4.2	<	2	<	2.5										

hmen																															
Data		Dimethyl Phthalate		Di-n-Butyl Phthalate		2,4-Dinitrotoluene		2,6-Dinitrotoluene		Di-n-Octyl Phthalate		1,2-Diphenylhydrazine		Fluoranthene		Fluorene		Hexachlorobenzene		Hexachlorobutadiene		Hexachlorocyclopentadiene		Hexachloroethane		Indeno(1,2,3-cd)Pyrene		Isophorone		Naphthalene	
Source data points		80		81		82		83		84		85		86		87		88		89		90		91		92		93		94	
		10		10		10		10		10		10		10		10		10		10		10		10		10		10		10	
SMR																															
SMR																															
SMR																															
SMR																															
SMR	<	1.6	<	2.5	<	5.7	<	1.9	<	2.5	<	10	<	2.2	<	1.9	<	1.9	<	0.9	<	1.9	<	1.6	<	3.7	<	2.2	<	10	<
SMR																															
SMR																															
SMR																															
SMR	<	1.6	<	2.5	<	5.7	<	1.9	<	2.5	<	10	<	2.2	<	1.9	<	1.9	<	0.9	<	1.9	<	1.6	<	3.7	<	2.2	<	10	<
SMR																															
SMR																															
SMR	<	1.6	<	2.5	<	5.7	<	1.9	<	2.5	<	10	<	2.2	<	1.9	<	1.9	<	0.9	<	1.9	<	1.6	<	3.7	<	2.2	<	10	<
SMR																															
SMR																															
SMR	<	1.6	<	2.5	<	5.7	<	1.9	<	2.5	<	10	<	2.2	<	1.9	<	1.9	<	0.9	<	1.9	<	1.6	<	3.7	<	2.2	<	10	<
SMR																															
SMR																															
SMR	<	1.6	<	2.5	<	5.7	<	1.9	<	2.5	<	10	<	2.2	<	1.9	<	1.9	<	0.9	<	1.9	<	1.6	<	3.7	<	2.2	<	10	<
SMR																															
SMR																															
SMR	<	1.6	<	2.5	<	5.7	<	1.9	<	2.5	<	10	<	2.2	<	1.9	<	1.9	<	0.9	<	1.9	<	1.6	<	3.7	<	2.2	<	10	<
SMR																															
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Attachment K-4 Non Priority Effluent Data

MBAS, Toxcity, Oil and Grease, BOD, Suspended Solids, Settleable Solids, Residual Chlorine, Coliform, Temperature

	Effluent Grease & Oil	Effluent MBAS	Effluent MBAS	Chronic Toxicity Selenastrum	Chronic Fathead Larvae - Growth *	Chronic Fathead Larvae - Survival *	Chronic Ceriodaphnia - Survival *	Chronic Ceriodaphnia - Reproduction *	Annual Test								
Month	MG/L	standard method	mg/L	TUc	TUc	TUc	TUc	TUc	Annual Test								
maximum	6.5	0	0.16	6.25	1	1	1	1.79	1.79								
Jan 2008	4.9075	<.5	0.086	1	1	1	1	1									
Feb 2008	4.685	<.5	0.01	1	1	1	1	1.79									
Mar 2008	3.2675	<.5	0.01	1													
Apr 2008	4.474	<.5	0.01	1													
May 2008	1.25	<.5	0.04	1.79													
Jun 2008	4.745	<.5	0.013	1													
Jul 2008	2.65	<.5	0.003	1													
Aug 2008	1.91	<.5		1													
Sep 2008	4.94	<.5	0.005	1													
Oct 2008	4.4425	<.5	0.098	1													
Nov 2008	3.71	<.5	0.065	1													
Dec 2008	3.01	<.5	0.14	1													
Jan 2009	4.575	<.5	0.02	1													
Feb 2009	2.7175	<.5	0.08	1													
Mar 2009	1.38	<.5	0.09	1													
Apr 2009	1.525	<.5	0.16	1													
May 2009	2.435	<.5		1													
Jun 2009	1.946	<.5		1													
Jul 2009	1.085	<.5		6.25													
Aug 2009	1.394	<.5		3.13													
Sep 2009	2.2175	<.5		1.395													
Oct 2009	1.8175	<.5		1													
Nov 2009	1.188	<.5		1													
Dec 2009	1.6475	<.5		1													
Jan 2010	1.605	<.5		1													
Feb 2010	2.135	<.5		1													
Mar 2010	<1.0	<.5		1													
Apr 2010	3.5	<.5		1													
May 2010	3.25	<.5		1													
Jun 2010	1.6575	<.5		1													
Jul 2010	<1.0	<.5		1													
Aug 2010	<1.0	<.5		1													
Sep 2010	2.475	<.5		1													
Oct 2010	<4.0	<.5		1													
Nov 2010	<5.0	<.5		1	1	1	1	1									
Dec 2010	1.2	<.5		1	1	1	1	1									
Jan 2011	<1.0	<.5	0.12	1	1	1	1	1									
Feb 2011	1.8	<.5	0.15	1													
Mar 2011	<1.0	<.5	0.05	1													
Apr 2011	1	<.5	0.06	1													
May 2011	<1.0	<.5	0.11	1													
Jun 2011	3.25	<.5	0.11	1													
Jul 2011	<1.0	<.5	0.07	1													
Aug 2011	<1.0	<.5	0.12	1													
Sep 2011	<1.0	<.5	0.07	1													
Oct 2011	<1.0	<.5	0.04	1													
Nov 2011	3.725	<.5	0.1	1													
Dec 2011	4.15	<.5	0.08	1													
Jan 2012	1.7	<0.50	0.14	1													
Feb 2012	1.3	<0.50	0.09	1													
Mar 2012	2.7	<0.50	0.06	1													
Apr 2012	3.4	<0.50	0.02	1													
May 2012	3.1	<0.50	0.06	1.79													
Jun 2012	2.5	<0.50	0.06	1													
Jul 2012	1.6	<0.50	0.07	1													
Aug 2012	2.2	<0.50	0.1	1													
Sep 2012	2.9	<0.50	0.06	1													
Oct 2012	5	<0.50	0.1	1													
Nov 2012	4.5	<0.50	0.1	1	1	1	1	1	1.79								
Dec 2012	4.3	<0.50	0.07	1													
Jan-2013	6.5	<0.50	0.09	1													
Feb-2013	4.8	<0.50	0.05	1													
Mar-2013	3.9			1													
Apr-2013	4.6			1													
maximum	6.5		0.16	6.25	1	1	1	1.79	1.79								
minimum	1		0.003	1	1	1	1	1	1.79								
average	2.935903846		0.072682927	1.146171875	1	1	1	1.131666667	1.79								
count	52		41	64	6	6	6	6	1								
stdev	1.348864833		0.041062973	0.708609853	0	0	0	0.294415617	0								
coefficient of variation	0.459437674		0.564960368	0.618240482	0	0	0	0.260161075	0								
Electronically reported after NDN	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C)				Chlorine, Total Residual		Settleable Solids		Total Suspended Solids (TSS)			Coliform		Temperature		Total Dissolved Solids	
	Monthly Average (Mean)	Min Percent Reduction	Single	Weekly Average (Mean)	Daily Maximum	Daily Maximum	Monthly Average (Mean)	Monthly Average of Daily	Min	Single	Weekly Average (Mean)	Fecal	Total	ave	max	ave	max

	mg/L	%	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	%	mg/L	mg/L	mg/L	mg/L	F	F	mg/L	mg/L
maximum	3.3	97.5	7	4	0.27	0.1	0.1	1.46	98.5	4.36	2	2	30	73.3	80.2	1470	1698
	=	=	=	=	0	<	<	=		=	=	=	=	=	=	=	=

Attachment K-6 Non Priority Pollutants Effluent Data Ammonia																						
Total Ammonia Data October 2011 to April 2013							Total Ammonia Data Summer October 2011 and May to October 2012							Total IAmmonia Winter Data Nov 2011 to Apr 2012, Nov to April 2013								
Location	Parameter	Result	Units	Sampling Date	Average monthly		Location	Parameter	Qual	Result	Units	Sampling Date	Average Monthly		Location	Parameter	Qual	Result	Units	Sampling Date	Average Monthly	
Effluent	Ammonia,	=	0.9 mg/L	10/5/2011	0.975		Effluent	Ammonia,	=	0.9 mg/L	10/5/2011	0.98			Effluent	Ammonia,	=	1.2 mg/L	11/1/2011	1.025		
Effluent	Ammonia,	=	1.1 mg/L	10/12/2011			Effluent	Ammonia,	=	1.1 mg/L	10/12/2011				Effluent	Ammonia,	=	0.6 mg/L	11/9/2011			
Effluent	Ammonia,	=	0.9 mg/L	10/18/2011			Effluent	Ammonia,	=	0.9 mg/L	10/18/2011				Effluent	Ammonia,	=	1.1 mg/L	11/17/2011			
Effluent	Ammonia,	=	1 mg/L	10/26/2011			Effluent	Ammonia,	=	1 mg/L	10/26/2011				Effluent	Ammonia,	=	1.2 mg/L	11/22/2011			
Effluent	Ammonia,	=	1.2 mg/L	11/1/2011	1.025		Effluent	Ammonia,	=	0.8 mg/L	5/2/2012	0.92			Effluent	Ammonia,	=	0.8 mg/L	12/1/2011	0.72		
Effluent	Ammonia,	=	0.6 mg/L	11/9/2011			Effluent	Ammonia,	=	0.9 mg/L	5/9/2012				Effluent	Ammonia,	=	0.7 mg/L	12/7/2011			
Effluent	Ammonia,	=	1.1 mg/L	11/17/2011			Effluent	Ammonia,	=	1.1 mg/L	5/16/2012				Effluent	Ammonia,	=	0.7 mg/L	12/14/2011			
Effluent	Ammonia,	=	1.2 mg/L	11/22/2011			Effluent	Ammonia,	=	1.1 mg/L	5/23/2012				Effluent	Ammonia,	=	0.9 mg/L	12/21/2011			
Effluent Transfer	Ammonia, Total (as N)	=	0.8 mg/L	12/1/2011	0.72		Effluent Transfer	Ammonia, Total (as N)	=	0.7 mg/L	5/30/2012				Effluent Transfer	Ammonia, Total (as N)	=	0.5 mg/L	12/28/2011			
Effluent Transfer	Ammonia, Total (as N)	=	0.7 mg/L	12/7/2011			Effluent Transfer	Ammonia, Total (as N)	=	0.8 mg/L	6/6/2012	0.63			Effluent Transfer	Ammonia, Total (as N)	=	0.8 mg/L	1/4/2012	0.85		
Effluent	Ammonia,	=	0.7 mg/L	12/14/2011			Effluent	Ammonia,	=	0.7 mg/L	6/13/2012				Effluent	Ammonia,	=	0.8 mg/L	1/11/2012			
Effluent	Ammonia,	=	0.9 mg/L	12/21/2011			Effluent	Ammonia,	=	0.7 mg/L	6/20/2012				Effluent	Ammonia,	=	0.9 mg/L	1/18/2012			
Effluent	Ammonia,	=	0.5 mg/L	12/28/2011			Effluent	Ammonia,	=	0.3 mg/L	6/27/2012				Effluent	Ammonia,	=	0.9 mg/L	1/25/2012			
Effluent	Ammonia,	=	0.8 mg/L	1/4/2012	0.85		Effluent	Ammonia,	=	0.8 mg/L	7/5/2012	0.88			Effluent	Ammonia,	=	1.2 mg/L	2/1/2012	0.825		
Effluent	Ammonia,	=	0.8 mg/L	1/11/2012			Effluent	Ammonia,	=	1.2 mg/L	7/11/2012				Effluent	Ammonia,	=	0.4 mg/L	2/8/2012			
Effluent	Ammonia,	=	0.9 mg/L	1/18/2012			Effluent	Ammonia,	=	0.8 mg/L	7/18/2012				Effluent	Ammonia,	=	0.8 mg/L	2/15/2012			
Effluent	Ammonia,	=	0.9 mg/L	1/25/2012			Effluent	Ammonia,	=	0.7 mg/L	7/25/2012				Effluent	Ammonia,	=	0.9 mg/L	2/22/2012			
Effluent	Ammonia,	=	1.2 mg/L	2/1/2012	0.825		Effluent	Ammonia,	=	0.8 mg/L	8/1/2012	0.80			Effluent	Ammonia,	=	0.8 mg/L	3/7/2012	1		
Effluent	Ammonia,	=	0.4 mg/L	2/8/2012			Effluent	Ammonia,	=	0.6 mg/L	8/8/2012				Effluent	Ammonia,	=	1 mg/L	3/14/2012			
Effluent	Ammonia,	=	0.8 mg/L	2/15/2012			Effluent	Ammonia,	=	0.9 mg/L	8/15/2012				Effluent	Ammonia,	=	0.9 mg/L	3/21/2012			
Effluent	Ammonia,	=	0.9 mg/L	2/22/2012			Effluent	Ammonia,	=	0.9 mg/L	8/22/2012				Effluent	Ammonia,	=	1.3 mg/L	3/28/2012			
Effluent	Ammonia,	=	0.8 mg/L	3/7/2012	1		Effluent	Ammonia,	=	0.8 mg/L	8/29/2012				Effluent	Ammonia,	=	0.9 mg/L	4/4/2012	0.933333		
Effluent																						

R1 to R5 Ammonia Summer Receiving and Projected Limits										R1 to R5 Ammonia Winter Receiving and Projected Limits							
Date	Station	Temp ¹	pH ²	Salinity	I	pKas	Average monthly Ammonia	1/2 Ammonia for ND		MDEL	0.772388	Date	Station	Temp ¹	pH ²	Salinity	I
Oct-11	R1	18.6	7.54	1.5	0.029936	9.248472586	0.5	0.5		AMEL	0.290793	Nov-11	R2	15.7	8.47	1.4	0.027937532
Oct-11	R2	19.3	7.9	1.4	0.027938	9.248240754	2.1	2.1				Nov-11	R3	14.9	8.11	1.3	0.025939383
Oct-11	R3	18.8	7.72	1.4	0.027938	9.248240754	0.5	0.5				Nov-11	R4	15.3	7.89	1.3	0.025939383
Oct-11	R4	19.1	7.49	1.4	0.027938	9.248240754	0.7	0.7				Dec-11	R2	12.1	7.97	2	0.039934878
Oct-11	R5	19.7	8.07	1.4	0.027938	9.248240754	0.5	0.5				Dec-11	R3	10.8	7.81	1.7	0.033934393
May-12	R1	19.8	7.84	3.1	0.061968	9.252188255	1.5	1.5				Dec-11	R4	10.9	7.77	1.7	0.033934393
May-12	R2	19.6	8.34	2.9	0.057958	9.25172314	0.3	0.3				Jan-12	R2	13.3	7.88	2.3	0.045938989
May-12	R3	19.72	8.31	2.8	0.055954	9.251490654	0.1	0.1				Jan-12	R3	12.51	7.72	1.7	0.033934393
May-12	R4	19.6	7.7	1.6	0.031935	9.248704464	0.1	0.1				Jan-12	R4	12.63	7.74	1.9	0.037934313
May-12	R5	19.26	8.43	2.3	0.045939	9.250328923	0.1	0.1				Feb-12	R2	14.2	7.93	2.3	0.045938989
Jun-12	R1	20.9	7.83	2.3	0.045939	9.250328923	1.4	1.4				Feb-12	R3	13.71	7.79	2.2	0.043937216
Jun-12	R2	22	8.53	2.3	0.045939	9.250328923	0.4	0.4				Feb-12	R4	13.8	7.62	2.1	0.041935845
Jun-12	R3	21.25	8.4	2.2	0.043937	9.250096717	0.4	0.4				Mar-12	R2	14.8	7.8	7	0.14047948
Jun-12	R4	21.21	7.55	1.5	0.029936	9.248472586	0.2	0.2				Mar-12	R3	14.65	7.82	8.3	0.166788003
Jun-12	R5	21.61	8.43	2.1	0.041936	9.249864558	0.4	0.4				Mar-12	R4	16.31	7.7	1.4	0.027937532
Jul-12	R1	21.68	8.25	1.7	0.033934	9.24893639	0.7	0.7				Apr-12	R2	15.5	7.9	21.1	0.4295764
Jul-12	R2	22.76	8.96	1.6	0.031935	9.248704464	0.2	0.2				Apr-12	R3	14.59	7.8	21.5	0.437899898
Jul-12	R3	22.28	8.93	1.7	0.033934	9.24893639	<1	0.05				Apr-12	R4	19.33	7.92	2.4	0.047941167
Jul-12	R4	21.9	7.59	1.3	0.025939	9.248008968	<1	0.05				Nov-12	R1	16.43	8.63	1.4	0.027937532
Jul-12	R5	22.69	8.41	1.6	0.031935	9.248704464	<1	0.05				Nov-12	R1	15.7	8.71	1.3	0.025939383
Aug-12	R1	22.6	7.87	1.8	0.158078	9.263337081	0.7	0.7				Nov-12	R2	18.03	8.93	1.3	0.025939383
Aug-12	R2	24.45	8.45	1.7	0.033934	9.24893639	0.6	0.6				Nov-12	R3	16.5	8.8	1.3	0.025939383
Aug-12	R3	24.88	8.41	1.7	0.033934	9.24893639	0.6	0.6				Nov-12	R4	16.59	7.9	1.3	0.025939383
Aug-12	R4	23.05	7.42	1.2	0.023942	9.24777723	0.2	0.2				Nov-12	R5	16.74	8.86	1.4	0.027937532
Aug-12	R5	23.94	7.89	1.8	0.035934	9.249168362	0.3	0.3				Dec-12	R2	15.175	8.78	1.9	0.037934313
Sep-12	R1											Dec-12	R3	15	8.613	2.1	0.041935845
Sep-12	R2	24.55	9.08	1.3	0.025939	9.248008968	0.1	0.1				Dec-12	R4	14.645	8.1125	1.9	0.037934313
Sep-12	R3	23.07	8.84	1.3	0.025939	9.248008968	0.1	0.1				Dec-12	R5	14.9225	8.44	1.925	0.038434417
Sep-12	R4	22.89	7.35	1.2	0.023942	9.24777723	0.1	0.1				Jan-13	R2				
Sep-12	R5	22.34	8.95	1.4	0.027938	9.248240754	0.1	0.1				Jan-13	R3	10.7825	8.78	1.9	0.037934313
Oct-12	R1	20.06	8.44	1.4	0.027938	9.248240754	0.3	0.3				Jan-13	R4	10.9025	8.66	1.95	0.038934545
Oct-12	R2	22.02	8.85	1.4	0.027938	9.248240754	<1	0.05				Feb-13	R2				
Oct-12	R3	20.52	8.65	1.4	0.027938	9.248240754	<1	0.05				Feb-13	R3	13.6425	8.825	1.65	0.032934665
Oct-12	R4	20.28	7.46	1.3	0.025939	9.248008968	<1	0.05				Feb-13	R4	13.7875	8.57	1.6	0.031935037
Oct-12	R5	20.71	8.88	1.4	0.027938	9.248240754	<1	0.05				Mar-13	R2				
Max		24.88	9.08	3.1			2.1	2.1				Mar-13	R3	16.3275	8.79	1.5	0.029936084
Mini		18.6	7.35	1.2			0.1	0.05				Mar-13	R4	16.0725	8.525	1.45	0.028936758
STDEV		1.727642	0.51772	0.49018				0.457533				Apr-13	R2				
Avg		21.38588	8.198824	1.717647				0.398529				Apr-13	R3	18.1875	8.4675	1.4	0.027937532
CV		0.080784	0.063146	0.285379				1.148052				Apr-13	R4	18.82	8.2225	1.325	0.026438883
90%		23.679	8.915	2.3	0.045939	9.250328923	0.789705		1-hour EPA Objective (total ammonia)								
50%		21.43	8.325	1.55	0.030936	9.248588519		0.418235	4-day EPA Objective (total ammonia)				Max		19.33	8.93	21.5
									Conc.	LTA		Mini		10.7825	7.62	1.3	
						ECA Multiplier 1-HR 99%		0.180439	0.789705005	0.142493		STDEV		2.153377	0.42745	4.62964106	
						ECA Multiplier 4-Day 95%		0.33323	0.418235307	0.139369		Avg		14.81347	8.229319	3.11666667	
						MDEL Multiplier 99%		5.542047		MDEL	0.772388	CV		0.145366	0.051942	1.48544633	
						AMEL Multiplier 95%		2.086503		AMEL	0.290793	90%		17.385	8.795	4.7	0.094102853
												50%		14.91125	8.11125	1.7	0.033934393
																	ECA M
																	ECA Mu
																	MDE
																	AME

[illegible]

Attachment N-1 Priority Non-Metal Reasonable Potential Analysis and Limit Derivation

CTR#		Units	CV	MEC (Ventura WRP effluent)	Saltwater		CTR CRITERIA				SWRCB min	REASONABLE POTENTIAL ANALYSIS (RPA)						Recommendation
					C acute = CMC tot	C chronic = CCC tot	C acute = CMC tot	C chronic = CCC tot	Not applicable C hh W&O	C hh O		Lowest C	Tier 1 MEC ≥ Lowest C	B			Tier 2 B>C & present in Effl.	
														R3 max	R4 max	R5 max		
14	Cyanide	µg/L	0.6	<5	1	1	22	5.2	700	220,000	5	1	NO	<5	<5	<5	NO	Interim Monitoring - No Limit- MDL>C
15	Asbestos	Fibers/ L		not measured	NONE	NONE	NONE	NONE	7,000,000	NONE		7x10 ⁶	NO	no sample	no sample	no sample		Interim Monitoring - No Limit
16	2,3,7,8-TCDD (Dioxin)	µg/L	0.6	<.000005	NONE	NONE	NONE	NONE	1.3E-08	1.4E-08		1.4E-08	NO	<5.2	<5.2	<5	NO	Interim Monitoring - No Limit-MDL for B>C
17	Acrolein	µg/L	0.6	<.6	NONE	NONE	NONE	NONE	320	780	2	780	NO	<.6	<.6	<.6	NO	Interim Monitoring - No Limit
18	Acrylonitrile	µg/L	0.6	<.5	NONE	NONE	NONE	NONE	0.059	0.66	2	0.66	NO	<.5	<.5	<.5	NO	Interim Monitoring - No Limit
19	Benzene	µg/L	0.6	<.2	NONE	NONE	NONE	NONE	1.2	71	0.5	71	NO	<.2	<.2	<.2	NO	Interim Monitoring - No Limit
20	Bromoform	µg/L	0.99	4.3	NONE	NONE	NONE	NONE	4.3	360	0.5	360	NO	1.44	2.96		NO	Interim Monitoring - No Limit
21	Carbon Tetrachloride	µg/L	0.6	<.12	NONE	NONE	NONE	NONE	0.25	4.4	0.5	4.4	NO	<.12	<.12	<.12	NO	Interim Monitoring - No Limit
22	Chlorobenzene	µg/L	0.6	<.2	NONE	NONE	NONE	NONE	680	21,000	1	21,000	NO	<.2	<.2	<.2	NO	Interim Monitoring - No Limit
23	Dibromochloromethane	µg/L	0.6	14.5	NONE	NONE	NONE	NONE	0.401	34	0.5	34	NO	2.84	3.63	0.71	NO	Interim Monitoring - No Limit
24	Chloroethane	µg/L	0.6	<.52	NONE	NONE	NONE	NONE	NONE	NONE	0.5	No criteria	No Criteria Available	<.52	<.52	<.52	No Criteria Available	No Limit - No Criteria Available
25	2-chloroethyl vinyl ether	µg/L	0.6	<.13	NONE	NONE	NONE	NONE	NONE	NONE	1	No criteria	No Criteria Available	<.13	<.13	<.13	No Criteria Available	No Limit - No Criteria Available
26	Chloroform	µg/L	0.6	36	NONE	NONE	NONE	NONE	Reserved	Reserved	0.5	Reserved	No Criteria Available	9.55	10.9	1.26	No Criteria Available	No Limit - No Criteria Available
27	Dichlorobromomethane	µg/L	0.6	25.6	NONE	NONE	NONE	NONE	0.56	46	0.5	46	NO	1.12	4.89	0.28	NO	Interim Monitoring - No Limit
28	1,1-Dichloroethane	µg/L	0.6	<.07	NONE	NONE	NONE	NONE	NONE	NONE	0.5	5	NO	<.07	<.07	<.07	NO	Interim Monitoring - No Limit
29	1,2-dichloroethane	µg/L	0.6	<.03	NONE	NONE	NONE	NONE	0.38	99	0.5	99	NO	<.03	<.03	<.03	NO	Interim Monitoring - No Limit
30	1,1-Dichloroethylene	µg/L	0.6	<.13	NONE	NONE	NONE	NONE	0.057	3.2		3.2	NO	<.13	<.13	<.13	NO	Interim Monitoring - No Limit
31	1,2-dichloropropane	µg/L	0.6	<.04	NONE	NONE	NONE	NONE	0.52	39		39	NO	<.04	<.04	<.04	NO	Interim Monitoring - No Limit
32	1,3-dichloropropylene	µg/L	0.6	<.68	NONE	NONE	NONE	NONE	10	1,700		1,700	NO	<.68	<.68	<.68	NO	Interim Monitoring - No Limit
33	Ethylbenzene	µg/L	0.6	<.34	NONE	NONE	NONE	NONE	3100	29,000		29,000	NO	<.34	<.34	<.34	NO	Interim Monitoring - No Limit
34	Methyl bromide	µg/L	0.6	<.5	NONE	NONE	NONE	NONE	48	4,000		4,000	NO	<.5	<.5	<.5	NO	Interim Monitoring - No Limit
35	Methyl chloride	µg/L	0.6	<.08	NONE	NONE	NONE	NONE	Narrative	Narrative		Narrative	No Criteria Available	<.08	<.08	<.08	No Criteria Available	No Limit - No Criteria Available
36	Methylene chloride	µg/L	0.6	<.25	NONE	NONE	NONE	NONE	4.7	1,600		1,600	NO	<.25	<.25	<.25	NO	Interim Monitoring - No Limit
37	1,1,2,2-tetrachloroethane	µg/L	0.6	<.03	NONE	NONE	NONE	NONE	0.17	11		11	NO	<.03	<.03	<.03	NO	Interim Monitoring - No Limit
38	Tetrachloroethylene	µg/L	0.6	<.03	NONE	NONE	NONE	NONE	0.8	8.85		8.85	NO	<.03	<.03	<.03	NO	Interim Monitoring - No Limit
39	Toluene	µg/L	0.6	<.2	NONE	NONE	NONE	NONE	6800	200,000		200,000	NO	<.2	<.2	<.2	NO	Interim Monitoring - No Limit
40	Trans 1,2-Dichloroethylene	µg/L	0.6	<.1	NONE	NONE	NONE	NONE	700	140,000		140,000	NO	<.1	<.1	<.1	NO	Interim Monitoring - No Limit
41	1,1,1-Trichloroethane	µg/L	0.6	<.03	NONE	NONE	NONE	NONE	Narrative	Narrative		Narrative	NO	<.0365	<.0365	<.0365	No Criteria Available	Interim Monitoring - No Limit
42	1,1,2-trichloroethane	µg/L	0.6	<.02	NONE	NONE	NONE	NONE	0.6	42		42	NO	<.031	<.031	<.031	NO	Interim Monitoring - No Limit
43	Trichloroethylene	µg/L	0.6	<.12	NONE	NONE	NONE	NONE	2.7	81		81	NO	<.0277	<.0277	<.0277	NO	Interim Monitoring - No Limit
44	Vinyl chloride	µg/L	0.6	<.18	NONE	NONE	NONE	NONE	2	525		525	NO	<.0983	<.0983	<.0983	NO	Interim Monitoring - No Limit
45	2-chlorophenol	µg/L	0.6	<.3	NONE	NONE	NONE	NONE	120	400		400	NO	<.3	<.3	<.3	NO	Interim Monitoring - No Limit
46	2,4-dichlorophenol	µg/L	0.6	<.7	NONE	NONE	NONE	NONE	93	790		790	NO	<.7	<.7	<.7	NO	Interim Monitoring - No Limit
47	2,4-dimethylphenol	µg/L	0.6	<.27	NONE	NONE	NONE	NONE	540	2,300		2,300	NO	<.27	<.27	<.27	NO	Interim Monitoring - No Limit
48	4,6-dinitro-o-resol (aka 2-methyl-4,6-Dinitrophenol)	µg/L	0.6	<.10	NONE	NONE	NONE	NONE	13.4	765		765	NO	<.10	<.10	<.10	NO	Interim Monitoring - No Limit
49	2,4-dinitrophenol	µg/L	0.6	<.42	NONE	NONE	NONE	NONE	70	14,000		14,000	NO	<.42	<.42	<.42	NO	Interim Monitoring - No Limit
50	2-nitrophenol	µg/L	0.6	<.3.6	NONE	NONE	NONE	NONE	NONE	NONE		None	No Criteria Available	<.3.6	<.3.6	<.3.6	No Criteria Available	No Criteria Available
51	4-nitrophenol	µg/L	0.6	<.2.4	NONE	NONE	NONE	NONE	NONE	NONE		None	No Criteria Available	<.2.4	<.2.4	<.2.4	No Criteria Available	No Criteria Available
52	3-Methyl-4-Chlorophenol (aka P-chloro-m-resol)	µg/L	0.6	<.3	NONE	NONE	NONE	NONE	NONE	NONE		None	No Criteria Available	<.3	<.3	<.3	No Criteria Available	No Criteria Available
53	Pentachlorophenol	µg/L	0.6	<.3.6	13	7.9	pH dependent	pH dependent	0.28	8.2		7.9	NO	<.6	<.6	<.6	NO	Interim Monitoring - No Limit
54	Phenol	µg/L	0.6	<.1.5	NONE	NONE	NONE	NONE	21,000	4,600,000		4.6x10 ⁶	NO	<.1.5	<.1.5	<.1.5	NO	Interim Monitoring - No Limit
55	2,4,6-trichlorophenol	µg/L	0.6	<.2.7	NONE	NONE	NONE	NONE	2.1	6.5		6.5	NO	<.2.7	<.2.7	<.2.7	NO	Interim Monitoring - No Limit
56	Acenaphthene	µg/L	0.6	<.2	NONE	NONE	NONE	NONE	1200	2,700		2,700	NO	<.2	<.2	<.2	NO	Interim Monitoring - No Limit
57	Acenaphthylene	µg/L	0.6	<.1.9	NONE	NONE	NONE	NONE	NONE	NONE		NONE	No Criteria Available	<.1.9	<.1.9	<.1.9	No Criteria Available	No Criteria Available
58	Anthracene	µg/L	0.6	<.1.9	NONE	NONE	NONE	NONE	9600	110,000		110,000	NO	<.1.9	<.1.9	<.1.9	NO	Interim Monitoring - No Limit
59	Benzidine	µg/L	0.6	<.4.4	NONE	NONE	NONE	NONE	0.00012	0.00054		0.00054	NO	<.4.4	<.4.4	<.4.4	NO	Interim Monitoring - No Limit-MDL >C
60	Benzo(a)Anthracene	µg/L	0.6	<.7.8	NONE	NONE	NONE	NONE	0.0044	0.049		0.049	NO	<.7.8	<.7.8	<.7.8	NO	Interim Monitoring - No Limit- MDL>C
61	Benzo(a)Pyrene	µg/L	0.6	<.2.5	NONE	NONE	NONE	NONE	0.0044	0.049		0.049	NO	<.2.5	<.2.5	<.2.5	NO	Interim Monitoring - No Limit- MDL>C
62	Benzo(b)Fluoranthene	µg/L	0.6	<.4.8	NONE	NONE	NONE	NONE	0.0044	0.049		0.049	NO	<.4.8	<.4.8	<.4.8	NO	Interim Monitoring - No Limit- MDL>C
Analysis 63	Benzo(ghi)Perylene	µg/L	0.6	<.4.1	NONE	NONE	NONE	NONE	1	NONE		NONE	No Criteria Available	<.4.1	<.4.1	<.4.1	No Criteria Available	Interim Monitoring - No Limit

CTR#		Units	CV	MEC (Ventura WRP)	Saltwater		CTR CRITERIA				SWRCB min	REASONABLE POTENTIAL ANALYSIS (RPA)						Recommendation
							Freshwater		Human Health			Tier 1	B			Tier 2 RPA		
					C acute =	C chronic	C acute =	C chronic =	Not									
123	Aroclor 1248	µg/L	0.6	<.065	NONE	0.03	NONE	0.014	0.00017	0.00017		0.00017	NO	<.065	<.065	<.065	NO	Interim Monitoring - No Limit- MDL>C
124	Aroclor 1254	µg/L	0.6	<.065	NONE	0.03	NONE	0.014	0.00017	0.00017		0.00017	NO	<.065	<.065	<.065	NO	Interim Monitoring - No Limit- MDL>C
125	Aroclor 1260	µg/L	0.6	<.065	NONE	0.03	NONE	0.014	0.00017	0.00017		0.00017	NO	<.065	<.065	<.065	NO	Interim Monitoring - No Limit- MDL>C
126	Toxaphene	µg/L	0.6	<10	0.21	0.0002	0.73	0.0002	0.0073	0.00075		0.0002	NO	<10	<10	<10	NO	Interim Monitoring - No Limit- MDL>C
These metals are hardness dependent. CTR criteria was calculated using a hardness capped at 400 mg/L, pursuant to CTR.																		

Attachment N-2 Priority Metal Criterion Calculation

Copper																			
Freshwater										Freshwater									
CMC or Acute										CCC or Chronic									
CMC = WER x Conversion Factor x (exp {mA [ln(Hardness)] + bA})										CCC = WER x Conversion Factor x (exp {mC [ln(Hardness)] + bC})									
HARDNESS	WER	Conversion Factor	mA	bA	Total Recoverable Limit	Dissolved Fraction Limit	WER	Conversion Factor	mC	bC	Total Recoverable Limit	Dissolved Fraction Limit	Conversion Factor	Total Recoverable Limit	Dissolved Fraction Limit	Conversion Factor	Total Recoverable Limit	Dissolved Fraction Limit	Human Health
(mg/L)						(µg/L)						(µg/L)							Organisms only
100	1	0.96	0.9422	-1.7	14.00	13.44	1	0.96	0.8545	-1.702	9.33	8.96	0.83	5.78313253	4.8	0.83	3.734939759	3.1	---
200	1	0.96	0.9422	-1.7	26.90	25.82	1	0.96	0.8545	-1.702	16.87	16.19	0.83	5.78313253	4.8	0.83	3.734939759	3.1	---
300	1	0.96	0.9422	-1.7	39.41	37.84	1	0.96	0.8545	-1.702	23.85	22.90	0.83	5.78313253	4.8	0.83	3.734939759	3.1	---
400	1	0.96	0.9422	-1.7	51.68	49.62	1	0.96	0.8545	-1.702	30.50	29.28	0.83	5.78313253	4.8	0.83	3.734939759	3.1	---
413	1	0.96	0.9422	-1.7	53.27	51.14	1	0.96	0.8545	-1.702	31.34	30.09	0.83	5.78313253	4.8	0.83	3.734939759	3.1	---
494	1	0.96	0.9422	-1.7	63.06	60.53	1	0.96	0.8545	-1.702	36.53	35.07	0.83	5.78313253	4.8	0.83	3.734939759	3.1	---
										Applied	Discharger 2002	Metal translator	0.86	6.724572709	6.72457271		4.342953208	4.34295321	
										Applied	Discharger 2009	WER total copper (geometric mean of 12 tests)	2.08	13.98711124	13.9871112		9.033342673	9.03334267	
Lead																			
CMC or Acute										CCC or Chronic									
CMC = WER x Conversion Factor x (exp {mA [ln(Hardness)] + bA})										CCC = WER x Conversion Factor x (exp {mC [ln(Hardness)] + bC})									
HARDNESS	WER	Conversion Factor*	mA	bA	Total Recoverable Limit	Dissolved Fraction Limit	WER	Conversion Factor	mC	bC	Total Recoverable Limit	Dissolved Fraction Limit	Conversion Factor	Total Recoverable Limit	Dissolved Fraction Limit	Conversion Factor	Total Recoverable Limit	Dissolved Fraction Limit	Organisms only
(mg/L)						(µg/L)						(µg/L)							
100	1	0.791	1.273	-1.46	81.65	64.58	1	0.791001442	1.273	-4.705	3.18	2.52	0.951	220.8201893	210	0.951	8.517350158	8.1	narrative
200	1	0.69000158	1.273	-1.46	197.31	136.14	1	0.69000158	1.273	-4.705	7.69	5.31	0.951	220.8201893	210	0.951	8.517350158	8.1	narrative
300	1	0.630920448	1.273	-1.46	330.60	208.58	1	0.630920448	1.273	-4.705	12.88	8.13	0.951	220.8201893	210	0.951	8.517350158	8.1	narrative
400	1	0.589001718	1.273	-1.46	476.82	280.85	1	0.589001718	1.273	-4.705	18.58	10.94	0.951	220.8201893	210	0.951	8.517350158	8.1	narrative
										Not applied	Discharger 2002	No Translator							
										Not applied	Discharger 2007	No WER							
Nickel																			
Freshwater										Freshwater									
CMC or Acute										CCC or Chronic									
CMC = WER x Conversion Factor x (exp {mA [ln(Hardness)] + bA})										CCC = WER x Conversion Factor x (exp {mC [ln(Hardness)] + bC})									
HARDNESS	WER	Conversion Factor	mA	bA	Total Recoverable Limit	Dissolved Fraction Limit	WER	Conversion Factor	mC	bC	Total Recoverable Limit	Dissolved Fraction Limit	Conversion Factor	Total Recoverable Limit	Dissolved Fraction Limit	Conversion Factor	Total Recoverable Limit	Dissolved Fraction Limit	Organisms only
(mg/L)						(µg/L)						(µg/L)							
100	1	0.998	0.846	2.255	469.17	468.24	1	0.997	0.846	0.0584	52.16	52.01	0.99	74.74747475	74	0.99	8.282828283	8.2	4600
200	1	0.998	0.846	2.255	843.35	841.66	1	0.997	0.846	0.0584	93.76	93.48	0.99	74.74747475	74	0.99	8.282828283	8.2	4600
300	1	0.998	0.846	2.255	1188.44	1186.07	1	0.997	0.846	0.0584	132.13	131.74	0.99	74.74747475	74	0.99	8.282828283	8.2	4600
400	1	0.998	0.846	2.255	1515.92	1512.89	1	0.997	0.846	0.0584	168.54	168.04	0.99	74.74747475	74	0.99	8.282828283	8.2	4600
										Applied	Discharger 2002	metal translator	0.81	91.35802469	91.3580247		10.22571393	10.2257139	
										Not applied	Discharger 2007	No WER							
Selenium																			
Freshwater										Freshwater									
CMC or Acute										CCC or Chronic									
Bioaccumulative										Bioaccumulative									
HARDNESS	WER	Conversion Factor	mA	bA	Total Recoverable Limit	Dissolved Fraction Limit	WER	Conversion Factor	mC	bC	Total Recoverable Limit	Dissolved Fraction Limit	Conversion Factor	Total Recoverable Limit	Dissolved Fraction Limit	Conversion Factor	Total Recoverable Limit	Dissolved Fraction Limit	Organisms only
(mg/L)						reserved													
1	1						1				5		0.998	290.58	290	0.998	71.14	71	none

Attachment N-3 Priority Reasonable Potential Analysis and Limit Derivation

CTR#		Units	CV	MEC (Ventura WRP effluent)	Saltwater		CTR CRITERIA				SWRC B min	REASONABLE POTENTIAL ANALYSIS (RPA)							AQUATIC LIFE CALCULATIONS (Freshwater or Saltwater)					AQUATIC LIFE CALCULATIONS (Freshwater or Saltwater)					PROPOSED LIMIT		Recommendation
					C acute = CMC tot	C chronic = CCC	C acute = CMC tot	C chronic = CCC	Not applicable C h h W&O	C h h O		Lowest C	Tier 1 MEC ≥ Lowest	B			Tier 2 B>C & present in	Need limit?	ECA acute multi	LTA acute	ECA chronic	LTA chronic	Lowest LTA	AMEL multiplier (n=4)	AMEL aqlife	MDEL multiplier	MDEL aqlife	Lowest AMEL	Lowest MDEL		
														R3 max	R4 max	R5 max															
1	Antimony	µg/L		<1	NONE	NONE	NONE	NONE	14	4300	0.5	4300	NO	<1		<1	NO	NO												Interim Monitoring - No CTR-based Limit	
2	Arsenic	µg/L		<2	69	36	340	150	Narrative	Narrative	1	36 no criteria	NO	7.7	7.35	6.26	NO	NO												Interim Monitoring - No CTR-based Limit	
3	Beryllium	µg/L		<2	NONE	NONE	NONE	NONE	Narrative	Narrative	0.5		NO	<2	<2	<2	NO	NO												Interim Monitoring - No CTR-based Limit	
4	Cadmium	µg/L		<4	42.25	9.36	21.58	7.31	Narrative	Narrative	0.25	7.31	NO	<4	<4	<4	NO	NO												Interim Monitoring - No CTR-based Limit	
5a	Chromium III*	µg/L		<7	Narrative	Narrative	5404.62	644.2	Narrative	Narrative		644.2	NO	9.05	<7	7.43	NO	NO												Interim Monitoring - No CTR-based Limit	
5b	Chromium VI	µg/L		<1	1107.75	50.35	16.3	11.4	Narrative	Narrative	5	11.4	NO	not sampled	not sampled	not sampled	NO	NO												Interim Monitoring - No CTR-based Limit	
6	Copper*	µg/L	0.79	8.88	13.99	9.033	51.68	30.5	1300	NONE	0.5	9.033	YES	3.01	4.97	3.54	NO	YES	0.252	3.529	0.443	4.006	3.529	1.740	6.140	3.964	13.990	6.1	14.0	New limit: Tier 1, MEC > C, with 2008 WER of 2.08 and dissolved conversion factor of .83. Correcting old MDEL = 8.8, AMEL = 4.2 with WER of 2007 1.58 and conversion factor.	
7	Lead	µg/L	0.6	9.44	220.8	8.5	476.82	18.58	Narrative	Narrative	0.5	8.5	YES	7.63	<5	5.54	NO	YES	0.321	70.895	0.527	4.483	4.483	1.552	6.960	3.114	13.963	7.0	14.0	Reinstate limit: Tier 1, MEC > C, MDEL = 14, AMEL = 7.	
8	Mercury	µg/L	0.6	<.02	-	-	-	-	0.05	0.051	0.2	0.05	NO	<2	<2	<2	NO	NO												Interim Monitoring - No CTR-based Limit	
9	Nickel*	µg/L	1.09	35.4	91.36	10.23	1515.92	168.54	610	4600	1	10.23	YES	50.2	30.2	42	YES	YES	0.189	17.254	0.348	3.558	3.558	2.032	7.229	5.295	18.839	7.2	18.8	New limit: Tier 1, MEC > C, Update from Order 00-143 MDEL = 15.2, AMEL = 5.3 without translator.	
10	Selenium	µg/L	0.6	13.7	290.58	71.14	-	5	Narrative	Narrative	5	5	YES	7.65	8.07	20.1	YES	YES			0.527	2.637	2.637	1.552	4.094	3.114	8.213	4.1	8.2	New limit for MDEL: Tier 1, MEC > C. Update from Order 00-143 MDEL = 8.8, AMEL = 2.9.	
11	Silver*	µg/L	0.6	<2	2.24	NONE	44.05	NONE	NONE	NONE	0.25	2.24	NO	<2	<2	<2	NO	NO												Interim Monitoring - No CTR Limit	
12	Thallium	µg/L	0.6	<1	NONE	NONE	NONE	NONE	1.7	6.3	1	6.3	NO	<1	<1	<1	NO	NO												Interim Monitoring - No CTR Limit	
13	Zinc	µg/L	0.33	68.8	107.1	96.4	387.83	388	NONE	NONE	1	96.4	NO	32.4	31.5	20.1	NO	NO												Interim Monitoring - No CTR Limit	

ATTACHMENT P – PRETREATMENT REPORTING REQUIREMENTS

The Discharger is required to submit annual Pretreatment Program Compliance Report (Report) to the Regional Water Board and USEPA Region 9. This Attachment outlines the minimum reporting requirements of the Report. If there is any conflict between requirements stated in this attachment and provisions stated in the Waste Discharge Requirements (WDR), those contained in the WDR will prevail.

A. Pretreatment Requirements

1. The City shall be responsible and liable for the performance of all Control Authority pretreatment requirements contained in 40 CFR Part 403, including any subsequent regulatory revisions to Part 403. Where Part 403 or subsequent revision places mandatory actions upon the City as Control Authority but does not specify a timetable for completion of the actions, the City shall complete the required actions within six months from the issuance date of this permit or the effective date of the Part 403 revisions, whichever comes later. For violations of pretreatment requirements, the City shall be subject to enforcement actions, penalties, fines and other remedies by the U.S. Environmental Protection Agency (EPA) or other appropriate parties, as provided in the Act. EPA may initiate enforcement action against a nondomestic user for noncompliance with applicable standards and requirements as provided in the act.
2. The City shall enforce the requirements promulgated under sections 307(b), 307(c), 307(d) and 402(b) of the Act with timely, appropriate and effective enforcement actions. The City shall cause all nondomestic users subject to federal categorical standards to achieve compliance no later than the date specified in those requirements or, in the case of a new nondomestic user, upon commencement of the discharge.
3. The City shall perform the pretreatment functions as required in 40 CFR Part 403 including, but not limited to:
 - a. Implement the necessary legal authorities as provided in 40 CFR Part 403.8(f)(1);
 - b. Enforce the pretreatment requirements under 40 CFR Part 403.5 and 403.6;
 - c. Implement the programmatic functions as provided in 40 CFR Part 403.8(f)(2); and
 - d. Provide the requisite funding and personnel to implement the pretreatment program as provided in 40 CFR Part 403.8(f)(3).
4. The City shall submit annually a report to EPA Pacific Southwest Region, and the State describing its pretreatment activities over the previous year. In the event the City is not in compliance with any conditions or requirements of this permit, then the City shall also include the reasons for noncompliance and state how and when the City shall comply with such conditions and requirements. This annual report shall cover operations from January 1 through December 31 and is due on March 1 of each year. The report shall contain, but not be limited to, the following information:
 - a. A summary of analytical results from representative, flow proportioned, 24-hour composite sampling of the POTW's influent and effluent for those pollutants EPA has identified under section 307(a) of the Act which are known or suspected to be discharged by nondomestic users. This will consist of an annual full priority pollutant scan, with quarterly samples analyzed only for those pollutants detected in the full scan. The City is not required to sample and analyze for asbestos. Sludge sampling and analysis are covered in the sludge

section of this permit. The City shall also provide any influent or effluent monitoring data for nonpriority pollutants which the City believes may be causing or contributing to interference or pass through. Sampling and analysis shall be performed with the techniques prescribed in 40 CFR Part 136;

- b. A discussion of Upset, Interference or Pass Through incidents, if any, at the treatment plant which the City knows or suspects were caused by nondomestic users of the POTW system. The discussion shall include the reasons why the incidents occurred, the corrective actions taken and, if known, the name and address of the nondomestic user(s) responsible. The discussion shall also include a review of the applicable pollutant limitations to determine whether any additional limitations, or changes to existing requirements, may be necessary to prevent pass through or interference;
- c. An updated list of the City's significant industrial users (SIUs) including their names and addresses, and a list of deletions, additions and SIU name changes keyed to the previously submitted list. The City shall provide a brief explanation for each change. The list shall identify the SIUs subject to federal categorical standards by specifying which set(s) of standards are applicable to each SIU. The list shall also indicate which SIUs are subject to local limitations;
- d. The City shall characterize the compliance status of each SIU by providing a list or table which includes the following information:
 - i. Name of the SIU;
 - ii. Category, if subject to federal categorical standards;
 - iii. The type of wastewater treatment or control processes in place;
 - iv. The number of samples taken by the POTW during the year;
 - v. The number of samples taken by the SIU during the year;
 - vi. For an SIU subject to discharge requirements for total toxic organics, whether all required certifications were provided;
 - vii. A list of the standards violated during the year. Identify whether the violations were for categorical standards or local limits;
 - viii. Whether the facility is in significant noncompliance (SNC) as defined at 40 CFR part 403.8(f)(2)(viii) at any time during the year; and
 - ix. A summary of enforcement or other actions taken during the year to return the SIU to compliance. Describe the type of action, final compliance date, and the amount of fines and penalties collected, if any. Describe any proposed actions for bringing the SIU into compliance.
- e. A brief description of any programs the POTW implements to reduce pollutants from nondomestic users that are not classified as SIUs;
- f. A brief description of any significant changes in operating the pretreatment program which differ from the previous year including, but not limited to, changes concerning the program's administrative structure, local limits, monitoring program or monitoring frequencies, legal authority, enforcement policy, funding levels, or staffing levels;
- g. A summary of the annual pretreatment budget, including the cost of pretreatment program functions and equipment purchases; and
- h. A summary of activities to involve and inform the public of the program including a copy of the newspaper notice, if any, required under 40 CFR 403.8(f)(2)(viii).

B. LOCAL LIMITS EVALUATION

1. In accordance with 40 CFR 122.44(j)(2)(ii), the POTW shall provide a written technical evaluation of the need to revise local limits under 40 CFR Part 4035.(c)(1) within 180 days of issuance or reissuance of the NPDES permit.

C. SIGNATORY REQUIREMENTS AND REPORT SUBMITTAL

1. Signatory Requirements.

The annual report must be signed by a principal executive officer, ranking elected official or other duly authorized employee if such employee is responsible for the overall operation of the POTW. Any person signing these reports must make the following certification [40 CFR 403.6(a)(2)(ii)]:

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

2. Report Submittal.

An original copy of the Annual Report must be sent to the Pretreatment Program Coordinator of the Regional Water Board and the duplicate copies of the Report must be sent to USEPA through the following addresses:

Information and Technology Unit
Attn: Pretreatment Program Coordinator
California Regional Water Quality Control Board, Los Angeles Region
320 West 4th Street, Suite 200
Los Angeles, CA 90013

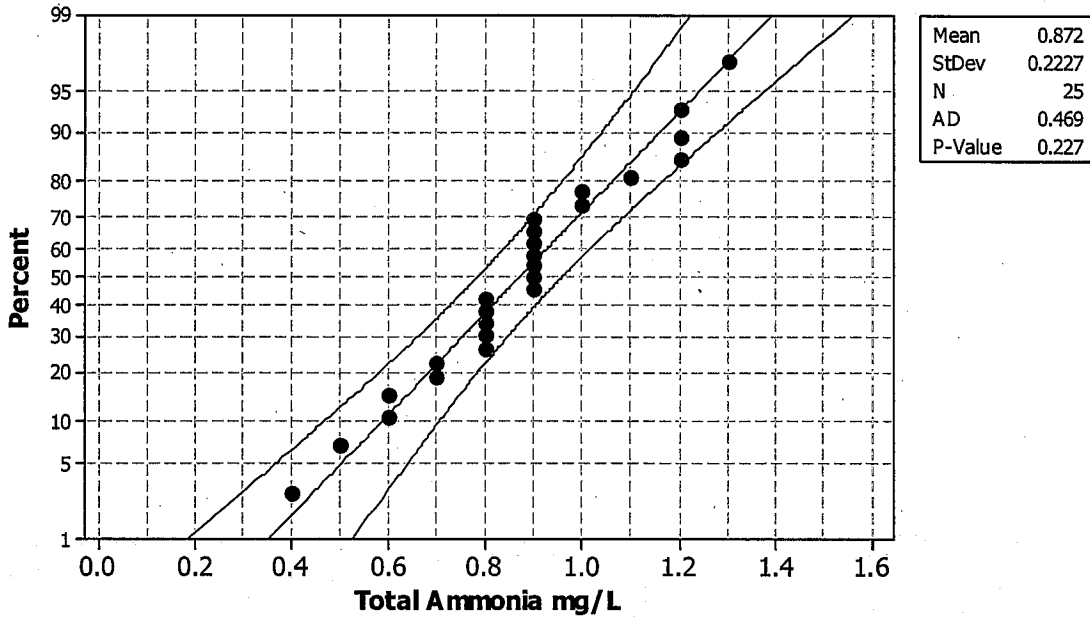
Pretreatment Program
CWA Compliance Office (WTR-7)
Water Division
U.S. Environmental Protection Agency, Region IX
75 Hawthorne Street
San Francisco, CA 94105-3901

ATTACHMENT Q Ammonia Effluent Limit Statistics

Minitab Project Report

Probability Plot of Winter Effluent Total Ammonia after NDN Implementation

Normal - 95% CI

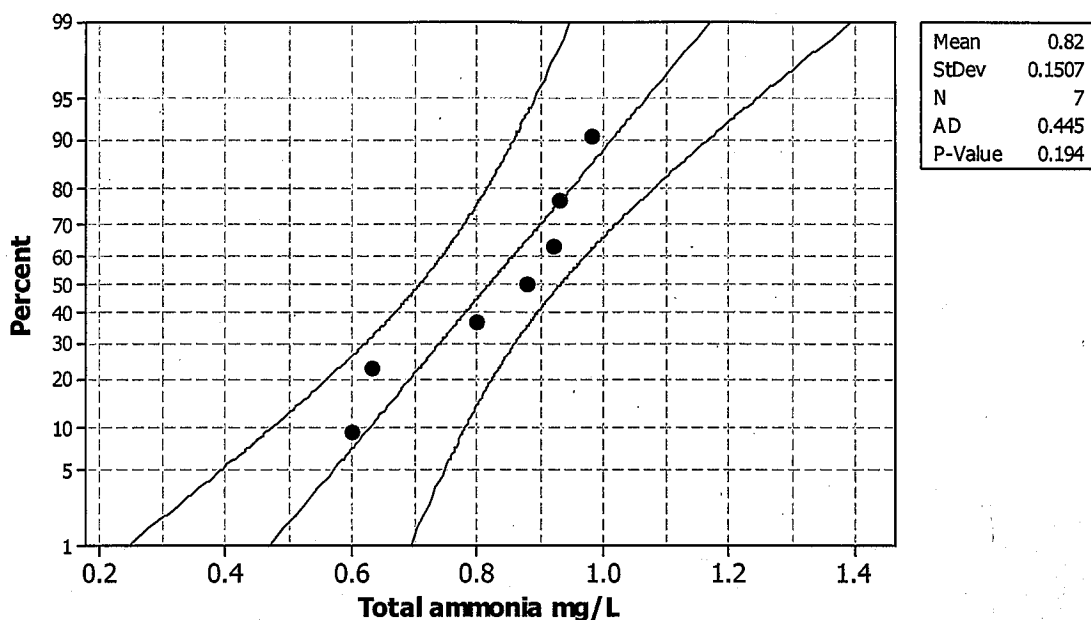


Percent	C2	Lower Bound	Upper Bound
1	0.353898	0.183969	0.523826
2	0.414608	0.259088	0.570128
3	0.453127	0.306451	0.599804
4	0.482104	0.341896	0.622312
5	0.505674	0.370592	0.640755
6	0.525735	0.394911	0.656560
7	0.543326	0.416145	0.670507
8	0.559076	0.435080	0.683071
9	0.573400	0.452233	0.694566
10	0.586585	0.467961	0.705209
20	0.684562	0.582566	0.786559
30	0.755210	0.661929	0.848492
40	0.815577	0.726844	0.904310
50	0.872	0.784699	0.959301
60	0.928423	0.839690	1.01716
70	0.988790	0.895508	1.08207
80	1.05944	0.957441	1.16143
90	1.15742	1.03879	1.27604
91	1.17060	1.04943	1.29177
92	1.18492	1.06093	1.30892
93	1.20067	1.07349	1.32786
94	1.21826	1.08744	1.34909
95	1.23833	1.10324	1.37341
96	1.26190	1.12169	1.40210
97	1.29087	1.14420	1.43755
98	1.32939	1.17387	1.48491
99	1.39010	1.22017	1.56003

Minitab Project Report

Probability Plot of Summer Total Effluent Ammonia after NDN

Normal - 95% CI



Percent C2 Lower Bound Upper Bound

1	0.469500	0.245831	0.693169
2	0.510572	0.306271	0.714872
3	0.536630	0.344238	0.729022
4	0.556233	0.372563	0.739902
5	0.572178	0.395430	0.748926
6	0.585750	0.414756	0.756743
7	0.597650	0.431586	0.763713
8	0.608305	0.446555	0.770054
9	0.617995	0.460081	0.775909
10	0.626915	0.472451	0.781378
20	0.693197	0.561384	0.825010
30	0.740991	0.621131	0.860851
40	0.781829	0.668239	0.895420
50	0.82	0.708388	0.931612
60	0.858171	0.744580	0.971761
70	0.899009	0.779149	1.01887
80	0.946803	0.814990	1.07862
90	1.01309	0.858622	1.16755
91	1.02201	0.864091	1.17992
92	1.03170	0.869946	1.19344
93	1.04235	0.876287	1.20841
94	1.05425	0.883257	1.22524
95	1.06782	0.891074	1.24457
96	1.08377	0.900098	1.26744
97	1.10337	0.910978	1.29576
98	1.12943	0.925128	1.33373
99	1.17050	0.946831	1.39417